

CIS Microsoft Azure Foundations Benchmark

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Overview

All CIS Benchmarks™ (Benchmarks) focus on technical configuration settings used to maintain and/or increase the security of the addressed technology, and they should be used in **conjunction** with other essential cyber hygiene tasks like:

- Monitoring the base operating system and applications for vulnerabilities and quickly updating with the latest security patches.
- End-point protection (Antivirus software, Endpoint Detection and Response (EDR), etc.).
- Logging and monitoring user and system activity.

In the end, the Benchmarks are designed to be a key **component** of a comprehensive cybersecurity program.

Important Usage Information

All Benchmarks are available free for non-commercial use from the [CIS Website](#). They can be used to manually assess and remediate systems and applications. In lieu of manual assessment and remediation, there are several tools available to assist with assessment:

- [CIS Configuration Assessment Tool \(CIS-CAT® Pro Assessor\)](#)
- [CIS Benchmarks™ Certified 3rd Party Tooling](#)

These tools make the hardening process much more scalable for large numbers of systems and applications.

NOTE: Some tooling focuses only on the Benchmark Recommendations that can be fully automated (skipping ones marked **Manual**). It is important that **ALL** Recommendations (**Automated** and **Manual**) be addressed since all are important for properly securing systems and are typically in scope for audits.

Key Stakeholders

Cybersecurity is a collaborative effort, and cross functional cooperation is imperative within an organization to discuss, test, and deploy Benchmarks in an effective and efficient way. The Benchmarks are developed to be best practice configuration guidelines applicable to a wide range of use cases. In some organizations, exceptions to specific Recommendations will be needed, and this team should work to prioritize the problematic Recommendations based on several factors like risk, time, cost, and labor. These exceptions should be properly categorized and documented for auditing purposes.

Apply the Correct Version of a Benchmark

Benchmarks are developed and tested for a specific set of products and versions and applying an incorrect Benchmark to a system can cause the resulting pass/fail score to be incorrect. This is due to the assessment of settings that do not apply to the target systems. To assure the correct Benchmark is being assessed:

- **Deploy the Benchmark applicable to the way settings are managed in the environment:** An example of this is the Microsoft Windows family of Benchmarks, which have separate Benchmarks for Group Policy, Intune, and Stand-alone systems based upon how system management is deployed. Applying the wrong Benchmark in this case will give invalid results.
- **Use the most recent version of a Benchmark:** This is true for all Benchmarks, but especially true for cloud technologies. Cloud technologies change frequently and using an older version of a Benchmark may have invalid methods for auditing and remediation.

Exceptions

The guidance items in the Benchmarks are called recommendations and not requirements, and exceptions to some of them are expected and acceptable. The Benchmarks strive to be a secure baseline, or starting point, for a specific technology, with known issues identified during Benchmark development are documented in the Impact section of each Recommendation. In addition, organizational, system specific requirements, or local site policy may require changes as well, or an exception to a Recommendation or group of Recommendations (e.g. A Benchmark could Recommend that a Web server not be installed on the system, but if a system's primary purpose is to function as a Webserver, there should be a documented exception to this Recommendation for that specific server).

In the end, exceptions to some Benchmark Recommendations are common and acceptable, and should be handled as follows:

- The reasons for the exception should be reviewed cross-functionally and be well documented for audit purposes.
- A plan should be developed for mitigating, or eliminating, the exception in the future, if applicable.
- If the organization decides to accept the risk of this exception (not work toward mitigation or elimination), this should be documented for audit purposes.

It is the responsibility of the organization to determine their overall security policy, and which settings are applicable to their unique needs based on the overall risk profile for the organization.

Remediation

CIS has developed [Build Kits](#) for many technologies to assist in the automation of hardening systems. Build Kits are designed to correspond to Benchmark's "Remediation" section, which provides the manual remediation steps necessary to make that Recommendation compliant to the Benchmark.

When remediating systems (changing configuration settings on deployed systems as per the Benchmark's Recommendations), please approach this with caution and test thoroughly.

The following is a reasonable remediation approach to follow:

- CIS Build Kits, or internally developed remediation methods should never be applied to production systems without proper testing.
- Proper testing consists of the following:
 - Understand the configuration (including installed applications) of the targeted systems. Various parts of the organization may need different configurations (e.g., software developers vs standard office workers).
 - Read the Impact section of the given Recommendation to help determine if there might be an issue with the targeted systems.
 - Test the configuration changes with representative lab system(s). If issues arise during testing, they can be resolved prior to deploying to any production systems.
 - When testing is complete, initially deploy to a small sub-set of production systems and monitor closely for issues. If there are issues, they can be resolved prior to deploying more broadly.
 - When the initial deployment above is completed successfully, iteratively deploy to additional systems and monitor closely for issues. Repeat this process until the full deployment is complete.

Summary

Using the Benchmarks Certified tools, working as a team with key stakeholders, being selective with exceptions, and being careful with remediation deployment, it is possible to harden large numbers of deployed systems in a cost effective, efficient, and safe manner.

NOTE: As previously stated, the PDF versions of the CIS Benchmarks™ are available for free, non-commercial use on the [CIS Website](#). All other formats of the CIS Benchmarks™ (MS Word, Excel, and [Build Kits](#)) are available for CIS [SecureSuite](#)® members.

CIS-CAT® Pro is also available to CIS [SecureSuite](#)® members.

Target Technology Details

This document, CIS Microsoft Azure Foundations Benchmark, provides prescriptive guidance for establishing a secure baseline configuration for Microsoft Azure. This Benchmark is scoped to establish foundational security for tenancy in the Microsoft Azure cloud services platform. For Cloud Service Providers, the "Foundations" Benchmark is meant to be used as a first step which is complimented by "Service Category" Benchmarks as a second step. This relationship is further explained in the "Introduction" section. Section overviews are used extensively in this document to provide specific and very important context - review section overviews diligently.

The sections of this document are titled to reflect the product category names found in the Microsoft Azure Product Directory. These categorical sections are then divided into subsections that will be titled to reflect the specific services being addressed. Recommendations will be found in these subsections if "Foundational" recommendations are available for the service, OR if a Service Category Benchmark has not yet been created to address the Service Category.

To obtain the latest version of this guide, please visit <https://benchmarks.cisecurity.org>. If you have questions, comments, or have identified ways to improve this guide, please write us at BenchmarkInfo@cisecurity.org.

Intended Audience

This document is intended for system and application administrators, security specialists, auditors, help desk, and platform deployment personnel who plan to develop, deploy, assess, or secure solutions that incorporate Microsoft Azure.

Consensus Guidance

This CIS Benchmark™ was created using a consensus review process comprised of a global community of subject matter experts. The process combines real world experience with data-based information to create technology specific guidance to assist users to secure their environments. Consensus participants provide perspective from a diverse set of backgrounds including consulting, software development, audit and compliance, security research, operations, government, and legal.

Each CIS Benchmark undergoes two phases of consensus review. The first phase occurs during initial Benchmark development. During this phase, subject matter experts convene to discuss, create, and test working drafts of the Benchmark. This discussion occurs until consensus has been reached on Benchmark recommendations. The second phase begins after the Benchmark has been published. During this phase, all feedback provided by the Internet community is reviewed by the consensus team for incorporation in the Benchmark. If you are interested in participating in the consensus process, please visit <https://workbench.cisecurity.org/>.

Typographical Conventions

The following typographical conventions are used throughout this guide:

Convention	Meaning
<code>Stylized Monospace font</code>	Used for blocks of code, command, and script examples. Text should be interpreted exactly as presented.
<code>Monospace font</code>	Used for inline code, commands, UI/Menu selections or examples. Text should be interpreted exactly as presented.
<code><Monospace font in brackets></code>	Text set in angle brackets denote a variable requiring substitution for a real value.
<i>Italic font</i>	Used to reference other relevant settings, CIS Benchmarks and/or Benchmark Communities. Also, used to denote the title of a book, article, or other publication.
Bold font	Additional information or caveats things like Notes , Warnings , or Cautions (usually just the word itself and the rest of the text normal).

Recommendation Definitions

The following defines the various components included in a CIS recommendation as applicable. If any of the components are not applicable it will be noted, or the component will not be included in the recommendation.

Title

Concise description for the recommendation's intended configuration.

Assessment Status

An assessment status is included for every recommendation. The assessment status indicates whether the given recommendation can be automated or requires manual steps to implement. Both statuses are equally important and are determined and supported as defined below:

Automated

Represents recommendations for which assessment of a technical control can be fully automated and validated to a pass/fail state. Recommendations will include the necessary information to implement automation.

Manual

Represents recommendations for which assessment of a technical control cannot be fully automated and requires all or some manual steps to validate that the configured state is set as expected. The expected state can vary depending on the environment.

Profile

A collection of recommendations for securing a technology or a supporting platform. Most benchmarks include at least a Level 1 and Level 2 Profile. Level 2 extends Level 1 recommendations and is not a standalone profile. The Profile Definitions section in the benchmark provides the definitions as they pertain to the recommendations included for the technology.

Description

Detailed information pertaining to the setting with which the recommendation is concerned. In some cases, the description will include the recommended value.

Rationale Statement

Detailed reasoning for the recommendation to provide the user a clear and concise understanding on the importance of the recommendation.

Impact Statement

Any security, functionality, or operational consequences that can result from following the recommendation.

Audit Procedure

Systematic instructions for determining if the target system complies with the recommendation.

Remediation Procedure

Systematic instructions for applying recommendations to the target system to bring it into compliance according to the recommendation.

Default Value

Default value for the given setting in this recommendation, if known. If not known, either not configured or not defined will be applied.

References

Additional documentation relative to the recommendation.

CIS Critical Security Controls® (CIS Controls®)

The mapping between a recommendation and the CIS Controls is organized by CIS Controls version, Safeguard, and Implementation Group (IG). The Benchmark in its entirety addresses the CIS Controls safeguards of (v7) "5.1 - Establish Secure Configurations" and (v8) "4.1 - Establish and Maintain a Secure Configuration Process" so individual recommendations will not be mapped to these safeguards.

Additional Information

Supplementary information that does not correspond to any other field but may be useful to the user.

Profile Definitions

The following configuration profiles are defined by this Benchmark:

- **Level 1**

Items in this profile intend to:

- be practical and prudent;
- provide security focused best practice hardening of a technology; and
- limit impact to the utility of the technology beyond acceptable means.

- **Level 2**

This profile extends the "Level 1" profile. Items in this profile exhibit one or more of the following characteristics:

- are intended for environments or use cases where security is more critical than manageability and usability
- acts as defense in depth measure
- may impact the utility or performance of the technology
- may include additional licensing, cost, or addition of third party software

Acknowledgements

This Benchmark exemplifies the great things a community of users, vendors, and subject matter experts can accomplish through consensus collaboration. The CIS community thanks the entire consensus team with special recognition to the following individuals who contributed greatly to the creation of this guide:

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Recommendations

1 Introduction

This introduction section and the subsections herein provide informative articles which instruct on the use of the CIS Foundations and Service Category Benchmarks. No recommendations will be found in this section, just articles of relevant information.

Please carefully review the articles in this introductory section and orient yourself with our structured approach to Benchmarking for Cloud Service Providers (CSPs). This approach differs from other CIS Benchmarks because:

- there are too many different products/services in CSP product directories to practically cover in any one Benchmark,
- architectural and design decisions will affect the scope and relevance of recommendations, and
- there are a variety of methods for interfacing with CSP products and services.

Cloud Benchmarks - A Two-Step Approach to Securing Your Cloud Environments:

- **Step 1:** Start with Foundations Benchmarks. Apply as many recommendations as **practical** for your environment; "100%" 'compliance' is not always possible. Not all Foundations Benchmark recommendations can be applied at the same time, and not all recommendations will be relevant to your environment. Use the recommendation Profile Levels and your understanding of your unique environment architecture to determine which recommendations are in scope.
- **Step 2:** Use the Service Category Benchmarks for service-specific defense-in-depth recommendations. Apply recommendations only for the services **IN USE** in your environment. Use the recommendation Profile Levels, and your understanding of your unique environment architecture to determine which recommendations are in scope.

1.1 CIS Microsoft Azure Foundations Benchmarks

The suggested approach for securing your Microsoft Azure cloud environment is to start with the **latest version** of the CIS Microsoft Azure Foundations Benchmark. Because CSP environments are constantly changing, previous versions of the Foundations Benchmarks should not be used. Previous releases may contain incorrect product names, outdated procedures, deprecated features, and other inaccuracies. The CIS Foundations Benchmark provides prescriptive guidance for configuring a subset of Microsoft Azure Services with an emphasis on foundational, testable, and architecture agnostic settings for services.

The Microsoft Azure Foundations Benchmark is what you should start with when beginning to secure your Azure environment. It is also the foundation for which all other Azure Service Category Benchmarks are built on so that as you grow your cloud presence and usage of the services offered you have the necessary guidance to securely configure your environment as it fits with your company's policy.

All CIS Benchmarks are created and maintained through consensus-based collaboration. Should you have feedback, suggested changes, or just like to get involved in the continued maintenance and development of CIS Microsoft Azure Benchmarks, please register on CIS WorkBench at <https://workbench.cisecurity.org> and join the CIS Microsoft Azure Benchmarks Community.

1.2 CIS Microsoft Azure Service Category Benchmarks

After configuring your environment with the CIS Microsoft Azure Foundations Benchmark, we suggest pursuing defense-in-depth and service-specific recommendations for your Azure Services by reviewing the Service Category Benchmarks. The Service Category Benchmarks are being produced with the vision that recommendations for all security-relevant products/services offered by a CSP should have a 'home,' but the Foundations Benchmarks should retain the most crucial recommendations and not be made vast, intimidating, and impractical.

The Service Category Benchmark recommendations should be applied **ONLY** for the CSP products and services that are actively **IN USE** in your environment. In each Service Category Benchmark, you may find that your environment uses none, or only a couple services from a list of many. Please review the services employed in your environment carefully to accurately scope the recommendations you apply. Failure to apply only the recommendations you need may introduce vulnerabilities, technical debt, and unnecessary expenses.

Using the Microsoft Azure Product Directory (<https://azure.microsoft.com/en-us/products/>) as a source of categorical grouping of these services, our vision is to produce a full set of CIS Microsoft Azure Service Category Benchmarks to cover all security-relevant services. A list of planned and published Service Category Benchmarks for the Azure Community can be found on the community dashboard here: <https://workbench.cisecurity.org/communities/72>.

Your help is needed to bring this vision to life! Please consider joining our CIS Microsoft Azure Community to contribute your expertise and knowledge in securing products and services from the Microsoft Azure product family.

All CIS Benchmarks are created and maintained through consensus-based collaboration. Should you have feedback, suggested changes, or just like to get involved in the continued maintenance and development of CIS Microsoft Azure Benchmarks, please register on CIS WorkBench at <https://workbench.cisecurity.org> and join the CIS Microsoft Azure Benchmarks community.

1.3 Multiple Methods of Audit and Remediation

Throughout the Benchmark, Audit and Remediation procedures are prescribed using up to five different methods. These multiple methods are presented for the convenience of readers who will be coming from different technical and experiential backgrounds. To perform any given Audit or Remediation, only one method needs to be performed. Not every method is available for every recommendation, and many that are available are not yet written for every recommendation. The methods presented in the Benchmark are formatted and titled as follows:

- **"From Azure Portal"** - This is the administrative GUI accessed at <https://portal.azure.com>.
- **"From Azure CLI"** - See additional detail in the next section.
- **"From PowerShell"** - See additional detail in the next section.
- **"From REST API"** - An Application Programming Interface (API) for HTTP operations on service endpoints.
- **"From Azure Policy"** - Azure Policy provides an object-based method of evaluating configuration states and other governance detail. Information for the purpose of automating Azure Policy evaluation can be found in the "Automating with Azure Policy" section below.

Setting Up PowerShell and Azure CLI

In order to use the Azure Command Line Interface (CLI) and the Azure PowerShell methods for audit and remediation procedures, the following permissions are required for the account running the procedures:

1. Global Reader
2. Security Reader
3. Subscription Contributor
4. Key Vault Get/List privileges on Keys, Secrets, Certificates, and Certificate Authorities
5. Network allow listing for any source IP address performing the audit activities
6. Permissions to use PowerShell and Azure CLI

These permissions can be directly assigned or assigned via Privileged Identity Management.

The Azure CLI tool can be installed from the following location:

<https://docs.microsoft.com/en-us/cli/azure/install-azure-cli-windows?tabs=azure-cli>

For PowerShell, the following cmdlets are required:

1. Azure PowerShell: <https://learn.microsoft.com/en-us/powershell/azure/install-azure-powershell?view=azps-15.0.0>
2. Microsoft Graph PowerShell: <https://learn.microsoft.com/en-us/powershell/microsoftgraph/installation?view=graph-powershell-1.0>

Authenticating with Azure CLI

Run the following command from either PowerShell or command prompt:

```
az login --tenant <tenant id> --subscription <subscription ID>
```

Authenticating with PowerShell

Login to the Azure tenant and subscription using the following command:

```
Connect-AzAccount
```

If you receive a message indicating **InteractiveBrowserCredential authentication failed**, disable web account manager (WAM) to force a browser authentication with the following command:

```
Update-AzConfig -EnableLoginByWam $false
```

Then attempt using 'Connect-AzAccount' again.

If the browser-based login is not available, you may need to use device-code authentication, but this should be avoided and is not recommended because it is a persistent authentication method and specifically blocked by recommendations found in the Azure Benchmarks. Instructions for this method will not be provided as it is not recommended.

For the Graph PowerShell module, the log in method is the same.

```
Connect-MgGraph
```

1.4 Automating with Azure Policy

Azure Policy provides built-in objects ("Policy Definitions") and a platform for "Policy as Code" that can be used to evaluate and/or enforce configuration states for individual resources or groups of resources. Where a relevant Azure Policy Definition or multiple Policy Definitions have been identified as applicable to a recommendation in this Benchmark, the Policy ID(s) and associated Policy Name will be listed in the "From Azure Policy" method header.

Interfacing with Azure Policy can be performed through the Azure Portal, Azure CLI, Azure PowerShell, REST API, Azure Resource Manager, Bicep, Terraform, or using GitHub Actions. Azure Policy Definitions can be batched together and structured using an "Initiative Definition" (aka: "Initiative" or "Compliance Initiative" or "Policy Set") which is constructed in a JSON file. Scoping and filtering an Azure Policy Initiative is necessary to ensure that the query is relevant to the architecture and requirements of an organization.

Microsoft provides some pre-built Initiative Definitions as a starting point intended to assist in achieving governance and compliance objectives - from Security Frameworks to CIS Benchmarks. These pre-built Initiative Definitions provide a structured set of Policy Definitions with additional metadata to evaluate resource states and configurations, then present the results in an easily readable format. While the Initiatives are a very valuable starting point, they will likely require additional attention for your specific environment, so it is a valuable investment in time to understand how to interact with them to create, remove, update, and delete policy definitions and modify enforcement effects. With some effort and care, policies within an initiative which start by default in an "audit" or "auditIfNotExists" effect can be matured into enforcement effects of "deny," "modify," or "deployIfNotExists" but each individual Policy Definition should be well understood before modifying the effect. In environments that have an advanced posture and custom policy needs, custom Policy Definitions can be created for specialized resource evaluations.

Building an Initiative Definition that fits the needs of your organization is generally a great way to improve overall security posture and has the potential to bring and keep your organization closer to compliance and best practice standards.

NOTE On Outdated Policy Initiatives of CIS Benchmarks:

CIS is aware that outdated versions of the CIS Benchmark are made available through the built-in Policy Initiatives offered in Azure. We always recommend using only the **latest available version** of the CIS Microsoft Azure Foundations Benchmark at all times and **using the latest published version of this Benchmark to update and maintain the policy initiative in your environments**. Cloud service provider environments are in a constant state of change, so best practice guidance - including Azure Policy recommendations - is also in a constant state of reconsideration and possible change.

General references for Azure Policy:

- Microsoft Learn - General Azure Policy Documentation: <https://learn.microsoft.com/en-us/azure/governance/policy/>
- Designing Policy as Code workflows: <https://learn.microsoft.com/en-us/azure/governance/policy/concepts/policy-as-code>
- Retrieving Azure Policy information: <https://learn.microsoft.com/en-us/azure/governance/policy/how-to/get-compliance-data>
- AzPolicyAdvertiser - a very well curated database of Azure Policy Definitions, Initiatives, and related material: https://www.azadvertizer.net/azpolicyadvertiser_all.html

Azure Policy GUI

- The Azure Policy GUI interface can be found in the Azure Admin Portal - simply search for **Policy** and select the Policy service.

Azure Policy from Azure CLI

- Azure CLI commands: <https://learn.microsoft.com/en-us/cli/azure/policy?view=azure-cli-latest>

Azure Policy from PowerShell

- PowerShell commands: <https://learn.microsoft.com/en-us/powershell/module/az.resources#policy>

Azure Policy from REST API

- REST API reference: <https://learn.microsoft.com/en-us/rest/api/policy/>

Azure Policy from GitHub Actions

- Azure Policy GitHub Actions: <https://github.com/marketplace/actions/azure-policy-compliance-scan>

Azure Policy from Terraform

- Provider: azurearm
- Reference: https://registry.terraform.io/providers/hashicorp/azurerm/latest/docs/resources/policy_definition

2 Analytics Services

To better understand the relationship between the Foundations Benchmark and Services Benchmarks, please read the "Introduction" section of this document.

Azure Product Directory Reference: <https://azure.microsoft.com/en-us/products#analytics>

FEEDBACK REQUEST: Is there a specific service or recommendation in this section that you'd like to see addressed or improved? Let us know by making a ticket or starting a discussion in the CIS Microsoft Azure Community (<https://workbench.cisecurity.org/communities/72>).

2.1 Azure Databricks

2.1.1 Ensure that Azure Databricks is deployed in a customer-managed virtual network (VNet) (Automated)

Profile Applicability:

- Level 1

Description:

Networking for Azure Databricks can be set up in a few different ways. Using a customer-managed Virtual Network (VNet) (also known as VNet Injection) ensures that compute clusters and control planes are securely isolated within the organization's network boundary. By default, Databricks creates a managed VNet, which provides limited control over network security policies, firewall configurations, and routing.

Rationale:

Using a customer-managed VNet ensures better control over network security and aligns with zero-trust architecture principles. It allows for:

- Restricted outbound internet access to prevent unauthorized data exfiltration.
- Integration with on-premises networks via VPN or ExpressRoute for hybrid connectivity.
- Fine-grained NSG policies to restrict access at the subnet level.
- Private Link for secure API access, avoiding public internet exposure.

Impact:

- Requires additional configuration during Databricks workspace deployment.
- Might increase operational overhead for network maintenance.
- May impact connectivity if misconfigured (e.g., restrictive NSG rules or missing routes).

Audit:

Audit from Azure Portal

1. Go to Azure Portal → Search for Databricks Workspaces.
2. Select the Databricks Workspace to audit.
3. Under Networking, check if the workspace is deployed in a Customer-Managed VNet.
4. If the Virtual Network field shows Databricks-Managed VNet, it is non-compliant.
5. Verify NSG rules and Private Endpoints for fine-grained access control.

Audit from Azure CLI

Run the following command to check if Databricks is using a customer-managed VNet:

```
az network vnet show --resource-group <resource-group-name> --name <vnet-name>
```

Ensure that Databricks subnets are present in the VNet configuration. Validate NSG rules attached to the Databricks subnets.

Audit from PowerShell

```
Get-AzDatabricksWorkspace -ResourceGroupName <resource-group-name> -Name <databricks-workspace-name> | Select-Object VirtualNetworkId
```

If VirtualNetworkId is null or shows a Databricks-Managed VNet, it is non-compliant.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [9c25c9e4-ee12-4882-afd2-11fb9d87893f](#) - **Name:** 'Azure Databricks Workspaces should be in a virtual network'

Remediation:

Remediate from Azure Portal

1. Delete the existing Databricks workspace (migration required).
2. Create a new Databricks workspace with VNet Injection:
3. Go to Azure Portal → Create Databricks Workspace.
4. Select Advanced Networking.
5. Choose Deploy into your own Virtual Network.
6. Specify a customer-managed VNet and associated subnets.
7. Enable Private Link for secure API access.

Remediate from Azure CLI

Deploy a new Databricks workspace in a custom VNet:

```
az databricks workspace create --name <databricks-workspace-name> \
  --resource-group <resource-group-name> \
  --location <region> \
  --managed-resource-group <managed-rg-name> \
  --enable-no-public-ip true \
  --network-security-group-rule "NoAzureServices" \
  --public-network-access Disabled \
  --custom-virtual-network-id /subscriptions/<subscription-id>/resourceGroups/<resource-group-name>/providers/Microsoft.Network/virtualNetworks/<vnet-name>
```

Ensure NSG Rules are correctly configured:

```
az network nsg rule create --resource-group <resource-group-name> \
  --nsg-name <nsg-name> \
  --name "DenyAllOutbound" \
  --direction Outbound \
  --access Deny \
  --priority 4096
```

Remediate from PowerShell

```
New-AzDatabricksWorkspace -ResourceGroupName <resource-group-name> -Name <databricks-workspace-name> -Location <region> -ManagedResourceGroupName <managed-rg-name> -CustomVirtualNetworkId "/subscriptions/<subscription-id>/resourceGroups/<resource-group-name>/providers/Microsoft.Network/virtualNetworks/<vnet-name>"
```

Default Value:

By default, Azure Databricks uses a Databricks-Managed VNet.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	12.2 <u>Establish and Maintain a Secure Network Architecture</u> Establish and maintain a secure network architecture. A secure network architecture must address segmentation, least privilege, and availability, at a minimum.			
v7	14.1 <u>Segment the Network Based on Sensitivity</u> Segment the network based on the label or classification level of the information stored on the servers, locate all sensitive information on separated Virtual Local Area Networks (VLANs).			

2.1.2 Ensure that Network Security Groups are Configured for Databricks Subnets (Automated)

Profile Applicability:

- Level 1

Description:

Network Security Groups (NSGs) should be implemented to control inbound and outbound traffic to Azure Databricks subnets, ensuring only authorized communication. NSGs operate using a rule-based model that includes both explicit allow/deny rules and an implicit deny at the end of the rule list. This means that any traffic not explicitly allowed is automatically denied. To ensure secure and predictable behavior, NSGs should be configured with explicit deny rules for known unwanted traffic, in addition to the default implicit deny, to improve visibility and auditability of blocked traffic. This approach helps enforce least privilege and minimizes the risk of unauthorized access to Databricks resources.

Rationale:

Using NSGs with both explicit allow and deny rules provides clear documentation and control over permitted and prohibited traffic. While Azure NSGs implicitly deny all traffic not explicitly allowed, defining explicit deny rules for known malicious or unnecessary sources enhances clarity, simplifies troubleshooting, and supports compliance audits. This layered approach strengthens the security posture of Databricks environments by ensuring only essential communication is permitted.

Impact:

- NSGs require ongoing maintenance to ensure rule accuracy and alignment with evolving business and security requirements.
- Misconfigured NSGs—especially overly broad allow rules or missing explicit denies—can inadvertently expose Databricks resources or block legitimate traffic.
- Relying solely on implicit deny may obscure the intent behind traffic restrictions, making it harder to audit or troubleshoot network behavior.

Audit:

Audit from Azure Portal

1. Navigate to Virtual Networks > Subnets, and review NSG assignments.

Audit from Azure CLI

```
az network nsg list --query "[].{Name:name, Rules:securityRules}"
```

Audit from PowerShell

```
Get-AzNetworkSecurityGroup -ResourceGroupName <resource-group-name>
```

Remediation:

Remediate from Azure Portal

1. Assign NSG to Databricks subnets under Networking > NSG Settings.

Default Value:

By default, Databricks subnets do not have NSGs assigned.

References:

1. <https://learn.microsoft.com/en-us/security/benchmark/azure/baselines/azure-databricks-security-baseline>
2. <https://learn.microsoft.com/en-us/azure/databricks/security/network/classic/vnet-inject#network-security-group-rules>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 Implement and Manage a Firewall on Servers Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v8	13.4 Perform Traffic Filtering Between Network Segments Perform traffic filtering between network segments, where appropriate.			
v7	9.4 Apply Host-based Firewalls or Port Filtering Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

2.1.3 Ensure that Traffic is Encrypted Between Cluster Worker Nodes (Manual)

Profile Applicability:

- Level 2

Description:

By default, data exchanged between worker nodes in an Azure Databricks cluster is not encrypted. To ensure that data is encrypted at all times, whether at rest or in transit, you can create an initialization script that configures your clusters to encrypt traffic between worker nodes using AES 256-bit encryption over a TLS 1.3 connection.

Rationale:

- Protects sensitive data during transit between cluster nodes, mitigating risks of data interception or unauthorized access.
- Aligns with organizational security policies and compliance requirements that mandate encryption of data in transit.
- Enhances overall security posture by ensuring that all inter-node communications within the cluster are encrypted.

Impact:

- Enabling encryption may introduce a performance penalty due to the computational overhead associated with encrypting and decrypting traffic. This can result in longer query execution times, especially for data-intensive operations.
- Implementing encryption requires creating and managing init scripts, which adds complexity to cluster configuration and maintenance.
- The shared encryption secret is derived from the hash of the keystore stored in DBFS. If the keystore is updated or rotated, all running clusters must be restarted to prevent authentication failures between Spark workers and drivers.

Audit:

Audit from Azure Portal

Review cluster init scripts:

1. Navigate to your Azure Databricks workspace, go to the "Clusters" section, select a cluster, and check the "Advanced Options" for any init scripts that configure encryption settings.

Verify spark configuration:

2. Ensure that the following Spark configurations are set:

```
spark.authenticate true
spark.authenticate.enableSaslEncryption true
spark.network.crypto.enabled true
spark.network.crypto.keyLength 256
spark.network.crypto.keyFactoryAlgorithm PBKDF2WithHmacSHA1
spark.io.encryption.enabled true
```

These settings can be found in the cluster's Spark configuration properties.

Check keystore management:

3. Verify that the Java KeyStore (JKS) file is securely stored in DBFS and that its integrity is maintained.
4. Ensure that the keystore password is securely managed and not hardcoded in scripts.

Remediation:

Create a JKS keystore:

1. Generate a Java KeyStore (JKS) file that will be used for SSL/TLS encryption.
2. Upload the keystore file to a secure directory in DBFS (e.g. /dbfs/<keystore-directory>/jetty_ssl_driver_keystore.jks).

Develop an init script:

3. Create an init script that performs the following tasks:
 - Retrieves the JKS keystore file and password.
 - Derives a shared encryption secret from the keystore.
 - Configures Spark driver and executor settings to enable encryption.
4. Example init script:

```

#!/bin/bash
set -euo pipefail
keystore_dbfs_file="/dbfs/<keystore-
directory>/jetty_ssl_driver_keystore.jks"
max_attempts=30
while [ ! -f ${keystore_dbfs_file} ]; do
    if [ "$max_attempts" == 0 ]; then
        echo "ERROR: Unable to find the file : $keystore_dbfs_file. Failing
the script."
        exit 1
    fi
    sleep 2s
    ((max_attempts--))
done
sasl_secret=$(sha256sum $keystore_dbfs_file | cut -d' ' -f1)
if [ -z "${sasl_secret}" ]; then
    echo "ERROR: Unable to derive the secret. Failing the script."
    exit 1
fi
local_keystore_file="$DB_HOME/keys/jetty_ssl_driver_keystore.jks"
local_keystore_password="gblgQqZ9ZIHS"
if [[ $DB_IS_DRIVER = "TRUE" ]]; then
    driver_conf=${DB_HOME}/driver/conf/spark-branch.conf
    echo "Configuring driver conf at $driver_conf"
    if [ ! -e $driver_conf ]; then
        echo "spark.authenticate true" >> $driver_conf
        echo "spark.authenticate.secret $sasl_secret" >> $driver_conf
        echo "spark.authenticate.enableSaslEncryption true" >> $driver_conf
        echo "spark.network.crypto.enabled true" >> $driver_conf
        echo "spark.network.crypto.keyLength 256" >> $driver_conf
        echo "spark.network.crypto.keyFactoryAlgorithm PBKDF2WithHmacSHA1"
    >> $driver_conf
        echo "spark.io.encryption.enabled true" >> $driver_conf
        echo "spark.ssl.enabled true" >> $driver_conf
        echo "spark.ssl.keyPassword $local_keystore_password" >>
$driver_conf
        echo "spark.ssl.keyStore $local_keystore_file" >> $driver_conf
        echo "spark.ssl.keyStorePassword $local_keystore_password" >>
$driver_conf
        echo "spark.ssl.protocol TLSv1.3" >> $driver_conf
    fi
fi
executor_conf=${DB_HOME}/conf/spark.executor.extraJavaOptions
echo "Configuring executor conf at $executor_conf"
if [ ! -e $executor_conf ]; then
    echo "-Dspark.authenticate=true" >> $executor_conf
    echo "-Dspark.authenticate.secret=$sasl_secret" >> $executor_conf
    echo "-Dspark.authenticate.enableSaslEncryption=true" >>
$executor_conf
        echo "-Dspark.network.crypto.enabled=true" >> $executor_conf
        echo "-Dspark.network.crypto.keyLength=256" >> $executor_conf
        echo "-Dspark.network.crypto.keyFactoryAlgorithm=PBKDF2WithHmacSHA1"
    >> $executor_conf
        echo "-Dspark.io.encryption.enabled=true" >> $executor_conf
        echo "-Dspark.ssl.enabled=true" >> $executor_conf
        echo "-Dspark.ssl.keyPassword=$local_keystore_password" >>
$executor_conf

```

```

echo "-Dspark.ssl.keyStore=$local_keystore_file" >> $executor_conf
echo "-Dspark.ssl.keyStorePassword=$local_keystore_password" >>
$executor_conf
echo "-Dspark.ssl.protocol=TLSv1.3" >> $executor_conf
fi

```

5. Save.

Default Value:

By default, traffic is not encrypted between cluster worker nodes.

References:

1. <https://learn.microsoft.com/en-us/azure/databricks/security/keys/encrypt-otw>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.10 <u>Encrypt Sensitive Data in Transit</u> Encrypt sensitive data in transit. Example implementations can include: Transport Layer Security (TLS) and Open Secure Shell (OpenSSH).			
v7	14.4 <u>Encrypt All Sensitive Information in Transit</u> Encrypt all sensitive information in transit.			

2.1.4 Ensure that Users and Groups are Synced from Microsoft Entra ID to Azure Databricks (Manual)

Profile Applicability:

- Level 1

Description:

To ensure centralized identity and access management, users and groups from Microsoft Entra ID should be synchronized with Azure Databricks. This is achieved through SCIM provisioning, which automates the creation, update, and deactivation of users and groups in Databricks based on Entra ID assignments. Enabling this integration ensures that access controls in Databricks remain consistent with corporate identity governance policies, reducing the risk of orphaned accounts, stale permissions, and unauthorized access.

Rationale:

Syncing users and groups from Microsoft Entra ID centralizes access control, enforces the least privilege principle by automatically revoking unnecessary access, reduces administrative overhead by eliminating manual user management, and ensures auditability and compliance with industry regulations.

Impact:

SCIM provisioning requires role mapping to avoid misconfigured user privileges.

Audit:

Audit from Azure Portal

Verify SCIM provisioning is enabled:

1. Go to **Microsoft Entra ID**.
2. Under **Manage**, click **Enterprise applications**.
3. Click the name of the Azure Databricks SCIM application.
4. Under **Provisioning**, confirm that SCIM provisioning is enabled and running.

Check user sync status in Azure Portal:

5. Under **Provisioning Logs**, verify the last successful sync and any failed entries.

Check user sync status in Databricks:

6. Go to **Admin Console > Identity and Access Management**.
7. Confirm that Users and Groups match those assigned in Microsoft Entra ID.

Ensure role-based access control (RBAC) mapping is correct:

8. Verify that users are assigned appropriate Databricks roles (e.g. Admin, User, Contributor).
9. Confirm that groups are mapped to workspace access roles.

Remediation:

Remediate from Azure Portal

Enable provisioning in Azure Portal:

1. Go to **Microsoft Entra ID**.
2. Under **Manage**, click **Enterprise applications**.
3. Click the name of the Azure Databricks SCIM application.
4. Under **Provisioning**, select **Automatic** and enter the SCIM endpoint and API token from Databricks.

Enable provisioning in Databricks:

5. Navigate to **Admin Console > Identity and Access Management**.
6. Enable SCIM provisioning and generate an API token.

Configure role assignments:

7. Ensure groups from Entra ID are mapped to appropriate Databricks roles.
8. Restrict administrative privileges to designated security groups.

Regularly monitor sync logs:

9. Periodically review sync logs in Microsoft Entra ID and Databricks Admin Console.
10. Configure Azure Monitor alerts for provisioning failures.

Disable manual user creation in Databricks:

11. Ensure that all user management is controlled via SCIM sync from Entra ID.
12. Disable personal access token usage for authentication.

Remediate from Azure CLI

Enable SCIM User and Group Provisioning in Azure Databricks:

```
az ad app update --id <databricks-app-id> --set  
provisioning.provisioningMode=Automatic
```

Default Value:

By default, Azure Databricks does not sync users and groups from Microsoft Entra ID.

References:

1. <https://learn.microsoft.com/en-us/azure/databricks/administration-guide/users-groups/scim/aad>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.6 Centralize Account Management Centralize account management through a directory or identity service.			
v7	16.2 Configure Centralized Point of Authentication Configure access for all accounts through as few centralized points of authentication as possible, including network, security, and cloud systems.			

2.1.5 Ensure that Unity Catalog is Configured for Azure Databricks (Manual)

Profile Applicability:

- Level 1

Description:

Unity Catalog is a centralized governance model for managing and securing data in Azure Databricks. It provides fine-grained access control to databases, tables, and views using Microsoft Entra ID identities. Unity Catalog also enhances data lineage, audit logging, and compliance monitoring, making it a critical component for security and governance.

Rationale:

- Enforces centralized access control policies and reduces data security risks.
- Enables identity-based authentication via Microsoft Entra ID.
- Improves compliance with industry regulations (e.g. GDPR, HIPAA, SOC 2) by providing audit logs and access visibility.
- Prevents unauthorized data access through table-, row-, and column-level security (RLS & CLS).

Impact:

- Improperly configured permissions may lead to data exfiltration or unauthorized access.
- Unity Catalog requires structured governance policies to be effective and prevent overly permissive access.

Audit:

Method 1: Verify unity catalog deployment:

1. As an Azure Databricks account admin, log into the account console.
2. Click Workspaces.
3. Find your workspace and check the Metastore column. If a metastore name is present, your workspace is attached to a Unity Catalog metastore and therefore enabled for Unity Catalog.

Method 2: Run a SQL query to confirm Unity Catalog enablement

Run the following SQL query in the SQL query editor or a notebook that is attached to a Unity Catalog-enabled compute resource. No admin role is required.


```
SELECT CURRENT_METASTORE();
```

If the query returns a metastore ID like the following, then your workspace is attached to a Unity Catalog metastore and therefore enabled for Unity Catalog.

Remediation:

Use the remediation procedure written in this article: <https://learn.microsoft.com/en-us/azure/databricks/data-governance/unity-catalog/get-started>.

Default Value:

New workspaces have Unity Catalog enabled by default. Existing workspaces may require manual enablement.

References:

1. <https://learn.microsoft.com/en-us/azure/databricks/data-governance/unity-catalog/>
2. <https://learn.microsoft.com/en-us/azure/databricks/admin/users-groups/>
3. <https://learn.microsoft.com/en-us/azure/databricks/data-governance/unity-catalog/enable-workspaces>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.6 Centralize Account Management Centralize account management through a directory or identity service.			
v8	6.7 Centralize Access Control Centralize access control for all enterprise assets through a directory service or SSO provider, where supported.			
v7	16.2 Configure Centralized Point of Authentication Configure access for all accounts through as few centralized points of authentication as possible, including network, security, and cloud systems.			

2.1.6 Ensure that Usage is Restricted and Expiry is Enforced for Databricks Personal Access Tokens (Manual)

Profile Applicability:

- Level 1

Description:

Databricks personal access tokens (PATs) provide API-based authentication for users and applications. By default, users can generate API tokens without expiration, leading to potential security risks if tokens are leaked, improperly stored, or not rotated regularly.

To mitigate these risks, administrators should:

- Restrict token creation to approved users and service principals.
- Enforce expiration policies to prevent long-lived tokens.
- Monitor token usage and revoke unused or compromised tokens.

Rationale:

Restricting usage and enforcing expiry for personal access tokens reduces exposure to long-lived tokens, minimizes the risk of API abuse if compromised, and aligns with security best practices through controlled issuance and enforced expiry.

Impact:

If revoked improperly, applications relying on these tokens may fail, requiring a remediation plan for token rotation. Increased administrative effort is required to track and manage API tokens effectively.

Audit:

Azure Databricks administrators can monitor and revoke personal access tokens within their workspace. Detailed instructions are available in the "Monitor and Revoke Personal Access Tokens" section of the Microsoft documentation:

<https://learn.microsoft.com/en-us/azure/databricks/admin/access-control/tokens>.

To evaluate the usage of personal access tokens in your Azure Databricks account, you can utilize the provided notebook that lists all PATs not rotated or updated in the last 90 days, allowing you to identify tokens that may require revocation. This process is detailed here: <https://learn.microsoft.com/en-us/azure/databricks/admin/access-control/tokens>.

Implementing diagnostic logging provides a comprehensive reference of audit log services and events, enabling you to track activities related to personal access tokens. More information can be found in the diagnostic log reference section:

<https://learn.microsoft.com/en-us/azure/databricks/admin/account-settings/audit-logs>.

Remediation:

Remediate from Azure Portal

Disable personal access tokens:

If your workspace does not require PATs, you can disable them entirely to prevent their use.

1. Navigate to your Azure Databricks workspace.
2. Click the **Settings** icon and select **Admin Console**.
3. Go to the **Advanced** tab.
4. Under **Personal Access Tokens**, toggle the setting to **Disabled**.

Databricks CLI:

```
databricks workspace-conf set-status --json '{"enableTokens": "false"}'
```

Control who can create and use personal access tokens:

Define which users or groups are authorized to create and utilize PATs.

1. Navigate to your Azure Databricks workspace.
2. Click the **Settings** icon and select **Admin Console**.
3. Go to the **Advanced** tab.
4. Click on **Personal Access Tokens** and then **Permissions**.
5. Assign the appropriate permissions (e.g. No Permissions, Can Use, Can Manage) to users or groups.

Set maximum lifetime for new personal access tokens:

Limit the validity period of new tokens to reduce potential misuse.

Databricks CLI:

```
databricks workspace-conf set-status --json '{"maxTokenLifetimeDays": "90"}'
```

Monitor and revoke personal access tokens:

Periodically review active tokens and revoke any that are unnecessary or potentially compromised.

Databricks CLI:

```
databricks token list  
databricks token delete --token-id <token-id>
```

Transition to OAuth for enhanced security:

Utilize OAuth tokens for authentication, offering improved security features over PATs.

Default Value:

By default, personal access tokens are enabled and users can create the Personal access token and their expiry time.

References:

1. <https://learn.microsoft.com/en-us/azure/databricks/administration-guide/access-control/tokens>
2. <https://learn.microsoft.com/en-us/azure/databricks/dev-tools/auth/>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	6.2 Establish an Access Revoking Process Establish and follow a process, preferably automated, for revoking access to enterprise assets, through disabling accounts immediately upon termination, rights revocation, or role change of a user. Disabling accounts, instead of deleting accounts, may be necessary to preserve audit trails.			
v7	16.7 Establish Process for Revoking Access Establish and follow an automated process for revoking system access by disabling accounts immediately upon termination or change of responsibilities of an employee or contractor . Disabling these accounts, instead of deleting accounts, allows preservation of audit trails.			

2.1.7 Ensure that Diagnostic Log Delivery is Configured for Azure Databricks (Automated)

Profile Applicability:

- Level 1

Description:

Azure Databricks Diagnostic Logging provides insights into system operations, user activities, and security events within a Databricks workspace. Enabling diagnostic logs helps organizations:

- Detect security threats by logging access, job executions, and cluster activities.
- Ensure compliance with industry regulations such as SOC 2, HIPAA, and GDPR.
- Monitor operational performance and troubleshoot issues proactively.

Rationale:

Diagnostic logging provides visibility into security and operational activities within Databricks workspaces while maintaining an audit trail for forensic investigations, and it supports compliance with regulatory standards that require logging and monitoring.

Impact:

Logs consume storage and may require additional monitoring tools, leading to increased operational overhead and costs. Incomplete log configurations may result in missing critical events, reducing monitoring effectiveness.

Audit:

Audit from Azure Portal

Check if diagnostic logging is enabled for the Databricks workspace:

1. Go to **Azure Databricks**.
2. Select a workspace.
3. In the left-hand menu, select **Monitoring > Diagnostic settings**.
4. Verify if a diagnostic setting is configured. If not, diagnostic logging is not enabled.

Ensure that logging is enabled for the following categories:

- **accounts**: User account activities.
- **Filesystem**: Databricks Filesystem Logs
- **clusters**: Cluster state changes and errors.
- **notebook**: Execution events.
- **jobs**: Job execution tracking.

Verify that logs are being sent to one or more of the following destinations:

- **Azure Log Analytics workspace**: For analysis and querying.
- **Azure Storage Account**: For long-term retention.
- **Azure Event Hubs**: For integration with SIEM tools.

Audit from Azure CLI

Check if diagnostic logging is enabled for the Databricks workspace:

```
az monitor diagnostic-settings list --resource <databricks-resource-id>
```

If the output is empty, no diagnostic settings are configured.

Verify log categories being collected:

```
az monitor diagnostic-settings show --name <setting-name> --resource  
<databricks-resource-id>
```

Review the output to confirm that the necessary log categories are enabled.

Check if logs are stored securely in an approved location:

```
az monitor diagnostic-settings list --resource <databricks-resource-id>
```

Review the storageAccountId, workspaceId, and eventHubAuthorizationRuleId fields in the output to confirm the log destinations.

Audit from PowerShell

Check if diagnostic logging is enabled for the Databricks workspace:

```
Get-AzDiagnosticSetting -ResourceId <databricks-resource-id>
```

An empty result indicates that diagnostic logging is not enabled.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~Definitions

- **Policy ID:** [138ff14d-b687-4faa-a81c-898c91a87fa2](#) - **Name:** 'Resource logs in Azure Databricks Workspaces should be enabled'

Remediation:

Remediate from Azure Portal

Enable diagnostic logging for Azure Databricks:

1. Navigate to your Azure Databricks workspace.
2. In the left-hand menu, select **Monitoring** > **Diagnostic settings**.

3. Click **+ Add diagnostic setting**.
4. Under **Category details**, select the log categories you wish to capture, such as AuditLogs, Clusters, Notebooks, and Jobs.
5. Choose a destination for the logs:
 - o **Log Analytics workspace**: For advanced querying and monitoring.
 - o **Storage account**: For long-term retention.
 - o **Event Hub**: For integration with third-party systems.
6. Provide a **Name** for the diagnostic setting.
7. Click **Save**.

Implement log retention policies:

1. Navigate to your Log Analytics workspace.
2. Under **General**, select **Usage and estimated costs**.
3. Click **Data Retention**.
4. Adjust the retention period slider to the desired number of days (up to 730 days).
5. Click **OK**.

Monitor logs for anomalies:

1. Navigate to **Azure Monitor**.
2. Select **Alerts > + New alert rule**.
3. Under **Scope**, specify the Databricks resource.
4. Define **Condition** based on log queries that identify anomalies (e.g. unauthorized access attempts).
5. Configure **Actions** to notify stakeholders or trigger automated responses.
6. Provide an Alert rule **name** and **description**.
7. Click **Create alert rule**.

Remediate from Azure CLI

Enable diagnostic logging for Azure Databricks:

```
az monitor diagnostic-settings create --name "DatabricksLogging" --resource <databricks-resource-id> --logs '[{"category": "accounts", "enabled": true}, {"category": "Clusters", "enabled": true}, {"category": "Notebooks", "enabled": true}, {"category": "Jobs", "enabled": true}]' --workspace <log-analytics-id>
```

Implement log retention policies:

```
az monitor log-analytics workspace update --resource-group <resource-group> --name <log-analytics-name> --retention-time 365
```

Monitor logs for anomalies:

```
az monitor activity-log alert create --name "DatabricksAnomalyAlert" --resource-group <resource-group> --scopes <databricks-resource-id> --condition "contains 'UnauthorizedAccess'"
```

References:

1. <https://learn.microsoft.com/en-us/azure/databricks/admin/account-settings/audit-log-delivery>
2. <https://learn.microsoft.com/en-us/troubleshoot/azure/azure-monitor/log-analytics/billing/configure-data-retention>
3. <https://docs.azure.cn/en-us/databricks/admin/account-settings/audit-logs>
4. <https://learn.microsoft.com/en-us/azure/azure-monitor/reference/supported-logs/microsoft-databricks-workspaces-logs>

Additional Information:

- Ensure that the Azure Databricks workspace is on the Premium plan to utilize diagnostic logging features.
- Regularly review and update alert rules to adapt to evolving security threats and operational requirements.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.2 <u>Collect Audit Logs</u> Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			

2.1.8 Ensure Critical Data in Azure Databricks is Encrypted with Customer-managed Keys (CMK) (Manual)

Profile Applicability:

- Level 2

Description:

Customer-managed keys introduce additional depth to security by providing a means to manage access control for encryption keys. Where compliance and security frameworks indicate the need, and organizational capacity allows, sensitive data at rest can be encrypted using customer-managed keys (CMK) rather than Microsoft-managed keys.

Rationale:

By default in Azure, data at rest tends to be encrypted using Microsoft-managed keys. If your organization wants to control and manage encryption keys for compliance and defense-in-depth, customer-managed keys can be established.

While it is possible to automate the assessment of this recommendation, the assessment status for this recommendation remains 'Manual' due to ideally limited scope. The scope of application—which workloads CMK is applied to—should be carefully considered to account for organizational capacity and targeted to workloads with specific need for CMK.

Impact:

If the key expires due to setting the 'activation date' and 'expiration date', the key must be rotated manually.

Using customer-managed keys may also incur additional man-hour requirements to create, store, manage, and protect the keys as needed.

Audit:

Audit from Azure Portal

1. Go to Azure Portal → Databricks Workspaces.
2. Select a Databricks Workspace and go to Encryption settings.
3. Check if customer-managed keys (CMK) are enabled under "Managed Disk Encryption".
4. If CMK is not enabled, the workspace is non-compliant.

Audit from Azure CLI

Run the following command to check encryption settings for Databricks workspace:

```
az databricks workspace show --name <databricks-workspace-name> --resource-group <resource-group-name> --query encryption
```

Ensure that keySource is set to **Microsoft.KeyVault**.

Audit from PowerShell

```
Get-AzDatabricksWorkspace -ResourceGroupName "<resource-group-name>" -Name "<databricks-workspace-name>" | Select-Object Encryption
```

Verify that encryption is set to **Customer-Managed Keys (CMK)**.

Audit from Databricks CLI

```
databricks workspace get-metadata --workspace-id <workspace-id>
```

Ensure that encryption settings reflect a CMK setup.

Remediation:

NOTE: These remediations assume that an Azure KeyVault already exists in the subscription.

Remediate from Azure CLI

1. Create a dedicated key:

```
az keyvault key create --vault-name <keyvault-name> --name <key-name> --protection <"software" or "hsm">
```

2. Assign permissions to Databricks:

```
az keyvault set-policy --name <keyvault-name> --resource-group <resource-group-name> --spn <databricks-spn> --key-permissions get wrapKey unwrapKey
```

3. Enable encryption with CMK:

```
az databricks workspace update --name <databricks-workspace-name> --resource-group <resource-group-name> --key-source "Microsoft.KeyVault" --key-name <key-name> --keyvault-uri <keyvault-uri>
```

Remediate from PowerShell

```
$Key = Add-AzKeyVaultKey -VaultName <keyvault-name> -Name <key-name> -Destination <"software" or "hsm">  
Set-AzDatabricksWorkspace -ResourceGroupName "<resource-group-name>" -WorkspaceName "<databricks-workspace-name>" -EncryptionKeySource "Microsoft.KeyVault" -KeyVaultUri $Key.Id
```

Default Value:

By default, encryption type is set to Microsoft-managed keys.

References:

1. <https://learn.microsoft.com/en-us/azure/security/fundamentals/data-encryption-best-practices#protect-data-at-rest>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-data-protection#dp-5-use-customer-managed-key-option-in-data-at-rest-encryption-when-required>
3. <https://learn.microsoft.com/en-us/azure/databricks/security/keys/cmk-managed-disks-azure/cmk-managed-disks-azure>

Additional Information:

This recommendation is based on the Common Reference Recommendation **Ensure critical data is encrypted with customer-managed keys (CMK)**, from the **Common Reference Recommendations > Secrets and Keys > Encryption Key Management > Customer Managed Keys** section.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.11 <u>Encrypt Sensitive Data at Rest</u> Encrypt sensitive data at rest on servers, applications, and databases containing sensitive data. Storage-layer encryption, also known as server-side encryption, meets the minimum requirement of this Safeguard. Additional encryption methods may include application-layer encryption, also known as client-side encryption, where access to the data storage device(s) does not permit access to the plain-text data.			
v7	14.8 <u>Encrypt Sensitive Information at Rest</u> Encrypt all sensitive information at rest using a tool that requires a secondary authentication mechanism not integrated into the operating system, in order to access the information.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1530	TA0009	M1041

2.1.9 Ensure 'No Public IP' is Set to 'Enabled' (Automated)

Profile Applicability:

- Level 1

Description:

Enable secure cluster connectivity (also known as no public IP) on Azure Databricks workspaces to ensure that clusters do not have public IP addresses and communicate with the control plane over a secure connection.

Rationale:

Enabling secure cluster connectivity limits exposure to the public internet, improving security and reducing the risk of external attacks.

Impact:

Enabling secure cluster connectivity requires careful network configuration. Before secure cluster connectivity can be enabled, Azure Databricks workspaces must be deployed in a customer-managed virtual network (VNet injection)—refer to the recommendation **Ensure that Azure Databricks is deployed in a customer-managed virtual network (VNet)**.

Audit:

Audit from Azure Portal

1. Go to **Azure Databricks**.
2. Click the name of a workspace.
3. Under **Settings**, click **Networking**.
4. Under **Network access**, ensure that **Deploy Azure Databricks workspace with Secure Cluster Connectivity (No Public IP)** is set to **Enabled**.
5. Repeat steps 1-4 for each workspace.

Audit from Azure CLI

Run the following command to list workspaces:

```
az databricks workspace list
```

For each workspace, run the following command to get the **enableNoPublicIp** setting:

```
az databricks workspace show --resource-group <resource-group> --name  
<workspace> --query parameters.enableNoPublicIp.value
```

Ensure that **true** is returned.

Audit from PowerShell

Run the following command to list workspaces:

```
Get-AzDatabricksWorkspace
```

Run the following command to get the workspace in a resource group with a given name:

```
$workspace = Get-AzDatabricksWorkspace -ResourceGroupName <resource-group> -  
Name <workspace>
```

Run the following command to get the **EnableNoPublicIp** setting:

```
$workspace.EnableNoPublicIP
```

Ensure that **True** is returned.

Repeat for each workspace.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~Definitions

- **Policy ID:** [51c1490f-3319-459c-bbbc-7f391bbed753](#) - **Name:** 'Azure Databricks Clusters should disable public IP'

Remediation:

Remediate from Azure Portal

1. Go to **Azure Databricks**.
2. Click the name of a workspace.
3. Under **Settings**, click **Networking**.
4. Under **Network access**, next to **Deploy Azure Databricks workspace with Secure Cluster Connectivity (No Public IP)**, click the radio button next to **Enabled**.
5. Click **Save**.
6. Repeat steps 1-5 for each workspace requiring remediation.

Remediate from Azure CLI

For each workspace requiring remediation, run the following command to set **enableNoPublicIp** to **true**:

```
az databricks workspace update --resource-group <resource-group> --name  
<workspace> --enable-no-public-ip true
```

Remediate from PowerShell

For each workspace requiring remediation, run the following command to set **EnableNoPublicIP** to **True**:

```
Update-AzDatabricksWorkspace -ResourceGroupName <resource-group> -Name  
<workspace> -EnableNoPublicIP
```

Default Value:

No **Public IP** is set to **Enabled** by default.

References:

1. <https://learn.microsoft.com/en-us/azure/databricks/security/network/classic/secure-cluster-connectivity>
2. <https://learn.microsoft.com/en-us/cli/azure/databricks/workspace>
3. <https://learn.microsoft.com/en-us/powershell/module/az.databricks>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1530	TA0009	M1037

2.1.10 Ensure 'Allow Public Network Access' is set to 'Disabled' (Automated)

Profile Applicability:

- Level 1

Description:

Disable public network access to prevent exposure to the internet and reduce the risk of unauthorized access. Use private endpoints to securely manage access within trusted networks.

Rationale:

Disabling public network access improves security by ensuring that Azure Databricks workspaces are not exposed on the public internet.

Impact:

NOTE: Prior to disabling public network access, it is strongly recommended that, for each workspace, either:

- virtual network integration is completed as described in **"Ensure that Azure Databricks is deployed in a customer-managed virtual network (VNet)"**

OR

- private endpoints/links are set up as described in **"Ensure private endpoints are used to access Azure Databricks workspaces."**

Disabling public network access restricts access to the service. This enhances security but will require the configuration of a virtual network and/or private endpoints for any services or users needing access within trusted networks.

Before public network access can be disabled, Azure Databricks workspaces must be deployed in a customer-managed virtual network (VNet injection)—refer to the recommendation **Ensure that Azure Databricks is deployed in a customer-managed virtual network (VNet)**, and **requiredNsgRules** must be set to a value other than **AllRules**.

Audit:

Audit from Azure Portal

1. Go to **Azure Databricks**.
2. Click the name of a workspace.
3. Under **Settings** click **Networking**.

4. Under **Network access**, ensure **Allow Public Network Access** is set to **Disabled**.
5. Repeat steps 1-4 for each workspace.

Audit from Azure CLI

Run the following command to list workspaces:

```
az databricks workspace list
```

For each workspace, run the following command to get the **publicNetworkAccess** setting:

```
az databricks workspace show --resource-group <resource-group> --name <workspace> --query publicNetworkAccess
```

Ensure that **"Disabled"** is returned.

Audit from PowerShell

Run the following command to list workspaces:

```
Get-AzDatabricksWorkspace
```

Run the following command to get the workspace in a resource group with a given name:

```
$workspace = Get-AzDatabricksWorkspace -ResourceGroupName <resource-group> -Name <workspace>
```

Run the following command to get the **PublicNetworkAccess** setting:

```
$workspace.PublicNetworkAccess
```

Ensure that **Disabled** is returned.

Repeat for each workspace.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

<https://portal.azure.com/#view/Microsoft Azure Policy/PolicyMenuBlade/~/Definitions>

- **Policy ID:** [0e7849de-b939-4c50-ab48-fc6b0f5eeba2](#) - **Name:** 'Azure Databricks Workspaces should disable public network access'

Remediation:

Remediate from Azure Portal

1. Go to **Azure Databricks**.
2. Click the name of a workspace.

3. Under **Settings** click **Networking**.
4. Under **Network access**, next to **Allow Public Network Access**, click the radio button next to **Disabled**.
5. Click **Save**.
6. Repeat steps 1-5 for each workspace requiring remediation.

Remediate from Azure CLI

For each workspace requiring remediation, run the following command to set **publicNetworkAccess** to **Disabled**:

```
az databricks workspace update --resource-group <resource-group> --name <workspace> --public-network-access Disabled
```

Remediate from PowerShell

For each workspace requiring remediation, run the following command to set **PublicNetworkAccess** to **Disabled**:

```
Update-AzDatabricksWorkspace -ResourceGroupName <resource-group> -Name <workspace> -PublicNetworkAccess Disabled
```

Default Value:

Allow Public Network Access is set to **Enabled** by default.

References:

1. <https://learn.microsoft.com/en-us/cli/azure/databricks/workspace>
2. <https://learn.microsoft.com/en-us/powershell/module/az.databricks>

Additional Information:

This recommendation is based on the Common Reference Recommendation **Ensure public network access is Disabled**, from the **Common Reference Recommendations > Networking > Virtual Networks (VNETs)** section.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			

Controls Version	Control	IG 1	IG 2	IG 3
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1530	TA0009	M1037

2.1.11 Ensure Private Endpoints are used to access Azure Databricks workspaces (Automated)

Profile Applicability:

- Level 2

Description:

Use private endpoints for Azure Databricks workspaces to allow clients and services to securely access data located over a network via an encrypted Private Link. To do this, the private endpoint uses an IP address from the VNet for each service. Network traffic between disparate services securely traverses encrypted over the VNet. This VNet can also link addressing space, extending your network and accessing resources on it. Similarly, it can be a tunnel through public networks to connect remote infrastructures together. This creates further security through segmenting network traffic and preventing outside sources from accessing it.

Rationale:

Using private endpoints for Azure Databricks workspaces ensures that all communication between clients, services, and data sources occurs over a secure, private IP space within an Azure Virtual Network (VNet). This approach eliminates exposure to the public internet, significantly reducing the attack surface and aligning with Zero Trust principles. Additionally, integrating Databricks with a VNet enables network segmentation, fine-grained access control, and hybrid connectivity through VNet peering or VPN/ExpressRoute.

Impact:

If an Azure Virtual Network is not implemented correctly, this may result in the loss of critical network traffic.

Private endpoints are charged per hour of use. Refer to <https://azure.microsoft.com/en-us/pricing/details/private-link/> and <https://azure.microsoft.com/en-us/pricing/calculator/> to estimate potential costs.

Before a private endpoint can be configured, Azure Databricks workspaces:

- must be deployed in a customer-managed virtual network (VNet injection)—refer to the recommendation **Ensure that Azure Databricks is deployed in a customer-managed virtual network (VNet)**
- must have secure cluster connectivity enabled—refer to the recommendation **Ensure 'Enable No Public IP' is set to 'Yes'**
- must be on the Premium pricing tier

Ensure the requirements and concepts are considered carefully before applying this recommendation. Refer to <https://learn.microsoft.com/en-us/azure/databricks/security/network/classic/private-link> for more information.

Audit:

Audit from Azure Portal

1. Go to **Azure Databricks**.
2. Click the name of a workspace.
3. Under **Settings**, click **Networking**.
4. Click **Private endpoint connections**.
5. Ensure a private endpoint connection exists with a connection state of **Approved**.
6. Repeat steps 1-5 for each workspace.

Audit from Azure CLI

Run the following command to list workspaces:

```
az databricks workspace list
```

For each workspace, run the following command to get the **privateEndpointConnections** configuration:

```
az databricks workspace show --resource-group <resource-group> --name  
<workspace> --query privateEndpointConnections
```

Ensure a private endpoint connection is returned with a **privateLinkServiceConnectionState status** of **Approved**.

Audit from PowerShell

Run the following command to list workspaces:

```
Get-AzDatabricksWorkspace
```

Run the following command to get the workspace in a resource group with a given name:

```
$workspace = Get-AzDatabricksWorkspace -ResourceGroupName <resource-group> -  
Name <workspace>
```

Run the following command to get the **PrivateEndpointConnection** configuration:

```
$workspace.PrivateEndpointConnection | Select-Object -Property  
Id,PrivateLinkServiceConnectionStateStatus
```

Ensure a private endpoint connection is returned with a **PrivateLinkServiceConnectionStateStatus** of **Approved**.

Repeat for each workspace.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [258823f2-4595-4b52-b333-cc96192710d8](#) - **Name:** 'Azure Databricks Workspaces should use private link'

Remediation:

Remediate from Azure Portal

1. Go to **Azure Databricks**.
2. Click the name of a workspace.
3. Under **Settings**, click **Networking**.
4. Click **Private endpoint connections**.
5. Click **+ Private endpoint**.
6. Under **Project details**, select a **Subscription** and a **Resource group**.
7. Under **Instance details**, provide a **Name**, **Network Interface Name**, and select a **Region**.
8. Click **Next : Resource >**.
9. Select a **Target sub-resource**.
10. Click **Next : Virtual Network >**.
11. Under **Networking**, select a **Virtual network** and a **Subnet**.
12. Optionally, configure **Private IP configuration** and **Application security group**.
13. Click **Next : DNS >**.
14. Optionally, configure **Private DNS integration**.
15. Click **Next : Tags >**.
16. Optionally, configure tags.
17. Click **Next : Review + create >**.
18. Click **Create**.
19. Repeat steps 1-18 for each workspace requiring remediation.

Remediate from Azure CLI

For each workspace requiring remediation, run the following command to create a private endpoint connection:

```
az network private-endpoint create --resource-group <resource-group> --name  
<private-endpoint> --location <location> --vnet-name <virtual-network> --  
subnet <subnet> --private-connection-resource-id <workspace> --connection-  
name <private-endpoint-connection> --group-id <browser_authentication|  
databricks_ui_api>
```

Default Value:

Private endpoints are not configured for Azure Databricks workspaces by default.

References:

1. <https://learn.microsoft.com/en-us/azure/databricks/security/network/classic/private-link>
2. <https://learn.microsoft.com/en-us/cli/azure/databricks/workspace>
3. <https://learn.microsoft.com/en-us/powershell/module/az.databricks>

Additional Information:

This recommendation is based on the Common Reference Recommendation **Ensure Private Endpoints are used to access {service}**, from the **Common Reference Recommendations > Networking > Private Endpoints** section.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	12.2 Establish and Maintain a Secure Network Architecture Establish and maintain a secure network architecture. A secure network architecture must address segmentation, least privilege, and availability, at a minimum.			
v7	14.1 Segment the Network Based on Sensitivity Segment the network based on the label or classification level of the information stored on the servers, locate all sensitive information on separated Virtual Local Area Networks (VLANs).			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1537	TA0010	M1037

2.1.12 Ensure Azure Databricks groups are reviewed periodically (Manual)

Profile Applicability:

- Level 1

Description:

Groups are used in Role Based Access Control to apply permissions to users and should be audited.

Rationale:

To ensure accurate privileges for your Azure Databricks implementation, your organization should review all users and permission assignments on a regular interval.

Impact:

Administrative overhead of user management.

Audit:

Audit from Azure Portal

1. From Azure Home select **Azure Databricks**.
2. Select the Databricks implementation you wish to audit.
3. Select **Access control (IAM)**.
4. In the horizontal menu select **Role assignments**.
5. Audit the list for each role and its assignment to each user.

Remediation:

Remediate from Azure Portal

1. From Azure Home select **Azure Databricks**.
2. Select the Databricks implementation you wish to audit.
3. Select **Access control (IAM)**.
4. Scroll down and select **Add role assignment**.
5. Search for the role you wish to add. Then select **Next**.
6. Select the group members you wish to add. Then select **Next**.
7. Review the info you have chosen, then select **Review + assign**.

Default Value:

By default Azure Databricks only has the **Owner** user and role assigned.

References:

1. <https://learn.microsoft.com/en-us/security/benchmark/azure/baselines/azure-databricks-security-baseline>
2. <https://learn.microsoft.com/en-us/azure/databricks/security/auth/>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	6.8 Define and Maintain Role-Based Access Control Define and maintain role-based access control, through determining and documenting the access rights necessary for each role within the enterprise to successfully carry out its assigned duties. Perform access control reviews of enterprise assets to validate that all privileges are authorized, on a recurring schedule at a minimum annually, or more frequently.			

3 Compute Services

SERVICE CATEGORY BENCHMARK AVAILABLE:

- "CIS Microsoft Azure Compute Services Benchmark"

To better understand the relationship between the Foundations Benchmark and Services Benchmarks, please read the "Introduction" section of this document.

PARTIAL RELOCATION - BE ADVISED!

Some recommendations previously covered in the CIS Microsoft Azure Foundations Benchmark (this document) related to Compute services have been relocated to a Benchmark titled **CIS Microsoft Azure Compute Services Benchmark**. This Compute Services section now provides a **foundational set** of secure configuration recommendations for products from Azure Product Directory's "Compute" category of services.

After applying foundational recommendations, take inventory of the entire set of Compute Services in use by your organization, then look to the Benchmarks section of the CIS Microsoft Azure Community (<https://workbench.cisecurity.org/communities/72>) for defense-in-depth guidance for those specific services in the **CIS Microsoft Azure Compute Services Benchmark**.

Services addressed in the **CIS Microsoft Azure Compute Services Benchmark**:

- App Service
- Azure Container Instances
- Azure CycleCloud
- Azure Dedicated Host
- Azure Functions
- Azure Kubernetes Service (AKS)
- Azure Quantum
- Azure Service Fabric
- Azure Spot Virtual Machines
- Azure Spring Apps
- Azure Virtual Desktop
- Azure VM Image Builder
- Azure VMware Solution
- Batch
- Cloud Services
- Linux Virtual Machines
- SQL Server on Azure Virtual Machines
- Static Web Apps
- Virtual Machine Scale Sets
- Virtual Machines

Azure Product Directory Reference: <https://azure.microsoft.com/en-us/products#compute>

FEEDBACK REQUEST: Is there a specific service or recommendation in this section that you'd like to see addressed or improved? Let us know by making a ticket or starting a discussion in the CIS Microsoft Azure Community (<https://workbench.cisecurity.org/communities/72>).

3.1 Virtual Machines

This section covers security recommendations to follow for the configuration of Virtual Machines on an Azure subscription.

3.1.1 Ensure only MFA Enabled Identities can Access Privileged Virtual Machine (Manual)

Profile Applicability:

- Level 2

Description:

Verify identities without MFA that can log in to a privileged virtual machine using separate login credentials. An adversary can leverage the access to move laterally and perform actions with the virtual machine's managed identity. Make sure the virtual machine only has necessary permissions, and revoke the admin-level permissions according to the principle of least privilege.

Rationale:

Integrating multi-factor authentication (MFA) as part of the organizational policy can greatly reduce the risk of an identity gaining control of valid credentials that may be used for additional tactics such as initial access, lateral movement, and collecting information. MFA can also be used to restrict access to cloud resources and APIs.

An Adversary may log into accessible cloud services within a compromised environment using Valid Accounts that are synchronized to move laterally and perform actions with the virtual machine's managed identity. The adversary may then perform management actions or access cloud-hosted resources as the logged-on managed identity.

Impact:

This recommendation requires the Entra ID P2 license to implement.

Ensure that identities provisioned to a virtual machine utilize an RBAC/ABAC group and are allocated a role using Azure PIM, and that the role settings require MFA or use another third-party PAM solution for accessing virtual machines.

Audit:

Audit from Azure Portal

1. Log in to the Azure portal.
2. Select the **Subscription**, then click on **Access control (IAM)**.
3. Click **Role : All** and click **All** to display the drop-down menu.
4. Type **Virtual Machine Administrator Login** and select **Virtual Machine Administrator Login**.
5. Review the list of identities that have been assigned the **Virtual Machine Administrator Login** role.
6. Go to **Microsoft Entra ID**.
7. For **Per-user MFA**:
 1. Under **Manage**, click **Users**.

2. Click **Per-user MFA**.
3. Ensure that none of the identities assigned the **Virtual Machine Administrator Login** role from step 4 have **Status** set to **disabled**.
8. For **Conditional Access**:
 1. Under **Manage**, click **Security**.
 2. Under **Protect**, click **Conditional Access**.
 3. Ensure that none of the identities assigned the **Virtual Machine Administrator Login** role from step 4 are exempt from a Conditional Access policy requiring MFA for all users.

Remediation:

Remediate from Azure Portal

1. Log in to the Azure portal.
 2. This can be remediated by enabling MFA for user, Removing user access or Reducing access of managed identities attached to virtual machines.
- Case I : Enable MFA for users having access on virtual machines.
 1. Go to **Microsoft Entra ID**.
 2. For **Per-user MFA**:
 1. Under **Manage**, click **Users**.
 2. Click **Per-user MFA**.
 3. For each user requiring remediation, check the box next to their name.
 4. Click **Enable MFA**.
 5. Click **Enable**.
 3. For **Conditional Access**:
 1. Under **Manage**, click **Security**.
 2. Under **Protect**, click **Conditional Access**.
 3. Update the Conditional Access policy requiring MFA for all users, removing each user requiring remediation from the **Exclude** list.
 - Case II : Removing user access on a virtual machine.
 1. Select the **Subscription**, then click on **Access control (IAM)**.
 2. Select **Role assignments** and search for **Virtual Machine Administrator Login** or **Virtual Machine User Login** or any role that provides access to log into virtual machines.
 3. Click on **Role Name**, Select **Assignments**, and remove identities with no MFA configured.
 - Case III : Reducing access of managed identities attached to virtual machines.
 1. Select the **Subscription**, then click on **Access control (IAM)**.
 2. Select **Role Assignments** from the top menu and apply filters on **Assignment type** as **Privileged administrator roles** and **Type** as **Virtual Machines**.
 3. Click on **Role Name**, Select **Assignments**, and remove identities access make sure this follows the least privileges principal.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	6.5 <u>Require MFA for Administrative Access</u> Require MFA for all administrative access accounts, where supported, on all enterprise assets, whether managed on-site or through a third-party provider.			
v7	4.5 <u>Use Multifactor Authentication For All Administrative Access</u> Use multi-factor authentication and encrypted channels for all administrative account access.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1078.002, T1078.004		

4 Database Services (reference)

SERVICE CATEGORY BENCHMARK AVAILABLE:

- "CIS Microsoft Azure Database Services Benchmark"

To better understand the relationship between the Foundations Benchmark and Services Benchmarks, please read the "Introduction" section of this document.

FULL RELOCATION - BE ADVISED!

- **ALL** - recommendations previously covered in the CIS Microsoft Azure Foundations Benchmark (this document) related to products in the Database category of services have been relocated to a Benchmark titled **CIS Microsoft Azure Database Services Benchmark**. This section now serves only as a reference.

Services addressed in the **CIS Microsoft Azure Database Services Benchmark**:

- Azure Cache for Redis
- Azure Cosmos DB
- Azure Data Factory
- Azure Database for MariaDB
- Azure Database for MySQL
- Azure Database for PostgreSQL
- Azure Database Migration Service
- Azure SQL
- Azure SQL Database
- Azure SQL Edge
- Azure SQL Managed Instance
- SQL Server on Azure Virtual Machines
- Table Storage
- Azure Managed Instance for Apache Cassandra
- Azure confidential ledger

Azure Product Directory Reference: <https://azure.microsoft.com/en-us/products#databases>

FEEDBACK REQUEST: Is there a specific service or recommendation in this section that you'd like to see addressed or improved? Let us know by making a ticket or starting a discussion in the CIS Microsoft Azure Community (<https://workbench.cisecurity.org/communities/72>).

5 Identity Services

SUBSTANTIAL RELOCATION - BE ADVISED!

MOST identity recommendations previously covered in the CIS Microsoft Azure Foundations Benchmark (this document) have been migrated or merged to the CIS Microsoft 365 Foundations Benchmark. This relocation includes all recommendations directly related to:

- Microsoft 365 tenancy
- Entra ID P1 and P2 licenses
- Conditional Access

The CIS Microsoft 365 Benchmarks community <https://workbench.cisecurity.org/communities/128> is now the authoritative source for Entra ID recommendations, and the **CIS Microsoft 365 Foundations Benchmark** is the document to reference for securing Microsoft 365 tenancy.

This relocation was performed in Q1 2026 in an effort to deduplicate security best practice recommendations related to Entra ID and centralize ongoing discussions about settings and features that exist within the Entra ID product space.

The identity recommendations that remain within the CIS Microsoft Azure Foundations Benchmark will be limited to the free tier of Entra ID (unlicensed) and discuss Azure-specific applications of identity settings and features which are not discussed in the Microsoft 365 Foundations Benchmark.

This section covers security best practice recommendations for products in the Azure Identity services category.

Azure Product Directory Reference: <https://azure.microsoft.com/en-us/products#identity>

FEEDBACK REQUEST: Is there a specific service or recommendation in this section that you'd like to see addressed or improved? Let us know by making a ticket or starting a discussion in the CIS Microsoft Azure Community (<https://workbench.cisecurity.org/communities/72>).

5.1 Security Defaults (Per-User MFA)

IMPORTANT: READ BELOW BEFORE PROCEEDING!

- If your organization pays for Microsoft Entra ID licensing (included in Microsoft 365 E3, E5, F5, or Business Premium, and EM&S E3 or E5 licenses) and **CAN** use Conditional Access, ignore the recommendations in this section and proceed to the Conditional Access section.
- If your organization is using the free tier of Entra ID (Office 365 E1, E3, or E5, and Microsoft 365 F1 or F3 licenses) and **CAN NOT** use Conditional Access, proceed with the Security Defaults guidance in this section, and ignore the recommendations in the Conditional Access section.

Conditional Access is preferred, but Security Defaults (Per-User MFA) is recommended only if Conditional Access isn't available.

Why is this **IMPORTANT**?

The Azure "Security Defaults" recommendations represent an entry-level set of recommendations (such as Multi-Factor Authentication) which will be relevant to organizations and tenants that are either just starting to use Azure, or are only utilizing a bare minimum feature set, and rely on the free license tier of Microsoft Entra ID. Security Defaults recommendations are intended to ensure that these use cases are still capable of establishing a strong baseline of secure configuration.

If your subscription is licensed to use Microsoft Entra ID P1 or P2, it is strongly recommended that the "Security Defaults" section (this section and the recommendations therein) be bypassed in favor of the use of "Conditional Access."

IMPORTANT: READ ABOVE BEFORE PROCEEDING!

5.1.1 Ensure that 'security defaults' is Enabled in Microsoft Entra ID (Automated)

Profile Applicability:

- Level 1

Description:

[IMPORTANT - Please read the section overview: If your organization pays for Microsoft Entra ID licensing (included in Microsoft 365 E3, E5, F5, or Business Premium, and EM&S E3 or E5 licenses) and **CAN** use Conditional Access, ignore the recommendations in this section and proceed to the Conditional Access section.]

Security defaults in Microsoft Entra ID make it easier to be secure and help protect your organization. Security defaults contain preconfigured security settings for common attacks.

Security defaults is available to everyone. The goal is to ensure that all organizations have a basic level of security enabled at no extra cost. You may turn on security defaults in the Azure portal.

Rationale:

Security defaults provide secure default settings that we manage on behalf of organizations to keep customers safe until they are ready to manage their own identity security settings.

For example, doing the following:

- Requiring all users and admins to register for MFA.
- Challenging users with MFA - when necessary, based on factors such as location, device, role, and task.
- Disabling authentication from legacy authentication clients, which can't do MFA.

Impact:

This recommendation should be implemented initially and then may be overridden by other service/product specific CIS Benchmarks. Administrators should also be aware that certain configurations in Microsoft Entra ID may impact other Microsoft services such as Microsoft 365.

Audit:

Audit from Azure Portal

To ensure security defaults is enabled in your directory:

1. From Azure Home select the Portal Menu.

2. Browse to **Microsoft Entra ID > Properties**.
3. Select **Manage security defaults**.
4. Under **Security defaults**, verify that **Enabled (recommended)** is selected.

Audit from Powershell

```
Connect-MgGraph -Scopes "Policy.Read.All"  
(Get-MgPolicyIdentitySecurityDefaultEnforcementPolicy).IsEnabled
```

Returned value should be **true**.

Audit from Azure CLI

```
az login  
az rest --method get --url  
'https://graph.microsoft.com/v1.0/policies/identitySecurityDefaultsEnforcemen  
tPolicy' --query "isEnabled"
```

Returned value should be **true**.

Remediation:

Remediate from Azure Portal

To enable security defaults in your directory:

1. From Azure Home select the Portal Menu.
2. Browse to **Microsoft Entra ID > Properties**.
3. Select **Manage security defaults**.
4. Under **Security defaults**, select **Enabled (recommended)**.
5. Select **Save**.

Remediate from Powershell

```
Connect-MgGraph -Scopes "Policy.ReadWrite.ApplicationConfiguration"  
Update-MgPolicyIdentitySecurityDefaultEnforcementPolicy -IsEnabled $true  
(Get-MgPolicyIdentitySecurityDefaultEnforcementPolicy).IsEnabled
```

Remediate from Azure CLI

```
az rest --method patch --url  
'https://graph.microsoft.com/v1.0/policies/identitySecurityDefaultsEnforcemen  
tPolicy' --body '{"isEnabled":true}'  
az rest --method get --url  
'https://graph.microsoft.com/v1.0/policies/identitySecurityDefaultsEnforcemen  
tPolicy' --query "isEnabled"
```

Default Value:

If your tenant was created on or after October 22, 2019, security defaults may already be enabled in your tenant.

References:

1. <https://learn.microsoft.com/en-us/entra/fundamentals/security-defaults>
2. <https://techcommunity.microsoft.com/t5/azure-active-directory-identity/introducing-security-defaults/ba-p/1061414>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-identity-management#im-2-protect-identity-and-authentication-systems>

Additional Information:

This recommendation differs from the [Microsoft 365 Benchmark](#). This is because the potential impact associated with disabling Security Defaults is dependent upon the security settings implemented in the environment. It is recommended that organizations disabling Security Defaults implement appropriate security settings to replace the settings configured by Security Defaults.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.8 Uninstall or Disable Unnecessary Services on Enterprise Assets and Software Uninstall or disable unnecessary services on enterprise assets and software, such as an unused file sharing service, web application module, or service function.		●	●
v8	6.3 Require MFA for Externally-Exposed Applications Require all externally-exposed enterprise or third-party applications to enforce MFA, where supported. Enforcing MFA through a directory service or SSO provider is a satisfactory implementation of this Safeguard.	●	●	●
v8	6.5 Require MFA for Administrative Access Require MFA for all administrative access accounts, where supported, on all enterprise assets, whether managed on-site or through a third-party provider.	●	●	●
v7	16.3 Require Multi-factor Authentication Require multi-factor authentication for all user accounts, on all systems, whether managed onsite or by a third-party provider.		●	●

5.1.2 Ensure that 'Require Multifactor Authentication to register or join devices with Microsoft Entra' is set to 'Yes' (Manual)

Profile Applicability:

- Level 1

Description:

NOTE: This recommendation is only relevant if your subscription is using Per-User MFA. If your organization is licensed to use Conditional Access, the preferred method of requiring MFA to join devices to Entra ID is to use a Conditional Access policy (see additional information below for link).

Joining or registering devices to Microsoft Entra ID should require multi-factor authentication.

Rationale:

Multi-factor authentication is recommended when adding devices to Microsoft Entra ID. When set to **Yes**, users who are adding devices from the internet must first use the second method of authentication before their device is successfully added to the directory. This ensures that rogue devices are not added to the domain using a compromised user account.

Impact:

A slight impact of additional overhead, as Administrators will now have to approve every access to the domain.

Audit:

Audit from Azure Portal

1. From Azure Home select the Portal Menu
2. Select **Microsoft Entra ID**
3. Under **Manage**, select **Devices**
4. Under **Manage**, select **Device settings**
5. Under **Microsoft Entra join and registration settings**, ensure that **Require Multifactor Authentication to register or join devices with Microsoft Entra** is set to **Yes**

Remediation:

Remediate from Azure Portal

1. From Azure Home select the Portal Menu
2. Select **Microsoft Entra ID**

3. Under **Manage**, select **Devices**
4. Under **Manage**, select **Device settings**
5. Under **Microsoft Entra join and registration settings**, set **Require Multifactor Authentication to register or join devices with Microsoft Entra** to **Yes**
6. Click **Save**

Default Value:

By default, **Require Multifactor Authentication to register or join devices with Microsoft Entra** is set to **No**.

References:

1. <https://learn.microsoft.com/en-us/entra/identity/conditional-access/policy-all-users-device-registration>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-identity-management#im-6-use-strong-authentication-controls>

Additional Information:

If Conditional Access is available, this recommendation should be bypassed in favor of the Conditional Access implementation of requiring Multifactor Authentication to register or join devices with Microsoft Entra.

<https://learn.microsoft.com/en-us/entra/identity/conditional-access/policy-all-users-device-registration>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	6.3 <u>Require MFA for Externally-Exposed Applications</u> Require all externally-exposed enterprise or third-party applications to enforce MFA, where supported. Enforcing MFA through a directory service or SSO provider is a satisfactory implementation of this Safeguard.			
v8	6.4 <u>Require MFA for Remote Network Access</u> Require MFA for remote network access.			
v7	16.3 <u>Require Multi-factor Authentication</u> Require multi-factor authentication for all user accounts, on all systems, whether managed onsite or by a third-party provider.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1078.004	TA0001	M1032

5.1.3 Ensure that 'multifactor authentication' is 'enabled' For All Users (Automated)

Profile Applicability:

- Level 1

Description:

[IMPORTANT - Please read the section overview: If your organization pays for Microsoft Entra ID licensing (included in Microsoft 365 E3, E5, F5, or Business Premium, and EM&S E3 or E5 licenses) and **CAN** use Conditional Access, ignore the recommendations in this section and proceed to the Conditional Access section.]

Enable multifactor authentication for all users.

Note: Since 2024, Azure has been rolling out mandatory multifactor authentication. For more information:

- <https://azure.microsoft.com/en-us/blog/announcing-mandatory-multi-factor-authentication-for-azure-sign-in>
- <https://learn.microsoft.com/en-us/entra/identity/authentication/concept-mandatory-multifactor-authentication>

Rationale:

Multifactor authentication requires an individual to present a minimum of two separate forms of authentication before access is granted. Multifactor authentication provides additional assurance that the individual attempting to gain access is who they claim to be. With multifactor authentication, an attacker would need to compromise at least two different authentication mechanisms, increasing the difficulty of compromise and thus reducing the risk.

Impact:

Users would require two forms of authentication before any access is granted. Additional administrative time will be required for managing dual forms of authentication when enabling multifactor authentication.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Entra ID**.
2. Under **Manage**, click **Users**.
3. Click **Per-user MFA** from the top menu.
4. Ensure that **Status** is **enabled** for all users.

Audit from PowerShell

Run the following Graph PowerShell command:

```
get-mguser -All | where {$_.StrongAuthenticationMethods.Count -eq 0} |  
Select-Object -Property UserPrincipalName
```

If the output contains any **UserPrincipalName**, then this recommendation is non-compliant.

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Entra ID**.
2. Under **Manage**, click **Users**.
3. Click **Per-user MFA** from the top menu.
4. Click the box next to a user with **Status disabled**.
5. Click **Enable MFA**.
6. Click **Enable**.
7. Repeat steps 1-6 for each user requiring remediation.

Other options within Azure Portal

- <https://learn.microsoft.com/en-us/entra/identity/authentication/tutorial-enable-azure-mfa>
- <https://learn.microsoft.com/en-us/entra/identity/authentication/howto-mfa-mfasettings>
- <https://learn.microsoft.com/en-us/entra/identity/conditional-access/policy-old-require-mfa-admin>
- <https://learn.microsoft.com/en-us/entra/identity/authentication/howto-mfa-getstarted#enable-multi-factor-authentication-with-conditional-access>

Default Value:

Multifactor authentication is not enabled for all users by default. Starting in 2024, multifactor authentication is enabled for administrative accounts by default.

References:

1. <https://learn.microsoft.com/en-us/entra/identity/authentication/concept-mfa-howitworks>
2. <https://learn.microsoft.com/en-us/entra/identity/authentication/concept-mandatory-multifactor-authentication>
3. <https://azure.microsoft.com/en-us/blog/announcing-mandatory-multi-factor-authentication-for-azure-sign-in/>
4. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-identity-management#im-4-authenticate-server-and-services>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	6.3 <u>Require MFA for Externally-Exposed Applications</u> Require all externally-exposed enterprise or third-party applications to enforce MFA, where supported. Enforcing MFA through a directory service or SSO provider is a satisfactory implementation of this Safeguard.			
v8	6.4 <u>Require MFA for Remote Network Access</u> Require MFA for remote network access.			
v7	16.3 <u>Require Multi-factor Authentication</u> Require multi-factor authentication for all user accounts, on all systems, whether managed onsite or by a third-party provider.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1098	TA0003	M1032

5.1.4 Ensure that 'Allow users to remember multifactor authentication on devices they trust' is Disabled (Manual)

Profile Applicability:

- Level 1

Description:

[IMPORTANT - Please read the section overview: If your organization pays for Microsoft Entra ID licensing (included in Microsoft 365 E3, E5, F5, or Business Premium, and EM&S E3 or E5 licenses) and **CAN** use Conditional Access, ignore the recommendations in this section and proceed to the Conditional Access section.]

Do not allow users to remember multi-factor authentication on devices.

Rationale:

Remembering Multi-Factor Authentication (MFA) for devices and browsers allows users to have the option to bypass MFA for a set number of days after performing a successful sign-in using MFA. This can enhance usability by minimizing the number of times a user may need to perform two-step verification on the same device. However, if an account or device is compromised, remembering MFA for trusted devices may affect security. Hence, it is recommended that users not be allowed to bypass MFA.

Impact:

For every login attempt, the user will be required to perform multi-factor authentication.

Audit:

Audit from Azure Portal

1. From Azure Home select the Portal Menu
2. Select **Microsoft Entra ID**
3. Under **Manage**, click **Users**
4. Click the **Per-user MFA** button on the top bar
5. Click on **Service settings**
6. Ensure that **Allow users to remember multi-factor authentication on devices they trust** is not enabled

Remediation:

Remediate from Azure Portal

1. From Azure Home select the Portal Menu
2. Select **Microsoft Entra ID**
3. Under **Manage**, click **Users**

4. Click the **Per-user MFA** button on the top bar
5. Click on **Service settings**
6. Uncheck the box next to **Allow users to remember multi-factor authentication on devices they trust**
7. Click **Save**

Default Value:

By default, **Allow users to remember multi-factor authentication on devices they trust** is disabled.

References:

1. <https://learn.microsoft.com/en-us/entra/identity/authentication/howto-mfa-mfasettings#remember-multi-factor-authentication-for-devices-that-users-trust>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/security-controls-v3-identity-management#im-4-use-strong-authentication-controls-for-all-azure-active-directory-based-access>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-identity-management#im-6-use-strong-authentication-controls>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	6.3 <u>Require MFA for Externally-Exposed Applications</u> Require all externally-exposed enterprise or third-party applications to enforce MFA, where supported. Enforcing MFA through a directory service or SSO provider is a satisfactory implementation of this Safeguard.			
v8	6.4 <u>Require MFA for Remote Network Access</u> Require MFA for remote network access.			
v7	16.3 <u>Require Multi-factor Authentication</u> Require multi-factor authentication for all user accounts, on all systems, whether managed onsite or by a third-party provider.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1098	TA0003	M1032

5.2 Conditional Access (reference)

SUBSTANTIAL RELOCATION - BE ADVISED!

MOST identity recommendations previously covered in the CIS Microsoft Azure Foundations Benchmark (this document) have been migrated or merged to the CIS Microsoft 365 Foundations Benchmark. This relocation includes all recommendations directly related to:

- Microsoft 365 tenancy
- Entra ID P1 and P2 licenses
- Conditional Access

The CIS Microsoft 365 Benchmarks community <https://workbench.cisecurity.org/communities/128> is now the authoritative source for Entra ID recommendations, and the **CIS Microsoft 365 Foundations Benchmark** is the document to reference for securing Microsoft 365 tenancy.

This relocation was performed in Q1 2026 in an effort to deduplicate security best practice recommendations related to Entra ID and centralize ongoing discussions about settings and features that exist within the Entra ID product space.

The identity recommendations that remain within the CIS Microsoft Azure Foundations Benchmark will be limited to the free tier of Entra ID (unlicensed) and discuss Azure-specific applications of identity settings and features which are not discussed in the Microsoft 365 Foundations Benchmark.

For most Azure tenants, and certainly for organizations with a significant use of Microsoft Entra ID, Conditional Access policies are recommended and preferred.

Conditional Access requires one of the following plans:

- Microsoft Entra ID P1 or P2
- Microsoft 365 Business Premium
- Microsoft 365 E3 or E5
- Microsoft 365 F1, F3, F5 Security and F5 Security + Compliance
- Enterprise Mobility & Security E3 or E5

5.3 Periodic Identity Reviews

Security Best Practices for Identity services should include operational reviews that periodically ensure the integrity and necessity of accounts and permissions. These operational practices should be performed regularly on a cadence that is based on your organization's policy or compliance requirements.

NOTE: The recommendations in this section may not have a precise audit or remediation procedure because they may not be a configurable setting as much as they are an operative task that should be performed on a periodic basis.

5.3.1 Ensure that Azure Admin Accounts Are Not Used for Daily Operations (Manual)

Profile Applicability:

- Level 1

Description:

Microsoft Azure admin accounts should not be used for routine, non-administrative tasks.

Rationale:

Using admin accounts for daily operations increases the risk of accidental misconfigurations and security breaches.

Impact:

Minor administrative overhead includes managing separate accounts, enforcing stricter access controls, and potential licensing costs for advanced security features.

Audit:

Audit from Azure Portal

Monitor:

1. Go to **Monitor**.
2. Click **Activity log**.
3. Review the activity log and ensure that admin accounts are not being used for daily operations.

Microsoft Entra ID:

1. Go to **Microsoft Entra ID**.
2. Under **Monitoring**, click **Sign-in logs**.
3. Review the sign-in logs and ensure that admin accounts are not being accessed more frequently than necessary.

Remediation:

If admin accounts are being used for daily operations, consider the following:

- Monitor and alert on unusual activity.
- Enforce the principle of least privilege.
- Revoke any unnecessary administrative access.
- Use Conditional Access to limit access to resources.
- Ensure that administrators have separate admin and user accounts.

- Use Microsoft Entra ID Protection to detect, investigate, and remediate identity-based risks.
- Use Privileged Identity Management (PIM) in Microsoft Entra ID to limit standing administrator access to privileged roles, discover who has access, and review privileged access.

References:

1. <https://learn.microsoft.com/en-us/security/privileged-access-workstations/critical-impact-accounts>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.4 Restrict Administrator Privileges to Dedicated Administrator Accounts Restrict administrator privileges to dedicated administrator accounts on enterprise assets. Conduct general computing activities, such as internet browsing, email, and productivity suite use, from the user's primary, non-privileged account.			
v7	4.3 Ensure the Use of Dedicated Administrative Accounts Ensure that all users with administrative account access use a dedicated or secondary account for elevated activities. This account should only be used for administrative activities and not internet browsing, email, or similar activities.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1098	TA0004	M1026

5.3.2 *Ensure that Guest Users are Reviewed on a Regular Basis (Manual)*

Profile Applicability:

- Level 1

Description:

Microsoft Entra ID has native and extended identity functionality allowing you to invite people from outside your organization to be guest users in your cloud account and sign in with their own work, school, or social identities.

While an automated assessment procedure exists for this recommendation, the assessment status remains manual. Evaluating the appropriateness of guest users requires a manual review, as it depends on the specific needs and context of each organization and environment.

Rationale:

Guest users are typically added outside your employee on-boarding/off-boarding process and could potentially be overlooked indefinitely. To prevent this, guest users should be reviewed on a regular basis. During this audit, guest users should also be determined to not have administrative privileges.

Impact:

Before removing guest users, determine their use and scope. Like removing any user, there may be unforeseen consequences to systems if an account is removed without careful consideration.

Audit:

Audit from Azure Portal

1. From Azure Home select the Portal Menu
2. Select **Microsoft Entra ID**
3. Under **Manage**, select **Users**
4. Click on **Add filter**
5. Select **User type**
6. Select **Guest** from the Value dropdown
7. Click **Apply**
8. Audit the listed guest users

Audit from Azure CLI

```
az ad user list --query "[?userType=='Guest']"
```

Ensure all users listed are still required and not inactive.

Audit from Azure PowerShell

```
Get-AzureADUser |Where-Object {$_.UserType -like "Guest"} |Select-Object  
DisplayName, UserPrincipalName, UserType -Unique
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [e9ac8f8e-ce22-4355-8f04-99b911d6be52](#) - **Name:** 'Guest accounts with read permissions on Azure resources should be removed'
- **Policy ID:** [94e1c2ac-cbbe-4cac-a2b5-389c812dee87](#) - **Name:** 'Guest accounts with write permissions on Azure resources should be removed'
- **Policy ID:** [339353f6-2387-4a45-abe4-7f529d121046](#) - **Name:** 'Guest accounts with owner permissions on Azure resources should be removed'

Remediation:

Remediate from Azure Portal

1. From Azure Home select the Portal Menu
2. Select **Microsoft Entra ID**
3. Under **Manage**, select **Users**
4. Click on **Add filter**
5. Select **User type**
6. Select **Guest** from the Value dropdown
7. Click **Apply**
8. Check the box next to all **Guest** users that are no longer required or are inactive
9. Click **Delete**
10. Click **OK**

Remediate from Azure CLI

Before deleting the user, set it to inactive using the ID from the Audit Procedure to determine if there are any dependent systems.

```
az ad user update --id <exampleaccountid@domain.com> --account-enabled  
{false}
```

After determining that there are no dependent systems, delete the user.

```
Remove-AzureADUser -ObjectId <exampleaccountid@domain.com>
```

Remediate from Azure PowerShell

Before deleting the user, set it to inactive using the ID from the Audit Procedure to determine if there are any dependent systems.

```
Set-AzureADUser -ObjectId "<exampleaccountid@domain.com>" -AccountEnabled false
```

After determining that there are no dependent systems, delete the user.

```
PS C:\>Remove-AzureADUser -ObjectId <exampleaccountid@domain.com>
```

Default Value:

By default no guest users are created.

References:

1. <https://learn.microsoft.com/en-us/entra/external-id/user-properties>
2. <https://learn.microsoft.com/en-us/entra/fundamentals/how-to-create-delete-users#delete-a-user>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-privileged-access#pa-4-review-and-reconcile-user-access-regularly>
4. <https://www.microsoft.com/en-us/security/business/microsoft-entra-pricing>
5. <https://learn.microsoft.com/en-us/entra/identity/monitoring-health/howto-manage-inactive-user-accounts>
6. <https://learn.microsoft.com/en-us/entra/fundamentals/users-restore>

Additional Information:

It is good practice to use a dynamic security group to manage guest users.

To create the dynamic security group:

1. Navigate to the 'Microsoft Entra ID' blade in the Azure Portal
2. Select the 'Groups' item
3. Create new
4. Type of 'dynamic'
5. Use the following dynamic selection rule. "(user.userType -eq "Guest")"
6. Once the group has been created, select access reviews option and create a new access review with a period of monthly and send to relevant administrators for review.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.1 Establish and Maintain an Inventory of Accounts Establish and maintain an inventory of all accounts managed in the enterprise. The inventory must include both user and administrator accounts. The inventory, at a minimum, should contain the person's name, username, start/stop dates, and department. Validate that all active accounts are authorized, on a recurring schedule at a minimum quarterly, or more frequently.			
v8	5.3 Disable Dormant Accounts Delete or disable any dormant accounts after a period of 45 days of inactivity, where supported.			
v7	16.6 Maintain an Inventory of Accounts Maintain an inventory of all accounts organized by authentication system.			
v7	16.8 Disable Any Unassociated Accounts Disable any account that cannot be associated with a business process or business owner.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1078.004	TA0001	M1018

5.3.3 Ensure That Use of the 'User Access Administrator' Role is Restricted (Automated)

Profile Applicability:

- Level 1

Description:

The User Access Administrator role grants the ability to view all resources and manage access assignments at any subscription or management group level within the tenant. Due to its high privilege level, this role assignment should be removed immediately after completing the necessary changes at the root scope to minimize security risks.

Rationale:

The User Access Administrator role provides extensive access control privileges. Unnecessary assignments heighten the risk of privilege escalation and unauthorized access. Removing the role immediately after use minimizes security exposure.

Impact:

Increased administrative effort to manage and remove role assignments appropriately.

Audit:

Audit from Azure Portal

1. From Azure Home select the Portal Menu.
2. Select **Subscriptions**.
3. Select a subscription.
4. Select **Access control (IAM)**.
5. In the horizontal menu select **Roles**.
6. Search for **User Access Administrator**.
7. In the **Details** column select **View**.
8. In the horizontal menu select **Assignments**.
9. Audit the list for assignments.

Audit from Azure CLI

Run the following command:

```
az role assignment list --role "User Access Administrator" --scope "/"
```

Ensure that the command does not return any **User Access Administrator** role assignment information.

Remediation:

Remediate from Azure Portal

1. From Azure Home select the Portal Menu.
2. Select **Subscriptions**.
3. Select a subscription.
4. Select **Access control (IAM)**.
5. Select **Access control (IAM)**.
6. In the horizontal menu select **Roles**.
7. Search for **User Access Administrator**.
8. In the **Details** column select **View**.
9. In the horizontal menu select **Assignments**.
10. Select the user you wish to remove.
11. Select **Remove**.

Remediate from Azure CLI

Run the following command:

```
az role assignment delete --role "User Access Administrator" --scope "/"
```

Default Value:

By default the **User Access Administrator** role is not assigned.

References:

1. <https://learn.microsoft.com/en-us/azure/role-based-access-control/built-in-roles>
2. <https://learn.microsoft.com/en-us/azure/role-based-access-control/elevate-access-global-admin>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	6.8 Define and Maintain Role-Based Access Control Define and maintain role-based access control, through determining and documenting the access rights necessary for each role within the enterprise to successfully carry out its assigned duties. Perform access control reviews of enterprise assets to validate that all privileges are authorized, on a recurring schedule at a minimum annually, or more frequently.			●
v7	4.1 Maintain Inventory of Administrative Accounts Use automated tools to inventory all administrative accounts, including domain and local accounts, to ensure that only authorized individuals have elevated privileges.		●	●

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1548	TA0004	M1018

5.3.4 Ensure that All 'Privileged' Role Assignments are Periodically Reviewed (Manual)

Profile Applicability:

- Level 1

Description:

Periodic review of privileged role assignments is performed to ensure that the privileged roles assigned to users are accurate and appropriate.

Rationale:

Privileged roles are crown jewel assets that can be used by malicious insiders, threat actors, and even through mistake to significantly damage an organization in numerous ways. These roles should be periodically reviewed to:

- identify lingering permissions assignment (e.g. an administrator has been terminated, the administrator account is being retained, but the permissions are no longer necessary and has not been properly addressed by process)
- detect lateral movement through privilege escalation (e.g. an account with administrative permission has been compromised and is elevating other accounts in an attempt to circumvent detection mechanisms)

Impact:

Increased administrative effort to manage and remove role assignments appropriately.

Audit:

Audit from Azure Portal

1. From Azure Home select the Portal Menu.
2. Select **Subscriptions**.
3. Select a subscription.
4. Select **Access control (IAM)**.
5. In the horizontal menu, select **Roles**.
6. In the second horizontal menu select **Privileged administrator roles**.
7. For each role, audit by selecting the **View** link under the **Details** column.
8. Once selected, choose **Assignments** to view the users and resources that are assigned the role.

NOTE: The judgement of what constitutes 'appropriate and current' assignments requires a clear understanding of your organization's personnel, systems, policy, and security requirements. This cannot be effectively prescribed in procedure.

Remediation:

References:

1. <https://learn.microsoft.com/en-us/azure/role-based-access-control/built-in-roles>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	6.8 Define and Maintain Role-Based Access Control Define and maintain role-based access control, through determining and documenting the access rights necessary for each role within the enterprise to successfully carry out its assigned duties. Perform access control reviews of enterprise assets to validate that all privileges are authorized, on a recurring schedule at a minimum annually, or more frequently.			●
v7	4.1 Maintain Inventory of Administrative Accounts Use automated tools to inventory all administrative accounts, including domain and local accounts, to ensure that only authorized individuals have elevated privileges.		●	●

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1548	TA0004	M1018

5.3.5 Ensure Disabled User Accounts do not Have Read, Write, or Owner Permissions (Manual)

Profile Applicability:

- Level 1

Description:

Ensure that any roles granting read, write, or owner permissions are removed from disabled Azure user accounts.

While an automated assessment procedure exists for this recommendation, the assessment status remains manual. Removing role assignments from disabled user accounts depends on the context and requirements of each organization and environment.

Rationale:

Disabled accounts should not retain access to resources, as this poses a security risk. Removing role assignments mitigates potential unauthorized access and enforces the principle of least privilege.

Impact:

Ensure disabled accounts are not relied on for break glass or automated processes before removing roles to avoid service disruption.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Entra ID**.
2. Under **Manage**, click **Users**.
3. Click **Add filter**.
4. Click **Account enabled**.
5. Click the toggle switch to set the value to **No**.
6. Click **Apply**.
7. Click the **Display name** of a disabled user account.
8. Click **Azure role assignments**.
9. Ensure that no read, write, or owner roles are assigned to the user account.
10. Repeat steps 7-9 for each disabled user account.

Audit from PowerShell

Run the following command to connect to Microsoft Entra ID:

```
Connect-AzureAD
```

Run the following command to list users:

```
Get-AzureADUser
```

Run the following command to get a user:

```
$user = Get-AzureADUser -ObjectId <object-id>
```

Run the following command to get the **AccountEnabled** setting for the user:

```
$user.AccountEnabled
```

If **AccountEnabled** is **False**, run the following command to get the role assignments for the user:

```
Get-AzRoleAssignment -ObjectId $user.ObjectId
```

Ensure that no read, write, or owner roles are assigned to the user.

Repeat for each user.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~Definitions

- **Policy ID:** [0cfea604-3201-4e14-88fc-fae4c427a6c5](#) - **Name:** 'Blocked accounts with owner permissions on Azure resources should be removed'
- **Policy ID:** [8d7e1fde-fe26-4b5f-8108-f8e432cbc2be](#) - **Name:** 'Blocked accounts with read and write permissions on Azure resources should be removed'

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Entra ID**.
2. Under **Manage**, click **Users**.
3. Click **Add filter**.
4. Click **Account enabled**.
5. Click the toggle switch to set the value to **No**.
6. Click **Apply**.
7. Click the **Display name** of a disabled user account with read, write, or owner roles assigned.
8. Click **Azure role assignments**.
9. Click the name of a read, write, or owner role.
10. Click **Assignments**.
11. Click **Remove** in the row for the disabled user account.
12. Click **Yes**.
13. Repeat steps 7-12 for disabled user accounts requiring remediation.

Remediate from PowerShell

For each account requiring remediation, run the following command to remove an assigned role:

```
Remove-AzRoleAssignment -ObjectId $user.ObjectId -RoleDefinitionName <role-definition-name>
```

Default Value:

Disabled user accounts retain their prior role assignments.

References:

1. <https://learn.microsoft.com/en-us/powershell/module/az.resources/get-azaduser>
2. <https://learn.microsoft.com/en-us/powershell/module/az.resources/get-azroleassignment>
3. <https://learn.microsoft.com/en-us/powershell/module/az.resources/remove-azroleassignment>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.1 Establish and Maintain an Inventory of Accounts Establish and maintain an inventory of all accounts managed in the enterprise. The inventory must include both user and administrator accounts. The inventory, at a minimum, should contain the person's name, username, start/stop dates, and department. Validate that all active accounts are authorized, on a recurring schedule at a minimum quarterly, or more frequently.			
v7	16.6 Maintain an Inventory of Accounts Maintain an inventory of all accounts organized by authentication system.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1078	TA0004	M1018

5.3.6 Ensure 'Tenant Creator' Role Assignments are Periodically Reviewed (Manual)

Profile Applicability:

- Level 1

Description:

Perform a periodic review of the Tenant Creator role assignment to ensure that the assignments are accurate and appropriate.

This recommendation should be applied alongside the recommendation **"Ensure that 'Restrict non-admin users from creating tenants' is set to 'Yes'"**.

Rationale:

Unnecessary assignments increase the risk of privilege escalation and unauthorized access.

Impact:

Verify that the Tenant Creator role is no longer required by any assignments before removal to avoid disruption of critical functions.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Entra ID**.
2. Under **Manage**, click **Roles and administrators**.
3. In the search bar, type **Tenant Creator**.
4. Click the role.
5. Review the assignments and ensure that they are appropriate.

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Entra ID**.
2. Under **Manage**, click **Roles and administrators**.
3. In the search bar, type **Tenant Creator**.
4. Click the role.
5. Click the name of an assignment.
6. Check the box next to the **Tenant Creator** role.
7. Click **X Remove assignments**.
8. Click **Yes**.
9. Repeat steps 1-8 for each assignment requiring remediation.

Default Value:

The Tenant Creator role is not assigned by default.

References:

1. <https://learn.microsoft.com/en-us/azure/active-directory-b2c/tenant-management-check-tenant-creation-permission>
2. <https://learn.microsoft.com/en-us/entra/identity/role-based-access-control/permissions-reference#tenant-creator>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.4 Restrict Administrator Privileges to Dedicated Administrator Accounts Restrict administrator privileges to dedicated administrator accounts on enterprise assets. Conduct general computing activities, such as internet browsing, email, and productivity suite use, from the user's primary, non-privileged account.			
v7	4.3 Ensure the Use of Dedicated Administrative Accounts Ensure that all users with administrative account access use a dedicated or secondary account for elevated activities. This account should only be used for administrative activities and not internet browsing, email, or similar activities.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1098	TA0004	M1026

5.3.7 Ensure All Non-privileged Role Assignments are Periodically Reviewed (Manual)

Profile Applicability:

- Level 1

Description:

Perform a periodic review of non-privileged role assignments to ensure that the non-privileged roles assigned to users are appropriate.

Note: Determining 'appropriate' assignments requires a clear understanding of your organization's personnel, systems, policies, and security requirements. This cannot be effectively prescribed in a procedure.

Rationale:

To ensure the principle of least privilege is followed, non-privileged role assignments should be reviewed periodically to confirm that users are granted only the minimum level of permissions they need to perform their tasks.

Impact:

Increased administrative effort to manage and remove role assignments appropriately.

Audit:

Audit from Azure Portal

1. Go to **Subscriptions**.
2. Click the name of a subscription.
3. Click **Access control (IAM)**.
4. Click **Role assignments**.
5. Click **Job function roles**.
6. For each role, ensure the assignments are appropriate.
7. Repeat steps 1-6 for each subscription.

Remediation:

Remediate from Azure Portal

1. Go to **Subscriptions**.
2. Click the name of a subscription.
3. Click **Access control (IAM)**.
4. Click **Role assignments**.
5. Click **Job function roles**.
6. Check the box next to any inappropriate assignments.

7. Click **Delete**.
8. Click **Yes**.
9. Repeat steps 1-8 for each subscription.

Default Value:

Users do not have non-privileged roles assigned to them by default.

References:

1. <https://learn.microsoft.com/en-us/azure/role-based-access-control/role-assignments>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	6.8 Define and Maintain Role-Based Access Control Define and maintain role-based access control, through determining and documenting the access rights necessary for each role within the enterprise to successfully carry out its assigned duties. Perform access control reviews of enterprise assets to validate that all privileges are authorized, on a recurring schedule at a minimum annually, or more frequently.			●
v7	16.6 Maintain an Inventory of Accounts Maintain an inventory of all accounts organized by authentication system.		●	●

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1548	TA0004	M1018

5.4 Ensure that No Custom Subscription Administrator Roles Exist (Automated)

Profile Applicability:

- Level 1

Description:

The principle of least privilege should be followed and only necessary privileges should be assigned instead of allowing full administrative access.

Rationale:

Custom roles in Azure with administrative access can obfuscate the permissions granted and introduce complexity and blind spots to the management of privileged identities. For less mature security programs without regular identity audits, the creation of Custom roles should be avoided entirely. For more mature security programs with regular identity audits, Custom Roles should be audited for use and assignment, used minimally, and the principle of least privilege should be observed when granting permissions

Impact:

Subscriptions will need to be handled by Administrators with permissions.

Audit:

Audit from Azure Portal

1. From Azure Home select the Portal Menu.
2. Select **Subscriptions**.
3. Select a subscription.
4. Select **Access control (IAM)**.
5. Select **Roles**.
6. Click **Type** and select **Custom role** from the drop-down menu.
7. Select **View** next to a role.
8. Select **JSON**.
9. Check for **assignableScopes** set to the subscription, and **actions** set to *****.
10. Repeat steps 7-9 for each custom role.

Audit from Azure CLI

List custom roles:

```
az role definition list --custom-role-only True
```

Check for entries with **assignableScope** of the **subscription**, and an action of *****

Audit from PowerShell

```
Connect-AzAccount
Get-AzRoleDefinition |Where-Object {($_.IsCustom -eq $true) -and
($_.Actions.contains('*'))}
```

Check the output for **AssignableScopes** value set to the subscription.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [a451c1ef-c6ca-483d-87ed-f49761e3ffb5](#) - **Name:** 'Audit usage of custom RBAC roles'

Remediation:

Remediate from Azure Portal

1. From Azure Home select the Portal Menu.
2. Select **Subscriptions**.
3. Select a subscription.
4. Select **Access control (IAM)**.
5. Select **Roles**.
6. Click **Type** and select **Custom role** from the drop-down menu.
7. Check the box next to each role which grants subscription administrator privileges.
8. Select **Delete**.
9. Select **Yes**.

Remediate from Azure CLI

List custom roles:

```
az role definition list --custom-role-only True
```

Check for entries with **assignableScope** of the **subscription**, and an action of *****.

To remove a violating role:

```
az role definition delete --name <role name>
```

Note that any role assignments must be removed before a custom role can be deleted. Ensure impact is assessed before deleting a custom role granting subscription administrator privileges.

Default Value:

By default, no custom owner roles are created.

References:

1. <https://learn.microsoft.com/en-us/azure/cost-management-billing/manage/add-change-subscription-administrator>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-privileged-access#pa-1-separate-and-limit-highly-privileged-administrative-users>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-privileged-access#pa-3-manage-lifecycle-of-identities-and-entitlements>
4. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-governance-strategy#gs-2-define-and-implement-enterprise-segmentation-separation-of-duties-strategy>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-governance-strategy#gs-6-define-and-implement-identity-and-privileged-access-strategy>
6. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-privileged-access#pa-7-follow-just-enough-administration-least-privilege-principle>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.4 Restrict Administrator Privileges to Dedicated Administrator Accounts Restrict administrator privileges to dedicated administrator accounts on enterprise assets. Conduct general computing activities, such as internet browsing, email, and productivity suite use, from the user's primary, non-privileged account.			
v8	6.8 Define and Maintain Role-Based Access Control Define and maintain role-based access control, through determining and documenting the access rights necessary for each role within the enterprise to successfully carry out its assigned duties. Perform access control reviews of enterprise assets to validate that all privileges are authorized, on a recurring schedule at a minimum annually, or more frequently.			
v7	4.1 Maintain Inventory of Administrative Accounts Use automated tools to inventory all administrative accounts, including domain and local accounts, to ensure that only authorized individuals have elevated privileges.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1098.003	TA0003	M1026

5.5 Ensure that a Custom Role is Assigned Permissions for Administering Resource Locks (Manual)

Profile Applicability:

- Level 2

Description:

Resource locking is a powerful protection mechanism that can prevent inadvertent modification or deletion of resources within Azure subscriptions and resource groups, and it is a recommended NIST configuration.

Rationale:

Given that the resource lock functionality is outside of standard Role-Based Access Control (RBAC), it would be prudent to create a resource lock administrator role to prevent inadvertent unlocking of resources.

Impact:

By adding this role, specific permissions may be granted for managing only resource locks rather than needing to provide the broad Owner or User Access Administrator role, reducing the risk of the user being able to cause unintentional damage.

Audit:

Audit from Azure Portal

1. In the Azure portal, navigate to a subscription or resource group.
2. Click **Access control (IAM)**.
3. Click **Roles**.
4. Click **Type : All**.
5. Click to view the drop-down menu.
6. Select **Custom role**.
7. Click **View** in the **Details** column of a custom role.
8. Review the role permissions.
9. Click **Assignments** and review the assignments.
10. Click the **X** to exit the custom role details page.
11. Repeat steps 7-10. Ensure that at least one custom role exists that assigns the **Microsoft.Authorization/locks** permission to appropriate members.
12. Repeat steps 1-11 for each subscription or resource group.

Remediation:

Remediate from Azure Portal

1. In the Azure portal, navigate to a subscription or resource group.
2. Click **Access control (IAM)**.
3. Click **+ Add**.
4. Click **Add custom role**.
5. In the **Custom role name** field enter **Resource Lock Administrator**.
6. In the **Description** field enter **Can Administer Resource Locks**.
7. For **Baseline permissions** select **Start from scratch**.
8. Click **Next**.
9. Click **Add permissions**.
10. In the **Search for a permission** box, type **Microsoft.Authorization/locks**.
11. Click the result.
12. Check the box next to **Permission**.
13. Click **Add**.
14. Click **Review + create**.
15. Click **Create**.
16. Click **OK**.
17. Click **+ Add**.
18. Click **Add role assignment**.
19. In the **Search by role name, description, permission, or ID** box, type **Resource Lock Administrator**.
20. Select the role.
21. Click **Next**.
22. Click **+ Select members**.
23. Select appropriate members.
24. Click **Select**.
25. Click **Review + assign**.
26. Click **Review + assign** again.
27. Repeat steps 1-26 for each subscription or resource group requiring remediation.

Remediate from PowerShell:

Below is a PowerShell definition for a resource lock administrator role created at an Azure Management group level

```
Import-Module Az.Accounts
Connect-AzAccount

$role = Get-AzRoleDefinition "User Access Administrator"
$role.Id = $null
$role.Name = "Resource Lock Administrator"
$role.Description = "Can Administer Resource Locks"
$role.Actions.Clear()
$role.Actions.Add("Microsoft.Authorization/locks/*")
$role.AssignableScopes.Clear()

* Scope at the Management group level Management group

$role.AssignableScopes.Add("/providers/Microsoft.Management/managementGroups/MG-Name")

New-AzRoleDefinition -Role $role
Get-AzureRmRoleDefinition "Resource Lock Administrator"
```

Default Value:

A role for administering resource locks does not exist by default.

References:

1. <https://learn.microsoft.com/en-us/azure/role-based-access-control/custom-roles>
2. <https://learn.microsoft.com/en-us/azure/role-based-access-control/check-access>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-privileged-access#pa-1-separate-and-limit-highly-privilegedadministrative-users>
4. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-privileged-access#pa-7-follow-just-enough-administration-least-privilege-principle>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-privileged-access#pa-3-manage-lifecycle-of-identities-and-entitlements>
6. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-governance-strategy#gs-2-define-and-implement-enterprise-segmentationseparation-of-duties-strategy>
7. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-governance-strategy#gs-6-define-and-implement-identity-and-privileged-access-strategy>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	6.8 <u>Define and Maintain Role-Based Access Control</u> Define and maintain role-based access control, through determining and documenting the access rights necessary for each role within the enterprise to successfully carry out its assigned duties. Perform access control reviews of enterprise assets to validate that all privileges are authorized, on a recurring schedule at a minimum annually, or more frequently.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

5.6 Ensure that 'Subscription leaving Microsoft Entra tenant' and 'Subscription entering Microsoft Entra tenant' is set to 'Permit no one' (Manual)

Profile Applicability:

- Level 2

Description:

Users who are set as subscription owners are able to make administrative changes to the subscriptions and move them into and out of Microsoft Entra ID.

Rationale:

Permissions to move subscriptions in and out of a Microsoft Entra tenant must only be given to appropriate administrative personnel. A subscription that is moved into a Microsoft Entra tenant may be within a folder to which other users have elevated permissions. This prevents loss of data or unapproved changes of the objects within by potential bad actors.

Impact:

Subscriptions will need to have these settings turned off to be moved.

Audit:

Audit from Azure Portal

1. From the Azure Portal Home select the portal menu
2. Select **Subscriptions**
3. In the horizontal menu select **Manage Policies**
4. Ensure **Subscription leaving Microsoft Entra tenant** and **Subscription entering Microsoft Entra tenant** are set to **Permit no one**

Remediation:

Remediate from Azure Portal

1. From the Azure Portal Home select the portal menu
2. Select **Subscriptions**
3. In the horizontal menu select **Manage Policies**
4. Set **Subscription leaving Microsoft Entra tenant** and **Subscription entering Microsoft Entra tenant** to **Permit no one**
5. Click **Save changes**

Default Value:

By default **Subscription leaving Microsoft Entra tenant** and **Subscription entering Microsoft Entra tenant** are set to **Allow everyone (default)**

References:

1. <https://learn.microsoft.com/en-us/azure/cost-management-billing/manage/manage-azure-subscription-policy>
2. <https://learn.microsoft.com/en-us/entra/fundamentals/how-subscriptions-associated-directory>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-identity-management#im-2-protect-identity-and-authentication-systems>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.4 Restrict Administrator Privileges to Dedicated Administrator Accounts Restrict administrator privileges to dedicated administrator accounts on enterprise assets. Conduct general computing activities, such as internet browsing, email, and productivity suite use, from the user's primary, non-privileged account.			
v8	6.1 Establish an Access Granting Process Establish and follow a process, preferably automated, for granting access to enterprise assets upon new hire, rights grant, or role change of a user.			
v8	6.2 Establish an Access Revoking Process Establish and follow a process, preferably automated, for revoking access to enterprise assets, through disabling accounts immediately upon termination, rights revocation, or role change of a user. Disabling accounts, instead of deleting accounts, may be necessary to preserve audit trails.			
v7	4.3 Ensure the Use of Dedicated Administrative Accounts Ensure that all users with administrative account access use a dedicated or secondary account for elevated activities. This account should only be used for administrative activities and not internet browsing, email, or similar activities.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1578	TA0040	M1018

5.7 Ensure there are between 2 and 3 Subscription Owners (Automated)

Profile Applicability:

- Level 1

Description:

The Owner role in Azure grants full control over all resources in a subscription, including the ability to assign roles to others.

Rationale:

Limit the number of security principals (users, groups, service principals, and managed identities) assigned the Owner role to between 2 and 3. If groups are used, ensure their membership is tightly controlled and regularly reviewed to avoid privilege sprawl.

Impact:

Implementation may require changes in administrative workflows or the redistribution of roles and responsibilities. The recommendation to have between 2 and 3 Owners per subscription must account for all security principals that can be assigned the Owner role, not just individual users. This includes:

- User accounts
- Entra ID groups
- Service principals (used by applications or automation)
- Managed identities (system-assigned or user-assigned)

Audit:

Audit from Azure Portal

1. Go to **Subscriptions**.
2. Click the name of a subscription.
3. Click **Access Controls (IAM)**.
4. Click **Role assignments**.
5. Click **Role : All**.
6. Click the arrow next to **All**.
7. Click **Owner**.
8. Ensure a minimum of 2 and a maximum of 3 members are returned.
9. Repeat steps 1-8 for each subscription.

Audit from Azure CLI

Run the following command to list members with role **Owner** at a given subscription scope:

```
az role assignment list --role Owner --scope /subscriptions/<subscription-id> --query "[].{PrincipalName:principalName, Type:principalType}"
```

Ensure a minimum of 2 and a maximum of 3 members are returned.

For each Owner assignment, check: If principalType == Group, review group membership. If principalType == ServicePrincipal or ManagedIdentity, validate necessity and scope.

Repeat for each subscription.

Audit from PowerShell

Run the following command to list members with role **Owner** at a given subscription scope:

```
Get-AzRoleAssignment -RoleDefinitionName Owner -Scope /subscriptions/<subscription-id>
```

Ensure a minimum of 2 and a maximum of 3 members are returned.

Repeat for each subscription.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~Definitions

- **Policy ID:** [09024ccc-0c5f-475e-9457-b7c0d9ed487b](#) - **Name:** 'There should be more than one owner assigned to your subscription'
- **Policy ID:** [4f11b553-d42e-4e3a-89be-32ca364cad4c](#) - **Name:** 'A maximum of 3 owners should be designated for your subscription'

Remediation:

Remediate from Azure Portal

1. Go to **Subscriptions**.
2. Click the name of a subscription.
3. Click **Access Controls (IAM)**.
4. Click **Role assignments**.
5. Click **Role : All**.
6. Click the arrow next to **All**.
7. Click **Owner**.
8. Check the box next to members from whom the owner role should be removed.

9. Click **Delete**.
10. Click **Yes**.
11. Repeat steps 1-10 for each subscription requiring remediation.

Remediate from Azure CLI

Run the following command to delete role assignments by role assignment id:

```
az role assignment delete --ids <role-assignment-ids>
```

Default Value:

A subscription has 1 owner by default.

References:

1. <https://learn.microsoft.com/en-us/cli/azure/role/assignment>
2. <https://learn.microsoft.com/en-us/powershell/module/az.resources/get-azroleassignment>
3. <https://learn.microsoft.com/en-us/azure/role-based-access-control/built-in-roles/privileged#owner>
4. <https://learn.microsoft.com/en-us/azure/role-based-access-control/role-assignments-portal-subscription-admin>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.1 <u>Establish and Maintain an Inventory of Accounts</u> Establish and maintain an inventory of all accounts managed in the enterprise. The inventory must include both user and administrator accounts. The inventory, at a minimum, should contain the person's name, username, start/stop dates, and department. Validate that all active accounts are authorized, on a recurring schedule at a minimum quarterly, or more frequently.			
v7	4.1 <u>Maintain Inventory of Administrative Accounts</u> Use automated tools to inventory all administrative accounts, including domain and local accounts, to ensure that only authorized individuals have elevated privileges.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1078	TA0004	M1026

6 Management and Governance Services

To better understand the relationship between the Foundations Benchmark and Services Benchmarks, please read the "Introduction" section of this document.

Azure Product Directory Reference: <https://azure.microsoft.com/en-us/products#management-and-governance>

FEEDBACK REQUEST: Is there a specific service or recommendation in this section that you'd like to see addressed or improved? Let us know by making a ticket or starting a discussion in the CIS Microsoft Azure Community (<https://workbench.cisecurity.org/communities/72>).

6.1 Logging and Monitoring

This section covers security recommendations to follow for logging and monitoring policies on an Azure Subscription.

Scoping: A necessary exercise for effective and efficient use of Logging and Monitoring

For recommendations contained in this section, it is crucial that your organization consider and settle on the scope of application for each recommendation individually. The scope of application cannot be realistically written in a generic prescriptive way within these recommendations, so a scoping exercise is strongly recommended. A scoping exercise will help you determine which resources are "in scope" and will receive partial or complete logging and monitoring treatment, and which resources are "out of scope" and will not receive any logging and monitoring treatment.

Your objectives with the scoping exercise should be to:

- Produce a clear classification of resources
- Understand the control requirements of any relevant security or compliance frameworks
- Ensure the appropriate personnel can detect and react to threats
- Ensure relevant resources have a historical register for accountability and investigation
- Minimize alert fatigue and cost

Release Environments provide a helpful context for understanding scope from a DevOps perspective. For example:

1. Production Environment
2. Staging Environment
3. Testing Environment
4. Development Environment

While resources considered in the scope of a Production Environment might have a full set of recommendations applied for logging and monitoring, other release environments might have a limited set of recommendations applied for the sake of accountability. The names of these environments and which resources are in the scope of each environment will vary from one organization to another.

6.1.1 Configuring Diagnostic Settings

The Azure Diagnostic Settings capture control/management activities performed on a subscription or Azure AD Tenant. By default, the Azure Portal retains activity logs only for 90 days. The Diagnostic Settings define the type of events that are stored or streamed and the outputs—storage account, log analytics workspace, event hub, and others. The Diagnostic Settings, if configured properly, can ensure that all logs are retained for longer duration. This section has recommendations for correctly configuring the Diagnostic Settings so that all logs captured are retained for longer periods.

Azure Subscriptions

When configuring Diagnostic Settings, you may choose to export in one of four ways in which you need to ensure appropriate data retention. The options are Log Analytics workspace, Event Hub, Storage Account, and Partner Solutions. It is important to ensure you are aware and have set retention as your organization sees fit.

Azure AD Logs

In order to retain sign in logs, user account changes, application provisioning logs, or other logs that are visible to only on the Tenant in Azure AD, separate Diagnostic settings must be specified.

Deployment by Policy

Deploying Azure diagnostics should ideally be done by policy to ensure a consistent configuration. Microsoft provides a full set of policies for all diagnostic capable resource types in their GitHub repository. If you chose to deploy by policy, it is best to route the diagnostics to a Log Analytics Workspace so that they can be used in Azure Monitor or Azure Sentinel. Be aware that this has a cost attached to it. Future versions of the CIS Azure Foundations Benchmark will aim to cover the use of policy in greater detail.

6.1.1.1 Ensure that a 'Diagnostic Setting' Exists for Subscription Activity Logs (Automated)

Profile Applicability:

- Level 1

Description:

Enable Diagnostic settings for exporting activity logs. Diagnostic settings are available for each individual resource within a subscription. Settings should be configured for all appropriate resources for your environment.

Rationale:

A diagnostic setting controls how a diagnostic log is exported. By default, logs are retained only for 90 days. Diagnostic settings should be defined so that logs can be exported and stored for a longer duration to analyze security activities within an Azure subscription.

Audit:

Audit from Azure Portal

To identify Diagnostic Settings on a subscription:

1. Go to **Monitor**
2. Click **Activity Log**
3. Click **Export Activity Logs**
4. Select a **Subscription**
5. Ensure a **Diagnostic setting** exists for the selected Subscription

To identify Diagnostic Settings on specific resources:

1. Go to **Monitoring**
2. Click **Diagnostic settings**
3. Ensure a **Diagnostic setting** exists for all appropriate resources.

Audit from Azure CLI

To identify Diagnostic Settings on a subscription:

```
az monitor diagnostic-settings subscription list --subscription  
<subscription-id>
```

To identify Diagnostic Settings on a resource

```
az monitor diagnostic-settings list --resource <resource-id>
```


Audit from PowerShell

To identify Diagnostic Settings on a Subscription:

```
Get-AzSubscriptionDiagnosticSetting -SubscriptionId <subscription-id>
```

To identify Diagnostic Settings on a specific resource:

```
Get-AzDiagnosticSetting -ResourceId <resource-id>
```

Remediation:

Remediate from Azure Portal

To enable Diagnostic Settings on a Subscription:

1. Go to **Monitor**
2. Click on **Activity log**
3. Click on **Export Activity Logs**
4. Click **+ Add diagnostic setting**
5. Enter a **Diagnostic setting name**
6. Select **Categories** for the diagnostic setting
7. Select the appropriate **Destination details** (this may be Log Analytics, Storage Account, Event Hub, or Partner solution)
8. Click **Save**

To enable Diagnostic Settings on a specific resource:

1. Go to **Monitoring**
2. Click **Diagnostic settings**
3. Select **Add diagnostic setting**
4. Enter a **Diagnostic setting name**
5. Select the appropriate log, metric, and destination (this may be Log Analytics, Storage Account, Event Hub, or Partner solution)
6. Click **Save**

Repeat these step for all resources as needed.

Remediate from Azure CLI

To configure Diagnostic Settings on a Subscription:

```
az monitor diagnostic-settings subscription create --subscription  
<subscription id> --name <diagnostic settings name> --location <location> <[-  
-event-hub <event hub ID> --event-hub-auth-rule <event hub auth rule ID>] [--  
storage-account <storage account ID>] [--workspace <log analytics workspace  
ID>] --logs "<JSON encoded categories>" (e.g.  
[{"category:Security,enabled:true},{category:Administrative,enabled:true},{cat  
egory:Alert,enabled:true},{category:Policy,enabled:true}])
```

To configure Diagnostic Settings on a specific resource:

```
az monitor diagnostic-settings create --subscription <subscription ID> --
resource <resource ID> --name <diagnostic settings name> <[--event-hub <event
hub ID> --event-hub-rule <event hub auth rule ID>] [--storage-account
<storage account ID>] [--workspace <log analytics workspace ID>] --logs
<resource specific JSON encoded log settings> --metrics <metric settings
(shorthand|json-file|yaml-file)>
```

Remediate from PowerShell

To configure Diagnostic Settings on a subscription:

```
$logCategories = @();
$logCategories += New-AzDiagnosticSettingSubscriptionLogSettingsObject -
Category Administrative -Enabled $true
$logCategories += New-AzDiagnosticSettingSubscriptionLogSettingsObject -
Category Security -Enabled $true
$logCategories += New-AzDiagnosticSettingSubscriptionLogSettingsObject -
Category ServiceHealth -Enabled $true
$logCategories += New-AzDiagnosticSettingSubscriptionLogSettingsObject -
Category Alert -Enabled $true
$logCategories += New-AzDiagnosticSettingSubscriptionLogSettingsObject -
Category Recommendation -Enabled $true
$logCategories += New-AzDiagnosticSettingSubscriptionLogSettingsObject -
Category Policy -Enabled $true
$logCategories += New-AzDiagnosticSettingSubscriptionLogSettingsObject -
Category Autoscale -Enabled $true
$logCategories += New-AzDiagnosticSettingSubscriptionLogSettingsObject -
Category ResourceHealth -Enabled $true

New-AzSubscriptionDiagnosticSetting -SubscriptionId <subscription ID> -Name
<Diagnostic settings name> <[--EventHubAuthorizationRule <event hub auth rule
ID> -EventHubName <event hub name>] [--StorageAccountId <storage account ID>]
[--WorkSpaceId <log analytics workspace ID>] [--MarketplacePartner ID <full ARM
Marketplace resource ID>]> -Log $logCategories
```

To configure Diagnostic Settings on a specific resource:

```
$logCategories = @()
$logCategories += New-AzDiagnosticSettingLogSettingsObject -Category
<resource specific log category> -Enabled $true

Repeat command and variable assignment for each Log category specific to the
resource where this Diagnostic Setting will get configured.

$metricCategories = @()
$metricCategories += New-AzDiagnosticSettingMetricSettingsObject -Enabled
$true [-Category <resource specific metric category | AllMetrics>] [-
RetentionPolicyDay <Integer>] [-RetentionPolicyEnabled $true]

Repeat command and variable assignment for each Metric category or use the
'AllMetrics' category.

New-AzDiagnosticSetting -ResourceId <resource ID> -Name <Diagnostic settings
name> -Log $logCategories -Metric $metricCategories [-
EventHubAuthorizationRuleId <event hub auth rule ID> -EventHubName <event hub
name>] [-StorageAccountId <storage account ID>] [-WorkspaceId <log analytics
workspace ID>] [-MarketplacePartnerId <full ARM marketplace resource ID>]>
```

Default Value:

By default, diagnostic setting is not set.

References:

1. <https://learn.microsoft.com/en-us/azure/azure-monitor/fundamentals/data-sources>
2. <https://learn.microsoft.com/en-us/cli/azure/monitor/diagnostic-settings?view=azure-cli-latest>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-3-enable-logging-for-security-investigation>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.9 Centralize Audit Logs Centralize, to the extent possible, audit log collection and retention across enterprise assets.			
v7	6.5 Central Log Management Ensure that appropriate logs are being aggregated to a central log management system for analysis and review.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1485	TA0040	M1053

6.1.1.2 Ensure Diagnostic Setting Captures Appropriate Categories (Automated)

Profile Applicability:

- Level 1

Description:

Prerequisite: A Diagnostic Setting must exist. If a Diagnostic Setting does not exist, the navigation and options within this recommendation will not be available. Please review the recommendation at the beginning of this subsection titled: "Ensure that a 'Diagnostic Setting' exists."

The diagnostic setting should be configured to log the appropriate activities from the control/management plane.

Rationale:

A diagnostic setting controls how the diagnostic log is exported. Capturing the diagnostic setting categories for appropriate control/management plane activities allows proper alerting.

Audit:

Audit from Azure Portal

1. Go to **Monitor**.
2. Click **Activity log**.
3. Click on **Export Activity Logs**.
4. Select the appropriate **Subscription**.
5. Click **Edit setting** next to a diagnostic setting.
6. Ensure that the following categories are checked: **Administrative, Alert, Policy, and Security**.

Audit from Azure CLI

Ensure the categories '**Administrative**', '**Alert**', '**Policy**', and '**Security**' set to: 'enabled: true'

```
az monitor diagnostic-settings subscription list --subscription <subscription ID>
```

Audit from PowerShell

Ensure the categories Administrative, Alert, Policy, and Security are set to Enabled:True

```
Get-AzSubscriptionDiagnosticSetting -Subscription <subscriptionID>
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [3b980d31-7904-4bb7-8575-5665739a8052](#) - **Name:** 'An activity log alert should exist for specific Security operations'
- **Policy ID:** [b954148f-4c11-4c38-8221-be76711e194a](#) - **Name:** 'An activity log alert should exist for specific Administrative operations'
- **Policy ID:** [c5447c04-a4d7-4ba8-a263-c9ee321a6858](#) - **Name:** 'An activity log alert should exist for specific Policy operations'

Remediation:

Remediate from Azure Portal

1. Go to **Monitor**.
2. Click **Activity log**.
3. Click on **Export Activity Logs**.
4. Select the **Subscription** from the drop down menu.
5. Click **Edit setting** next to a diagnostic setting.
6. Check the following categories: **Administrative, Alert, Policy, and Security**.
7. Choose the destination details according to your organization's needs.
8. Click **Save**.

Remediate from Azure CLI

```
az monitor diagnostic-settings subscription create --subscription
<subscription id> --name <diagnostic settings name> --location <location> <[-
-event-hub <event hub ID> --event-hub-auth-rule <event hub auth rule ID>] [--
storage-account <storage account ID>] [--workspace <log analytics workspace
ID>] --logs
"[{category:Security,enabled:true},{category:Administrative,enabled:true},{ca
tegory:Alert,enabled:true},{category:Policy,enabled:true}]"
```

Remediate from PowerShell

```
$logCategories = @();
$logCategories += New-AzDiagnosticSettingSubscriptionLogSettingsObject -
Category Administrative -Enabled $true
$logCategories += New-AzDiagnosticSettingSubscriptionLogSettingsObject -
Category Security -Enabled $true
$logCategories += New-AzDiagnosticSettingSubscriptionLogSettingsObject -
Category Alert -Enabled $true
$logCategories += New-AzDiagnosticSettingSubscriptionLogSettingsObject -
Category Policy -Enabled $true

New-AzSubscriptionDiagnosticSetting -SubscriptionId <subscription ID> -Name
<Diagnostic settings name> <[-EventHubAuthorizationRule <event hub auth rule
ID> -EventHubName <event hub name>] [-StorageAccountId <storage account ID>]
[-WorkSpaceId <log analytics workspace ID>] [-MarketplacePartner ID <full ARM
Marketplace resource ID>]> -Log $logCategories
```

Default Value:

When the diagnostic setting is created using Azure Portal, by default no categories are selected.

References:

1. <https://learn.microsoft.com/en-us/azure/azure-monitor/essentials/diagnostic-settings>
2. <https://learn.microsoft.com/en-us/azure/azure-monitor/essentials/resource-manager-diagnostic-settings>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-3-enable-logging-for-security-investigation>
4. <https://learn.microsoft.com/en-us/cli/azure/monitor/diagnostic-settings?view=azure-cli-latest>
5. <https://learn.microsoft.com/en-us/powershell/module/az.monitor/new-azsubscriptiondiagnosticsetting?view=azps-13.4.0>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 Collect Detailed Audit Logs Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.3 Enable Detailed Logging Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1485	TA0040	M1053

6.1.1.3 Ensure the Storage Account Containing the Container with Activity Logs is Encrypted with Customer-managed Key (CMK) (Manual)

Profile Applicability:

- Level 2

Description:

Customer-managed keys introduce additional depth to security by providing a means to manage access control for encryption keys. Where compliance and security frameworks indicate the need, and organizational capacity allows, sensitive data at rest can be encrypted using customer-managed keys (CMK) rather than Microsoft-managed keys.

Rationale:

By default in Azure, data at rest tends to be encrypted using Microsoft-managed keys. If your organization wants to control and manage encryption keys for compliance and defense-in-depth, customer-managed keys can be established.

Configuring the storage account with the activity log export container to use CMKs provides additional confidentiality controls on log data, as a given user must have read permission on the corresponding storage account and must be granted decrypt permission by the CMK.

While it is possible to automate the assessment of this recommendation, the assessment status for this recommendation remains 'Manual' due to ideally limited scope. The scope of application—which workloads CMK is applied to—should be carefully considered to account for organizational capacity and targeted to workloads with specific need for CMK.

Impact:

If the key expires due to setting the 'activation date' and 'expiration date', the key must be rotated manually.

Using customer-managed keys may also incur additional man-hour requirements to create, store, manage, and protect the keys as needed.

Audit:

Audit from Azure Portal

1. Go to **Monitor**.
2. Select **Activity log**.
3. Select **Export Activity Logs**.
4. Select a **Subscription**.
5. Note the name of the **Storage Account** for the diagnostic setting.

6. Navigate to **Storage accounts**.
7. Click on the storage account name noted in Step 5.
8. Under **Security + networking**, click **Encryption**.
9. Ensure **Customer-managed keys** is selected and a key is set.

Audit from Azure CLI

1. Get storage account id configured with log profile:

```
az monitor diagnostic-settings subscription list --subscription <subscription id> --query 'value[*].storageAccountId'
```

2. Ensure the storage account is encrypted with CMK:

```
az storage account list --query "[?name=='<Storage Account Name>']"
```

In command output ensure **keySource** is set to **Microsoft.Keyvault** and **keyVaultProperties** is not set to **null**

Audit from PowerShell

```
Get-AzStorageAccount -ResourceGroupName <resource group name> -Name <storage account name>|select-object -ExpandProperty encryption|format-list
```

Ensure the value of **KeyVaultProperties** is not **null** or empty, and ensure **KeySource** is not set to **Microsoft.Storage**.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [fbb99e8e-e444-4da0-9ff1-75c92f5a85b2](#) - **Name:** 'Storage account containing the container with activity logs must be encrypted with BYOK'

Remediation:

Remediate from Azure Portal

1. Go to **Monitor**.
2. Select **Activity log**.
3. Select **Export Activity Logs**.
4. Select a **Subscription**.
5. Note the name of the **Storage Account** for the diagnostic setting.
6. Navigate to **Storage accounts**.
7. Click on the storage account.
8. Under **Security + networking**, click **Encryption**.

9. Next to **Encryption type**, select **Customer-managed keys**.
10. Complete the steps to configure a customer-managed key for encryption of the storage account.

Remediate from Azure CLI

```
az storage account update --name <name of the storage account> --resource-group <resource group for a storage account> --encryption-key-source=Microsoft.Keyvault --encryption-key-vault <Key Vault URI> --encryption-key-name <KeyName> --encryption-key-version <Key Version>
```

Remediate from PowerShell

```
Set-AzStorageAccount -ResourceGroupName <resource group name> -Name <storage account name> -KeyvaultEncryption -KeyVaultUri <key vault URI> -KeyName <key name>
```

Default Value:

By default, encryption type is set to Microsoft-managed keys.

References:

1. <https://learn.microsoft.com/en-us/azure/security/fundamentals/data-encryption-best-practices#protect-data-at-rest>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-data-protection#dp-5-use-customer-managed-key-option-in-data-at-rest-encryption-when-required>
3. <https://learn.microsoft.com/en-us/azure/azure-monitor/essentials/activity-log?tabs=cli#managing-legacy-log-profiles>

Additional Information:

This recommendation is based on the Common Reference Recommendation **Ensure critical data is encrypted with customer-managed keys (CMK)**, from the **Common Reference Recommendations > Secrets and Keys > Encryption Key Management > Customer Managed Keys** section.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.11 Encrypt Sensitive Data at Rest Encrypt sensitive data at rest on servers, applications, and databases containing sensitive data. Storage-layer encryption, also known as server-side encryption, meets the minimum requirement of this Safeguard. Additional encryption methods may include application-layer encryption, also known as client-side encryption, where access to the data storage device(s) does not permit access to the plain-text data.			

Controls Version	Control	IG 1	IG 2	IG 3
v7	14.8 Encrypt Sensitive Information at Rest Encrypt all sensitive information at rest using a tool that requires a secondary authentication mechanism not integrated into the operating system, in order to access the information.			●

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1530	TA0009	M1041

6.1.1.4 Ensure that Logging for Azure Key Vault is 'Enabled' (Automated)

Profile Applicability:

- Level 1

Description:

Enable AuditEvent logging for key vault instances to ensure interactions with key vaults are logged and available.

Rationale:

Monitoring how and when key vaults are accessed, and by whom, enables an audit trail of interactions with confidential information, keys, and certificates managed by Azure Key Vault. Enabling logging for Key Vault saves information in a user provided destination of either an Azure storage account or Log Analytics workspace. The same destination can be used for collecting logs for multiple Key Vaults.

Impact:

There may be increased storage costs depending upon how many metrics are selected for logging.

Audit:

Audit from Azure Portal

1. Go to **Key vaults**.
2. For each Key vault, under **Monitoring**, go to **Diagnostic settings**.
3. Click **Edit setting** next to a diagnostic setting.
4. Ensure that a destination is configured.
5. Under **Category groups**, ensure that **audit** and **allLogs** are checked.

Audit from Azure CLI

List all key vaults

```
az keyvault list
```

For each keyvault **id**

```
az monitor diagnostic-settings list --resource <id>
```

Ensure that **storageAccountId** reflects your desired destination and that **categoryGroup** and **enabled** are set as follows in the sample outputs below.

```
"logs": [
{
  "categoryGroup": "audit",
  "enabled": true,
},
{
  "categoryGroup": "allLogs",
  "enabled": true,
}
]
```

Audit from PowerShell

List the key vault(s) in the subscription

```
Get-AzKeyVault
```

For each key vault, run the following:

```
Get-AzDiagnosticSetting -ResourceId <key_vault_id>
```

Ensure that **StorageAccountId**, **ServiceBusRuleId**, **MarketplacePartnerId**, or **WorkspaceId** is set as appropriate. Also, ensure that **enabled** is set to **true**, and that **categoryGroup** reflects both **audit** and **allLogs** category groups.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/%2FDefinitions

- **Policy ID:** [cf820ca0-f99e-4f3e-84fb-66e913812d21](#) - **Name:** 'Resource logs in Key Vault should be enabled'

Remediation:

Remediate from Azure Portal

1. Go to **Key vaults**.
2. Select a Key vault.
3. Under **Monitoring**, select **Diagnostic settings**.
4. Click **Edit setting** to update an existing diagnostic setting, or **Add diagnostic setting** to create a new one.
5. If creating a new diagnostic setting, provide a name.
6. Configure an appropriate destination.
7. Under **Category groups**, check **audit** and **allLogs**.
8. Click **Save**.

Remediate from Azure CLI

To update an existing **Diagnostic Settings**

```
az monitor diagnostic-settings update --name "<diagnostic_setting_name>" --resource <key_vault_id>
```

To create a new **Diagnostic Settings**

```
az monitor diagnostic-settings create --name "<diagnostic_setting_name>" --resource <key_vault_id> --logs "[{category:audit,enabled:true},{category:allLogs,enabled:true}]" <[--event-hub <event_hub_ID> --event-hub-rule <event_hub_auth_rule_ID> | --storage-account <storage_account_ID> | --workspace <log_analytics_workspace_ID> | --marketplace-partner-id <solution_resource_ID>]>
```

Remediate from PowerShell

Create the **Log** settings object

```
$logSettings = @()
$logSettings += New-AzDiagnosticSettingLogSettingsObject -Enabled $true -Category audit
$logSettings += New-AzDiagnosticSettingLogSettingsObject -Enabled $true -Category allLogs
```

Create the **Metric** settings object

```
$metricSettings = @()
$metricSettings += New-AzDiagnosticSettingMetricSettingsObject -Enabled $true -Category AllMetrics
```

Create the **Diagnostic Settings** for each **Key Vault**

```
New-AzDiagnosticSetting -Name "<diagnostic_setting_name>" -ResourceId <key_vault_id> -Log $logSettings -Metric $metricSettings [-StorageAccountId <storage_account_ID> | -EventHubName <event_hub_name> -EventHubAuthorizationRuleId <event_hub_auth_rule_ID> | -WorkSpaceId <log analytics workspace ID> | -MarketPlacePartnerId <full resource ID for third-party solution>]
```

Default Value:

By default, Diagnostic AuditEvent logging is not enabled for Key Vault instances.

References:

1. <https://learn.microsoft.com/en-us/azure/key-vault/general/howto-logging>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-data-protection#dp-8-ensure-security-of-key-and-certificate-repository>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-3-enable-logging-for-security-investigation>
4. <https://learn.microsoft.com/en-us/powershell/module/az.monitor/get-azeventcategory?view=azps-15.3.0>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1485	TA0040	M1053

6.1.1.5 Ensure that Network Security Group Flow Logs are Captured and Sent to Log Analytics (Manual)

Profile Applicability:

- Level 2

Description:

Ensure that network flow logs are captured and fed into a central log analytics workspace.

Retirement Notice

On September 30, 2027, network security group (NSG) flow logs will be retired. As of June 30, 2025, creating new NSG flow logs is no longer possible. Azure recommends migrating to virtual network flow logs. Review <https://azure.microsoft.com/en-us/updates?id=Azure-NSG-flow-logs-Retirement> for more information.

For virtual network flow logs, consider applying the recommendation, **Ensure that virtual network flow logs are captured and sent to Log Analytics**, from this section.

Rationale:

Network Flow Logs provide valuable insight into the flow of traffic around your network and feed into both Azure Monitor and Azure Sentinel (if in use), permitting the generation of visual flow diagrams to aid with analyzing for lateral movement, etc.

Impact:

The impact of configuring NSG Flow logs is primarily one of cost and configuration. If deployed, it will create storage accounts that hold minimal amounts of data on a 5-day lifecycle before feeding to Log Analytics Workspace. This will increase the amount of data stored and used by Azure Monitor.

Audit:

Audit from Azure Portal

1. Navigate to **Network Watcher**.
2. Under **Logs**, select **Flow logs**.
3. Click **Add filter**.
4. From the **Filter** drop-down, select **Flow log type**.
5. From the **Value** drop-down, check **Network security group** only.
6. Click **Apply**.
7. Ensure that at least one network security group flow log is listed and is configured to send logs to a **Log Analytics Workspace**.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [27960feb-a23c-4577-8d36-ef8b5f35e0be](#) - **Name:** 'All flow log resources should be in enabled state'
- **Policy ID:** [c251913d-7d24-4958-af87-478ed3b9ba41](#) - **Name:** 'Flow logs should be configured for every network security group'
- **Policy ID:** [4c3c6c5f-0d47-4402-99b8-aa543dd8bcee](#) - **Name:** 'Flow logs should be configured for every virtual network'

Remediation:

As of June 30, 2025, creating new NSG flow logs is no longer possible. Azure recommends migrating to virtual network flow logs. Consider applying the recommendation, **Ensure that virtual network flow logs are captured and sent to Log Analytics**, from this section.

Default Value:

By default Network Security Group logs are not sent to Log Analytics.

References:

1. <https://learn.microsoft.com/en-us/azure/network-watcher/nsg-flow-logs-tutorial>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-4-enable-network-logging-for-security-investigation>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	13.6 Collect Network Traffic Flow Logs Collect network traffic flow logs and/or network traffic to review and alert upon from network devices.			
v7	12.8 Deploy NetFlow Collection on Networking Boundary Devices Enable the collection of NetFlow and logging data on all network boundary devices.			

6.1.1.6 Ensure that Virtual Network Flow Logs are Captured and Sent to Log Analytics (Manual)

Profile Applicability:

- Level 2

Description:

Ensure that virtual network flow logs are captured and fed into a central log analytics workspace.

Rationale:

Virtual network flow logs provide critical visibility into traffic patterns. Sending logs to a Log Analytics workspace enables centralized analysis, correlation, and alerting for faster threat detection and response.

Impact:

- Virtual network flow logs are charged per gigabyte of network flow logs collected and come with a free tier of 5 GB/month per subscription.
- If traffic analytics is enabled with virtual network flow logs, traffic analytics pricing applies at per gigabyte processing rates.
- The storage of logs is charged separately.

Audit:

Audit from Azure Portal

1. Go to **Network Watcher**.
2. Under **Logs**, select **Flow logs**.
3. Click **Add filter**.
4. From the **Filter** drop-down menu, select **Flow log type**.
5. From the **Value** drop-down menu, check **Virtual network** only.
6. Click **Apply**.
7. Ensure that at least one virtual network flow log is listed and is configured to send logs to a **Log Analytics Workspace**.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [2f080164-9f4d-497e-9db6-416dc9f7b48a](#) - **Name:** 'Network Watcher flow logs should have traffic analytics enabled'
- **Policy ID:** [4c3c6c5f-0d47-4402-99b8-aa543dd8bcee](#) - **Name:** 'Audit flow logs configuration for every virtual network'

Remediation:

Remediate from Azure Portal

1. Go to **Network Watcher**.
2. Under **Logs**, click **Flow logs**.
3. Click **+ Create**.
4. Select a subscription.
5. Next to **Flow log type**, select **Virtual network**.
6. Click **+ Select target resource**.
7. Select **Virtual network**.
8. Select a virtual network.
9. Click **Confirm selection**.
10. Select a storage account, or create a new storage account.
11. Set the retention in days for the storage account.
12. Click **Next**.
13. Under **Analytics**, for **Flow logs version**, select **Version 2**.
14. Check the box next to **Enable traffic analytics**.
15. Select a processing interval.
16. Select a **Log Analytics Workspace**.
17. Click **Next**.
18. Optionally, add **Tags**.
19. Click **Review + create**.
20. Click **Create**.
21. Repeat steps 1-20 for each subscription or virtual network requiring remediation.

References:

1. <https://learn.microsoft.com/en-us/azure/network-watcher/vnet-flow-logs-overview>
2. <https://learn.microsoft.com/en-us/azure/network-watcher/vnet-flow-logs-cli>

Additional Information:

On September 30, 2027, network security group (NSG) flow logs will be retired. As of June 30, 2025, creating new NSG flow logs is no longer possible. Azure recommends migrating to virtual network flow logs. After retirement, traffic analytics using NSG flow logs will no longer be supported, and existing NSG flow log resources will be deleted. Previously collected NSG flow log records will remain available per their retention policies. Review <https://azure.microsoft.com/en-us/updates?id=Azure-NSG-flow-logs-Retirement> for more information.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	13.6 <u>Collect Network Traffic Flow Logs</u> Collect network traffic flow logs and/or network traffic to review and alert upon from network devices.			
v7	12.8 <u>Deploy NetFlow Collection on Networking Boundary Devices</u> Enable the collection of NetFlow and logging data on all network boundary devices.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
		M1047

6.1.1.7 Ensure that a Microsoft Entra Diagnostic Setting Exists to Send Microsoft Graph Activity Logs to an Appropriate Destination (Manual)

Profile Applicability:

- Level 2

Description:

Ensure that a Microsoft Entra diagnostic setting is configured to send Microsoft Graph activity logs to a suitable destination, such as a Log Analytics workspace, storage account, or event hub. This enables centralized monitoring and analysis of all HTTP requests that the Microsoft Graph service receives and processes for a tenant.

Rationale:

Microsoft Graph activity logs provide visibility into HTTP requests made to the Microsoft Graph service, helping detect unauthorized access, suspicious activity, and security threats. Configuring diagnostic settings in Microsoft Entra ensures these logs are collected and sent to an appropriate destination for monitoring, analysis, and retention.

Impact:

A Microsoft Entra ID P1 or P2 tenant license is required to access the Microsoft Graph activity logs.

The amount of data logged and, thus, the cost incurred can vary significantly depending on the tenant size and the applications in your tenant that interact with the Microsoft Graph APIs.

See the following pricing calculations for respective services:

- Log Analytics: <https://learn.microsoft.com/en-us/azure/azure-monitor/logs/cost-logs#pricing-model>.
- Azure Storage: <https://azure.microsoft.com/en-us/pricing/details/storage/blobs/>.
- Event Hubs: <https://azure.microsoft.com/en-us/pricing/details/event-hubs/>.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Entra ID**.
2. Under **Monitoring**, click **Diagnostic settings**.
3. Next to each diagnostic setting, click **Edit setting**, and review the selected log categories and destination details.
4. Ensure that at least one diagnostic setting is configured to send **MicrosoftGraphActivityLogs** to an appropriate destination.

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Entra ID**.
2. Under **Monitoring**, click **Diagnostic settings**.
3. Click **+ Add diagnostic setting**.
4. Provide a **Diagnostic setting name**.
5. Under **Logs > Categories**, check the box next to **MicrosoftGraphActivityLogs**.
6. Configure an appropriate destination for the logs.
7. Click **Save**.

Default Value:

By default, Microsoft Entra diagnostic settings do not exist.

References:

1. <https://learn.microsoft.com/en-us/entra/identity/monitoring-health/howto-configure-diagnostic-settings>
2. <https://learn.microsoft.com/en-us/graph/microsoft-graph-activity-logs-overview>
3. <https://learn.microsoft.com/en-us/azure/azure-monitor/logs/cost-logs#pricing-model>
4. <https://azure.microsoft.com/en-us/pricing/details/storage/blobs/>
5. <https://azure.microsoft.com/en-us/pricing/details/event-hubs/>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.2 <u>Collect Audit Logs</u> Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
		M1047

6.1.1.8 Ensure that a Microsoft Entra Diagnostic Setting Exists to Send Microsoft Entra Activity Logs to an Appropriate Destination (Manual)

Profile Applicability:

- Level 2

Description:

Ensure that a Microsoft Entra diagnostic setting is configured to send Microsoft Entra activity logs to a suitable destination, such as a Log Analytics workspace, storage account, or event hub. This enables centralized monitoring and analysis of Microsoft Entra activity logs.

Rationale:

Microsoft Entra activity logs enables you to assess many aspects of your Microsoft Entra tenant. Configuring diagnostic settings in Microsoft Entra ensures these logs are collected and sent to an appropriate destination for monitoring, analysis, and retention.

Impact:

To export sign-in data, your organization needs an Azure AD P1 or P2 license.

The amount of data logged and, thus, the cost incurred can vary significantly depending on the tenant size.

See the following pricing calculations for respective services:

- Log Analytics: <https://learn.microsoft.com/en-us/azure/azure-monitor/logs/cost-logs#pricing-model>.
- Azure Storage: <https://azure.microsoft.com/en-us/pricing/details/storage/blobs/>.
- Event Hubs: <https://azure.microsoft.com/en-us/pricing/details/event-hubs/>.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Entra ID**.
2. Under **Monitoring**, click **Diagnostic settings**.
3. Next to each diagnostic setting, click **Edit setting**, and review the selected log categories and destination details.
4. Ensure that at least one diagnostic setting is configured to send the following logs to an appropriate destination:
 - **AuditLogs**
 - **SignInLogs**
 - **NonInteractiveUserSignInLogs**

- ServicePrincipalSignInLogs
- ManagedIdentitySignInLogs
- ProvisioningLogs
- ADFSSignInLogs
- RiskyUsers
- UserRiskEvents
- NetworkAccessTrafficLogs
- RiskyServicePrincipals
- ServicePrincipalRiskEvents
- EnrichedOffice365AuditLogs
- MicrosoftGraphActivityLogs
- RemoteNetworkHealthLogs
- NetworkAccessAlerts

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Entra ID**.
2. Under **Monitoring**, click **Diagnostic settings**.
3. Click **+ Add diagnostic setting**.
4. Provide a **Diagnostic setting name**.
5. Under **Logs > Categories**, check the box next to each of the following logs:
 - AuditLogs
 - SignInLogs
 - NonInteractiveUserSignInLogs
 - ServicePrincipalSignInLogs
 - ManagedIdentitySignInLogs
 - ProvisioningLogs
 - ADFSSignInLogs
 - RiskyUsers
 - UserRiskEvents
 - NetworkAccessTrafficLogs
 - RiskyServicePrincipals
 - ServicePrincipalRiskEvents
 - EnrichedOffice365AuditLogs
 - MicrosoftGraphActivityLogs
 - RemoteNetworkHealthLogs
 - NetworkAccessAlerts
6. Configure an appropriate destination for the logs.
7. Click **Save**.

Default Value:

By default, Microsoft Entra diagnostic settings do not exist.

References:

1. <https://learn.microsoft.com/en-us/entra/identity/monitoring-health/howto-configure-diagnostic-settings>
2. <https://learn.microsoft.com/en-us/entra/identity/monitoring-health/howto-access-activity-logs>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.2 Collect Audit Logs Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.			
v7	6.2 Activate audit logging Ensure that local logging has been enabled on all systems and networking devices.			
v7	6.3 Enable Detailed Logging Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
		M1047

6.1.1.9 Ensure that Intune Logs are Captured and Sent to Log Analytics (Manual)

Profile Applicability:

- Level 2

Description:

Ensure that Intune logs are captured and fed into a central log analytics workspace.

Rationale:

Intune includes built-in logs that provide information about your environments. Sending logs to a Log Analytics workspace enables centralized analysis, correlation, and alerting for faster threat detection and response.

Impact:

A Microsoft Intune plan is required to access Intune: <https://www.microsoft.com/en-us/security/business/microsoft-intune-pricing>.

The amount of data logged and, thus, the cost incurred can vary significantly depending on the tenant size.

For information on Log Analytics workspace costs, visit: <https://learn.microsoft.com/en-us/azure/azure-monitor/logs/cost-logs>.

Audit:

Audit from Azure Portal

1. Go to **Intune**.
2. Click **Reports**.
3. Under **Azure monitor**, click **Diagnostic settings**.
4. Next to each diagnostic setting, click **Edit setting**, and review the selected log categories and destination details.
5. Ensure that at least one diagnostic setting is configured to send the following logs to a Log Analytics workspace:
 - **AuditLogs**
 - **OperationalLogs**
 - **DeviceComplianceOrg**
 - **Devices**
 - **Windows365AuditLogs**

Remediation:

Remediate from Azure Portal

1. Go to **Intune**.
2. Click **Reports**.
3. Under **Azure monitor**, click **Diagnostic settings**.
4. Click **+ Add diagnostic setting**.
5. Provide a **Diagnostic setting name**.
6. Under **Logs > Categories**, check the box next to each of the following logs:
 - **AuditLogs**
 - **OperationalLogs**
 - **DeviceComplianceOrg**
 - **Devices**
 - **Windows365AuditLogs**
7. Under **Destination details**, check the box next to **Send to Log Analytics workspace**.
8. Select a **Subscription**.
9. Select a **Log Analytics workspace**.
10. Click **Save**.

Default Value:

By default, Intune diagnostic settings do not exist.

References:

1. <https://learn.microsoft.com/en-us/intune/intune-service/fundamentals/review-logs-using-azure-monitor>
2. <https://www.microsoft.com/en-us/security/business/microsoft-intune-pricing>
3. <https://learn.microsoft.com/en-us/azure/azure-monitor/logs/cost-logs>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.2 <u>Collect Audit Logs</u> Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
		M1047

6.1.2 Monitoring Using Activity Log Alerts

The recommendations provided in this section are intended to provide entry-level alerting for crucial activities on a tenant account. These recommended activities **should** be tuned to your needs. By default, each of these Activity Log Alerts tends to guide the reader to alerting at the "Subscription-wide" level which will capture and alert on rules triggered by all resources and resource groups contained within a subscription. This is not an ideal rule set for Alerting within larger and more complex organizations.

While this section provides recommendations for the creation of **Activity Log Alerts** specifically, Microsoft Azure supports four different types of alerts:

- Metric Alerts
- Log Alerts
- Activity Log Alerts
- Smart Detection Alerts

All Azure services (Microsoft provided or otherwise) that can generate alerts are assigned a "Resource provider namespace" when they are registered in an Azure tenant. The recommendations in this section are in no way exhaustive of the plethora of available "Providers" or "Resource Types." The Resource Providers that are registered in your Azure Tenant can be located in your Subscription. Each registered Provider in your environment **may** have available "Conditions" to raise alerts via Activity Log Alerts. These providers should be considered for inclusion in Activity Log Alert rules of your own making.

To view the registered resource providers in your Subscription(s), use this guide:

- <https://learn.microsoft.com/en-us/azure/azure-resource-manager/management/resource-providers-and-types>

If you wish to create custom alerting rules for Activity Log Alerts or other alert types, please refer to Microsoft documentation:

- <https://learn.microsoft.com/en-us/azure/azure-monitor/alerts/alerts-create-metric-alert-rule>

6.1.2.1 Ensure that Activity Log Alert Exists for Create Policy Assignment (Automated)

Profile Applicability:

- Level 1

Description:

Create an activity log alert for the Create Policy Assignment event.

Rationale:

Monitoring for create policy assignment events gives insight into changes done in "Azure policy - assignments" and can reduce the time it takes to detect unsolicited changes.

Audit:

Audit from Azure Portal

1. Navigate to the **Monitor** blade.
2. Click on **Alerts**.
3. In the Alerts window, click on **Alert rules**.
4. Ensure an alert rule exists where the Condition column contains **Operation name=Microsoft.Authorization/policyAssignments/write**.
5. Click on the Alert **Name** associated with the previous step.
6. Ensure the **Condition** panel displays the text **Whenever the Activity Log has an event with Category='Administrative', Operation name='Create policy assignment'** and does not filter on **Level, Status** or **Caller**.
7. Ensure the **Actions** panel displays an Action group is assigned to notify the appropriate personnel in your organization.

Audit from Azure CLI

```
az monitor activity-log alert list --subscription <subscription ID> --query "[[] . {Name:name, Enabled:enabled, Condition:condition.allOf, Actions:actions}]"
```

Look for **Microsoft.Authorization/policyAssignments/write** in the output. If it's missing, generate a finding.

Audit from PowerShell

```
Get-AzActivityLogAlert -SubscriptionId <subscription ID>|where-object  
{$_ .ConditionAllOf.Equal -match  
"Microsoft.Authorization/policyAssignments/write"}|select-object  
Location,Name,Enabled,ResourceGroupName,ConditionAllOf
```

If the output is empty, an **alert rule** for **Create Policy Assignments** is not configured.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/%2FDefinitions

- **Policy ID:** [c5447c04-a4d7-4ba8-a263-c9ee321a6858](#) - **Name:** 'An activity log alert should exist for specific Policy operations'

Remediation:

Remediate from Azure Portal

1. Navigate to the **Monitor** blade.
2. Select **Alerts**.
3. Select **Create**.
4. Select **Alert rule**.
5. Choose a subscription.
6. Select **Apply**.
7. Select the **Condition** tab.
8. Click **See all signals**.
9. Select **Create policy assignment (Policy assignment)**.
10. Click **Apply**.
11. Select the **Actions** tab.
12. Click **Select action groups** to select an existing action group, or **Create action group** to create a new action group.
13. Follow the prompts to choose or create an action group.
14. Select the **Details** tab.
15. Select a **Resource group**, provide an **Alert rule name** and an optional **Alert rule description**.
16. Click **Review + create**.
17. Click **Create**.

Remediate from Azure CLI

```
az monitor activity-log alert create --resource-group "<resource group name>"
--condition category=Administrative and
operationName=Microsoft.Authorization/policyAssignments/write> --scope
"/subscriptions/<subscription ID>" --name "<activity log rule name>" --
subscription <subscription ID> --action-group <action group ID>
```

Remediate from PowerShell

Create the **conditions** object.

```
$conditions = @()
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Administrative -Field category
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Microsoft.Authorization/policyAssignments/write -Field operationName
```

Get the **Action Group** information and store it in a variable, then create a new **Action** object.

```
$actionGroup = Get-AzActionGroup -ResourceGroupName <resource group name> -
Name <action group name>
$actionObject = New-AzActivityLogAlertActionGroupObject -Id $actionGroup.Id
```

Create the **Scope** variable.

```
$scope = "/subscriptions/<subscription ID>"
```

Create the **Activity Log Alert Rule** for
Microsoft.Authorization/policyAssignments/write

```
New-AzActivityLogAlert -Name "<activity alert rule name>" -ResourceGroupName
"<resource group name>" -Condition $conditions -Scope $scope -Location global
-Action $actionObject -Subscription <subscription ID> -Enabled $true
```

Default Value:

By default, no monitoring alerts are created.

References:

1. <https://azure.microsoft.com/en-us/updates?id=classic-alerting-monitoring-retirement>
2. <https://learn.microsoft.com/en-us/azure/azure-monitor/alerts/alerts-create-activity-log-alert-rule>
3. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/create-or-update>
4. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/list-by-subscription-id>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-3-enable-logging-for-security-investigation>
6. <https://learn.microsoft.com/en-us/rest/api/policy/policy-assignments>

7. <https://learn.microsoft.com/en-us/azure/azure-monitor/alerts/alerts-create-log-alert-rule>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

6.1.2.2 Ensure that Activity Log Alert exists for Delete Policy Assignment (Automated)

Profile Applicability:

- Level 1

Description:

Create an activity log alert for the Delete Policy Assignment event.

Rationale:

Monitoring for delete policy assignment events gives insight into changes done in "azure policy - assignments" and can reduce the time it takes to detect unsolicited changes.

Audit:

Audit from Azure Portal

1. Navigate to the **Monitor** blade.
2. Click on **Alerts**.
3. In the Alerts window, click on **Alert rules**.
4. Ensure an alert rule exists where the Condition column contains **Operation name=Microsoft.Authorization/policyAssignments/delete**.
5. Click on the Alert **Name** associated with the previous step.
6. Ensure the **Condition** panel displays the text **Whenever the Activity Log has an event with Category='Administrative', Operation name='Delete policy assignment'** and does not filter on **Level**, **Status** or **Caller**.
7. Ensure the **Actions** panel displays an Action group is assigned to notify the appropriate personnel in your organization.

Audit from Azure CLI

```
az monitor activity-log alert list --subscription <subscription ID> --query "[].{Name:name,Enabled:enabled,Condition:condition.allOf,Actions:actions}"
```

Look for **Microsoft.Authorization/policyAssignments/delete** in the output

Audit from PowerShell

```
Get-AzActivityLogAlert -SubscriptionId <subscription ID>|where-object  
{$_ .ConditionAllOf.Equal -match  
"Microsoft.Authorization/policyAssignments/delete"}|select-object  
Location,Name,Enabled,ResourceGroupName,ConditionAllOf
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [c5447c04-a4d7-4ba8-a263-c9ee321a6858](#) - **Name:** 'An activity log alert should exist for specific Policy operations'

Remediation:

Remediate from Azure Portal

1. Navigate to the **Monitor** blade.
2. Select **Alerts**.
3. Select **Create**.
4. Select **Alert rule**.
5. Choose a subscription.
6. Select **Apply**.
7. Select the **Condition** tab.
8. Click **See all signals**.
9. Select **Delete policy assignment (Policy assignment)**.
10. Click **Apply**.
11. Select the **Actions** tab.
12. Click **Select action groups** to select an existing action group, or **Create action group** to create a new action group.
13. Follow the prompts to choose or create an action group.
14. Select the **Details** tab.
15. Select a **Resource group**, provide an **Alert rule name** and an optional **Alert rule description**.
16. Click **Review + create**.
17. Click **Create**.

Remediate from Azure CLI

```
az monitor activity-log alert create --resource-group "<resource group name>"
--condition category=Administrative and
operationName=Microsoft.Authorization/policyAssignments/delete and
level=<verbose | information | warning | error | critical> --scope
"/subscriptions/<subscription ID>" --name "<activity log rule name>" --
subscription <subscription id> --action-group <action group ID>
```

Remediate from PowerShell

Create the conditions object

```
$conditions = @()  
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -  
Equal Administrative -Field category  
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -  
Equal Microsoft.Authorization/policyAssignments/delete -Field operationName  
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -  
Equal Verbose -Field level
```

Retrieve the **Action Group** information and store in a variable, then create the **Action** object.

```
$actionGroup = Get-AzActionGroup -ResourceGroupName <resource group name> -  
Name <action group name>  
$actionObject = New-AzActivityLogAlertActionGroupObject -Id $actionGroup.Id
```

Create the **Scope** variable.

```
$scope = "/subscriptions/<subscription id>"
```

Create the **Activity Log Alert Rule** for
Microsoft.Authorization/policyAssignments/delete.

```
New-AzActivityLogAlert -Name "<activity log alert rule name>" -  
ResourceGroupName "<resource group name>" -Condition $conditions -Scope  
$scope -Location global -Action $actionObject -Subscription <subscription ID>  
-Enabled $true
```

Default Value:

By default, no monitoring alerts are created.

References:

1. <https://learn.microsoft.com/en-us/azure/azure-monitor/alerts/alerts-create-activity-log-alert-rule>
2. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/create-or-update>
3. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/list-by-subscription-id>
4. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-3-enable-logging-for-security-investigation>
5. <https://azure.microsoft.com/en-us/products/blueprints/>

Additional Information:

This log alert also applies for Azure Blueprints.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

6.1.2.3 Ensure that Activity Log Alert Exists for Create or Update Network Security Group (Automated)

Profile Applicability:

- Level 1

Description:

Create an Activity Log Alert for the Create or Update Network Security Group event.

Rationale:

Monitoring for Create or Update Network Security Group events gives insight into network access changes and may reduce the time it takes to detect suspicious activity.

Audit:

Audit from Azure Portal

1. Navigate to the **Monitor** blade.
2. Click on **Alerts**.
3. In the Alerts window, click on **Alert rules**.
4. Ensure an alert rule exists where the Condition column contains **Operation name=Microsoft.Network/networkSecurityGroups/write**.
5. Click on the Alert **Name** associated with the previous step.
6. Ensure the **Condition** panel displays the text **Whenever the Activity Log has an event with Category='Administrative', Operation name='Create or Update Network Security Group'** and does not filter on **Level, Status** or **Caller**.
7. Ensure the **Actions** panel displays an Action group is assigned to notify the appropriate personnel in your organization.

Audit from Azure CLI

```
az monitor activity-log alert list --subscription <subscription ID> --query "[].{Name:name,Enabled:enabled,Condition:condition.allOf,Actions:actions}"
```

Look for **Microsoft.Network/networkSecurityGroups/write** in the output

Audit from PowerShell

```
Get-AzActivityLogAlert -SubscriptionId <subscription ID>|where-object {$_.ConditionAllOf.Equal -match "Microsoft.Network/networkSecurityGroups/write"}|select-object Location,Name,Enabled,ResourceGroupName,ConditionAllOf
```


Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [b954148f-4c11-4c38-8221-be76711e194a](#) - **Name:** 'An activity log alert should exist for specific Administrative operations'

Remediation:

Remediate from Azure Portal

1. Navigate to the **Monitor** blade.
2. Select **Alerts**.
3. Select **Create**.
4. Select **Alert rule**.
5. Choose a subscription.
6. Select **Apply**.
7. Select the **Condition** tab.
8. Click **See all signals**.
9. Select **Create or Update Network Security Group (Network Security Group)**.
10. Click **Apply**.
11. Select the **Actions** tab.
12. Click **Select action groups** to select an existing action group, or **Create action group** to create a new action group.
13. Follow the prompts to choose or create an action group.
14. Select the **Details** tab.
15. Select a **Resource group**, provide an **Alert rule name** and an optional **Alert rule description**.
16. Click **Review + create**.
17. Click **Create**.

Remediate from Azure CLI

```
az monitor activity-log alert create --resource-group "<resource group name>"
--condition category=Administrative and
operationName=Microsoft.Network/networkSecurityGroups/write and level=verbose
--scope "/subscriptions/<subscription ID>" --name "<activity log rule name>"
--subscription <subscription id> --action-group <action group ID>
```

Remediate from PowerShell

Create the **Conditions** object.

```
$conditions = @()
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Administrative -Field category
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Microsoft.Network/networkSecurityGroups/write -Field operationName
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Verbose -Field level
```

Retrieve the **Action Group** information and store in a variable, then create the **Actions** object.

```
$actionGroup = Get-AzActionGroup -ResourceGroupName <resource group name> -
Name <action group name>
$actionObject = New-AzActivityLogAlertActionGroupObject -Id $actionGroup.Id
```

Create the **Scope** object

```
$scope = "/subscriptions/<subscription id>"
```

Create the **Activity Log Alert Rule** for
Microsoft.Network/networkSecurityGroups/write

```
New-AzActivityLogAlert -Name "<activity log alert rule name>" -
ResourceGroupName "<resource group name>" -Condition $conditions -Scope
$scope -Location global -Action $actionObject -Subscription <subscription ID>
-Enabled $true
```

Default Value:

By default, no monitoring alerts are created.

References:

1. <https://azure.microsoft.com/en-us/updates?id=classic-alerting-monitoring-retirement>
2. <https://learn.microsoft.com/en-us/azure/azure-monitor/alerts/alerts-create-activity-log-alert-rule>
3. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/create-or-update>
4. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/list-by-subscription-id>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-3-enable-logging-for-security-investigation>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

6.1.2.4 Ensure that Activity Log Alert Exists for Delete Network Security Group (Automated)

Profile Applicability:

- Level 1

Description:

Create an activity log alert for the Delete Network Security Group event.

Rationale:

Monitoring for "Delete Network Security Group" events gives insight into network access changes and may reduce the time it takes to detect suspicious activity.

Audit:

Audit from Azure Portal

1. Navigate to the **Monitor** blade.
2. Click on **Alerts**.
3. In the Alerts window, click on **Alert rules**.
4. Ensure an alert rule exists where the Condition column contains **Operation name=Microsoft.Network/networkSecurityGroups/delete**.
5. Click on the Alert **Name** associated with the previous step.
6. Ensure the **Condition** panel displays the text **Whenever the Activity Log has an event with Category='Administrative', Operation name='Delete Network Security Group'** and does not filter on **Level**, **Status** or **Caller**.
7. Ensure the **Actions** panel displays an Action group is assigned to notify the appropriate personnel in your organization.

Audit from Azure CLI

```
az monitor activity-log alert list --subscription <subscription ID> --query "[].{Name:name,Enabled:enabled,Condition:condition.allOf,Actions:actions}"
```

Look for **Microsoft.Network/networkSecurityGroups/delete** in the output

Audit from PowerShell

```
Get-AzActivityLogAlert -SubscriptionId <subscription ID>|where-object  
{$_ .ConditionAllOf.Equal -match  
"Microsoft.Network/networkSecurityGroups/delete"}|select-object  
Location,Name,Enabled,ResourceGroupName,ConditionAllOf
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [b954148f-4c11-4c38-8221-be76711e194a](#) - **Name:** 'An activity log alert should exist for specific Administrative operations'

Remediation:

Remediate from Azure Portal

1. Navigate to the **Monitor** blade.
2. Select **Alerts**.
3. Select **Create**.
4. Select **Alert rule**.
5. Choose a subscription.
6. Select **Apply**.
7. Select the **Condition** tab.
8. Click **See all signals**.
9. Select **Delete Network Security Group (Network Security Group)**.
10. Click **Apply**.
11. Select the **Actions** tab.
12. Click **Select action groups** to select an existing action group, or **Create action group** to create a new action group.
13. Follow the prompts to choose or create an action group.
14. Select the **Details** tab.
15. Select a **Resource group**, provide an **Alert rule name** and an optional **Alert rule description**.
16. Click **Review + create**.
17. Click **Create**.

Remediate from Azure CLI

```
az monitor activity-log alert create --resource-group "<resource group name>"
--condition category=Administrative and
operationName=Microsoft.Network/networkSecurityGroups/delete and
level=<verbose | information | warning | error | critical> --scope
"/subscriptions/<subscription ID>" --name "<activity log rule name>" --
subscription <subscription id> --action-group <action group ID>
```

Remediate from PowerShell

Create the **Conditions** object.

```
$conditions = @()
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Administrative -Field category
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Microsoft.Network/networkSecurityGroups/delete -Field operationName
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Verbose -Field level
```

Retrieve the **Action Group** information and store in a variable, then create the **Actions** object.

```
$actionGroup = Get-AzActionGroup -ResourceGroupName <resource group name> -
Name <action group name>
$actionObject = New-AzActivityLogAlertActionGroupObject -Id $actionGroup.Id
```

Create the **Scope** object

```
$scope = "/subscriptions/<subscription id>"
```

Create the **Activity Log Alert Rule** for
Microsoft.Network/networkSecurityGroups/delete

```
New-AzActivityLogAlert -Name "<activity log alert rule name>" -
ResourceGroupName "<resource group name>" -Condition $conditions -Scope
$scope -Location global -Action $actionObject -Subscription <subscription ID>
-Enabled $true
```

Default Value:

By default, no monitoring alerts are created.

References:

1. <https://azure.microsoft.com/en-us/updates?id=classic-alerting-monitoring-retirement>
2. <https://learn.microsoft.com/en-us/azure/azure-monitor/alerts/alerts-create-activity-log-alert-rule>
3. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/create-or-update>
4. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/list-by-subscription-id>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-3-enable-logging-for-security-investigation>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

6.1.2.5 Ensure that Activity Log Alert Exists for Create or Update Security Solution (Automated)

Profile Applicability:

- Level 1

Description:

Create an activity log alert for the Create or Update Security Solution event.

Rationale:

Monitoring for Create or Update Security Solution events gives insight into changes to the active security solutions and may reduce the time it takes to detect suspicious activity.

Audit:

Audit from Azure Portal

1. Navigate to the **Monitor** blade.
2. Click on **Alerts**.
3. In the Alerts window, click on **Alert rules**.
4. Ensure an alert rule exists where the Condition column contains **Operation name=Microsoft.Security/securitySolutions/write**.
5. Click on the Alert **Name** associated with the previous step.
6. Ensure the **Condition** panel displays the text **Whenever the Activity Log has an event with Category='Administrative', Operation name='Create or Update Security Solutions'** and does not filter on **Level, Status** or **Caller**.
7. Ensure the **Actions** panel displays an Action group is assigned to notify the appropriate personnel in your organization.

Audit from Azure CLI

```
az monitor activity-log alert list --subscription <subscription Id> --query "[].{Name:name,Enabled:enabled,Condition:condition.allOf,Actions:actions}"
```

Look for **Microsoft.Security/securitySolutions/write** in the output

Audit from PowerShell

```
Get-AzActivityLogAlert -SubscriptionId <subscription ID>|where-object  
{$_ .ConditionAllOf.Equal -match  
"Microsoft.Security/securitySolutions/write"}|select-object  
Location,Name,Enabled,ResourceGroupName,ConditionAllOf
```


Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [3b980d31-7904-4bb7-8575-5665739a8052](#) - **Name:** 'An activity log alert should exist for specific Security operations'

Remediation:

Remediate from Azure Portal

1. Navigate to the **Monitor** blade.
2. Select **Alerts**.
3. Select **Create**.
4. Select **Alert rule**.
5. Choose a subscription.
6. Select **Apply**.
7. Select the **Condition** tab.
8. Click **See all signals**.
9. Select **Create or Update Security Solutions (Security Solutions)**.
10. Click **Apply**.
11. Select the **Actions** tab.
12. Click **Select action groups** to select an existing action group, or **Create action group** to create a new action group.
13. Follow the prompts to choose or create an action group.
14. Select the **Details** tab.
15. Select a **Resource group**, provide an **Alert rule name** and an optional **Alert rule description**.
16. Click **Review + create**.
17. Click **Create**.

Remediate from Azure CLI

```
az monitor activity-log alert create --resource-group "<resource group name>"
--condition category=Administrative and
operationName=Microsoft.Security/securitySolutions/write and level=<verbose |
information | warning | error | critical> --scope
"/subscriptions/<subscription ID>" --name "<activity log rule name>" --
subscription <subscription id> --action-group <action group ID>
```

Remediate from PowerShell

Create the **Conditions** object.

```
$conditions = @()
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Administrative -Field category
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Microsoft.Security/securitySolutions/write -Field operationName
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Verbose -Field level
```

Retrieve the **Action Group** information and store in a variable, then create the **Actions** object.

```
$actionGroup = Get-AzActionGroup -ResourceGroupName <resource group name> -
Name <action group name>
$actionObject = New-AzActivityLogAlertActionGroupObject -Id $actionGroup.Id
```

Create the **Scope** object

```
$scope = "/subscriptions/<subscription ID>"
```

Create the **Activity Log Alert Rule** for
Microsoft.Security/securitySolutions/write

```
New-AzActivityLogAlert -Name "<activity log alert rule name>" -
ResourceGroupName "<resource group name>" -Condition $conditions -Scope
$scope -Location global -Action $actionObject -Subscription <subscription ID>
-Enabled $true
```

Default Value:

By default, no monitoring alerts are created.

References:

1. <https://azure.microsoft.com/en-us/updates?id=classic-alerting-monitoring-retirement>
2. <https://learn.microsoft.com/en-us/azure/azure-monitor/alerts/alerts-create-activity-log-alert-rule>
3. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/create-or-update>
4. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/list-by-subscription-id>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-3-enable-logging-for-security-investigation>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

6.1.2.6 Ensure that Activity Log Alert Exists for Delete Security Solution (Automated)

Profile Applicability:

- Level 1

Description:

Create an activity log alert for the Delete Security Solution event.

Rationale:

Monitoring for Delete Security Solution events gives insight into changes to the active security solutions and may reduce the time it takes to detect suspicious activity.

Audit:

Audit from Azure Portal

1. Navigate to the **Monitor** blade.
2. Click on **Alerts**.
3. In the Alerts window, click on **Alert rules**.
4. Ensure an alert rule exists where the Condition column contains **Operation name=Microsoft.Security/securitySolutions/delete**.
5. Click on the Alert **Name** associated with the previous step.
6. Ensure the **Condition** panel displays the text **Whenever the Activity Log has an event with Category='Administrative', Operation name='Delete Security Solutions'** and does not filter on **Level**, **Status** or **Caller**.
7. Ensure the **Actions** panel displays an Action group is assigned to notify the appropriate personnel in your organization.

Audit from Azure CLI

```
az monitor activity-log alert list --subscription <subscription Id> --query "[].{Name:name,Enabled:enabled,Condition:condition.allOf,Actions:actions}"
```

Look for **Microsoft.Security/securitySolutions/delete** in the output

Audit from PowerShell

```
Get-AzActivityLogAlert -SubscriptionId <subscription ID>|where-object  
{$_ .ConditionAllOf.Equal -match  
"Microsoft.Security/securitySolutions/delete"}|select-object  
Location,Name,Enabled,ResourceGroupName,ConditionAllOf
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [3b980d31-7904-4bb7-8575-5665739a8052](#) - **Name:** 'An activity log alert should exist for specific Security operations'

Remediation:

Remediate from Azure Portal

1. Navigate to the **Monitor** blade.
2. Select **Alerts**.
3. Select **Create**.
4. Select **Alert rule**.
5. Choose a subscription.
6. Select **Apply**.
7. Select the **Condition** tab.
8. Click **See all signals**.
9. Select **Delete Security Solutions (Security Solutions)**.
10. Click **Apply**.
11. Select the **Actions** tab.
12. Click **Select action groups** to select an existing action group, or **Create action group** to create a new action group.
13. Follow the prompts to choose or create an action group.
14. Select the **Details** tab.
15. Select a **Resource group**, provide an **Alert rule name** and an optional **Alert rule description**.
16. Click **Review + create**.
17. Click **Create**.

Remediate from Azure CLI

```
az monitor activity-log alert create --resource-group "<resource group name>"
--condition category=Administrative and
operationName=Microsoft.Security/securitySolutions/delete and level=<verbose
| information | warning | error | critical> --scope
"/subscriptions/<subscription ID>" --name "<activity log rule name>" --
subscription <subscription id> --action-group <action group ID>
```

Remediate from PowerShell

Create the **Conditions** object.

```
$conditions = @()
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Administrative -Field category
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Microsoft.Security/securitySolutions/delete -Field operationName
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Verbose -Field level
```

Retrieve the **Action Group** information and store in a variable, then create the **Actions** object.

```
$actionGroup = Get-AzActionGroup -ResourceGroupName <resource group name> -
Name <action group name>
$actionObject = New-AzActivityLogAlertActionGroupObject -Id $actionGroup.Id
```

Create the **Scope** object

```
$scope = "/subscriptions/<subscription ID>"
```

Create the **Activity Log Alert Rule** for
Microsoft.Security/securitySolutions/delete

```
New-AzActivityLogAlert -Name "<activity log alert rule name>" -
ResourceGroupName "<resource group name>" -Condition $conditions -Scope
$scope -Location global -Action $actionObject -Subscription <subscription ID>
-Enabled $true
```

Default Value:

By default, no monitoring alerts are created.

References:

1. <https://azure.microsoft.com/en-us/updates?id=classic-alerting-monitoring-retirement>
2. <https://learn.microsoft.com/en-us/azure/azure-monitor/alerts/alerts-create-activity-log-alert-rule>
3. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/create-or-update>
4. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/list-by-subscription-id>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-3-enable-logging-for-security-investigation>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

6.1.2.7 Ensure that Activity Log Alert Exists for Create or Update SQL Server Firewall Rule (Automated)

Profile Applicability:

- Level 1

Description:

Create an activity log alert for the Create or Update SQL Server Firewall Rule event.

Rationale:

Monitoring for Create or Update SQL Server Firewall Rule events gives insight into network access changes and may reduce the time it takes to detect suspicious activity.

Impact:

There will be a substantial increase in log size if there are a large number of administrative actions on a server.

Audit:

Audit from Azure Portal

1. Navigate to the **Monitor** blade.
2. Click on **Alerts**.
3. In the Alerts window, click on **Alert rules**.
4. Ensure an alert rule exists where the Condition column contains **Operation name=Microsoft.Sql/servers/firewallRules/write**.
5. Click on the Alert **Name** associated with the previous step.
6. Ensure the **Condition** panel displays the text **Whenever the Activity Log has an event with Category='Administrative', Operation name='Create/Update server firewall rule'** and does not filter on **Level**, **Status** or **Caller**.
7. Ensure the **Actions** panel displays an Action group is assigned to notify the appropriate personnel in your organization.

Audit from Azure CLI

```
az monitor activity-log alert list --subscription <subscription Id> --query "[].{Name:name,Enabled:enabled,Condition:condition.allOf,Actions:actions}"
```

Look for **Microsoft.Sql/servers/firewallRules/write** in the output

Audit from PowerShell

```
Get-AzActivityLogAlert -SubscriptionId <subscription ID>|where-object  
{$_ .ConditionAllOf.Equal -match  
"Microsoft.Sql/servers/firewallRules/write"}|select-object  
Location,Name,Enabled,ResourceGroupName,ConditionAllOf
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [b954148f-4c11-4c38-8221-be76711e194a](#) - **Name:** 'An activity log alert should exist for specific Administrative operations'

Remediation:

Remediate from Azure Portal

1. Navigate to the **Monitor** blade.
2. Select **Alerts**.
3. Select **Create**.
4. Select **Alert rule**.
5. Choose a subscription.
6. Select **Apply**.
7. Select the **Condition** tab.
8. Click **See all signals**.
9. Select **Create/Update server firewall rule (Server Firewall Rule)**.
10. Click **Apply**.
11. Select the **Actions** tab.
12. Click **Select action groups** to select an existing action group, or **Create action group** to create a new action group.
13. Follow the prompts to choose or create an action group.
14. Select the **Details** tab.
15. Select a **Resource group**, provide an **Alert rule name** and an optional **Alert rule description**.
16. Click **Review + create**.
17. Click **Create**.

Remediate from Azure CLI

```
az monitor activity-log alert create --resource-group "<resource group name>"
--condition category=Administrative and
operationName=Microsoft.Sql/servers/firewallRules/write and level=<verbose |
information | warning | error | critical> --scope
"/subscriptions/<subscription ID>" --name "<activity log rule name>" --
subscription <subscription id> --action-group <action group ID>
```

Remediate from PowerShell

Create the **Conditions** object.

```
$conditions = @()
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Administrative -Field category
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Microsoft.Sql/servers/firewallRules/write -Field operationName
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Verbose -Field level
```

Retrieve the **Action Group** information and store in a variable, then create the **Actions** object.

```
$actionGroup = Get-AzActionGroup -ResourceGroupName <resource group name> -
Name <action group name>
$actionObject = New-AzActivityLogAlertActionGroupObject -Id $actionGroup.Id
```

Create the **Scope** object

```
$scope = "/subscriptions/<subscription ID>"
```

Create the **Activity Log Alert Rule** for
Microsoft.Sql/servers/firewallRules/write

```
New-AzActivityLogAlert -Name "<activity log alert rule name>" -
ResourceGroupName "<resource group name>" -Condition $conditions -Scope
$scope -Location global -Action $actionObject -Subscription <subscription ID>
-Enabled $true
```

Default Value:

By default, no monitoring alerts are created.

References:

1. <https://azure.microsoft.com/en-us/updates?id=classic-alerting-monitoring-retirement>
2. <https://learn.microsoft.com/en-us/azure/azure-monitor/alerts/alerts-create-activity-log-alert-rule>
3. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/create-or-update>

4. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/list-by-subscription-id>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-3-enable-logging-for-security-investigation>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

6.1.2.8 Ensure that Activity Log Alert Exists for Delete SQL Server Firewall Rule (Automated)

Profile Applicability:

- Level 1

Description:

Create an activity log alert for the "Delete SQL Server Firewall Rule."

Rationale:

Monitoring for Delete SQL Server Firewall Rule events gives insight into SQL network access changes and may reduce the time it takes to detect suspicious activity.

Impact:

There will be a substantial increase in log size if there are a large number of administrative actions on a server.

Audit:

Audit from Azure Portal

1. Navigate to the **Monitor** blade.
2. Click on **Alerts**.
3. In the Alerts window, click on **Alert rules**.
4. Ensure an alert rule exists where the Condition column contains **Operation name=Microsoft.Sql/servers/firewallRules/delete**.
5. Click on the Alert **Name** associated with the previous step.
6. Ensure the **Condition** panel displays the text **Whenever the Activity Log has an event with Category='Administrative', Operation name='Delete server firewall rule'** and does not filter on **Level**, **Status** or **Caller**.
7. Ensure the **Actions** panel displays an Action group is assigned to notify the appropriate personnel in your organization.

Audit from Azure CLI

```
az monitor activity-log alert list --subscription <subscription Id> --query "[].{Name:name,Enabled:enabled,Condition:condition.allOf,Actions:actions}"
```

Look for **Microsoft.Sql/servers/firewallRules/delete** in the output

Audit from PowerShell

```
Get-AzActivityLogAlert -SubscriptionId <subscription ID>|where-object  
{$_ .ConditionAllOf.Equal -match  
"Microsoft.Sql/servers/firewallRules/delete"}|select-object  
Location,Name,Enabled,ResourceGroupName,ConditionAllOf
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~Definitions

- **Policy ID:** [b954148f-4c11-4c38-8221-be76711e194a](#) - **Name:** 'An activity log alert should exist for specific Administrative operations'

Remediation:

Remediate from Azure Portal

1. Navigate to the **Monitor** blade.
2. Select **Alerts**.
3. Select **Create**.
4. Select **Alert rule**.
5. Choose a subscription.
6. Select **Apply**.
7. Select the **Condition** tab.
8. Click **See all signals**.
9. Select **Delete server firewall rule (Server Firewall Rule)**.
10. Click **Apply**.
11. Select the **Actions** tab.
12. Click **Select action groups** to select an existing action group, or **Create action group** to create a new action group.
13. Follow the prompts to choose or create an action group.
14. Select the **Details** tab.
15. Select a **Resource group**, provide an **Alert rule name** and an optional **Alert rule description**.
16. Click **Review + create**.
17. Click **Create**.

Remediate from Azure CLI

```
az monitor activity-log alert create --resource-group "<resource group name>"
--condition category=Administrative and
operationName=Microsoft.Sql/servers/firewallRules/delete and level=<verbose |
information | warning | error | critical> --scope
"/subscriptions/<subscription ID>" --name "<activity log rule name>" --
subscription <subscription id> --action-group <action group ID>
```

Remediate from PowerShell

Create the **Conditions** object.

```
$conditions = @()
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Administrative -Field category
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Microsoft.Sql/servers/firewallRules/delete -Field operationName
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Verbose -Field level
```

Retrieve the **Action Group** information and store in a variable, then create the **Actions** object.

```
$actionGroup = Get-AzActionGroup -ResourceGroupName <resource group name> -
Name <action group name>
$actionObject = New-AzActivityLogAlertActionGroupObject -Id $actionGroup.Id
```

Create the **Scope** object

```
$scope = "/subscriptions/<subscription ID>"
```

Create the **Activity Log Alert Rule** for
Microsoft.Sql/servers/firewallRules/delete

```
New-AzActivityLogAlert -Name "<activity log alert rule name>" -
ResourceGroupName "<resource group name>" -Condition $conditions -Scope
$scope -Location global -Action $actionObject -Subscription <subscription ID>
-Enabled $true
```

Default Value:

By default, no monitoring alerts are created.

References:

1. <https://azure.microsoft.com/en-us/updates?id=classic-alerting-monitoring-retirement>
2. <https://learn.microsoft.com/en-us/azure/azure-monitor/alerts/alerts-create-activity-log-alert-rule>
3. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/create-or-update>
4. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/list-by-subscription-id>

5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-3-enable-logging-for-security-investigation>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

6.1.2.9 Ensure that Activity Log Alert Exists for Create or Update Public IP Address rule (Automated)

Profile Applicability:

- Level 1

Description:

Create an activity log alert for the Create or Update Public IP Addresses rule.

Rationale:

Monitoring for Create or Update Public IP Address events gives insight into network access changes and may reduce the time it takes to detect suspicious activity.

Impact:

There will be a substantial increase in log size if there are a large number of administrative actions on a server.

Audit:

Audit from Azure Portal

1. Navigate to the **Monitor** blade.
2. Click on **Alerts**.
3. In the Alerts window, click on **Alert rules**.
4. Ensure an alert rule exists where the Condition column contains **Operation name=Microsoft.Network/publicIPAddresses/write**.
5. Click on the Alert **Name** associated with the previous step.
6. Ensure the **Condition** panel displays the text **Whenever the Activity Log has an event with Category='Administrative', Operation name='Create or Update Public Ip Address'** and does not filter on **Level**, **Status** or **Caller**.
7. Ensure the **Actions** panel displays an Action group is assigned to notify the appropriate personnel in your organization.

Audit from Azure CLI

```
az monitor activity-log alert list --subscription <subscription Id> --query "[].{Name:name,Enabled:enabled,Condition:condition.allOf,Actions:actions}"
```

Look for **Microsoft.Network/publicIPAddresses/write** in the output

Audit from PowerShell

```
Get-AzActivityLogAlert -SubscriptionId <subscription ID>|where-object  
{$_ .ConditionAllOf.Equal -match  
"Microsoft.Network/publicIPAddresses/write"}|select-object  
Location,Name,Enabled,ResourceGroupName,ConditionAllOf
```

Remediation:

Remediate from Azure Portal

1. Navigate to the **Monitor** blade.
2. Select **Alerts**.
3. Select **Create**.
4. Select **Alert rule**.
5. Choose a subscription.
6. Select **Apply**.
7. Select the **Condition** tab.
8. Click **See all signals**.
9. Select **Create or Update Public Ip Address (Public Ip Address)**.
10. Click **Apply**.
11. Select the **Actions** tab.
12. Click **Select action groups** to select an existing action group, or **Create action group** to create a new action group.
13. Follow the prompts to choose or create an action group.
14. Select the **Details** tab.
15. Select a **Resource group**, provide an **Alert rule name** and an optional **Alert rule description**.
16. Click **Review + create**.
17. Click **Create**.

Remediate from Azure CLI

```
az monitor activity-log alert create --resource-group "<resource group name>"  
--condition category=Administrative and  
operationName=Microsoft.Network/publicIPAddresses/write and level=<verbose |  
information | warning | error | critical> --scope  
"/subscriptions/<subscription ID>" --name "<activity log rule name>" --  
subscription <subscription id> --action-group <action group ID>
```

Remediate from PowerShell

Create the **Conditions** object.

```
$conditions = @()
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Administrative -Field category
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Microsoft.Network/publicIPAddresses/write -Field operationName
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Verbose -Field level
```

Retrieve the **Action Group** information and store in a variable, then create the **Actions** object.

```
$actionGroup = Get-AzActionGroup -ResourceGroupName <resource group name> -
Name <action group name>
$actionObject = New-AzActivityLogAlertActionGroupObject -Id $actionGroup.Id
```

Create the **Scope** object

```
$scope = "/subscriptions/<subscription ID>"
```

Create the **Activity Log Alert Rule** for
Microsoft.Network/publicIPAddresses/write

```
New-AzActivityLogAlert -Name "<activity log alert rule name>" -
ResourceGroupName "<resource group name>" -Condition $conditions -Scope
$scope -Location global -Action $actionObject -Subscription <subscription ID>
-Enabled $true
```

Default Value:

By default, no monitoring alerts are created.

References:

1. <https://azure.microsoft.com/en-us/updates?id=classic-alerting-monitoring-retirement>
2. <https://learn.microsoft.com/en-us/azure/azure-monitor/alerts/alerts-create-activity-log-alert-rule>
3. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/create-or-update>
4. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/list-by-subscription-id>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-3-enable-logging-for-security-investigation>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

6.1.2.10 Ensure that Activity Log Alert Exists for Delete Public IP Address rule (Automated)

Profile Applicability:

- Level 1

Description:

Create an activity log alert for the Delete Public IP Address rule.

Rationale:

Monitoring for Delete Public IP Address events gives insight into network access changes and may reduce the time it takes to detect suspicious activity.

Impact:

There will be a substantial increase in log size if there are a large number of administrative actions on a server.

Audit:

Audit from Azure Portal

1. Navigate to the **Monitor** blade.
2. Click on **Alerts**.
3. In the Alerts window, click on **Alert rules**.
4. Ensure an alert rule exists where the Condition column contains **Operation name=Microsoft.Network/publicIPAddresses/delete**.
5. Click on the Alert **Name** associated with the previous step.
6. Ensure the **Condition** panel displays the text **Whenever the Activity Log has an event with Category='Administrative', Operation name='Delete Public Ip Address'** and does not filter on **Level**, **Status** or **Caller**.
7. Ensure the **Actions** panel displays an Action group is assigned to notify the appropriate personnel in your organization.

Audit from Azure CLI

```
az monitor activity-log alert list --subscription <subscription Id> --query "[].{Name:name,Enabled:enabled,Condition:condition.allOf,Actions:actions}"
```

Look for **Microsoft.Network/publicIPAddresses/delete** in the output

Audit from PowerShell

```
Get-AzActivityLogAlert -SubscriptionId <subscription ID>|where-object  
{$_ .ConditionAllOf.Equal -match  
"Microsoft.Network/publicIPAddresses/delete"}|select-object  
Location,Name,Enabled,ResourceGroupName,ConditionAllOf
```

Remediation:

Remediate from Azure Portal

1. Navigate to the **Monitor** blade.
2. Select **Alerts**.
3. Select **Create**.
4. Select **Alert rule**.
5. Choose a subscription.
6. Select **Apply**.
7. Select the **Condition** tab.
8. Click **See all signals**.
9. Select **Delete Public Ip Address (Public Ip Address)**.
10. Click **Apply**.
11. Select the **Actions** tab.
12. Click **Select action groups** to select an existing action group, or **Create action group** to create a new action group.
13. Follow the prompts to choose or create an action group.
14. Select the **Details** tab.
15. Select a **Resource group**, provide an **Alert rule name** and an optional **Alert rule description**.
16. Click **Review + create**.
17. Click **Create**.

Remediate from Azure CLI

```
az monitor activity-log alert create --resource-group "<resource group name>"  
--condition category=Administrative and  
operationName=Microsoft.Network/publicIPAddresses/delete and level=<verbose |  
information | warning | error | critical> --scope  
"/subscriptions/<subscription ID>" --name "<activity log rule name>" --  
subscription <subscription id> --action-group <action group ID>
```

Remediate from PowerShell

Create the **Conditions** object.

```
$conditions = @()
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Administrative -Field category
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Microsoft.Network/publicIPAddresses/delete -Field operationName
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Equal Verbose -Field level
```

Retrieve the **Action Group** information and store in a variable, then create the **Actions** object.

```
$actionGroup = Get-AzActionGroup -ResourceGroupName <resource group name> -
Name <action group name>
$actionObject = New-AzActivityLogAlertActionGroupObject -Id $actionGroup.Id
```

Create the **Scope** object

```
$scope = "/subscriptions/<subscription ID>"
```

Create the **Activity Log Alert Rule** for
Microsoft.Network/publicIPAddresses/delete

```
New-AzActivityLogAlert -Name "<activity log alert rule name>" -
ResourceGroupName "<resource group name>" -Condition $conditions -Scope
$scope -Location global -Action $actionObject -Subscription <subscription ID>
-Enabled $true
```

Default Value:

By default, no monitoring alerts are created.

References:

1. <https://azure.microsoft.com/en-us/updates?id=classic-alerting-monitoring-retirement>
2. <https://learn.microsoft.com/en-us/azure/azure-monitor/alerts/alerts-create-activity-log-alert-rule>
3. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/create-or-update>
4. <https://learn.microsoft.com/en-us/rest/api/monitor/activity-log-alerts/list-by-subscription-id>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-3-enable-logging-for-security-investigation>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

6.1.2.11 Ensure that an Activity Log Alert Exists for Service Health (Automated)

Profile Applicability:

- Level 1

Description:

Create an activity log alert for Service Health.

Rationale:

Monitoring for Service Health events provides insight into service issues, planned maintenance, security advisories, and other changes that may affect the Azure services and regions in use.

Impact:

There is no charge for creating activity log alert rules.

Audit:

Audit from Azure Portal

1. Go to **Monitor**.
2. Click **Alerts**.
3. Click **Alert rules**.
4. Ensure an alert rule exists for a subscription with **Condition** set to **Service names=All, Event types=All** and **Target resource type** set to **Subscription**.
5. If an alert rule is found for step 4, click the name of the alert rule.
6. Ensure the **Actions** panel displays an action group configured to notify appropriate personnel.
7. Repeat steps 1-6 for each subscription.

Audit from Azure CLI

Run the following command to list activity log alerts:

```
az monitor activity-log alert list --subscription <subscription-id>
```

For each activity log alert, run the following command:

```
az monitor activity-log alert show --subscription <subscription-id> --resource-group <resource-group> --activity-log-alert-name <activity-log-alert>
```

Ensure an alert exists for **ServiceHealth** with **scopes** set to a subscription ID.

Repeat for each subscription.

Audit from PowerShell

Run the following command to locate **ServiceHealth** alert rules for a subscription:

```
Get-AzActivityLogAlert -SubscriptionId <subscription-id> | where-object  
{$_ .ConditionAllOf.Equal -match "ServiceHealth"} | select-object  
Location,Name,Enabled,ResourceGroupName,ConditionAllOf
```

Ensure that at least one **ServiceHealth** alert rule is returned.

Repeat for each subscription.

Remediation:

Remediate from Azure Portal

1. Go to **Monitor**.
2. Click **Alerts**.
3. Click **+ Create**.
4. Select **Alert rule** from the drop-down menu.
5. Choose a subscription.
6. Click **Apply**.
7. Select the **Condition** tab.
8. Click **See all signals**.
9. Select **Service health**.
10. Click **Apply**.
11. Open the drop-down menu next to **Event types**.
12. Check the box next to **Select all**.
13. Select the **Actions** tab.
14. Click **Select action groups** to select an existing action group, or **Create action group** to create a new action group.
15. Follow the prompts to choose or create an action group.
16. Select the **Details** tab.
17. Select a **Resource group**, provide an **Alert rule name** and an optional **Alert rule description**.
18. Click **Review + create**.
19. Click **Create**.
20. Repeat steps 1-19 for each subscription requiring remediation.

Remediate from Azure CLI

For each subscription requiring remediation, run the following command to create a **ServiceHealth** alert rule for a subscription:

```
az monitor activity-log alert create --subscription <subscription-id> --  
resource-group <resource-group> --name <alert-rule> --condition  
category=ServiceHealth and properties.incidentType=Incident --scope  
/subscriptions/<subscription-id> --action-group <action-group>
```

Remediate from PowerShell

Create the **Conditions** object:

```
$conditions = @()
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Field category -Equal ServiceHealth
$conditions += New-AzActivityLogAlertAlertRuleAnyOfOrLeafConditionObject -
Field properties.incidentType -Equal Incident
```

Retrieve the **Action Group** information and store in a variable:

```
$actionGroup = Get-AzActionGroup -ResourceGroupName <resource-group> -Name
<action-group>
```

Create the **Actions** object:

```
$actionObject = New-AzActivityLogAlertActionGroupObject -Id $actionGroup.Id
```

Create the **Scope** object:

```
$scope = "/subscriptions/<subscription-id>"
```

Create the activity log alert rule:

```
New-AzActivityLogAlert -Name <alert-rule> -ResourceGroupName <resource-group>
-Condition $conditions -Scope $scope -Location global -Action $actionObject -
Subscription <subscription-id> -Enabled $true
```

Repeat for each subscription requiring remediation.

Default Value:

By default, no monitoring alerts are created.

References:

1. <https://learn.microsoft.com/en-us/azure/service-health/overview>
2. <https://learn.microsoft.com/en-us/azure/service-health/alerts-activity-log-service-notifications-portal>
3. <https://azure.microsoft.com/en-us/pricing/details/monitor/#faq>
4. <https://learn.microsoft.com/en-us/cli/azure/monitor/activity-log/alert>
5. <https://learn.microsoft.com/en-us/powershell/module/az.monitor/get-azactivitylogalert>
6. <https://learn.microsoft.com/en-us/powershell/module/az.monitor/new-azactivitylogalert>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
		M1047

6.1.3 Configuring Application Insights

6.1.3.1 Ensure Application Insights are Configured (Automated)

Profile Applicability:

- Level 2

Description:

Application Insights within Azure act as an Application Performance Monitoring solution providing valuable data into how well an application performs and additional information when performing incident response. The types of log data collected include application metrics, telemetry data, and application trace logging data providing organizations with detailed information about application activity and application transactions. Both data sets help organizations adopt a proactive and retroactive means to handle security and performance related metrics within their modern applications.

Rationale:

Configuring Application Insights provides additional data not found elsewhere within Azure as part of a much larger logging and monitoring program within an organization's Information Security practice. The types and contents of these logs will act as both a potential cost saving measure (application performance) and a means to potentially confirm the source of a potential incident (trace logging). Metrics and Telemetry data provide organizations with a proactive approach to cost savings by monitoring an application's performance, while the trace logging data provides necessary details in a reactive incident response scenario by helping organizations identify the potential source of an incident within their application.

Impact:

Because Application Insights relies on a Log Analytics Workspace, an organization will incur additional expenses when using this service.

Audit:

Audit from Azure Portal

1. Navigate to **Application Insights**.
2. Ensure an **Application Insights** service is configured and exists.

Audit from Azure CLI

```
az monitor app-insights component show --query "[].{ID:appId, Name:name, Tenant:tenantId, Location:location, Provisioning_State:provisioningState}"
```

Ensure the above command produces output, otherwise **Application Insights** has not been configured.

Audit from PowerShell

```
Get-AzApplicationInsights|select  
location,name,appid,provisioningState,tenantid
```

Remediation:

Remediate from Azure Portal

1. Navigate to **Application Insights**.
2. Under the **Basics** tab within the **PROJECT DETAILS** section, select the **Subscription**.
3. Select the **Resource group**.
4. Within the **INSTANCE DETAILS**, enter a **Name**.
5. Select a **Region**.
6. Next to **Resource Mode**, select **Workspace-based**.
7. Within the **WORKSPACE DETAILS**, select the **Subscription** for the log analytics workspace.
8. Select the appropriate **Log Analytics Workspace**.
9. Click **Next:Tags >**.
10. Enter the appropriate **Tags** as **Name, Value** pairs.
11. Click **Next:Review+Create**.
12. Click **Create**.

Remediate from Azure CLI

```
az monitor app-insights component create --app <app name> --resource-group  
<resource group name> --location <location> --kind "web" --retention-time  
<INT days to retain logs> --workspace <log analytics workspace ID> --  
subscription <subscription ID>
```

Remediate from PowerShell

```
New-AzApplicationInsights -Kind "web" -ResourceGroupName <resource group  
name> -Name <app insights name> -location <location> -RetentionInDays <INT  
days to retain logs> -SubscriptionID <subscription ID> -WorkspaceResourceId  
<log analytics workspace ID>
```

Default Value:

Application Insights are not enabled by default.

References:

1. <https://learn.microsoft.com/en-us/azure/azure-monitor/app/app-insights-overview>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.2 <u>Collect Audit Logs</u> Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.			
v7	6.2 <u>Activate audit logging</u> Ensure that local logging has been enabled on all systems and networking devices.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1606	TA0006	M1047

6.1.4 Ensure that Azure Monitor Resource Logging is Enabled for All Services that Support it (Manual)

Profile Applicability:

- Level 1

Description:

Resource Logs capture activity to the data access plane while the Activity log is a subscription-level log for the control plane. Resource-level diagnostic logs provide insight into operations that were performed within that resource itself; for example, reading or updating a secret from a Key Vault. Currently, 95 Azure resources support Azure Monitoring (See the more information section for a complete list), including Network Security Groups, Load Balancers, Key Vault, AD, Logic Apps, and CosmosDB. The content of these logs varies by resource type.

A number of back-end services were not configured to log and store Resource Logs for certain activities or for a sufficient length. It is crucial that monitoring is correctly configured to log all relevant activities and retain those logs for a sufficient length of time. Given that the mean time to detection in an enterprise is 240 days, a minimum retention period of two years is recommended.

While an automated assessment procedure exists for this recommendation, the assessment status remains manual. Determining whether resource logging should be enabled for specific resources depends on the context and requirements of each organization and environment.

Rationale:

A lack of monitoring reduces the visibility into the data plane, and therefore an organization's ability to detect reconnaissance, authorization attempts or other malicious activity. Unlike Activity Logs, Resource Logs are not enabled by default. Specifically, without monitoring it would be impossible to tell which entities had accessed a data store that was breached. In addition, alerts for failed attempts to access APIs for Web Services or Databases are only possible when logging is enabled.

This recommendation's Level 1 profile is dependent upon the free level of retention of 30 days. More than 30 days retention of resource logs will increase the cost of this recommendation.

Impact:

Costs for monitoring varies with Log Volume. Not every resource needs to have logging enabled. It is important to determine the security classification of the data being processed by the given resource and adjust the logging based on which events need to be tracked. This is typically determined by governance and compliance requirements.

Audit:

Audit from Azure Portal

The specific steps for configuring resources within the Azure console vary depending on resource, but typically the steps are:

1. Go to the resource
2. Click on **Diagnostic settings**
3. In the blade that appears, click **Add diagnostic setting**
4. Configure the diagnostic settings
5. Click on **Save**

Audit from Azure CLI

List all **resources** for a **subscription**

```
az resource list --subscription <subscription id>
```

For each **resource** run the following

```
az monitor diagnostic-settings list --resource <resource ID>
```

An empty result means a **diagnostic settings** is not configured for that resource. An error message means a **diagnostic settings** is not supported for that resource.

Audit from PowerShell

Get a list of **resources** in a **subscription** context and store in a variable

```
$resources = Get-AzResource
```

Loop through each **resource** to determine if a diagnostic setting is configured or not.

```
foreach ($resource in $resources) {$diagnosticSetting = Get-AzDiagnosticSetting -ResourceId $resource.id -ErrorAction "SilentlyContinue";  
if ([string]::IsNullOrEmpty($diagnosticSetting)) {$message = "Diagnostic Settings not configured for resource: " + $resource.Name;Write-Output $message}else{$diagnosticSetting}}
```

A result of **Diagnostic Settings not configured for resource: <resource name>** means a **diagnostic settings** is not configured for that resource. Otherwise, the output of the above command will show configured **Diagnostic Settings** for a resource.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [cf820ca0-f99e-4f3e-84fb-66e913812d21](#) - **Name:** 'Resource logs in Key Vault should be enabled'
- **Policy ID:** [91a78b24-f231-4a8a-8da9-02c35b2b6510](#) - **Name:** 'App Service apps should have resource logs enabled'
- **Policy ID:** [428256e6-1fac-4f48-a757-df34c2b3336d](#) - **Name:** 'Resource logs in Batch accounts should be enabled'
- **Policy ID:** [057ef27e-665e-4328-8ea3-04b3122bd9fb](#) - **Name:** 'Resource logs in Azure Data Lake Store should be enabled'
- **Policy ID:** [c95c74d9-38fe-4f0d-af86-0c7d626a315c](#) - **Name:** 'Resource logs in Data Lake Analytics should be enabled'
- **Policy ID:** [83a214f7-d01a-484b-91a9-ed54470c9a6a](#) - **Name:** 'Resource logs in Event Hub should be enabled'
- **Policy ID:** [383856f8-de7f-44a2-81fc-e5135b5c2aa4](#) - **Name:** 'Resource logs in IoT Hub should be enabled'
- **Policy ID:** [34f95f76-5386-4de7-b824-0d8478470c9d](#) - **Name:** 'Resource logs in Logic Apps should be enabled'
- **Policy ID:** [b4330a05-a843-4bc8-bf9a-cacce50c67f4](#) - **Name:** 'Resource logs in Search services should be enabled'
- **Policy ID:** [f8d36e2f-389b-4ee4-898d-21aeb69a0f45](#) - **Name:** 'Resource logs in Service Bus should be enabled'
- **Policy ID:** [f9be5368-9bf5-4b84-9e0a-7850da98bb46](#) - **Name:** 'Resource logs in Azure Stream Analytics should be enabled'
- **Policy ID:** [8a04f872-51e9-4313-97fb-fc1c3543011c](#) - **Name:** 'Azure Application Gateway should have Resource logs enabled'

Remediation:

Azure Subscriptions should log every access and operation for all resources. Logs should be sent to Storage and a Log Analytics Workspace or equivalent third-party system. Logs should be kept in readily-accessible storage for a minimum of one year, and then moved to inexpensive cold storage for a duration of time as necessary. If retention policies are set but storing logs in a Storage Account is disabled (for example, if only Event Hubs or Log Analytics options are selected), the retention policies have no effect. Enable all monitoring at first, and then be more aggressive moving data to cold storage if the volume of data becomes a cost concern.

Remediate from Azure Portal

The specific steps for configuring resources within the Azure console vary depending on resource, but typically the steps are:

1. Go to the resource
2. Click on **Diagnostic settings**
3. In the blade that appears, click **Add diagnostic setting**
4. Configure the diagnostic settings
5. Click on **Save**

Remediate from Azure CLI

For each **resource**, run the following making sure to use a **resource** appropriate JSON encoded **category** for the **--logs** option.

```
az monitor diagnostic-settings create --name <diagnostic settings name> --
resource <resource ID> --logs "[{category:<resource specific
category>,enabled:true,retention-policy:{enabled:true,days:180}}]" --metrics
"[{category:AllMetrics,enabled:true,retention-
policy:{enabled:true,days:180}}]" <[--event-hub <event hub ID> --event-hub-
rule <event hub auth rule ID> | --storage-account <storage account ID> |--
workspace <log analytics workspace ID> | --marketplace-partner-id <full
resource ID of third-party solution>]>
```

Remediate from PowerShell

Create the **log** settings object

```
$logSettings = @()
$logSettings += New-AzDiagnosticSettingLogSettingsObject -Enabled $true -
RetentionPolicyDay 180 -RetentionPolicyEnabled $true -Category <resource
specific category>
$logSettings += New-AzDiagnosticSettingLogSettingsObject -Enabled $true -
RetentionPolicyDay 180 -RetentionPolicyEnabled $true -Category <resource
specific category number 2>
```

Create the **metric** settings object

```
$metricSettings = @()
$metricSettings += New-AzDiagnosticSettingMetricSettingsObject -Enabled $true
-RetentionPolicyDay 180 -RetentionPolicyEnabled $true -Category AllMetrics
```

Create the diagnostic setting for a specific resource

```
New-AzDiagnosticSetting -Name "<diagnostic settings name>" -ResourceId
<resource ID> -Log $logSettings -Metric $metricSettings
```

Default Value:

By default, Azure Monitor Resource Logs are 'Disabled' for all resources.

References:

1. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-3-enable-logging-for-security-investigation>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-5-centralize-security-log-management-and-analysis>
3. <https://learn.microsoft.com/en-us/azure/azure-monitor/essentials/monitor-azure-resource>
4. Supported Log Categories: <https://learn.microsoft.com/en-us/azure/azure-monitor/reference/logs-index>
5. Logs and Audit - Fundamentals: <https://learn.microsoft.com/en-us/azure/security/fundamentals/log-audit>
6. Collecting Logs: <https://learn.microsoft.com/en-us/azure/azure-monitor/essentials/activity-log>
7. Key Vault Logging: <https://learn.microsoft.com/en-us/azure/key-vault/general/logging>
8. Monitor Diagnostic Settings: <https://learn.microsoft.com/en-us/cli/azure/monitor/diagnostic-settings>
9. Overview of Diagnostic Logs: <https://learn.microsoft.com/en-us/azure/azure-monitor/fundamentals/data-sources>
10. Supported Services for Diagnostic Logs: <https://learn.microsoft.com/en-us/azure/azure-monitor/essentials/resource-logs-schema>
11. Diagnostic Logs for CDNs: <https://learn.microsoft.com/en-us/azure/cdn/cdn-azure-diagnostic-logs>

Additional Information:

For an up-to-date list of Azure resources which support Azure Monitor, refer to the "Supported Log Categories" reference.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v8	8.9 <u>Centralize Audit Logs</u> Centralize, to the extent possible, audit log collection and retention across enterprise assets.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

Controls Version	Control	IG 1	IG 2	IG 3
v7	6.5 Central Log Management Ensure that appropriate logs are being aggregated to a central log management system for analysis and review.		●	●

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1485	TA0040	M1053

6.1.5 Ensure Basic, Free, and Consumption SKUs are not used on Production artifacts requiring monitoring and SLA (Manual)

Profile Applicability:

- Level 2

Description:

The use of **Basic**, **Free**, or **Consumption** SKUs in Azure, whilst cost effective, have significant limitations in terms of what can be monitored and what support can be realized from Microsoft. Typically, these SKUs do not have a service SLA and Microsoft may refuse to provide support for them. Furthermore, they offer reduced or no high availability guarantees and have limited observability and diagnostic capabilities. As a result, Microsoft may limit or refuse support for workloads running on these SKUs. Consequently **Basic**, **Free**, or **Consumption** SKUs should **never** be used for production workloads unless your organization explicitly approved it as an exception.

While an automated assessment procedure exists for this recommendation, the assessment status remains manual. Determining appropriate SKUs depends on the context and requirements of each organization and environment.

Rationale:

Typically, production workloads need to be continuously monitored and have high availability and resilience, and they should have an SLA with Microsoft. Using Basic, Free, or Consumption SKUs for deployed Azure services generally prevents these requirements from being met.

Therefore, production workloads must use **Standard** or higher SKUs to ensure operational reliability, and supportability. The following Azure resource types must use **Standard** SKUs (or higher) at a minimum when deployed in production environments:

- Public IP Addresses
- Network Load Balancers
- Azure Cache for Redis
- Azure SQL (PaaS Databases)
- VPN Gateways
- Azure Bastion
- Application Gateway
- Virtual Network Gateway (VPN)
- Azure Container Registry
- Event Hubs
- Service Bus
- API Management
- Azure Monitor (Log Analytics)

Using Basic, Free, or Consumption SKUs for any of the above services typically removes SLA guarantees, high availability, and operational visibility such as monitoring and diagnostics, which are usually considered mandatory requirements for production environments.

Impact:

Enforcing Standard (or higher) SKUs for production environments has the following impacts:

1. There will be a cost increase compared to Basic or Free SKUs.
2. Improved reliability, monitoring, and Microsoft SLA coverage will be available and will provide enterprise-grade support for the production service and compliance with production readiness standards.

All Azure resources must be clearly classified as Production or Non-Production by one or more of the following mechanisms:

- Mandatory tags (e.g., **Environment=Production**)
- Separation via Management Groups
- Dedicated Subscriptions

Audit:

This needs to be audited by Azure Policy (one for each resource type) and denied for each artifact that is production.

Audit from Azure Portal

1. Open **Azure Resource Graph Explorer**
2. Click **New query**
3. Paste the following into the query window:

```
Resources
| where isnotnull(sku)
| where sku.name =~ 'Basic'
    or sku.tier =~ 'Basic'
    or sku.name =~ 'Free'
    or sku.tier =~ 'Free'
    or sku.name =~ 'Consumption'
    or sku.tier =~ 'Consumption'
| order by type
```

4. Click **Run query** then evaluate the results in the results window.
5. Ensure that no production artifacts are returned.

If production workload or environment tags exist, use the following query and edit the tag name and value in **Environment** and **Production** to reflect your tagging standard:

```
Resources
| where tostring(tags.Environment) =~ 'Production'
| where isnotnull(sku)
| where sku.name =~ 'Basic'
    or sku.tier =~ 'Basic'
    or sku.name =~ 'Free'
    or sku.tier =~ 'Free'
    or sku.name =~ 'Consumption'
    or sku.tier =~ 'Consumption'
| order by type
```

Audit from Azure CLI

```
az graph query -q "
Resources
| where isnotnull(sku)
| where sku.name =~ 'Basic'
    or sku.tier =~ 'Basic'
    or sku.name =~ 'Free'
    or sku.tier =~ 'Free'
    or sku.name =~ 'Consumption'
    or sku.tier =~ 'Consumption'
| order by type
"
```

Alternatively, to filter on a specific resource group:

```
az graph query -q "
Resources
| where resourceGroup == '<resourceGroupName>'
| where isnotnull(sku)
| where sku.name =~ 'Basic'
    or sku.tier =~ 'Basic'
    or sku.name =~ 'Free'
    or sku.tier =~ 'Free'
    or sku.name =~ 'Consumption'
    or sku.tier =~ 'Consumption'
| order by type
"
```

Ensure that no production artifacts are returned.

If production workload or environment tags exist, use the following query and edit the tag name and value in **Environment** and **Production** to reflect your tagging standard:


```
az graph query -q "
Resources
| where tostring(tags.Environment) =~ 'Production'
| where isnotnull(sku)
| where sku.name =~ 'Basic'
    or sku.tier =~ 'Basic'
    or sku.name =~ 'Free'
    or sku.tier =~ 'Free'
    or sku.name =~ 'Consumption'
    or sku.tier =~ 'Consumption'
| order by type
"
```

Audit from PowerShell

```
Get-AzResource |
Where-Object {
    $_.Sku -and (
        $_.Sku.Name -in @("Basic","Free","Consumption") -or
        $_.Sku.Tier -in @("Basic","Free","Consumption")
    )
}
```

If production workload or environment tags exist, use the following query and edit the tag name and value in **Environment** and **Production** to reflect your tagging standard:

```
Get-AzResource |
Where-Object {
    $_.Tags.Environment -eq "Production" -and
    $_.Sku -and (
        $_.Sku.Name -in @("Basic","Free","Consumption") -or
        $_.Sku.Tier -in @("Basic","Free","Consumption")
    )
}
```

Ensure that no production artifacts are returned.

Remediation:

Each resource has its own process for upgrading from Basic/Free/Consumption to Standard (or higher) SKUs. Follow the official Microsoft documentation for the relevant service.

- Public IP Address: <https://learn.microsoft.com/en-us/azure/virtual-network/ip-services/public-ip-upgrade>.
- Basic Load Balancer: <https://learn.microsoft.com/en-us/azure/load-balancer/load-balancer-basic-upgrade-guidance>.
- Azure Cache for Redis: <https://learn.microsoft.com/en-us/azure/azure-cache-for-redis/cache-how-to-scale>.
- Azure SQL Database: <https://learn.microsoft.com/en-us/azure/azure-sql/database/scale-resources>.
- VPN Gateway: <https://learn.microsoft.com/en-us/azure/vpn-gateway/gateway-sku-resize>.

- Virtual Network Gateway (VPN): <https://learn.microsoft.com/azure/vpn-gateway/gateway-sku-resize>
- Azure Bastion: <https://learn.microsoft.com/azure/bastion/bastion-faq#is-there-a-basic-sku> (No Basic SKU exists; upgrade = redeploy with Standard SKU. Developer tier is not supported for production; deploy using Standard SKU.)
- Application Gateway: <https://learn.microsoft.com/azure/application-gateway/upgrade-application-gateway>
- Azure Container Registry: <https://learn.microsoft.com/azure/container-registry/container-registry-skus>
- Event Hubs: <https://learn.microsoft.com/azure/event-hubs/event-hubs-scale>
- Service Bus: <https://learn.microsoft.com/azure/service-bus-messaging/service-bus-scaling>
- API Management: <https://learn.microsoft.com/azure/api-management/upgrade-and-scale>
- Azure Monitor (Log Analytics Workspace): <https://learn.microsoft.com/azure/azure-monitor/logs/cost-logs#pricing-tiers> (The legacy Free tier cannot be upgraded. A new workspace must be created using the Pay-As-You-Go or Commitment Tier pricing model.)

Default Value:

Policy should enforce Standard SKUs (or higher) for the following artifacts:

- Public IP Addresses
- Network Load Balancers
- Azure Cache for Redis
- SQL PaaS Databases
- VPN Gateways
- Azure Bastion
- Application Gateway
- Virtual Network Gateway (VPN)
- Azure Container Registry
- Event Hubs
- Service Bus
- API Management
- Azure Monitor (Log Analytics Workspace)

References:

1. <https://azure.microsoft.com/en-us/support/plans>
2. <https://azure.microsoft.com/en-us/support/plans/response/>
3. <https://learn.microsoft.com/en-us/azure/virtual-network/ip-services/public-ip-upgrade>
4. <https://learn.microsoft.com/en-us/azure/load-balancer/load-balancer-basic-upgrade-guidance>
5. <https://learn.microsoft.com/en-us/azure/azure-cache-for-redis/cache-how-to-scale>

6. <https://learn.microsoft.com/en-us/azure/azure-sql/database/scale-resources>
7. <https://learn.microsoft.com/en-us/azure/vpn-gateway/gateway-sku-resize>
8. <https://learn.microsoft.com/azure/bastion/bastion-faq#is-there-a-basic-sku>
9. <https://learn.microsoft.com/azure/application-gateway/upgrade-application-gateway>
10. <https://learn.microsoft.com/azure/container-registry/container-registry-skus>
11. <https://learn.microsoft.com/azure/event-hubs/event-hubs-scale>
12. <https://learn.microsoft.com/azure/service-bus-messaging/service-bus-scaling>
13. <https://learn.microsoft.com/azure/api-management/upgrade-and-scale>
14. <https://learn.microsoft.com/azure/azure-monitor/logs/cost-logs#pricing-tiers>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	2.2 Ensure Authorized Software is Currently Supported Ensure that only currently supported software is designated as authorized in the software inventory for enterprise assets. If software is unsupported, yet necessary for the fulfillment of the enterprise's mission, document an exception detailing mitigating controls and residual risk acceptance. For any unsupported software without an exception documentation, designate as unauthorized. Review the software list to verify software support at least monthly, or more frequently.			
v7	2.2 Ensure Software is Supported by Vendor Ensure that only software applications or operating systems currently supported by the software's vendor are added to the organization's authorized software inventory. Unsupported software should be tagged as unsupported in the inventory system.			

6.2 Ensure that Resource Locks are set for Mission-Critical Azure Resources (Manual)

Profile Applicability:

- Level 2

Description:

Resource Manager Locks provide a way for administrators to lock down Azure resources to prevent deletion of, or modifications to, a resource. These locks sit outside of the Role Based Access Controls (RBAC) hierarchy and, when applied, will place restrictions on the resource for all users. These locks are very useful when there is an important resource in a subscription that users should not be able to delete or change. Locks can help prevent accidental and malicious changes or deletion.

While an automated assessment procedure exists for this recommendation, the assessment status remains manual. Determining resources that require resource locks depends on the context and requirements of each organization and environment.

Rationale:

As an administrator, it may be necessary to lock a subscription, resource group, or resource to prevent other users in the organization from accidentally deleting or modifying critical resources. The lock level can be set to **CanNotDelete** or **ReadOnly** to achieve this purpose.

- **CanNotDelete** means authorized users can still read and modify a resource, but they cannot delete the resource.
- **ReadOnly** means authorized users can read a resource, but they cannot delete or update the resource. Applying this lock is similar to restricting all authorized users to the permissions granted by the Reader role.

Impact:

There can be unintended outcomes of locking a resource. Applying a lock to a parent service will cause it to be inherited by all resources within. Conversely, applying a lock to a resource may not apply to connected storage, leaving it unlocked. Please see the documentation for further information.

Audit:

Audit from Azure Portal

1. Navigate to the specific Azure Resource or Resource Group.
2. Click on **Locks**.
3. Ensure the lock is defined with name and description, with type **Read-only** or **Delete** as appropriate.

Audit from Azure CLI

Review the list of all locks set currently:

```
az lock list --resource-group <resourcegroupname> --resource-name
<resourcename> --namespace <Namespace> --resource-type <type> --parent ""
```

Audit from PowerShell

Run the following command to list all resources and loop through those resources to check for applied locks. Requires Az PowerShell Module (not supported in Graph PowerShell at this time).

```
Connect-AzAccount
$resources = Get-AzResource
$complianceResults = @()
foreach ($resource in $resources) {
    $locks = Get-AzResourceLock -ResourceName $resource.Name `
                                -ResourceType $resource.ResourceType `
                                -ResourceGroupName
$resource.ResourceGroupName
    foreach ($lock in $locks) {
        $isCompliant = $false
        if ($lock.Properties.Level -eq "CanNotDelete" -or
$lock.Properties.Level -eq "ReadOnly") {
            $isCompliant = $true
        }
        $complianceResults += [PSCustomObject]@{
            ResourceName      = $resource.Name
            ResourceGroup      = $resource.ResourceGroupName
            ResourceType       = $resource.ResourceType
            LockName           = $lock.Name
            LockLevel           = $lock.Properties.Level
            Compliant           = $isCompliant
        }
    }
}
$complianceResults | Format-Table -AutoSize
```

Review the output of the **Properties** setting. Compliant settings will have the **CanNotDelete** or **ReadOnly** value.

Remediation:

Remediate from Azure Portal

1. Navigate to the specific Azure Resource or Resource Group.
2. For each mission critical resource, click on **Locks**.
3. Click **Add**.
4. Give the lock a name and a description, then select the type, **Read-only** or **Delete** as appropriate.
5. Click OK.

Remediate from Azure CLI

To lock a resource, provide the name of the resource, its resource type, and its resource group name.

```
az lock create --name <LockName> --lock-type <CanNotDelete/Read-only> --resource-group <resourceGroupName> --resource-name <resourceName> --resource-type <resourceType>
```

Remediate from PowerShell

```
Get-AzResourceLock -ResourceName <Resource Name> -ResourceType <Resource Type> -ResourceGroupName <Resource Group Name> -Locktype <CanNotDelete/Read-only>
```

Default Value:

By default, no locks are set.

References:

1. <https://learn.microsoft.com/en-us/azure/azure-resource-manager/management/lock-resources>
2. <https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/landing-zone/design-area/management-platform#inventory-and-visibility-recommendations>
3. <https://learn.microsoft.com/en-us/azure/governance/blueprints/concepts/resource-locking>
4. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-asset-management#am-4-limit-access-to-asset-management>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 Configure Data Access Control Lists Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 Protect Information through Access Control Lists Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

7 Networking Services

To better understand the relationship between the Foundations Benchmark and Services Benchmarks, please read the "Introduction" section of this document.

This section covers security recommendations to follow in order to set networking policies on an Azure subscription.

Azure Product Directory Reference: <https://azure.microsoft.com/en-us/products#networking>

FEEDBACK REQUEST: Is there a specific service or recommendation in this section that you'd like to see addressed or improved? Let us know by making a ticket or starting a discussion in the CIS Microsoft Azure Community (<https://workbench.cisecurity.org/communities/72>).

7.1 Ensure that RDP Access from the Internet is Evaluated and Restricted (Automated)

Profile Applicability:

- Level 1

Description:

Network security groups should be periodically evaluated for port misconfigurations. Where RDP is not explicitly required and narrowly configured for resources attached to a network security group, Internet-level access to Azure resources should be restricted or eliminated.

Rationale:

The potential security problem with using RDP over the Internet is that attackers can use various brute force techniques to gain access to Azure Virtual Machines. Once the attackers gain access, they can use a virtual machine as a launch point for compromising other machines on an Azure Virtual Network or even attack networked devices outside of Azure.

Audit:

Audit from Azure Portal

1. Go to **Network security groups**.
2. Click the name of a network security group.
3. Under **Settings**, click **Inbound security rules**.
4. Ensure that no inbound security rule exists that matches the following:
 - Port: **3389** or range including 3389
 - Protocol: **TCP** or **Any**
 - Source: **0.0.0.0/0**, **Internet**, or **Any**
 - Action: **Allow**
5. Repeat steps 1-4 for each network security group.

To audit from Azure Resource Graph:

1. Go to **Resource Graph Explorer**.
2. Click **New query**.
3. Paste the following into the query window:


```
resources | where type =~ "microsoft.network/networksecuritygroups" |
project id, name, securityRule = properties.securityRules | mv-expand
securityRule | extend access = securityRule.properties.access,
direction = securityRule.properties.direction, protocol =
securityRule.properties.protocol, destinationPort =
case(isempty(securityRule.properties.destinationPortRange),
securityRule.properties.destinationPortRanges,
securityRule.properties.destinationPortRange), sourceAddress =
case(isempty(securityRule.properties.sourceAddressPrefix),
securityRule.properties.sourceAddressPrefixes,
securityRule.properties.sourceAddressPrefix) | where access =~ "Allow"
and direction =~ "Inbound" and protocol in~ ("tcp", "*") | mv-expand
destinationPort | mv-expand sourceAddress | extend destinationPortMin =
toint(split(destinationPort, "-")[0]), destinationPortMax =
toint(split(destinationPort, "-")[-1]) | where (destinationPortMin <=
3389 and destinationPortMax >= 3389) or destinationPort == "" | where
sourceAddress in~ ("*", "0.0.0.0", "internet", "any") or sourceAddress
endswith "/0"
```

4. Click **Run query**.
5. Ensure that no results are returned.

Audit from Azure CLI

List network security groups with non-default security rules:

```
az network nsg list --query [*].[name,securityRules]
```

Ensure that no network security group has an inbound security rule that matches the following:

```
"access" : "Allow"
"destinationPortRange" : "3389", "*", or "<range-including-3389>"
"direction" : "Inbound"
"protocol" : "TCP" or "*"
"sourceAddressPrefix" : "0.0.0.0/0", "Internet", or "*"
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

<https://portal.azure.com/#view/Microsoft Azure Policy/PolicyMenuBlade/~ /Definitions>

- **Policy ID:** [22730e10-96f6-4aac-ad84-9383d35b5917](#) - **Name:** 'Management ports should be closed on your virtual machines'

Remediation:

Remediate from Azure Portal

1. Go to **Network security groups**.
2. Click the name of a network security group.
3. Under **Settings**, click **Inbound security rules**.
4. Check the box next to any inbound security rule matching:
 - Port: **3389** or range including 3389
 - Protocol: **TCP** or **Any**
 - Source: **0.0.0.0/0**, **Internet**, or **Any**
 - Action: **Allow**
5. Click **Delete**.
6. Click **Yes**.
7. Repeat steps 1-6 for each network security group requiring remediation.

Remediate from Azure CLI

For each network security group rule requiring remediation, run the following command to delete the rule:

```
az network nsg rule delete --resource-group <resource-group> --nsg-name  
<network-security-group> --name <rule>
```

Default Value:

By default, RDP access from internet is not **enabled**.

References:

1. <https://learn.microsoft.com/en-us/azure/security/fundamentals/network-best-practices#disable-rdpssh-access-to-virtual-machines>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-network-security#ns-1-establish-network-segmentation-boundaries>
3. Express Route: <https://learn.microsoft.com/en-us/azure/expressroute/>
4. Site-to-Site VPN: <https://learn.microsoft.com/en-us/azure/vpn-gateway/tutorial-site-to-site-portal>
5. Point-to-Site VPN: <https://learn.microsoft.com/en-us/azure/vpn-gateway/point-to-site-certificate-gateway>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>13.4 Perform Traffic Filtering Between Network Segments</u> Perform traffic filtering between network segments, where appropriate.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1021.001	TA0001	M1035

7.2 Ensure that SSH Access from the Internet is Evaluated and Restricted (Automated)

Profile Applicability:

- Level 1

Description:

Network security groups should be periodically evaluated for port misconfigurations. Where SSH is not explicitly required and narrowly configured for resources attached to a network security group, Internet-level access to Azure resources should be restricted or eliminated.

Rationale:

The potential security problem with using SSH over the Internet is that attackers can use various brute force techniques to gain access to Azure Virtual Machines. Once the attackers gain access, they can use a virtual machine as a launch point for compromising other machines on the Azure Virtual Network or even attack networked devices outside of Azure.

Audit:

Audit from Azure Portal

1. Go to **Network security groups**.
2. Click the name of a network security group.
3. Under **Settings**, click **Inbound security rules**.
4. Ensure that no inbound security rule exists that matches the following:
 - Port: **22** or range including 22
 - Protocol: **TCP** or **Any**
 - Source: **0.0.0.0/0**, **Internet**, or **Any**
 - Action: **Allow**
5. Repeat steps 1-4 for each network security group.

To audit from Azure Resource Graph:

1. Go to **Resource Graph Explorer**.
2. Click **New query**.
3. Paste the following into the query window:

```
resources | where type =~ "microsoft.network/networksecuritygroups" |
project id, name, securityRule = properties.securityRules | mv-expand
securityRule | extend access = securityRule.properties.access,
direction = securityRule.properties.direction, protocol =
securityRule.properties.protocol, destinationPort =
case(isempty(securityRule.properties.destinationPortRange),
securityRule.properties.destinationPortRanges,
securityRule.properties.destinationPortRange), sourceAddress =
case(isempty(securityRule.properties.sourceAddressPrefix),
securityRule.properties.sourceAddressPrefixes,
securityRule.properties.sourceAddressPrefix) | where access =~ "Allow"
and direction =~ "Inbound" and protocol in~ ("tcp", "*") | mv-expand
destinationPort | mv-expand sourceAddress | extend destinationPortMin =
toint(split(destinationPort, "-")[0]), destinationPortMax =
toint(split(destinationPort, "-")[-1]) | where (destinationPortMin <=
22 and destinationPortMax >= 22) or destinationPort == "" | where
sourceAddress in~ ("*", "0.0.0.0", "internet", "any") or sourceAddress
endswith "/0"
```

4. Click **Run query**.
5. Ensure that no results are returned.

Audit from Azure CLI

List network security groups with non-default security rules:

```
az network nsg list --query [*].[name,securityRules]
```

Ensure that no network security group has an inbound security rule that matches the following:

```
"access" : "Allow"
"destinationPortRange" : "22", "*", or "<range-including-22>"
"direction" : "Inbound"
"protocol" : "TCP" or "*"
"sourceAddressPrefix" : "0.0.0.0/0", "Internet", or "*"
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [22730e10-96f6-4aac-ad84-9383d35b5917](#) - **Name:** 'Management ports should be closed on your virtual machines'

Remediation:

Remediate from Azure Portal

1. Go to **Network security groups**.
2. Click the name of a network security group.
3. Under **Settings**, click **Inbound security rules**.
4. Check the box next to any inbound security rule matching:
 - o Port: **22** or range including 22
 - o Protocol: **TCP** or **Any**
 - o Source: **0.0.0.0/0**, **Internet**, or **Any**
 - o Action: **Allow**
5. Click **Delete**.
6. Click **Yes**.
7. Repeat steps 1-6 for each network security group requiring remediation.

Remediate from Azure CLI

For each network security group rule requiring remediation, run the following command to delete the rule:

```
az network nsg rule delete --resource-group <resource-group> --nsg-name  
<network-security-group> --name <rule>
```

Default Value:

By default, SSH access from internet is not **enabled**.

References:

1. <https://learn.microsoft.com/en-us/azure/security/fundamentals/network-best-practices#disable-rdpssh-access-to-virtual-machines>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-network-security#ns-1-establish-network-segmentation-boundaries>
3. Express Route: <https://learn.microsoft.com/en-us/azure/expressroute/>
4. Site-to-Site VPN: <https://learn.microsoft.com/en-us/azure/vpn-gateway/tutorial-site-to-site-portal>
5. Point-to-Site VPN: <https://learn.microsoft.com/en-us/azure/vpn-gateway/point-to-site-certificate-gateway>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>13.4 Perform Traffic Filtering Between Network Segments</u> Perform traffic filtering between network segments, where appropriate.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1021.004	TA0001	M1035

7.3 Ensure that UDP Port Access from the Internet is Evaluated and Restricted (Automated)

Profile Applicability:

- Level 1

Description:

Network security groups should be periodically evaluated for port misconfigurations. Where UDP is not explicitly required and narrowly configured for resources attached to a network security group, Internet-level access to Azure resources should be restricted or eliminated.

Rationale:

The potential security problem with broadly exposing UDP services over the Internet is that attackers can use DDoS amplification techniques to reflect spoofed UDP traffic from Azure Virtual Machines. The most common types of these attacks exploit exposed DNS, NTP, SSDP, SNMP, CLDAP, and other UDP-based services as amplification sources to disrupt services on other machines within the Azure Virtual Network, or even attack networked devices outside of Azure.

Audit:

Audit from Azure Portal

1. Go to **Network security groups**.
2. Click the name of a network security group.
3. Under **Settings**, click **Inbound security rules**.
4. Ensure that no inbound security rule exists that matches the following:
 - Port: **53, 123, 161, 389, or 1900**, or range including **53, 123, 161, 389, or 1900**.
 - Protocol: **UDP** or **Any**
 - Source: **0.0.0.0/0, Internet, or Any**
 - Action: **Allow**
5. Repeat steps 1-4 for each network security group.

To audit from Azure Resource Graph:

1. Go to **Resource Graph Explorer**.
2. Click **New query**.
3. Paste the following into the query window:


```
resources | where type =~ "microsoft.network/networksecuritygroups" |
project id, name, securityRule = properties.securityRules | mv-expand
securityRule | extend access = securityRule.properties.access,
direction = securityRule.properties.direction, protocol =
securityRule.properties.protocol, destinationPort =
case(isempty(securityRule.properties.destinationPortRange),
securityRule.properties.destinationPortRanges,
securityRule.properties.destinationPortRange), sourceAddress =
case(isempty(securityRule.properties.sourceAddressPrefix),
securityRule.properties.sourceAddressPrefixes,
securityRule.properties.sourceAddressPrefix) | where access =~ "Allow"
and direction =~ "Inbound" and protocol in~ ("udp", "*") | mv-expand
destinationPort | mv-expand sourceAddress | extend destinationPortMin =
toint(split(destinationPort, "-")[0]), destinationPortMax =
toint(split(destinationPort, "-")[-1]) | where (destinationPortMin <=
53 and destinationPortMax >= 53) or (destinationPortMin <= 123 and
destinationPortMax >= 123) or (destinationPortMin <= 161 and
destinationPortMax >= 161) or (destinationPortMin <= 389 and
destinationPortMax >= 389) or (destinationPortMin <= 1900 and
destinationPortMax >= 1900) or destinationPort == "" | where
sourceAddress in~ ("*", "0.0.0.0", "internet", "any") or sourceAddress
endswith "/0"
```

4. Click **Run query**.
5. Ensure that no results are returned.

Audit from Azure CLI

List network security groups with non-default security rules:

```
az network nsg list --query [*].[name,securityRules]
```

Ensure that no network security group has an inbound security rule that matches the following:

```
"access" : "Allow"
"destinationPortRange" : "53", "123", "161", "389", "1900", "*" or "<range-
including-53-123-161-389-1900-or-other-vulnerable-udp-based-services>"
"direction" : "Inbound"
"protocol" : "UDP" or "*"
"sourceAddressPrefix" : "0.0.0.0/0", "Internet", or "*"
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [9daedab3-fb2d-461e-b861-71790eead4f6](#) - **Name:** 'All network ports should be restricted on network security groups associated to your virtual machine'

Remediation:

Remediate from Azure Portal

1. Go to **Network security groups**.
2. Click the name of a network security group.
3. Under **Settings**, click **Inbound security rules**.
4. Check the box next to any inbound security rule matching:
 - o Port: Port: **53, 123, 161, 389**, or **1900**, or range including **53, 123, 161, 389**, or **1900**.
 - o Protocol: **UDP** or **Any**
 - o Source: **0.0.0.0/0**, **Internet**, or **Any**
 - o Action: **Allow**
5. Click **Delete**.
6. Click **Yes**.
7. Repeat steps 1-6 for each network security group requiring remediation.

Remediate from Azure CLI

For each network security group rule requiring remediation, run the following command to delete the rule:

```
az network nsg rule delete --resource-group <resource-group> --nsg-name <network-security-group> --name <rule>
```

Default Value:

By default, UDP access from internet is not **enabled**.

References:

1. <https://learn.microsoft.com/en-us/azure/security/fundamentals/network-best-practices#secure-your-critical-azure-service-resources-to-only-your-virtual-networks>
2. <https://learn.microsoft.com/en-us/azure/ddos-protection/fundamental-best-practices>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-network-security#ns-1-establish-network-segmentation-boundaries>
4. ExpressRoute: <https://learn.microsoft.com/en-us/azure/expressroute/>
5. Site-to-site VPN: <https://learn.microsoft.com/en-us/azure/vpn-gateway/tutorial-site-to-site-portal>
6. Point-to-site VPN: <https://learn.microsoft.com/en-us/azure/vpn-gateway/point-to-site-certificate-gateway>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>13.4 Perform Traffic Filtering Between Network Segments</u> Perform traffic filtering between network segments, where appropriate.			
v7	<u>9.2 Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1095	TA0011	M1037

7.4 Ensure that HTTP(S) Access from the Internet is Evaluated and Restricted (Automated)

Profile Applicability:

- Level 1

Description:

Network security groups should be periodically evaluated for port misconfigurations. Where HTTP(S) is not explicitly required and narrowly configured for resources attached to a network security group, Internet-level access to Azure resources should be restricted or eliminated.

Rationale:

The potential security problem with using HTTP(S) over the Internet is that attackers can use various brute force techniques to gain access to Azure resources. Once the attackers gain access, they can use the resource as a launch point for compromising other resources within the Azure tenant.

Audit:

Audit from Azure Portal

1. Go to **Network security groups**.
2. Click the name of a network security group.
3. Under **Settings**, click **Inbound security rules**.
4. Ensure that no inbound security rule exists that matches the following:
 - Port: **80, 443**, or range including 80 or 443
 - Protocol: **TCP** or **Any**
 - Source: **0.0.0.0/0, Internet**, or **Any**
 - Action: **Allow**
5. Repeat steps 1-4 for each network security group.

To audit from Azure Resource Graph:

1. Go to **Resource Graph Explorer**.
2. Click **New query**.
3. Paste the following into the query window:

```
resources | where type =~ "microsoft.network/networksecuritygroups" |
project id, name, securityRule = properties.securityRules | mv-expand
securityRule | extend access = securityRule.properties.access,
direction = securityRule.properties.direction, protocol =
securityRule.properties.protocol, destinationPort =
case(isempty(securityRule.properties.destinationPortRange),
securityRule.properties.destinationPortRanges,
securityRule.properties.destinationPortRange), sourceAddress =
case(isempty(securityRule.properties.sourceAddressPrefix),
securityRule.properties.sourceAddressPrefixes,
securityRule.properties.sourceAddressPrefix) | where access =~ "Allow"
and direction =~ "Inbound" and protocol in~ ("tcp", "*") | mv-expand
destinationPort | mv-expand sourceAddress | extend destinationPortMin =
toint(split(destinationPort, "-")[0]), destinationPortMax =
toint(split(destinationPort, "-")[-1]) | where (destinationPortMin <=
80 and destinationPortMax >= 80) or (destinationPortMin <= 443 and
destinationPortMax >= 443) or destinationPort == "" | where
sourceAddress in~ ("*", "0.0.0.0", "internet", "any") or sourceAddress
endswith "/0"
```

4. Click **Run query**.
5. Ensure that no results are returned.

Audit from Azure CLI

List network security groups non-default security rules:

```
az network nsg list --query [*].[name,securityRules]
```

Ensure that no network security group has an inbound security rule that matches the following:

```
"access" : "Allow"
"destinationPortRange" : "80", "443", "*", or "<range-including-80-or-443>"
"direction" : "Inbound"
"protocol" : "TCP" or "*"
"sourceAddressPrefix" : "0.0.0.0/0", "Internet", or "*"
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

<https://portal.azure.com/#view/Microsoft Azure Policy/PolicyMenuBlade/~/Definitions>

- **Policy ID:** [9daedab3-fb2d-461e-b861-71790eead4f6](#) - **Name:** 'All network ports should be restricted on network security groups associated to your virtual machine'

Remediation:

Remediate from Azure Portal

1. Go to **Network security groups**.
2. Click the name of a network security group.
3. Under **Settings**, click **Inbound security rules**.
4. Check the box next to any inbound security rule matching:
 - o Port: **80**, **443**, or range including 80 or 443
 - o Protocol: **TCP** or **Any**
 - o Source: **0.0.0.0/0**, **Internet**, or **Any**
 - o Action: **Allow**
5. Click **Delete**.
6. Click **Yes**.
7. Repeat steps 1-6 for each network security group requiring remediation.

Remediate from Azure CLI

For each network security group rule requiring remediation, run the following command to delete the rule:

```
az network nsg rule delete --resource-group <resource-group> --nsg-name  
<network-security-group> --name <rule>
```

Default Value:

By default, HTTP(S) access from internet is not **enabled**.

References:

1. Express Route: <https://learn.microsoft.com/en-us/azure/expressroute/>
2. Site-to-Site VPN: <https://learn.microsoft.com/en-us/azure/vpn-gateway/tutorial-site-to-site-portal>
3. Point-to-Site VPN: <https://learn.microsoft.com/en-us/azure/vpn-gateway/point-to-site-certificate-gateway>
4. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-network-security#ns-1-establish-network-segmentation-boundaries>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	13.4 Perform Traffic Filtering Between Network Segments Perform traffic filtering between network segments, where appropriate.			

Controls Version	Control	IG 1	IG 2	IG 3
v7	9.2 <u>Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1190	TA0001	M1050

7.5 Ensure that Network Security Group Flow Log Retention Days is Set to Greater than or equal to 90 (Automated)

Profile Applicability:

- Level 2

Description:

Network security group flow logs should be enabled and the retention period set to greater than or equal to 90 days.

Retirement Notice

On September 30, 2027, network security group (NSG) flow logs will be retired. As of June 30, 2025, creating new NSG flow logs is no longer possible. Azure recommends migrating to virtual network flow logs. Review <https://azure.microsoft.com/en-us/updates?id=Azure-NSG-flow-logs-Retirement> for more information.

For virtual network flow logs, consider applying the recommendation **Ensure that virtual network flow log retention days is set to greater than or equal to 90** in this section.

Rationale:

Flow logs enable capturing information about IP traffic flowing in and out of network security groups. Logs can be used to check for anomalies and give insight into suspected breaches.

Impact:

This will keep IP traffic logs for 90 days or longer. As a level 2, first determine your need to retain data, then apply your selection here. As this is data stored for a longer period, your monthly storage costs will increase depending on your data use.

Audit:

Audit from Azure Portal

1. Go to **Network Watcher**.
2. Under **Logs**, select **Flow logs**.
3. Click the name of a network security group flow log.
4. Ensure that **Status** is set to **On**.
5. Ensure that **Retention days** is set to **0**, **90**, or a number greater than 90. If **Retention days** is set to **0**, the logs are retained indefinitely with no retention policy.
6. Repeat steps 1-5 for each network security group flow log.

Audit from Azure CLI

Run the following command to list network watchers:

```
az network watcher list
```

Run the following command to list the name and retention policy of flow logs in a network watcher:

```
az network watcher flow-log list --location <location> --query  
[*].[name,retentionPolicy]
```

For each network security group flow log, ensure that **enabled** is set to **true**, and **days** is set to **0**, **90**, or a number greater than 90. If **days** is set to **0**, the logs are retained indefinitely with no retention policy.

Remediation:

Remediate from Azure Portal

1. Go to **Network Watcher**.
2. Under **Logs**, select **Flow logs**.
3. Click the name of a network security group flow log.
4. Set **Status** to **On**.
5. Set **Retention days** to **0**, **90**, or a number greater than 90. If **Retention days** is set to **0**, the logs are retained indefinitely with no retention policy.
6. Click **Save**.
7. Repeat steps 1-6 for each network security flow log requiring remediation.

Remediate from Azure CLI

For each network security group flow log requiring remediation, run the following command to enable the flow log and set **retention** to **0**, **90**, or a number greater than 90:

```
az network watcher flow-log configure --nsg <network-security-group> --  
enabled true --resource-group <resource-group> --retention <number-of-days> -  
-storage-account <storage-account>
```

Default Value:

By default, network security group flow logs are **disabled**.

References:

1. <https://learn.microsoft.com/en-us/azure/network-watcher/nsg-flow-logs-overview>
2. <https://learn.microsoft.com/en-us/cli/azure/network/watcher/flow-log>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-6-configure-log-storage-retention>
4. <https://learn.microsoft.com/en-gb/azure/network-watcher/nsg-flow-logs-migrate>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.3 <u>Ensure Adequate Audit Log Storage</u> Ensure that logging destinations maintain adequate storage to comply with the enterprise's audit log management process.	●	●	●
v8	8.10 <u>Retain Audit Logs</u> Retain audit logs across enterprise assets for a minimum of 90 days.		●	●
v7	6.4 <u>Ensure adequate storage for logs</u> Ensure that all systems that store logs have adequate storage space for the logs generated.		●	●

7.6 Ensure that Network Watcher is 'Enabled' for Azure Regions That are in Use (Automated)

Profile Applicability:

- Level 2

Description:

Enable Network Watcher for physical regions in Azure subscriptions.

Rationale:

Network diagnostic and visualization tools available with Network Watcher help users understand, diagnose, and gain insights to the network in Azure.

Impact:

There are additional costs per transaction to run and store network data. For high-volume networks these charges will add up quickly.

Audit:

Audit from Azure Portal

1. Use the Search bar to search for and click on the **Network Watcher** service.
2. From the Overview menu item, review each Network Watcher listed, and ensure that a network watcher is listed for each region in use by the subscription.

Audit from Azure CLI

```
az network watcher list --query  
"[].{Location:location,State:provisioningState}" -o table
```

This will list all network watchers and their provisioning state. Ensure **provisioningState** is **Succeeded** for each network watcher.

```
az account list-locations --query  
"[?metadata.regionType=='Physical'].{Name:name,DisplayName:regionalDisplayNam  
e}" -o table
```

This will list all physical regions that exist in the subscription. Compare this list to the previous one to ensure that for each region in use, a network watcher exists with **provisioningState** set to **Succeeded**.

Audit from PowerShell

Get a list of Network Watchers

```
Get-AzNetworkWatcher
```

Make sure each watcher is set with the **ProvisioningState** setting set to **Succeeded** and all **Locations** that are in use by the subscription are using a watcher.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/%2FDefinitions

- **Policy ID:** [b6e2945c-0b7b-40f5-9233-7a5323b5cdc6](#) - **Name:** 'Network Watcher should be enabled'

Remediation:

Opting out of Network Watcher automatic enablement is a permanent change. Once you opt-out you cannot opt-in without contacting support.

To manually enable Network Watcher in each region where you want to use Network Watcher capabilities, follow the steps below.

Remediate from Azure Portal

1. Use the Search bar to search for and click on the **Network Watcher** service.
2. Click **Create**.
3. Select a **Region** from the drop-down menu.
4. Click **Add**.

Remediate from Azure CLI

```
az network watcher configure --locations <region> --enabled true --resource-group <resource_group>
```

Default Value:

Network Watcher is automatically enabled. When you create or update a virtual network in your subscription, Network Watcher will be enabled automatically in your Virtual Network's region. There is no impact to your resources or associated charge for automatically enabling Network Watcher.

References:

1. <https://learn.microsoft.com/en-us/azure/network-watcher/network-watcher-overview>
2. <https://learn.microsoft.com/en-us/cli/azure/network/watcher>
3. <https://learn.microsoft.com/en-us/cli/azure/network/watcher#az-network-watcher-configure>
4. <https://learn.microsoft.com/en-us/azure/network-watcher/network-watcher-create>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-4-enable-network-logging-for-security-investigation>
6. <https://azure.microsoft.com/en-us/pricing/details/network-watcher/>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	12.2 <u>Establish and Maintain a Secure Network Architecture</u> Establish and maintain a secure network architecture. A secure network architecture must address segmentation, least privilege, and availability, at a minimum.			
v8	12.4 <u>Establish and Maintain Architecture Diagram(s)</u> Establish and maintain architecture diagram(s) and/or other network system documentation. Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	11.2 <u>Document Traffic Configuration Rules</u> All configuration rules that allow traffic to flow through network devices should be documented in a configuration management system with a specific business reason for each rule, a specific individual's name responsible for that business need, and an expected duration of the need.			
v7	12.1 <u>Maintain an Inventory of Network Boundaries</u> Maintain an up-to-date inventory of all of the organization's network boundaries.			

7.7 Ensure that Public IP Addresses are Evaluated on a Periodic Basis (Manual)

Profile Applicability:

- Level 1

Description:

Public IP Addresses provide tenant accounts with Internet connectivity for resources contained within the tenant. During the creation of certain resources in Azure, a Public IP Address may be created. All Public IP Addresses within the tenant should be periodically reviewed for accuracy and necessity.

While an automated assessment procedure exists for this recommendation, the assessment status remains manual. Evaluating the appropriateness of public IP addresses requires a manual review, as it depends on the specific needs and context of each organization and environment.

Rationale:

Public IP Addresses allocated to the tenant should be periodically reviewed for necessity. Public IP Addresses that are not intentionally assigned and controlled present a publicly facing vector for threat actors and significant risk to the tenant.

Audit:

Audit from Azure Portal

1. Open the **All Resources** blade
2. Click on **Add Filter**
3. In the Add Filter window, select the following:

Filter: **Type** Operator: **Equals** Value: **Public IP address**

4. Click the **Apply** button
5. For each Public IP address in the list, use Overview (or Properties) to review the **"Associated to:"** field and determine if the associated resource is still relevant to your tenant environment. If the associated resource is relevant, ensure that additional controls exist to mitigate risk (e.g. Firewalls, VPNs, Traffic Filtering, Virtual Gateway Appliances, Web Application Firewalls, etc.) on all subsequently attached resources.

Audit from Azure CLI

List all Public IP addresses:

```
az network public-ip list
```

For each Public IP address in the output, review the **"name"** property and determine if the associated resource is still relevant to your tenant environment. If the associated resource is relevant, ensure that additional controls exist to mitigate risk (e.g. Firewalls, VPNs, Traffic Filtering, Virtual Gateway Appliances, Web Application Firewalls, etc.) on all subsequently attached resources.

Remediation:

Remediation will vary significantly depending on your organization's security requirements for the resources attached to each individual Public IP address.

Default Value:

During Virtual Machine and Application creation, a setting may create and attach a public IP.

References:

1. <https://learn.microsoft.com/en-us/cli/azure/network/public-ip>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-network-security>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	12.1 <u>Ensure Network Infrastructure is Up-to-Date</u> Ensure network infrastructure is kept up-to-date. Example implementations include running the latest stable release of software and/or using currently supported network-as-a-service (NaaS) offerings. Review software versions monthly, or more frequently, to verify software support.			
v7	12.1 <u>Maintain an Inventory of Network Boundaries</u> Maintain an up-to-date inventory of all of the organization's network boundaries.			

7.8 Ensure that Virtual Network Flow Log Retention Days is Set to Greater than or Equal to 90 (Automated)

Profile Applicability:

- Level 2

Description:

Ensure that virtual network flow logs are retained for greater than or equal to 90 days.

Rationale:

Virtual network flow logs provide critical visibility into traffic patterns. Logs can be used to check for anomalies and give insight into suspected breaches.

Impact:

- Virtual network flow logs are charged per gigabyte of network flow logs collected and come with a free tier of 5 GB/month per subscription.
- If traffic analytics is enabled with virtual network flow logs, traffic analytics pricing applies at per gigabyte processing rates.
- The storage of logs is charged separately, and the cost will depend on the amount of logs and the retention period.

Audit:

Audit from Azure Portal

1. Go to **Network Watcher**.
2. Under **Logs**, select **Flow logs**.
3. Click **Add filter**.
4. From the **Filter** drop-down menu, select **Flow log type**.
5. From the **Value** drop-down menu, check **Virtual network** only.
6. Click **Apply**.
7. Click the name of a virtual network flow log.
8. Under **Storage Account**, ensure that **Retention days** is set to **0**, **90**, or a number greater than 90. If **Retention days** is set to **0**, the logs are retained indefinitely with no retention policy.
9. Repeat steps 7 and 8 for each virtual network flow log.

Audit from Azure CLI

Run the following command to list network watchers:

```
az network watcher list
```

Run the following command to list the name and retention policy of flow logs in a network watcher:

```
az network watcher flow-log list --location <location> --query  
[*].[name,retentionPolicy]
```

For each flow log, ensure that **days** is set to **0**, **90**, or a number greater than 90. If **days** is set to **0**, the logs are retained indefinitely with no retention policy.

Repeat for each network watcher.

Remediation:

Remediate from Azure Portal

1. Go to **Network Watcher**.
2. Under **Logs**, select **Flow logs**.
3. Click **Add filter**.
4. From the **Filter** drop-down menu, select **Flow log type**.
5. From the **Value** drop-down menu, check **Virtual network** only.
6. Click **Apply**.
7. Click the name of a virtual network flow log.
8. Under **Storage Account**, set **Retention days** to **0**, **90**, or a number greater than 90. If **Retention days** is set to **0**, the logs are retained indefinitely with no retention policy.
9. Repeat steps 7 and 8 for each virtual network flow log requiring remediation.

Remediate from Azure CLI

Run the following command update the retention policy for a flow log in a network watcher, setting **retention** to **0**, **90**, or a number greater than 90:

```
az network watcher flow-log update --location <location> --name <flow-log> --  
retention <number-of-days>
```

Repeat for each virtual network flow log requiring remediation.

Default Value:

When a virtual network flow log is created using the Azure CLI, retention days is set to 0 by default. When creating via the Azure Portal, retention days must be specified by the creator.

References:

1. <https://learn.microsoft.com/en-us/azure/network-watcher/vnet-flow-logs-manage>
2. <https://learn.microsoft.com/en-us/cli/azure/network/watcher/flow-log>

Additional Information:

As network security group flow logs are on the retirement path, Azure recommends migrating to virtual network flow logs.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.3 <u>Ensure Adequate Audit Log Storage</u> Ensure that logging destinations maintain adequate storage to comply with the enterprise's audit log management process.			
v8	8.10 <u>Retain Audit Logs</u> Retain audit logs across enterprise assets for a minimum of 90 days.			
v7	6.4 <u>Ensure adequate storage for logs</u> Ensure that all systems that store logs have adequate storage space for the logs generated.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
		M1047

7.9 Ensure 'Authentication type' is Set to 'Azure Active Directory' only for Azure VPN Gateway Point-to-Site Configuration (Automated)

Profile Applicability:

- Level 2

Description:

Enable only 'Azure Active Directory' (Microsoft Entra ID) authentication for Azure VPN Gateway point-to-site connections.

Rationale:

Microsoft Entra ID authentication provides strong security and centralized identity management, and reduces risks associated with static credentials and certificate management.

Impact:

Azure VPN Gateways incur hourly charges, with additional costs for point-to-site connections and data transfer. Pricing varies by SKU and usage. Refer to <https://azure.microsoft.com/en-us/pricing/details/vpn-gateway/> for details.

Audit:

Audit from Azure Portal

1. Go to **Virtual network gateways**.
2. Under **VPN gateway**, click **VPN gateways**.
3. Click the name of a VPN gateway.
4. Under **Settings**, click **Point-to-site configuration**.
5. Ensure **Authentication type** is set to **Azure Active Directory** only.
6. Repeat steps 1-5 for each VPN gateway.

Audit from CLI

```
az network vnet-gateway show -g <rg> -n <name> --query  
"vpnClientConfiguration.vpnAuthenticationTypes"
```

Each gateway's vpnClientConfiguration.vpnAuthenticationTypes array tells you the auth method. The check passes when it equals ["AAD"] (Azure Active Directory only).

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [21a6bc25-125e-4d13-b82d-2e19b7208ab7](#) - **Name:** 'VPN gateways should use only Azure Active Directory (Azure AD) authentication for point-to-site users'

Remediation:

Remediate from Azure Portal

1. Go to **Virtual network gateways**.
2. Under **VPN gateway**, click **VPN gateways**.
3. Click the name of a VPN gateway.
4. Under **Settings**, click **Point-to-site configuration**.
5. Ensure **Authentication type** click to expand the drop-down menu.
6. Check the box next to **Azure Active Directory**, and uncheck the boxes next to **Azure certificate** and **RADIUS authentication**.
7. Provide a **Tenant**, **Audience**, and **Issuer** for the **Azure Active Directory** configuration.
8. Click **Save**.
9. Repeat steps 1-8 for each VPN gateway requiring remediation.

Default Value:

'Authentication type' is selected during creation of point-to-site configuration.

References:

1. <https://learn.microsoft.com/en-us/azure/vpn-gateway/vpn-gateway-about-vpngateways>
2. <https://learn.microsoft.com/en-us/azure/vpn-gateway/point-to-site-entra-gateway>
3. <https://learn.microsoft.com/en-us/azure/vpn-gateway/openvpn-azure-ad-tenant>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.6 <u>Centralize Account Management</u> Centralize account management through a directory or identity service.		●	●
v7	16.2 <u>Configure Centralized Point of Authentication</u> Configure access for all accounts through as few centralized points of authentication as possible, including network, security, and cloud systems.		●	●

7.10 Ensure Azure Web Application Firewall (WAF) is Enabled on Azure Application Gateway (Automated)

Profile Applicability:

- Level 2

Description:

Azure Web Application Firewall helps protect applications from common exploits and attacks by inspecting and filtering incoming traffic.

Rationale:

Using Azure Web Application Firewall with Azure Application Gateway reduces exposure to external threats by mitigating attacks on public facing applications.

Impact:

The **WAF V2** tier for Azure Application Gateways costs more than the **Basic** and **Standard V2** tiers. Pricing includes a fixed hourly charge plus a charge per capacity-unit hour. Refer to <https://azure.microsoft.com/en-gb/pricing/details/application-gateway/> for details.

Audit:

Audit from Azure Portal

1. Go to **Application gateways**.
2. Click the name of an application gateway.
3. In the **Overview**, under **Essentials**, ensure **Tier** is set to **WAF V2**.
4. Repeat steps 1-3 for each application gateway.

Audit from Azure CLI

Run the following command to list application gateways:

```
az network application-gateway list
```

For each application gateway, run the following command to get the firewall policy id:

```
az network application-gateway show --resource-group <resource-group> --name <application-gateway> --query firewallPolicy.id
```

Ensure a firewall policy id is returned.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [564feb30-bf6a-4854-b4bb-0d2d2d1e6c66](#) - **Name:** 'Web Application Firewall (WAF) should be enabled for Application Gateway'

Remediation:

Note: **Basic** tier application gateways cannot be upgraded to the **WAF V2** tier. Create a new **WAF V2** tier application gateway to replace a **Basic** tier application gateway.

Remediate from Azure Portal

To remediate a **Standard V2** tier application gateway:

1. Go to **Application gateways**.
2. Click **Add filter**.
3. From the **Filter** drop-down menu, select **SKU size**.
4. Check the box next to **Standard_v2** only.
5. Click **Apply**.
6. Click the name of an application gateway.
7. Under **Settings**, click **Web application firewall**.
8. Under **Configure**, next to **Tier**, click **WAF V2**.
9. Select an existing or create a new **WAF policy**.
10. Click **Save**.
11. Repeat steps 1-10 for each **Standard V2** tier application gateway requiring remediation.

Default Value:

Azure Web Application Firewall is enabled by default for the **WAF V2** tier of Azure Application Gateway. It is not available in the **Basic** tier. Application gateways deployed using the **Standard V2** tier can be upgraded to the **WAF V2** tier to enable Azure Web Application Firewall.

References:

1. <https://learn.microsoft.com/en-us/azure/application-gateway/features>
2. <https://learn.microsoft.com/en-us/cli/azure/network/application-gateway>
3. <https://azure.microsoft.com/en-us/pricing/details/application-gateway>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	13.10 Perform Application Layer Filtering Perform application layer filtering. Example implementations include a filtering proxy, application layer firewall, or gateway.			●
v7	12.9 Deploy Application Layer Filtering Proxy Server Ensure that all network traffic to or from the Internet passes through an authenticated application layer proxy that is configured to filter unauthorized connections.			●

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1190	TA0001	M1030

7.11 Ensure Subnets Are Associated with Network Security Groups (Automated)

Profile Applicability:

- Level 1

Description:

Protect subnet resources by ensuring subnets are associated with network security groups, which can filter inbound and outbound traffic using security rules.

Rationale:

Unprotected subnets can expose resources to unauthorized access.

Impact:

Minor administrative effort is required to ensure subnets are associated with network security groups. There is no cost to create or use network security groups.

Audit:

Audit from Azure Portal

1. Go to **Virtual networks**.
2. Click the name of a virtual network.
3. Under **Settings**, click **Subnets**.
4. Click the name of a subnet.
5. Under **Security**, ensure **Network security group** is not set to **None**.
6. Repeat steps 1-5 for each virtual network and subnet.

Audit from Azure CLI

Run the following command to list virtual networks:

```
az network vnet list
```

For each virtual network, run the following command to list subnets:

```
az network vnet show --resource-group <resource-group> --name <virtual-network> --query subnets
```

For each subnet, run the following command to get the network security group id:

```
az network vnet subnet show --resource-group <resource-group> --vnet-name <virtual-network> --name <subnet> --query networkSecurityGroup.id
```

Ensure a network security group id is returned.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [e71308d3-144b-4262-b144-efdc3cc90517](#) - **Name:** 'Subnets should be associated with a Network Security Group'

Remediation:

Remediate from Azure Portal

1. Go to **Virtual networks**.
2. Click the name of a virtual network.
3. Under **Settings**, click **Subnets**.
4. Click the name of a subnet.
5. Under **Security**, next to **Network security group**, click **None** to display the drop-down menu.
6. Select a network security group.
7. Click **Save**.
8. Repeat steps 1-7 for each virtual network and subnet requiring remediation.

Remediate from Azure CLI

For each subnet requiring remediation, run the following command to associate it with a network security group:

```
az network vnet subnet update --resource-group <resource-group> --vnet-name <virtual-network> --name <subnet> --network-security-group <network-security-group>
```

Default Value:

By default, a subnet is not associated with a network security group.

References:

1. <https://learn.microsoft.com/en-us/azure/virtual-network/network-security-groups-overview>
2. <https://learn.microsoft.com/en-us/cli/azure/network/vnet>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 Implement and Manage a Firewall on Servers Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.			
v7	9.4 Apply Host-based Firewalls or Port Filtering Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1190	TA0001	M1030

7.12 Ensure the SSL Policy's 'Min protocol version' is Set to 'TLSv1_2' or Higher on Azure Application Gateway (Automated)

Profile Applicability:

- Level 1

Description:

The TLS (Transport Layer Security) protocol secures data in transit using asymmetric encryption. Application gateways use TLS 1.2 for the **Min protocol version** by default but can also be configured to use TLS 1.3.

According to Microsoft, "Starting August 31, 2025, all clients and backend servers interacting with Azure Application Gateway must use Transport Layer Security (TLS) 1.2 or higher, as support for TLS 1.0 and 1.1 will be discontinued."

Rationale:

In NIST SP 800-52 Rev. 2, NIST has mandated that servers that support both government-only applications as well as citizen or business-facing applications **"shall** be configured to use [or negotiate] TLS 1.2 and **should** be configured to use [or negotiate] TLS 1.3 as well." Further, NIST specifies that "agencies **shall** support TLS 1.3 by January 1, 2024 [for both government-only and citizen or business-facing applications]." In general, NIST indicates that "servers that support TLS 1.3 **should** be configured to use TLS 1.2 as well. However, TLS 1.2 may be disabled on servers that support TLS 1.3 [if TLS 1.2 is not needed for interoperability]."

POST-QUANTUM CRYPTOGRAPHY: TLS v1.3 introduces cryptographic suites which make Post-Quantum Cryptography (PQC) algorithms accessible. Where possible, TLS v1.3 should be implemented as the sole protocol in use for the security of data in transit, but only if TLS 1.2 interoperability is not required.

These protocols are outdated, vulnerable, and **shall not** be used: TLS 1.0, TLS 1.1, SSL 2.0, SSL 3.0.

Impact:

Using the latest TLS version may affect compatibility with clients and backend services.

Audit:

Audit from Azure Portal

1. Go to **Application gateways**.
2. Click the name of an application gateway.
3. Under **Settings**, click **Listeners**.
4. Under **SSL Policy**, ensure **Min protocol version** is set to **TLSv1_2** or higher.
5. Repeat steps 1-4 for each application gateway.

Audit from Azure CLI

Run the following command to list application gateways:

```
az network application-gateway list
```

For each application gateway, run the following command to get the SSL policy:

```
az network application-gateway ssl-policy show --resource-group <resource-group> --gateway-name <application-gateway>
```

For each SSL policy, run the following command to get the **minProtocolVersion**:

```
az network application-gateway ssl-policy predefined show --name <ssl-policy> --query minProtocolVersion
```

The recommended policy names for Azure Application Gateway are:

- 20220101S (TLS 1.2 and 1.3), and
- 20220101 (TLS 1.2 and 1.3)

PQC Compliant Cipher: TLS_256_AES_GCM_SHA384 (with ML-KEM, ML-DSA, SLH-DSA for key establishment algorithms approved by NIST currently)

Remediation:

Remediate from Azure Portal

1. Go to **Application gateways**.
2. Click the name of an application gateway.
3. Under **Settings**, click **Listeners**.
4. Under **SSL Policy**, next to the **Selected SSL Policy** name, click **change**.
5. Select an appropriate SSL policy with a **Min protocol version** of **TLSv1_2** or higher.
6. Click **Save**.
7. Repeat steps 1-6 for each application gateway requiring remediation.

Remediate from Azure CLI

Run the following command to list available SSL policy options:

```
az network application-gateway ssl-policy list-options
```

Run the following command to list available predefined SSL policies:

```
az network application-gateway ssl-policy predefined list
```

For each application gateway requiring remediation, run the following command to set a predefined SSL policy:

```
az network application-gateway ssl-policy set --resource-group <resource-group> --gateway-name <application-gateway> --name <ssl-policy> --policy-type Predefined
```

Alternatively, run the following command to set a custom SSL policy:

```
az network application-gateway ssl-policy set --resource-group <resource-group> --gateway-name <application-gateway> --policy-type Custom --min-protocol-version <min-protocol-version> --cipher-suites <cipher-suites>
```

Default Value:

Min protocol version is set to **TLSv1_2** by default.

References:

1. <https://learn.microsoft.com/en-us/azure/application-gateway/application-gateway-ssl-policy-overview>
2. <https://learn.microsoft.com/en-us/cli/azure/network/application-gateway>
3. <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-52r2.pdf>
4. <https://learn.microsoft.com/en-us/azure/application-gateway/application-gateway-tls-version-retirement>
5. NIST FAQ: <https://csrc.nist.gov/projects/post-quantum-cryptography/faqs>
6. PQC Standards: <https://csrc.nist.gov/Projects/post-quantum-cryptography#pqc-standards>

Additional Information:

The recommended policy names for Azure Application Gateway are:

- 20220101S (TLS 1.2 and 1.3), and
- 20220101 (TLS 1.2 and 1.3)

PQC Compliant Cipher: TLS_256_AES_GCM_SHA384 (with ML-KEM, ML-DSA, SLH-DSA for key establishment algorithms approved by NIST currently)

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.10 <u>Encrypt Sensitive Data in Transit</u> Encrypt sensitive data in transit. Example implementations can include: Transport Layer Security (TLS) and Open Secure Shell (OpenSSH).		●	●
v7	14.4 <u>Encrypt All Sensitive Information in Transit</u> Encrypt all sensitive information in transit.		●	●

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1040	TA0007	M1041

7.13 Ensure 'HTTP2' is Set to 'Enabled' on Azure Application Gateway (Automated)

Profile Applicability:

- Level 1

Description:

Enable HTTP/2 for improved performance, efficiency, and security.

HTTP/2 protocol support is available to clients that connect to application gateway listeners only. Communication with backend server pools is always HTTP/1.1.

Rationale:

Enabling HTTP/2 supports use of modern encrypted connections.

Impact:

Clients and backend services that do not support HTTP/2 will fall back to HTTP/1.1.

Audit:

Audit from Azure Portal

1. Go to **Application gateways**.
2. Click the name of an application gateway.
3. Under **Settings**, click **Configuration**.
4. Ensure **HTTP2** is set to **Enabled**.
5. Repeat steps 1-4 for each application gateway.

Audit from Azure CLI

Run the following command to list application gateways:

```
az network application-gateway list
```

For each application gateway, run the following command to get the HTTP2 setting:

```
az network application-gateway show --resource-group <resource-group> --name <application-gateway> --query enableHttp2
```

Ensure **true** is returned.

Audit from PowerShell

Run the following command to list application gateways:

```
Get-AzApplicationGateway
```

Run the following command to get the application gateway in a resource group with a given name:


```
$gateway = Get-AzApplicationGateway -ResourceGroupName <resource-group> -Name <application-gateway>
```

Run the following command to get the HTTP2 setting:

```
$gateway.EnableHttp2
```

Ensure **True** is returned.

Repeat for each application gateway.

Remediation:

Remediate from Azure Portal

1. Go to **Application gateways**.
2. Click the name of an application gateway.
3. Under **Settings**, click **Configuration**.
4. Under **HTTP2**, click **Enabled**.
5. Click **Save**.
6. Repeat steps 1-5 for each application gateway requiring remediation.

Remediate from Azure CLI

For each application gateway requiring remediation, run the following command to enable HTTP2:

```
az network application-gateway update --resource-group <resource-group> --name <application-gateway> --http2 Enabled
```

Remediate from PowerShell

Run the following command to get the application gateway in a resource group with a given name:

```
$gateway = Get-AzApplicationGateway -ResourceGroupName <resource-group> -Name <application-gateway>
```

Run the following command to enable HTTP2:

```
$gateway.EnableHttp2 = $true
```

Run the following command to apply the update:

```
Set-AzApplicationGateway -ApplicationGateway $gateway
```

Repeat for each application gateway requiring remediation.

Default Value:

HTTP2 is enabled by default.

References:

1. <https://learn.microsoft.com/en-us/azure/application-gateway/features#websocket-and-http2-traffic>
2. <https://learn.microsoft.com/en-us/cli/azure/network/application-gateway>
3. <https://learn.microsoft.com/en-us/powershell/module/az.network/get-azapplicationgateway>
4. <https://learn.microsoft.com/en-us/powershell/module/az.network/set-azapplicationgateway>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	2.2 Ensure Authorized Software is Currently Supported Ensure that only currently supported software is designated as authorized in the software inventory for enterprise assets. If software is unsupported, yet necessary for the fulfillment of the enterprise's mission, document an exception detailing mitigating controls and residual risk acceptance. For any unsupported software without an exception documentation, designate as unauthorized. Review the software list to verify software support at least monthly, or more frequently.			
v7	2.2 Ensure Software is Supported by Vendor Ensure that only software applications or operating systems currently supported by the software's vendor are added to the organization's authorized software inventory. Unsupported software should be tagged as unsupported in the inventory system.			

7.14 Ensure Request Body Inspection is Enabled in Azure Web Application Firewall policy on Azure Application Gateway (Automated)

Profile Applicability:

- Level 2

Description:

Enable request body inspection so that the Web Application Firewall evaluates the contents of HTTP message bodies for potential threats.

Rationale:

Enabling request body inspection strengthens security by allowing the Web Application Firewall to detect common attacks, such as SQL injection and cross-site scripting.

Impact:

Minor performance impact on the Web Application Firewall. Additional effort may be required to monitor findings.

Audit:

Audit from Azure Portal

1. Go to **Application gateways**.
2. Click the name of an application gateway.
3. Under **Settings**, click **Web application firewall**.
4. Under **Associated web application firewall policy**, click the policy name.
5. Under **Settings**, click **Policy settings**.
6. Ensure the box next to **Enforce request body inspection** is checked.
7. Repeat steps 1-6 for each application gateway.

Audit from Azure CLI

Run the following command to list application gateways:

```
az network application-gateway list
```

For each application gateway, run the following command to get the firewall policy id:

```
az network application-gateway show --resource-group <resource-group> --name <application-gateway> --query firewallPolicy.id
```

For each firewall policy, run the following command to get the request body inspection setting:

```
az network application-gateway waf-policy show --ids <firewall-policy> --query policySettings.requestBodyCheck
```

Ensure **true** is returned.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [ca85ef9a-741d-461d-8b7a-18c2da82c666](#) - **Name:** 'Azure Web Application Firewall on Azure Application Gateway should have request body inspection enabled'

Remediation:

Remediate from Azure Portal

1. Go to **Application gateways**.
2. Click the name of an application gateway.
3. Under **Settings**, click **Web application firewall**.
4. Under **Associated web application firewall policy**, click the policy name.
5. Under **Settings**, click **Policy settings**.
6. Check the box next to **Enforce request body inspection**.
7. Click **Save**.
8. Repeat steps 1-7 for each application gateway and firewall policy requiring remediation.

Remediate from Azure CLI

For each firewall policy requiring remediation, run the following command to enable request body inspection:

```
az network application-gateway waf-policy update --ids <firewall-policy> --policy-settings request-body-check=true
```

Default Value:

Request body inspection is enabled by default on Azure Application Gateways with Web Application Firewall.

References:

1. <https://learn.microsoft.com/en-gb/azure/web-application-firewall/ag/application-gateway-waf-request-size-limits#request-body-inspection>
2. <https://learn.microsoft.com/en-us/cli/azure/network/application-gateway>
3. <https://learn.microsoft.com/en-us/cli/azure/network/application-gateway/waf-policy>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	13.10 Perform Application Layer Filtering Perform application layer filtering. Example implementations include a filtering proxy, application layer firewall, or gateway.			●
v7	12.9 Deploy Application Layer Filtering Proxy Server Ensure that all network traffic to or from the Internet passes through an authenticated application layer proxy that is configured to filter unauthorized connections.			●

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1190	TA0001	M1030

7.15 Ensure Bot Protection is Enabled in Azure Web Application Firewall Policy on Azure Application Gateway (Automated)

Profile Applicability:

- Level 2

Description:

Enable bot protection on the Web Application Firewall to block or log requests from known malicious IP addresses identified through the Microsoft Threat Intelligence feed.

Rationale:

Internet traffic from bots can scrape, scan, and search for application vulnerabilities. Enabling bot protection stops requests from known malicious IP addresses and enhances the overall security of your application by reducing exposure to automated attacks.

Impact:

May require monitoring to identify false positives.

Audit:

Audit from Azure Portal

1. Go to **Application gateways**.
2. Click the name of an application gateway.
3. Under **Settings**, click **Web application firewall**.
4. Under **Associated web application firewall policy**, click the policy name.
5. Under **Settings**, click **Managed rules**.
6. Ensure a **Rule Id** containing **Microsoft_BotManagerRuleSet** is listed.
7. Click the **>** to expand the row.
8. Ensure the **Status** for **Malicious Bots** is set to **Enabled**.
9. Repeat steps 1-8 for each application gateway.

Audit from Azure CLI

Run the following command to list application gateways:

```
az network application-gateway list
```

For each application gateway, run the following command to get the firewall policy id:

```
az network application-gateway show --resource-group <resource-group> --name <application-gateway> --query firewallPolicy.id
```

For each firewall policy, run the following command to get the managed rule sets:

```
az network application-gateway waf-policy show --ids <firewall-policy> --query managedRules.managedRuleSets
```

Ensure a managed rule set with **ruleSetType** of **Microsoft_BotManagerRuleSet** is returned, and that no **ruleGroupOverrides** for **ruleGroupName** **KnownBadBots** with **state** **Disabled** are returned.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [ebea0d86-7fbd-42e3-8a46-27e7568c2525](#) - **Name:** 'Bot Protection should be enabled for Azure Application Gateway WAF'

Remediation:

Remediate from Azure Portal

1. Go to **Application gateways**.
2. Click the name of an application gateway.
3. Under **Settings**, click **Web application firewall**.
4. Under **Associated web application firewall policy**, click the policy name.
5. Under **Settings**, click **Managed rules**.
6. Click **Assign**.
7. Under **Bot Management ruleset**, click to display the drop-down menu.
8. Select a **Microsoft_BotManagerRuleSet**.
9. Click **Save**.
10. Click **X** to close the panel.
11. Repeat steps 1-10 for each application gateway and firewall policy requiring remediation.

Remediate from Azure CLI

For each firewall policy requiring remediation, run the following command to enable bot protection:

```
az network application-gateway waf-policy managed-rule rule-set add --resource-group <resource-group> --policy-name <firewall-policy> --type Microsoft_BotManagerRuleSet --version <0.1|1.0|1.1>
```

Default Value:

Bot protection is disabled by default on Azure Application Gateways with Web Application Firewall.

References:

1. <https://learn.microsoft.com/en-us/azure/web-application-firewall/ag/bot-protection-overview>
2. <https://learn.microsoft.com/en-us/azure/web-application-firewall/ag/bot-protection>
3. <https://learn.microsoft.com/en-us/cli/azure/network/application-gateway>
4. <https://learn.microsoft.com/en-us/cli/azure/network/application-gateway/waf-policy>
5. <https://learn.microsoft.com/en-us/cli/azure/network/application-gateway/waf-policy/managed-rule/rule-set>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	13.10 Perform Application Layer Filtering Perform application layer filtering. Example implementations include a filtering proxy, application layer firewall, or gateway.			●
v7	12.9 Deploy Application Layer Filtering Proxy Server Ensure that all network traffic to or from the Internet passes through an authenticated application layer proxy that is configured to filter unauthorized connections.			●

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1190	TA0001	M1030

7.16 Ensure Azure Network Security Perimeter is Used to Secure Azure Platform-as-a-service Resources (Manual)

Profile Applicability:

- Level 2

Description:

Azure Network Security Perimeter creates a logical boundary around Azure platform-as-a-service (PaaS) resources outside of virtual networks. By default, the network security perimeter denies public access to associated PaaS resources, with the ability to define explicit rules for inbound and outbound traffic.

While an automated assessment procedure exists for this recommendation, the assessment status remains manual. Determining appropriate network security perimeter profiles and resource assignments depends on the context and requirements of each organization and environment.

Rationale:

Network security perimeter denies public access to PaaS resources, reducing exposure and mitigating data exfiltration risks.

Impact:

Implementation requires administrative effort to configure and maintain network security perimeter profiles and resource assignments. Azure does not list any additional charges for using network security perimeters.

Audit:

Audit from Azure Portal

1. Go to **Resource groups**.
2. Click the name of a resource group.
3. Take note of PaaS resources.
4. Go to **Network Security Perimeters**.
5. Click the name of a network security perimeter.
6. Under **Settings**, click **Associated resources**.
7. Take note of the associated resources.
8. Repeat steps 1-7 and ensure each PaaS resource is associated with a network security perimeter.

Audit from Azure CLI

Run the following command to list resource groups:

```
az group list
```

For each resource group, run the following command to list resources:

```
az resource list --resource-group <resource-group>
```

Take note of PaaS resources.

For each resource group, run the following command to list network security perimeters:

```
az network perimeter list --resource-group <resource-group>
```

For each network security perimeter, run the following command to list resources:

```
az network perimeter association list --resource-group <resource-group> --  
perimeter-name <network-security-perimeter>
```

Ensure each PaaS resource is associated with a network security perimeter.

Remediation:

Remediate from Azure Portal

Create and associate PaaS resources with a new network security perimeter:

1. Go to **Network Security Perimeters**.
2. Click **+ Create**.
3. Select a **Subscription** and **Resource group**, provide a **Name**, select a **Region**, and provide a **Profile name**.
4. Click **Next**.
5. Click **+ Add**.
6. Check the box next to a PaaS resource to associate it with the network security perimeter.
7. Click **Select**.
8. Click **Next**.
9. Configure appropriate **Inbound access rules** for your organization.
10. Click **Next**.
11. Configure appropriate **Outbound access rules** for your organization.
12. Click **Review + create**.
13. Click **Create**.

Associate PaaS resources with an existing network security perimeter:

1. Go to **Network Security Perimeters**.
2. Click the name of a network security perimeter.
3. Under **Settings**, click **Associated resources**.
4. Click **+ Add**.
5. Select **Associate resources with a new profile** or **Associate resources with an existing profile**.
6. To associate resources with a new profile:
 - a) Provide a **Name**.

- b) Click **Next**.
 - c) Click **+ Add**.
 - d) Check the box next to a PaaS resource to associate it with the network security perimeter.
 - e) Click **Select**.
 - f) Click **Next**.
 - g) Configure appropriate **Inbound access rules** for your organization.
 - h) Click **Next**.
 - i) Configure appropriate **Outbound access rules** for your organization.
 - j) Click **Review + create**.
 - k) Click **Create**.
7. To associate resources with an existing profile:
- a) Next to **Profile**, click **Select** to display the drop-down menu.
 - b) Select a profile.
 - c) Click **+ Add**.
 - d) Check the box next to a PaaS resource to associate it with the network security perimeter.
 - e) Click **Select**.
 - f) Click **Associate**.

Remediate from Azure CLI

Use **az network perimeter profile list** or **az network perimeter profile create** to list existing or create a new network security perimeter profile.

For each PaaS resource requiring association with a network security perimeter, run the following command:

```
az network perimeter association create --resource-group <resource-group> --
perimeter-name <network-security-perimeter> --association-name <association>
--private-link-resource "{id:<paas-resource-id>}" --profile "{<profile-id>}"
```

Default Value:

PaaS resources are not associated with a network security perimeter by default.

References:

1. <https://learn.microsoft.com/en-us/azure/private-link/network-security-perimeter-concepts>
2. <https://learn.microsoft.com/en-us/azure/private-link/create-network-security-perimeter-portal>
3. <https://learn.microsoft.com/en-us/cli/azure/group>
4. <https://learn.microsoft.com/en-us/cli/azure/resource>
5. <https://learn.microsoft.com/en-us/cli/azure/network/perimeter>

Additional Information:

The current list of resources that can be associated with a network security perimeter are as follows:

- Azure Monitor
- Azure AI Search
- Cosmos DB
- Event Hubs
- Key Vault
- SQL DB
- Storage
- Azure OpenAI Service

While network security perimeter is generally available, Cosmos DB, SQL DB, and Azure OpenAI Service are in public preview.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	13.4 Perform Traffic Filtering Between Network Segments Perform traffic filtering between network segments, where appropriate.			
v7	9.4 Apply Host-based Firewalls or Port Filtering Apply host-based firewalls or port filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1190	TA0001	M1030

8 Security Services

To better understand the relationship between the Foundations Benchmark and Services Benchmarks, please read the "Introduction" section of this document.

This section covers security best practice recommendations for products in the Azure Security services category.

Azure Product Directory Reference: <https://azure.microsoft.com/en-us/products#security>

FEEDBACK REQUEST: Is there a specific service or recommendation in this section that you'd like to see addressed or improved? Let us know by making a ticket or starting a discussion in the CIS Microsoft Azure Community (<https://workbench.cisecurity.org/communities/72>).

8.1 Microsoft Defender for Cloud

This subsection provides guidance on the use of Microsoft Defender for Cloud and associated product plans. This guidance is intended to ensure that—at a minimum—the protective measures offered by these plans are being considered. Organizations may find that they have existing products or services that provide the same utility as some Microsoft Defender for Cloud products. Security and Administrative personnel need to make the determination on their organization's behalf regarding which—if any—of these recommendations are relevant to their organization's needs. In consideration of the above, and because of the potential for increased cost and complexity, please be aware that all Microsoft Defender for Cloud and associated plan recommendations are profiled as "Level 2" recommendations.

8.1.1 Microsoft Cloud Security Posture Management (CSPM)

Microsoft Defender for Cloud offers foundational and advanced Cloud Security Posture Management (CSPM) solutions to protect across multi-cloud and hybrid environments. Both solutions cover PaaS as well as IaaS. CSPM provides reporting functionality on security and regulatory frameworks including NIST 800 series, ISO 27001, PCI-DSS, CIS Benchmarks and Controls, and many more. CSPM also provides the ability to create your own custom framework, but this will require significant work. Regulatory standards are reported in a compliance dashboard which offers a summarized view against deployed standards and presents the ability to download compliance reports in various formats.

CSPM has two types of implementations:

1. Foundational (Free): This implementation is free and enabled by default with a limited set of features including:
 - Continuous assessment of the security configuration of cloud resources
 - Security recommendations to fix misconfigurations and weaknesses
 - Secure score summarizing current overall security posture
2. Full CSPM (Paid): Full CSPM is a paid product offering additional functionality including:
 - Identity and role assignments discovery
 - Network exposure detection
 - Attack path analysis
 - Cloud security explorer for risk hunting
 - Agentless vulnerability scanning
 - Agentless secrets scanning
 - Governance rules to drive timely remediation and accountability
 - Regulatory compliance and industry best practices
 - Data-aware security posture
 - Agentless discovery for Kubernetes
 - Agentless container vulnerability assessment

It is recommended that for full CSPM a cost review is undertaken particularly if your tenant is heavy on IaaS prior to implementing and matched to security requirements.

8.1.1.1 Ensure Microsoft Defender CSPM is Set to 'On' (Automated)

Profile Applicability:

- Level 2

Description:

Enable Microsoft Defender CSPM to continuously assess cloud resources for security misconfigurations, compliance risks, and exposure to threats.

Rationale:

Microsoft Defender CSPM provides detailed visibility into the security state of assets and workloads and offers hardening guidance to help improve security posture.

Impact:

Enabling Microsoft Defender CSPM incurs hourly charges for each billable compute, database, and storage resource. This can lead to significant costs in larger environments. Careful planning and cost analysis are recommended before enabling the service. Refer to <https://azure.microsoft.com/en-us/pricing/details/defender-for-cloud/#pricing> for pricing information.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, click **Environment settings**.
3. Click the name of a subscription.
4. Select the **Defender plans** blade.
5. Under **Cloud Security Posture Management (CSPM)**, in the row for **Defender CSPM**, ensure **Status** is set to **On**.

Audit from Azure CLI

Run the following command to get the **CloudPosture** plan pricing tier:

```
az security pricing show --name CloudPosture --query pricingTier
```

Ensure **"Standard"** is returned.

Audit from PowerShell

Run the following command to get the **CloudPosture** plan pricing tier:

```
Get-AzSecurityPricing -Name CloudPosture | Select-Object PricingTier
```

Ensure **Standard** is returned.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [1f90fc71-a595-4066-8974-d4d0802e8ef0](#) - **Name:** 'Microsoft Defender CSPM should be enabled'

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, click **Environment settings**.
3. Click the name of a subscription.
4. Select the **Defender plans** blade.
5. Under **Cloud Security Posture Management (CSPM)**, in the row for **Defender CSPM**, set the toggle switch for **Status** to **On**.
6. Click **Save**.

Remediate from Azure CLI

Run the following command to enable Defender CSPM:

```
az security pricing create --name CloudPosture --tier Standard --extensions  
name=ApiPosture isEnabled=true
```

Remediate from PowerShell

Run the following command to enable Defender CSPM:

```
Set-AzSecurityPricing -Name CloudPosture -PricingTier Standard -Extension  
'[{"name":"ApiPosture","isEnabled":"True"}]'
```

Default Value:

Defender CSPM is disabled by default.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/concept-cloud-security-posture-management>
2. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/tutorial-enable-cspm-plan>
3. <https://azure.microsoft.com/en-us/pricing/details/defender-for-cloud/#pricing>
4. <https://learn.microsoft.com/en-us/cli/azure/security/pricing>
5. <https://learn.microsoft.com/en-us/powershell/module/az.security/get-azsecuritypricing>

6. <https://learn.microsoft.com/en-us/powershell/module/az.security/set-azsecuritypricing>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>7.5 Perform Automated Vulnerability Scans of Internal Enterprise Assets</u> Perform automated vulnerability scans of internal enterprise assets on a quarterly, or more frequent, basis. Conduct both authenticated and unauthenticated scans, using a SCAP-compliant vulnerability scanning tool.			
v8	<u>7.6 Perform Automated Vulnerability Scans of Externally-Exposed Enterprise Assets</u> Perform automated vulnerability scans of externally-exposed enterprise assets using a SCAP-compliant vulnerability scanning tool. Perform scans on a monthly, or more frequent, basis.			
v7	<u>3.1 Run Automated Vulnerability Scanning Tools</u> Utilize an up-to-date SCAP-compliant vulnerability scanning tool to automatically scan all systems on the network on a weekly or more frequent basis to identify all potential vulnerabilities on the organization's systems.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1580	TA0007	M1018

8.1.2 Defender Plan: APIs

Defender for APIs in Microsoft Defender for Cloud offers full lifecycle protection, detection, and response coverage for APIs published in Azure API Management. Defender for APIs helps you to gain visibility into business-critical APIs. You can investigate and improve your API security posture, prioritize vulnerability fixes, and quickly detect active real-time threats. Defender for APIs requires additional configuration in the Microsoft API portal.

Note: There is a cost attached to using Defender for API.

8.1.2.1 Ensure Microsoft Defender for APIs is Set to 'On' (Automated)

Profile Applicability:

- Level 2

Description:

Microsoft Defender for APIs offers full lifecycle protection, detection, and response coverage for APIs.

While an automated assessment procedure exists for this recommendation, the assessment status remains manual. Due to its potentially high cost, Microsoft Defender for APIs may not be suitable for all environments and should be evaluated carefully before implementation.

Rationale:

Microsoft Defender for APIs helps provide visibility into business-critical APIs, assess and improve their security posture, prioritize vulnerability remediation, and detect threats in real time.

Impact:

Microsoft Defender for APIs uses a tiered pricing model, billed per subscription per hour, with each tier allowing a set limit of API calls. In high-traffic environments, this may result in significant or prohibitive costs. Careful evaluation of API usage patterns and pricing tiers is essential before enabling the service. Refer to <https://azure.microsoft.com/en-us/pricing/details/defender-for-cloud/#pricing> for pricing information.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, click **Environment settings**.
3. Click the name of a subscription.
4. Select the **Defender plans** blade.
5. Under **Cloud Workload Protection (CWP)**, in the row for **APIs**, ensure the **Status** is set to **On**.

Audit from Azure CLI

Run the following command to get the **Api** plan pricing tier:

```
az security pricing show --name Api --query pricingTier
```

Ensure **"Standard"** is returned.

Audit from PowerShell

Run the following command to get the **Api** plan pricing tier:

```
Get-AzSecurityPricing -Name Api | Select-Object PricingTier
```

Ensure **Standard** is returned.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [7926a6d1-b268-4586-8197-e8ae90c877d7](#) - **Name:** 'Microsoft Defender for APIs should be enabled'

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, click **Environment settings**.
3. Click the name of a subscription.
4. Select the **Defender plans** blade.
5. Under **Cloud Workload Protection (CWP)**, in the row for **APIs**, set the toggle switch for **Status** to **On**.
6. Select a plan.
7. Click **Save** to save the plan selection.
8. Click **Save** to enable Defender for APIs.

Remediate from Azure CLI

Run the following command to enable Defender for APIs:

```
az security pricing create --name Api --tier Standard --subplan <subplan>
```

Valid subplan values: **P1**, **P2**, **P3**, **P4**, and **P5**.

Remediate from PowerShell

Run the following command to enable Defender for APIs:

```
Set-AzSecurityPricing -Name Api -PricingTier Standard -SubPlan <subplan>
```

Valid SubPlan values: **P1**, **P2**, **P3**, **P4**, and **P5**.

Default Value:

Defender for APIs is disabled by default.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/defender-for-apis-introduction>
2. <https://azure.microsoft.com/en-us/pricing/details/defender-for-cloud/#pricing>
3. <https://learn.microsoft.com/en-us/cli/azure/security/pricing>
4. <https://learn.microsoft.com/en-us/powershell/module/az.security/get-azsecuritypricing>
5. <https://learn.microsoft.com/en-us/powershell/module/az.security/set-azsecuritypricing>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	16.2 <u>Establish and Maintain a Process to Accept and Address Software Vulnerabilities</u> Establish and maintain a process to accept and address reports of software vulnerabilities, including providing a means for external entities to report. The process is to include such items as: a vulnerability handling policy that identifies reporting process, responsible party for handling vulnerability reports, and a process for intake, assignment, remediation, and remediation testing. As part of the process, use a vulnerability tracking system that includes severity ratings, and metrics for measuring timing for identification, analysis, and remediation of vulnerabilities. Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard. Third-party application developers need to consider this an externally-facing policy that helps to set expectations for outside stakeholders.			
v7	18.8 <u>Establish a Process to Accept and Address Reports of Software Vulnerabilities</u> Establish a process to accept and address reports of software vulnerabilities, including providing a means for external entities to contact your security group.			

8.1.3 Defender Plan: Servers

8.1.3.1 Ensure that Defender for Servers is Set to 'On' (Automated)

Profile Applicability:

- Level 2

Description:

The Defender for Servers plan in Microsoft Defender for Cloud reduces security risk by providing actionable recommendations to improve and remediate machine security posture. Defender for Servers also helps to protect machines against real-time security threats and attacks.

Defender for Servers offers two paid plans:

Plan 1

The following components are enabled by default:

- Log Analytics agent (deprecated)
- Endpoint protection

Plan 1 also offers the following components, disabled by default:

- Vulnerability assessment for machines
- Guest Configuration agent (preview)

Plan 2

The following components are enabled by default:

- Log Analytics agent (deprecated)
- Vulnerability assessment for machines
- Endpoint protection
- Agentless scanning for machines

Plan 2 also offers the following components, disabled by default:

- Guest Configuration agent (preview)
- File Integrity Monitoring

Rationale:

Enabling Defender for Servers allows for greater defense-in-depth, with threat detection provided by the Microsoft Security Response Center (MSRC).

Impact:

Enabling Defender for Servers in Microsoft Defender for Cloud incurs an additional cost per resource. Refer to <https://azure.microsoft.com/en-us/pricing/details/defender-for-cloud/> and <https://azure.microsoft.com/en-us/pricing/calculator/> to estimate potential costs.

- Plan 1: Subscription only
- Plan 2: Subscription and workspace

Audit:

Audit from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, select **Environment settings**.
3. Click on a subscription name.
4. Select **Defender plans** in the left pane.
5. Under **Cloud Workload Protection (CWP)**, locate **Servers** in the Plan column, ensure Status is set to **On**.
6. Repeat steps 1-5 for each subscription.

Audit from Azure CLI

Run the following command:

```
az security pricing show -n VirtualMachines --query pricingTier
```

If the tenant is licensed and enabled, the output will indicate **Standard**.

Audit from PowerShell

Run the following command:

```
Get-AzSecurityPricing -Name 'VirtualMachines' |Select-Object Name,PricingTier
```

If the tenant is licensed and enabled, the **-PricingTier** parameter will indicate **Standard**.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [4da35fc9-c9e7-4960-aec9-797fe7d9051d](#) - **Name:** 'Azure Defender for servers should be enabled'

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, select **Environment settings**.
3. Click on a subscription name.
4. Click **Defender plans** in the left pane.
5. Under **Cloud Workload Protection (CWP)**, locate **Servers** in the Plan column, set Status to **On**.
6. Select **Save**.
7. Repeat steps 1-6 for each subscription requiring remediation.

Remediate from Azure CLI

Run the following command:

```
az security pricing create -n VirtualMachines --tier 'standard'
```

Remediate from PowerShell

Run the following command:

```
Set-AzSecurityPricing -Name 'VirtualMachines' -PricingTier 'Standard'
```

Default Value:

By default, the Defender for Servers plan is disabled.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/defender-for-servers-overview>
2. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/plan-defender-for-servers>
3. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/pricings/list>
4. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/pricings/update>
5. <https://learn.microsoft.com/en-us/powershell/module/az.security/get-azsecuritypricing>
6. <https://learn.microsoft.com/en-us/powershell/module/az.security/set-azsecuritypricing>
7. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-endpoint-security#es-1-use-endpoint-detection-and-response-edr>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>7.5 Perform Automated Vulnerability Scans of Internal Enterprise Assets</u> Perform automated vulnerability scans of internal enterprise assets on a quarterly, or more frequent, basis. Conduct both authenticated and unauthenticated scans, using a SCAP-compliant vulnerability scanning tool.		●	●
v8	<u>10.1 Deploy and Maintain Anti-Malware Software</u> Deploy and maintain anti-malware software on all enterprise assets.	●	●	●
v7	<u>3.1 Run Automated Vulnerability Scanning Tools</u> Utilize an up-to-date SCAP-compliant vulnerability scanning tool to automatically scan all systems on the network on a weekly or more frequent basis to identify all potential vulnerabilities on the organization's systems.		●	●
v7	<u>8.1 Utilize Centrally Managed Anti-malware Software</u> Utilize centrally managed anti-malware software to continuously monitor and defend each of the organization's workstations and servers.		●	●

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1210	TA0008	M1016, M1050

8.1.3.2 Ensure that 'Vulnerability assessment for machines' Component Status is set to 'On' (Manual)

Profile Applicability:

- Level 2

Description:

Enable vulnerability assessment for machines on both Azure and hybrid (Arc enabled) machines.

Rationale:

Vulnerability assessment for machines scans for various security-related configurations and events such as system updates, OS vulnerabilities, and endpoint protection, then produces alerts on threat and vulnerability findings.

Impact:

Microsoft Defender for Servers plan 2 licensing is required, and configuration of Azure Arc introduces complexity beyond this recommendation.

Audit:

Audit from Azure Portal

1. From Azure Home select the Portal Menu
2. Select **Microsoft Defender for Cloud**
3. Under **Management**, select **Environment Settings**
4. Select a subscription
5. Click on **Settings & monitoring**
6. Ensure that **Vulnerability assessment for machines** is set to **On**

Repeat the above for any additional subscriptions.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/_Definitions

- **Policy ID:** [501541f7-f7e7-4cd6-868c-4190fdad3ac9](#) - **Name:** 'A vulnerability assessment solution should be enabled on your virtual machines'

Remediation:

Remediate from Azure Portal

1. From Azure Home select the Portal Menu
2. Select **Microsoft Defender for Cloud**
3. Under **Management**, select **Environment Settings**
4. Select a subscription
5. Click on **Settings & Monitoring**
6. Set the **Status of Vulnerability assessment for machines** to **On**
7. Click **Continue**

Repeat the above for any additional subscriptions.

Default Value:

By default, **Automatic provisioning of monitoring agent** is set to **Off**.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/monitoring-components>
2. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/auto-provisioning-settings/list>
3. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/auto-provisioning-settings/create>
4. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-posture-vulnerability-management#pv-5-perform-vulnerability-assessments>

Additional Information:

While this feature is generally available as of publication, it is not yet available for Azure Government tenants.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.5 Perform Automated Vulnerability Scans of Internal Enterprise Assets Perform automated vulnerability scans of internal enterprise assets on a quarterly, or more frequent, basis. Conduct both authenticated and unauthenticated scans, using a SCAP-compliant vulnerability scanning tool.			

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>7.6 Perform Automated Vulnerability Scans of Externally-Exposed Enterprise Assets</u> Perform automated vulnerability scans of externally-exposed enterprise assets using a SCAP-compliant vulnerability scanning tool. Perform scans on a monthly, or more frequent, basis.			
v7	<u>3.1 Run Automated Vulnerability Scanning Tools</u> Utilize an up-to-date SCAP-compliant vulnerability scanning tool to automatically scan all systems on the network on a weekly or more frequent basis to identify all potential vulnerabilities on the organization's systems.			

8.1.3.3 Ensure that 'Endpoint protection' Component Status is set to 'On' (Automated)

Profile Applicability:

- Level 2

Description:

The Endpoint protection component enables Microsoft Defender for Endpoint (formerly 'Advanced Threat Protection' or 'ATP' or 'WDATP' - see additional info) to communicate with Microsoft Defender for Cloud.

IMPORTANT: When enabling integration between DfE & DfC it needs to be taken into account that this will have some side effects that may be undesirable.

1. For server 2019 & above if defender is installed (default for these server SKUs) this will trigger a deployment of the new unified agent and link to any of the extended configuration in the Defender portal.
2. If the new unified agent is required for server SKUs of Win 2016 or Linux and lower there is additional integration that needs to be switched on and agents need to be aligned.

Rationale:

Microsoft Defender for Endpoint integration brings comprehensive Endpoint Detection and Response (EDR) capabilities within Microsoft Defender for Cloud. This integration helps to spot abnormalities, as well as detect and respond to advanced attacks on endpoints monitored by Microsoft Defender for Cloud.

MDE works only with Standard Tier subscriptions.

Impact:

Endpoint protection requires licensing and is included in these plans:

- Defender for Servers plan 1
- Defender for Servers plan 2

Audit:

Audit from Azure Portal

1. From Azure Home select the Portal Menu.
2. Select **Microsoft Defender for Cloud**.
3. Under **Management**, select **Environment Settings**.
4. Click on the subscription name.
5. Click **Settings & monitoring**.

6. Ensure the **Status** for **Endpoint protection** is set to **On**.

Audit from Azure CLI

Ensure the output of the below command is **True**

```
az account get-access-token --query  
"{subscription:subscription,accessToken:accessToken}" --out tsv | xargs -L1  
bash -c 'curl -X GET -H "Authorization: Bearer $1" -H "Content-Type:  
application/json"  
https://management.azure.com/subscriptions/<subscriptionID>/providers/Microso  
ft.Security/settings?api-version=2022-05-01' | jq '.|.value[] |  
select(.name=="WDATP")'|jq '.properties.enabled'
```

Audit from PowerShell

Run the following commands to login and audit this check

```
Connect-AzAccount  
Set-AzContext -Subscription <subscriptionID>  
Get-AzSecuritySetting | Select-Object name,enabled |where-object {$_.name -eq  
"WDATP"}
```

PowerShell Output - Non-Compliant

Name	Enabled
WDATP	False

PowerShell Output - Compliant

Name	Enabled
WDATP	True

Remediation:

Remediate from Azure Portal

1. From Azure Home select the Portal Menu.
2. Go to **Microsoft Defender for Cloud**.
3. Under **Management**, select **Environment Settings**.
4. Click on the subscription name.
5. Click **Settings & monitoring**.
6. Set the **Status** for **Endpoint protection** to **On**.
7. Click **Continue**.

Remediate from Azure CLI

Use the command below to enable Standard pricing tier for Storage Accounts


```
az account get-access-token --query  
"{subscription:subscription,accessToken:accessToken}" --out tsv | xargs -L1  
bash -c 'curl -X PUT -H "Authorization: Bearer $1" -H "Content-Type:  
application/json"  
https://management.azure.com/subscriptions/<subscriptionID>/providers/Microsoft.Security/settings/WDATP?api-version=2022-05-01 -d@input.json'
```

Where input.json contains the Request body json data as mentioned below.

```
{  
  "id":  
  "/subscriptions/<Your_Subscription_Id>/providers/Microsoft.Security/settings/  
WDATP",  
  "kind": "DataExportSettings",  
  "type": "Microsoft.Security/settings",  
  "properties": {  
    "enabled": true  
  }  
}
```

Default Value:

By default, Endpoint protection is **off**.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/integration-defender-for-endpoint>
2. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/settings/list>
3. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/settings/update>
4. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-endpoint-security#es-1-use-endpoint-detection-and-response-edr>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-endpoint-security#es-2-use-modern-anti-malware-software>

Additional Information:

IMPORTANT: When enabling integration between DfE & DfC it needs to be taken into account that this will have some side effects that may be undesirable.

1. For server 2019 & above if defender is installed (default for these server SKUs) this will trigger a deployment of the new unified agent and link to any of the extended configuration in the Defender portal.
2. If the new unified agent is required for server SKUs of Win 2016 or Linux and lower there is additional integration that needs to be switched on and agents need to be aligned.

NOTE: "Microsoft Defender for Endpoint (MDE)" was formerly known as "Windows Defender Advanced Threat Protection (WDATP)." There are a number of places (e.g. Azure CLI) where the "WDATP" acronym is still used within Azure.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.5 <u>Perform Automated Vulnerability Scans of Internal Enterprise Assets</u> Perform automated vulnerability scans of internal enterprise assets on a quarterly, or more frequent, basis. Conduct both authenticated and unauthenticated scans, using a SCAP-compliant vulnerability scanning tool.			
v8	10.1 <u>Deploy and Maintain Anti-Malware Software</u> Deploy and maintain anti-malware software on all enterprise assets.			
v8	13.2 <u>Deploy a Host-Based Intrusion Detection Solution</u> Deploy a host-based intrusion detection solution on enterprise assets, where appropriate and/or supported.			
v7	3.1 <u>Run Automated Vulnerability Scanning Tools</u> Utilize an up-to-date SCAP-compliant vulnerability scanning tool to automatically scan all systems on the network on a weekly or more frequent basis to identify all potential vulnerabilities on the organization's systems.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1562	TA0005	

8.1.3.4 Ensure that 'Agentless scanning for machines' Component Status is Set to 'On' (Manual)

Profile Applicability:

- Level 2

Description:

Using disk snapshots, the agentless scanner scans for installed software, vulnerabilities, and plain text secrets.

Rationale:

The Microsoft Defender for Cloud agentless machine scanner provides threat detection, vulnerability detection, and discovery of sensitive information.

Impact:

Agentless scanning for machines requires licensing and is included in these plans:

- Defender CSPM
- Defender for Servers plan 2

Audit:

Audit from Azure Portal

1. From the Azure Portal **Home** page, select **Microsoft Defender for Cloud**
2. Under **Management** select **Environment Settings**
3. Select a subscription
4. Under **Settings > Defender Plans**, click **Settings & monitoring**
5. Under the Component column, locate the row for **Agentless scanning for machines**
6. Ensure that **On** is selected

Repeat the above for any additional subscriptions.

Remediation:

Audit from Azure Portal

1. From the Azure Portal **Home** page, select **Microsoft Defender for Cloud**
2. Under **Management** select **Environment Settings**
3. Select a subscription
4. Under **Settings > Defender Plans**, click **Settings & monitoring**
5. Under the Component column, locate the row for **Agentless scanning for machines**

6. Select **On**
7. Click **Continue** in the top left

Repeat the above for any additional subscriptions.

Default Value:

By default, Agentless scanning for machines is **off**.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/concept-agentless-data-collection>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-incident-response#ir-2-preparation---setup-incident-notification>
3. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/enable-agentless-scanning-vms>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.5 Perform Automated Vulnerability Scans of Internal Enterprise Assets Perform automated vulnerability scans of internal enterprise assets on a quarterly, or more frequent, basis. Conduct both authenticated and unauthenticated scans, using a SCAP-compliant vulnerability scanning tool.			
v8	7.6 Perform Automated Vulnerability Scans of Externally-Exposed Enterprise Assets Perform automated vulnerability scans of externally-exposed enterprise assets using a SCAP-compliant vulnerability scanning tool. Perform scans on a monthly, or more frequent, basis.			
v7	3.1 Run Automated Vulnerability Scanning Tools Utilize an up-to-date SCAP-compliant vulnerability scanning tool to automatically scan all systems on the network on a weekly or more frequent basis to identify all potential vulnerabilities on the organization's systems.			

8.1.3.5 Ensure that 'File Integrity Monitoring' Component Status is Set to 'On' (Manual)

Profile Applicability:

- Level 2

Description:

File Integrity Monitoring (FIM) is a feature that monitors critical system files in Windows or Linux for potential signs of attack or compromise.

Rationale:

FIM provides a detection mechanism for compromised files. When FIM is enabled, critical system files are monitored for changes that might indicate a threat actor is attempting to modify system files for lateral compromise within a host operating system.

Impact:

File Integrity Monitoring requires licensing and is included in the following plan:

- Defender for Servers plan 2

Audit:

Audit from Azure Portal

1. From the Azure Portal **Home** page, select **Microsoft Defender for Cloud**
2. Under **Management** select **Environment Settings**
3. Select a subscription
4. Under **Settings > Defender Plans**, click **Settings & monitoring**
5. Under the Component column, locate the row for **File Integrity Monitoring**
6. Ensure that **On** is selected

Repeat the above for any additional subscriptions.

Remediation:

Remediate from Azure Portal

1. From the Azure Portal **Home** page, select **Microsoft Defender for Cloud**
2. Under **Management** select **Environment Settings**
3. Select a subscription
4. Under **Settings > Defender Plans**, click **Settings & monitoring**
5. Under the Component column, locate the row for **File Integrity Monitoring**
6. Select **On**
7. Click **Continue** in the top left

Repeat the above for any additional subscriptions.

Default Value:

By default, File Integrity Monitoring is **Off**.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/file-integrity-monitoring-overview>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-incident-response#ir-2-preparation---setup-incident-notification>
3. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/file-integrity-monitoring-enable-defender-endpoint>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>7.5 Perform Automated Vulnerability Scans of Internal Enterprise Assets</u> Perform automated vulnerability scans of internal enterprise assets on a quarterly, or more frequent, basis. Conduct both authenticated and unauthenticated scans, using a SCAP-compliant vulnerability scanning tool.			
v8	<u>7.6 Perform Automated Vulnerability Scans of Externally-Exposed Enterprise Assets</u> Perform automated vulnerability scans of externally-exposed enterprise assets using a SCAP-compliant vulnerability scanning tool. Perform scans on a monthly, or more frequent, basis.			
v7	<u>3.1 Run Automated Vulnerability Scanning Tools</u> Utilize an up-to-date SCAP-compliant vulnerability scanning tool to automatically scan all systems on the network on a weekly or more frequent basis to identify all potential vulnerabilities on the organization's systems.			

8.1.4 Defender Plan: Containers

8.1.4.1 Ensure That Microsoft Defender for Containers Is Set To 'On' (Automated)

Profile Applicability:

- Level 2

Description:

Microsoft Defender for Containers helps improve, monitor, and maintain the security of containerized assets—including Kubernetes clusters, nodes, workloads, container registries, and images—across multi-cloud and on-premises environments.

By default, when enabling the plan through the Azure Portal, Microsoft Defender for Containers automatically configures the following components:

- **Agentless scanning for machines**
- **Defender sensor** for runtime protection
- **Azure Policy** for enforcing security best practices
- **K8S API access** for monitoring and threat detection
- **Registry access** for vulnerability assessment

Note: Microsoft Defender for Container Registries ('ContainerRegistry') is deprecated and has been replaced by Microsoft Defender for Containers ('Containers').

Rationale:

Enabling Microsoft Defender for Containers enhances defense-in-depth by providing advanced threat detection, vulnerability assessment, and security monitoring for containerized environments, leveraging insights from the Microsoft Security Response Center (MSRC).

Impact:

Microsoft Defender for Containers incurs a charge per vCore. Refer to <https://azure.microsoft.com/en-us/pricing/details/defender-for-cloud/> and <https://azure.microsoft.com/en-us/pricing/calculator/> to estimate potential costs.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, click **Environment settings**.
3. Click the name of a subscription.
4. Under **Settings**, click **Defender plans**.
5. Under **Cloud Workload Protection (CWP)**, in the row for **Containers**, ensure that the **Status** is set to **On** and **Monitoring coverage** displays **Full**.

6. Repeat steps 1-5 for each subscription.

Audit from Azure CLI

For Microsoft Defender for Container Registries (deprecated), run the following command:

```
az security pricing show --name "ContainerRegistry" --query pricingTier
```

Ensure that the command returns **Standard**.

For Microsoft Defender for Containers, run the following command:

```
az security pricing show --name "Containers" --query [pricingTier,extensions[*].[name,isEnabled]]
```

Ensure that the command returns **Standard**, and that each of the extensions (ContainerRegistriesVulnerabilityAssessments, AgentlessDiscoveryForKubernetes, AgentlessVmScanning, ContainerSensor) returns **True**.

Repeat for each subscription.

Audit from PowerShell

For Microsoft Defender for Container Registries (deprecated), run the following command:

```
Get-AzSecurityPricing -Name 'ContainerRegistry' | Select-Object Name,PricingTier
```

Ensure the command returns **PricingTier Standard**.

For Microsoft Defender for Containers, run the following command:

```
Get-AzSecurityPricing -Name 'Containers'
```

Ensure that **PricingTier** is set to **Standard**, and that each of the extensions (ContainerRegistriesVulnerabilityAssessments, AgentlessDiscoveryForKubernetes, AgentlessVmScanning, ContainerSensor) has **isEnabled** set to **True**.

Repeat for each subscription.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [1c988dd6-ade4-430f-a608-2a3e5b0a6d38](#) - **Name:** 'Microsoft Defender for Containers should be enabled'

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, click **Environment settings**.
3. Click the name of a subscription.
4. Under **Settings**, click **Defender plans**.
5. Under **Cloud Workload Protection (CWP)**, in the row for **Containers**, click **On** in the **Status** column.
6. If **Monitoring coverage** displays **Partial**, click **Settings** under **Partial**.
7. Set the status of each of the components to **On**.
8. Click **Continue**.
9. Click **Save**.
10. Repeat steps 1-9 for each subscription.

Remediate from Azure CLI

Note: Microsoft Defender for Container Registries ('ContainerRegistry') is deprecated and has been replaced by Microsoft Defender for Containers ('Containers').

Run the below command to enable the Microsoft Defender for Containers plan and its components:

```
az security pricing create -n 'Containers' --tier 'standard' --extensions  
name=ContainerRegistriesVulnerabilityAssessments isEnabled=True --extensions  
name=AgentlessDiscoveryForKubernetes isEnabled=True --extensions  
name=AgentlessVmScanning isEnabled=True --extensions name=ContainerSensor  
isEnabled=True
```

Remediate from PowerShell

Note: Microsoft Defender for Container Registries ('ContainerRegistry') is deprecated and has been replaced by Microsoft Defender for Containers ('Containers').

Run the below command to enable the Microsoft Defender for Containers plan and its components:

```
Set-AzSecurityPricing -Name 'Containers' -PricingTier 'Standard' -Extension  
'[{"name":"ContainerRegistriesVulnerabilityAssessments","isEnabled":"True"},{  
"name":"AgentlessDiscoveryForKubernetes","isEnabled":"True"}, {"name":"Agentle  
ssVmScanning","isEnabled":"True"}, {"name":"ContainerSensor","isEnabled":"True  
"}]'
```

Default Value:

The Microsoft Defender for Containers plan is disabled by default.

References:

1. <https://learn.microsoft.com/en-us/cli/azure/security/pricing>

2. <https://learn.microsoft.com/en-us/powershell/module/az.security/get-azsecuritypricing>
3. <https://learn.microsoft.com/en-us/powershell/module/az.security/set-azsecuritypricing>
4. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/defender-for-containers-introduction>
5. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/tutorial-enable-containers-azure>
6. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-1-enable-threat-detection-capabilities>

Additional Information:

The Azure Policy 'Microsoft Defender for Containers should be enabled' checks only that the **pricingTier** for **Containers** is set to **Standard**. It does not check the status of the plan's components.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.5 Perform Automated Vulnerability Scans of Internal Enterprise Assets Perform automated vulnerability scans of internal enterprise assets on a quarterly, or more frequent, basis. Conduct both authenticated and unauthenticated scans, using a SCAP-compliant vulnerability scanning tool.			
v8	7.6 Perform Automated Vulnerability Scans of Externally-Exposed Enterprise Assets Perform automated vulnerability scans of externally-exposed enterprise assets using a SCAP-compliant vulnerability scanning tool. Perform scans on a monthly, or more frequent, basis.			
v7	3.1 Run Automated Vulnerability Scanning Tools Utilize an up-to-date SCAP-compliant vulnerability scanning tool to automatically scan all systems on the network on a weekly or more frequent basis to identify all potential vulnerabilities on the organization's systems.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1210	TA0008	M1016, M1050

8.1.5 Defender Plan: Storage

8.1.5.1 Ensure That Microsoft Defender for Storage Is Set To 'On' (Automated)

Profile Applicability:

- Level 2

Description:

Turning on Microsoft Defender for Storage enables threat detection for Storage, providing threat intelligence, anomaly detection, and behavior analytics in the Microsoft Defender for Cloud.

Rationale:

Enabling Microsoft Defender for Storage allows for greater defense-in-depth, with threat detection provided by the Microsoft Security Response Center (MSRC).

Impact:

Turning on Microsoft Defender for Storage incurs an additional cost per resource.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, select **Environment Settings**.
3. Click on the subscription name.
4. Select the **Defender plans** blade.
5. Ensure **Status** is set to **On** for **Storage**.

Audit from Azure CLI

Ensure the output of the below command is Standard

```
az security pricing show -n StorageAccounts
```

Audit from PowerShell

```
Get-AzSecurityPricing -Name 'StorageAccounts' | Select-Object  
Name,PricingTier
```

Ensure output for **Name PricingTier** is **StorageAccounts Standard**

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [640d2586-54d2-465f-877f-9ffc1d2109f4](#) - **Name:** 'Microsoft Defender for Storage should be enabled'

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, select **Environment Settings**.
3. Click on the subscription name.
4. Select the **Defender plans** blade.
5. Set **Status** to **On** for **Storage**.
6. Select **Save**.

Remediate from Azure CLI

Ensure the output of the below command is Standard

```
az security pricing create -n StorageAccounts --tier 'standard'
```

Remediate from PowerShell

```
Set-AzSecurityPricing -Name 'StorageAccounts' -PricingTier 'Standard'
```

Default Value:

By default, Microsoft Defender plan is off.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/alerts-overview>
2. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/pricings/list>
3. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/pricings/update>
4. <https://learn.microsoft.com/en-us/powershell/module/az.security/get-azsecuritypricing>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-1-enable-threat-detection-capabilities>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.5 Perform Automated Vulnerability Scans of Internal Enterprise Assets Perform automated vulnerability scans of internal enterprise assets on a quarterly, or more frequent, basis. Conduct both authenticated and unauthenticated scans, using a SCAP-compliant vulnerability scanning tool.			

Controls Version	Control	IG 1	IG 2	IG 3
v7	3.1 <u>Run Automated Vulnerability Scanning Tools</u> Utilize an up-to-date SCAP-compliant vulnerability scanning tool to automatically scan all systems on the network on a weekly or more frequent basis to identify all potential vulnerabilities on the organization's systems.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1210	TA0008	M1016, M1050

8.1.5.2 Ensure Advanced Threat Protection Alerts for Storage Accounts Are Monitored (Manual)

Profile Applicability:

- Level 2

Description:

After enabling Microsoft Defender for Storage, configure an alert monitoring and response process to ensure that alerts are actioned in a timely manner. Integrate with SIEM solutions like Microsoft Sentinel, or configure email/webhook notifications to security teams.

Rationale:

Enabling Microsoft Defender for Storage without a monitoring process limits its value. Continuous monitoring and alert triage ensure that detected threats are acted upon quickly, reducing risk exposure.

Impact:

Requires integration effort with SIEM or alerting tools and a defined incident response process.

The amount of data logged and, thus, the cost incurred can vary significantly depending on the tenant size and the applications in your tenant that interact with the Microsoft Graph APIs.

See the following pricing calculations for respective services:

- Log Analytics: <https://learn.microsoft.com/en-us/azure/azure-monitor/logs/cost-logs#pricing-model>.
- Azure Storage: <https://azure.microsoft.com/en-us/pricing/details/storage/blobs/>.
- Event Hubs: <https://azure.microsoft.com/en-us/pricing/details/event-hubs/>.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, click **Environment Settings**.
3. Expand the Tenant Root Group(s) to reveal subscriptions.

For each subscription listed:

1. Click the subscription name to open the **Defender Plans** settings
2. In the settings on the left, click **Continuous Export**

Ensure that **Export enabled** is set to **On** and delivering **at least Security Alerts (Medium and High)** to an Event Hub or Log Analytics Workspace which is tied to a SIEM that is configured to monitor and alert for security alerts.

Remediation:

Connect Microsoft Defender for Cloud to a SIEM such as Microsoft Sentinel or another log analytics solution.

Remediate from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, click **Environment Settings**.
3. Expand the Tenant Root Group(s) to reveal subscriptions.

For each subscription listed:

1. Click the subscription name to open the **Defender Plans** settings
2. In the settings on the left, click **Continuous Export**
3. Select either **Event Hub, Log Analytics Workspace**, or both depending on your environment.
4. Set **Export enabled** to **On**
5. Under Exported data types, ensure that **at least Security Alerts (Medium and High)** is checked.
6. Under Export target, set the target Event Hub or Log Analytics Workspace which is tied to a SIEM that is configured to monitor and alert for security alerts.

Ensure security alerts are included in the security operations workflow and incident response plan.

Default Value:

By default, continuous export is **off**.

References:

1. <https://learn.microsoft.com/azure/defender-for-cloud/alerts-overview>
2. <https://learn.microsoft.com/azure/sentinel/connect-defender-for-cloud>
3. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/continuous-export>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 <u>Collect Detailed Audit Logs</u> Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.			
v7	6.3 <u>Enable Detailed Logging</u> Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.			

8.1.6 Defender Plan: App Service

8.1.6.1 Ensure That Microsoft Defender for App Services Is Set To 'On' (Automated)

Profile Applicability:

- Level 2

Description:

Turning on Microsoft Defender for App Service enables threat detection for App Service, providing threat intelligence, anomaly detection, and behavior analytics in the Microsoft Defender for Cloud.

Rationale:

Enabling Microsoft Defender for App Service allows for greater defense-in-depth, with threat detection provided by the Microsoft Security Response Center (MSRC).

Impact:

Turning on Microsoft Defender for App Service incurs an additional cost per resource.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Defender for Cloud**
2. Under **Management**, select **Environment Settings**
3. Click on the subscription name
4. Select **Defender plans**
5. Ensure Status is **On** for **App Service**

Audit from Azure CLI

Run the following command:

```
az security pricing show -n AppServices
```

Ensure **-PricingTier** is set to **Standard**

Audit from PowerShell

Run the following command:

```
Get-AzSecurityPricing -Name 'AppServices' | Select-Object Name, PricingTier
```

Ensure the **-PricingTier** is set to **Standard**

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [2913021d-f2fd-4f3d-b958-22354e2bdbcb](#) - **Name:** 'Azure Defender for App Service should be enabled'

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Defender for Cloud**
2. Under **Management**, select **Environment Settings**
3. Click on the subscription name
4. Select **Defender plans**
5. Set **App Service** Status to **On**
6. Select **Save**

Remediate from Azure CLI

Run the following command:

```
az security pricing create -n Appservices --tier 'standard'
```

Remediate from PowerShell

Run the following command:

```
Set-AzSecurityPricing -Name "AppServices" -PricingTier "Standard"
```

Default Value:

By default, Microsoft Defender plan is off.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/alerts-overview>
2. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/pricings/list>
3. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/pricings/update>
4. <https://learn.microsoft.com/en-us/powershell/module/az.security/get-azsecuritypricing>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-1-enable-threat-detection-capabilities>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>7.5 Perform Automated Vulnerability Scans of Internal Enterprise Assets</u> Perform automated vulnerability scans of internal enterprise assets on a quarterly, or more frequent, basis. Conduct both authenticated and unauthenticated scans, using a SCAP-compliant vulnerability scanning tool.			
v8	<u>7.6 Perform Automated Vulnerability Scans of Externally-Exposed Enterprise Assets</u> Perform automated vulnerability scans of externally-exposed enterprise assets using a SCAP-compliant vulnerability scanning tool. Perform scans on a monthly, or more frequent, basis.			
v8	<u>16.11 Leverage Vetted Modules or Services for Application Security Components</u> Leverage vetted modules or services for application security components, such as identity management, encryption, and auditing and logging. Using platform features in critical security functions will reduce developers' workload and minimize the likelihood of design or implementation errors. Modern operating systems provide effective mechanisms for identification, authentication, and authorization and make those mechanisms available to applications. Use only standardized, currently accepted, and extensively reviewed encryption algorithms. Operating systems also provide mechanisms to create and maintain secure audit logs.			
v7	<u>3.1 Run Automated Vulnerability Scanning Tools</u> Utilize an up-to-date SCAP-compliant vulnerability scanning tool to automatically scan all systems on the network on a weekly or more frequent basis to identify all potential vulnerabilities on the organization's systems.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1210	TA0008	M1016, M1050

8.1.7 Defender Plan: Databases

8.1.7.1 Ensure That Microsoft Defender for Azure Cosmos DB Is Set To 'On' (Automated)

Profile Applicability:

- Level 2

Description:

Microsoft Defender for Azure Cosmos DB scans all incoming network requests for threats to your Azure Cosmos DB resources.

Rationale:

In scanning Azure Cosmos DB requests within a subscription, requests are compared to a heuristic list of potential security threats. These threats could be a result of a security breach within your services, thus scanning for them could prevent a potential security threat from being introduced.

Impact:

Enabling Microsoft Defender for Azure Cosmos DB requires enabling Microsoft Defender for your subscription. Both will incur additional charges.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, select **Environment Settings**.
3. Click on the subscription name.
4. Select the **Defender plans** blade.
5. On the **Databases** row click on **Select types >**.
6. Ensure the toggle switch next to **Azure Cosmos DB** is set to **On**.

Audit from Azure CLI

Ensure the output of the below command is Standard

```
az security pricing show -n CosmosDbs --query pricingTier
```

Audit from PowerShell

```
Get-AzSecurityPricing -Name 'CosmosDbs' | Select-Object Name,PricingTier
```

Ensure output of **-PricingTier** is **Standard**

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [adbe85b5-83e6-4350-ab58-bf3a4f736e5e](#) - **Name:** 'Microsoft Defender for Azure Cosmos DB should be enabled'

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, select **Environment Settings**.
3. Click on the subscription name.
4. Select the **Defender plans** blade.
5. On the **Databases** row click on **Select types >**.
6. Set the toggle switch next to **Azure Cosmos DB** to **On**.
7. Click **Continue**.
8. Click **Save**.

Remediate from Azure CLI

Run the following command:

```
az security pricing create -n 'CosmosDbs' --tier 'standard'
```

Remediate from PowerShell

Use the below command to enable Standard pricing tier for Azure Cosmos DB

```
Set-AzSecurityPricing -Name 'CosmosDbs' -PricingTier 'Standard'
```

Default Value:

By default, Microsoft Defender for Azure Cosmos DB is not enabled.

References:

1. <https://azure.microsoft.com/en-us/pricing/details/defender-for-cloud/>
2. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/connect-azure-subscription>
3. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/alerts-overview>
4. <https://learn.microsoft.com/en-us/security/benchmark/azure/baselines/azure-cosmos-db-security-baseline>
5. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/tutorial-enable-databases-plan>

6. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#it-1-enable-threat-detection-capabilities>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.5 <u>Perform Automated Vulnerability Scans of Internal Enterprise Assets</u> Perform automated vulnerability scans of internal enterprise assets on a quarterly, or more frequent, basis. Conduct both authenticated and unauthenticated scans, using a SCAP-compliant vulnerability scanning tool.			
v8	16.11 <u>Leverage Vetted Modules or Services for Application Security Components</u> Leverage vetted modules or services for application security components, such as identity management, encryption, and auditing and logging. Using platform features in critical security functions will reduce developers' workload and minimize the likelihood of design or implementation errors. Modern operating systems provide effective mechanisms for identification, authentication, and authorization and make those mechanisms available to applications. Use only standardized, currently accepted, and extensively reviewed encryption algorithms. Operating systems also provide mechanisms to create and maintain secure audit logs.			
v7	3.1 <u>Run Automated Vulnerability Scanning Tools</u> Utilize an up-to-date SCAP-compliant vulnerability scanning tool to automatically scan all systems on the network on a weekly or more frequent basis to identify all potential vulnerabilities on the organization's systems.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1210	TA0008	M1016, M1050

8.1.7.2 Ensure That Microsoft Defender for Open-Source Relational Databases Is Set To 'On' (Automated)

Profile Applicability:

- Level 2

Description:

Turning on Microsoft Defender for Open-source relational databases enables threat detection for Open-source relational databases, providing threat intelligence, anomaly detection, and behavior analytics in the Microsoft Defender for Cloud.

Rationale:

Enabling Microsoft Defender for Open-source relational databases allows for greater defense-in-depth, with threat detection provided by the Microsoft Security Response Center (MSRC).

Impact:

Turning on Microsoft Defender for Open-source relational databases incurs an additional cost per resource.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, select **Environment Settings**.
3. Click on the subscription name.
4. Select the **Defender plans** blade.
5. Click **Select types >** in the row for **Databases**.
6. Ensure the toggle switch next to **Open-source relational databases** is set to **On**.

Audit from Azure CLI

Run the following command:

```
az security pricing show -n OpenSourceRelationalDatabases --query pricingTier
```

Audit from PowerShell

```
Get-AzSecurityPricing | Where-Object {$_.Name -eq 'OpenSourceRelationalDatabases'} | Select-Object Name, PricingTier
```

Ensure output for **Name PricingTier** is **OpenSourceRelationalDatabases Standard**

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/%2FDefinitions

- **Policy ID:** [0a9fbe0d-c5c4-4da8-87d8-f4fd77338835](#) - **Name:** 'Azure Defender for open-source relational databases should be enabled'

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, select **Environment Settings**.
3. Click on the subscription name.
4. Select the **Defender plans** blade.
5. Click **Select types >** in the row for **Databases**.
6. Set the toggle switch next to **Open-source relational databases** to **On**.
7. Select **Continue**.
8. Select **Save**.

Remediate from Azure CLI

Run the following command:

```
az security pricing create -n 'OpenSourceRelationalDatabases' --tier 'standard'
```

Remediate from PowerShell

Use the below command to enable Standard pricing tier for Open-source relational databases

```
set-azsecuritypricing -name "OpenSourceRelationalDatabases" -pricingtier "Standard"
```

Default Value:

By default, Microsoft Defender plan is off.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/alerts-overview>
2. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/pricings/update>
3. <https://learn.microsoft.com/en-us/powershell/module/az.security/get-azsecuritypricing>
4. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-data-protection#dp-2-monitor-anomalies-and-threats-targeting-sensitive-data>

5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#it-1-enable-threat-detection-capabilities>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.5 <u>Perform Automated Vulnerability Scans of Internal Enterprise Assets</u> Perform automated vulnerability scans of internal enterprise assets on a quarterly, or more frequent, basis. Conduct both authenticated and unauthenticated scans, using a SCAP-compliant vulnerability scanning tool.			
v8	16.11 <u>Leverage Vetted Modules or Services for Application Security Components</u> Leverage vetted modules or services for application security components, such as identity management, encryption, and auditing and logging. Using platform features in critical security functions will reduce developers' workload and minimize the likelihood of design or implementation errors. Modern operating systems provide effective mechanisms for identification, authentication, and authorization and make those mechanisms available to applications. Use only standardized, currently accepted, and extensively reviewed encryption algorithms. Operating systems also provide mechanisms to create and maintain secure audit logs.			
v7	3.1 <u>Run Automated Vulnerability Scanning Tools</u> Utilize an up-to-date SCAP-compliant vulnerability scanning tool to automatically scan all systems on the network on a weekly or more frequent basis to identify all potential vulnerabilities on the organization's systems.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1210	TA0008	M1016, M1050

8.1.7.3 Ensure That Microsoft Defender for (Managed Instance) Azure SQL Databases Is Set To 'On' (Automated)

Profile Applicability:

- Level 2

Description:

Turning on Microsoft Defender for Azure SQL Databases enables threat detection for Managed Instance Azure SQL databases, providing threat intelligence, anomaly detection, and behavior analytics in Microsoft Defender for Cloud.

Rationale:

Enabling Microsoft Defender for Azure SQL Databases allows for greater defense-in-depth, includes functionality for discovering and classifying sensitive data, surfacing and mitigating potential database vulnerabilities, and detecting anomalous activities that could indicate a threat to your database.

Impact:

Turning on Microsoft Defender for Azure SQL Databases incurs an additional cost per resource.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, select **Environment Settings**.
3. Click on the subscription name.
4. Select the **Defender plans** blade.
5. Click **Select types >** in the row for **Databases**.
6. Ensure the toggle switch next to **Azure SQL Databases** is set to **On**.

Audit from Azure CLI

Run the following command:

```
az security pricing show -n SqlServers
```

Ensure **-PricingTier** is set to **Standard**

Audit from PowerShell

Run the following command:

```
Get-AzSecurityPricing -Name 'SqlServers' | Select-Object Name,PricingTier
```

Ensure the **-PricingTier** is set to **Standard**

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [7fe3b40f-802b-4cdd-8bd4-fd799c948cc2](#) - **Name:** 'Azure Defender for Azure SQL Database servers should be enabled'
- **Policy ID:** [abfb7388-5bf4-4ad7-ba99-2cd2f41cebb9](#) - **Name:** 'Azure Defender for SQL should be enabled for unprotected SQL Managed Instances'
- **Policy ID:** [3bc8a0d5-38e0-4a3d-a657-2cb64468fc34](#) - **Name:** 'Azure Defender for SQL should be enabled for unprotected MySQL flexible servers'
- **Policy ID:** [d38668f5-d155-42c7-ab3d-9b57b50f8fbf](#) - **Name:** 'Azure Defender for SQL should be enabled for unprotected PostgreSQL flexible servers'
- **Policy ID:** [d31e5c31-63b2-4f12-887b-e49456834fa1](#) - **Name:** 'Microsoft Defender for SQL should be enabled for unprotected Synapse workspaces'
- **Policy ID:** [938c4981-c2c9-4168-9cd6-972b8675f906](#) - **Name:** 'Microsoft Defender for SQL status should be protected for Arc-enabled SQL Servers'

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, select **Environment Settings**.
3. Click on the subscription name.
4. Select the **Defender plans** blade.
5. Click **Select types >** in the row for **Databases**.
6. Set the toggle switch next to **Azure SQL Databases** to **On**.
7. Select **Continue**.
8. Select **Save**.

Remediate from Azure CLI

Run the following command:

```
az security pricing create -n SqlServers --tier 'standard'
```

Remediate from PowerShell

Run the following command:

```
Set-AzSecurityPricing -Name 'SqlServers' -PricingTier 'Standard'
```

Default Value:

By default, Microsoft Defender plan is off.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/alerts-overview>
2. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/pricings/list>
3. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/pricings/update>
4. <https://learn.microsoft.com/en-us/powershell/module/az.security/get-azsecuritypricing>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-data-protection#dp-2-monitor-anomalies-and-threats-targeting-sensitive-data>
6. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-1-enable-threat-detection-capabilities>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.5 <u>Perform Automated Vulnerability Scans of Internal Enterprise Assets</u> Perform automated vulnerability scans of internal enterprise assets on a quarterly, or more frequent, basis. Conduct both authenticated and unauthenticated scans, using a SCAP-compliant vulnerability scanning tool.			
v8	16.11 <u>Leverage Vetted Modules or Services for Application Security Components</u> Leverage vetted modules or services for application security components, such as identity management, encryption, and auditing and logging. Using platform features in critical security functions will reduce developers' workload and minimize the likelihood of design or implementation errors. Modern operating systems provide effective mechanisms for identification, authentication, and authorization and make those mechanisms available to applications. Use only standardized, currently accepted, and extensively reviewed encryption algorithms. Operating systems also provide mechanisms to create and maintain secure audit logs.			
v7	3.1 <u>Run Automated Vulnerability Scanning Tools</u> Utilize an up-to-date SCAP-compliant vulnerability scanning tool to automatically scan all systems on the network on a weekly or more frequent basis to identify all potential vulnerabilities on the organization's systems.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1210	TA0008	M1016, M1050

8.1.7.4 Ensure That Microsoft Defender for SQL Servers on Machines Is Set To 'On' (Automated)

Profile Applicability:

- Level 2

Description:

Turning on Microsoft Defender for SQL servers on machines enables threat detection for SQL servers on machines, providing threat intelligence, anomaly detection, and behavior analytics in Microsoft Defender for Cloud.

Rationale:

Enabling Microsoft Defender for SQL servers on machines allows for greater defense-in-depth, functionality for discovering and classifying sensitive data, surfacing and mitigating potential database vulnerabilities, and detecting anomalous activities that could indicate a threat to your database.

Impact:

Turning on Microsoft Defender for SQL servers on machines incurs an additional cost per resource.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, select **Environment Settings**.
3. Click on the subscription name.
4. Select the **Defender plans** blade.
5. Click **Select types >** in the row for **Databases**.
6. Ensure the toggle switch next to **SQL servers on machines** is set to **On**.

Audit from Azure CLI

Ensure Defender for SQL is licensed with the following command:

```
az security pricing show -n SqlServerVirtualMachines
```

Ensure the 'PricingTier' is set to 'Standard'

Audit from PowerShell

Run the following command:

```
Get-AzSecurityPricing -Name 'SqlServerVirtualMachines' | Select-Object Name,PricingTier
```

Ensure the 'PricingTier' is set to 'Standard'

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [6581d072-105e-4418-827f-bd446d56421b](#) - **Name:** 'Azure Defender for SQL servers on machines should be enabled'
- **Policy ID:** [abfb4388-5bf4-4ad7-ba82-2cd2f41ceae9](#) - **Name:** 'Azure Defender for SQL should be enabled for unprotected Azure SQL servers'

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, select **Environment Settings**.
3. Click on the subscription name.
4. Select the **Defender plans** blade.
5. Click **Select types >** in the row for **Databases**.
6. Set the toggle switch next to **SQL servers on machines** to **On**.
7. Select **Continue**.
8. Select **Save**.

Remediate from Azure CLI

Run the following command:

```
az security pricing create -n SqlServerVirtualMachines --tier 'standard'
```

Remediate from PowerShell

Run the following command:

```
Set-AzSecurityPricing -Name 'SqlServerVirtualMachines' -PricingTier 'Standard'
```

Default Value:

By default, Microsoft Defender plan is off.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/defender-for-sql-usage>
2. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/alerts-overview>
3. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/pricings/update>
4. <https://learn.microsoft.com/en-us/powershell/module/az.security/get-azsecuritypricing>

5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-data-protection#dp-2-monitor-anomalies-and-threats-targeting-sensitive-data>
6. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-1-enable-threat-detection-capabilities>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.5 <u>Perform Automated Vulnerability Scans of Internal Enterprise Assets</u> Perform automated vulnerability scans of internal enterprise assets on a quarterly, or more frequent, basis. Conduct both authenticated and unauthenticated scans, using a SCAP-compliant vulnerability scanning tool.			
v8	16.11 <u>Leverage Vetted Modules or Services for Application Security Components</u> Leverage vetted modules or services for application security components, such as identity management, encryption, and auditing and logging. Using platform features in critical security functions will reduce developers' workload and minimize the likelihood of design or implementation errors. Modern operating systems provide effective mechanisms for identification, authentication, and authorization and make those mechanisms available to applications. Use only standardized, currently accepted, and extensively reviewed encryption algorithms. Operating systems also provide mechanisms to create and maintain secure audit logs.			
v7	3.1 <u>Run Automated Vulnerability Scanning Tools</u> Utilize an up-to-date SCAP-compliant vulnerability scanning tool to automatically scan all systems on the network on a weekly or more frequent basis to identify all potential vulnerabilities on the organization's systems.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1210	TA0008	M1016, M1050

8.1.8 Defender Plan: Key Vault

8.1.8.1 Ensure That Microsoft Defender for Key Vault Is Set To 'On' (Automated)

Profile Applicability:

- Level 2

Description:

Turning on Microsoft Defender for Key Vault enables threat detection for Key Vault, providing threat intelligence, anomaly detection, and behavior analytics in the Microsoft Defender for Cloud.

Rationale:

Enabling Microsoft Defender for Key Vault allows for greater defense-in-depth, with threat detection provided by the Microsoft Security Response Center (MSRC).

Impact:

Turning on Microsoft Defender for Key Vault incurs an additional cost per resource.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, select **Environment Settings**.
3. Click on the subscription name.
4. Select the **Defender plans** blade.
5. Ensure **Status** is set to **On** for **Key Vault**.

Audit from Azure CLI

Ensure the output of the below command is Standard

```
az security pricing show -n 'KeyVaults' --query 'pricingTier'
```

Audit from PowerShell

```
Get-AzSecurityPricing -Name 'KeyVaults' | Select-Object Name, PricingTier
```

Ensure output for **PricingTier** is **Standard**

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [0e6763cc-5078-4e64-889d-ff4d9a839047](#) - **Name:** 'Azure Defender for Key Vault should be enabled'

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, select **Environment Settings**.
3. Click on the subscription name.
4. Select the **Defender plans** blade.
5. Select **On** under **Status** for **Key Vault**.
6. Select **Save**.

Remediate from Azure CLI

Enable Standard pricing tier for Key Vault:

```
az security pricing create -n 'KeyVaults' --tier 'Standard'
```

Remediate from PowerShell

Enable Standard pricing tier for Key Vault:

```
Set-AzSecurityPricing -Name 'KeyVaults' -PricingTier 'Standard'
```

Default Value:

By default, Microsoft Defender plan is off.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/alerts-overview>
2. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/pricings/list>
3. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/pricings/update>
4. <https://learn.microsoft.com/en-us/powershell/module/az.security/get-azsecuritypricing>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-1-enable-threat-detection-capabilities>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.5 Perform Automated Vulnerability Scans of Internal Enterprise Assets Perform automated vulnerability scans of internal enterprise assets on a quarterly, or more frequent, basis. Conduct both authenticated and unauthenticated scans, using a SCAP-compliant vulnerability scanning tool.			

Controls Version	Control	IG 1	IG 2	IG 3
v7	3.1 <u>Run Automated Vulnerability Scanning Tools</u> Utilize an up-to-date SCAP-compliant vulnerability scanning tool to automatically scan all systems on the network on a weekly or more frequent basis to identify all potential vulnerabilities on the organization's systems.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1210	TA0008	M1016, M1050

8.1.9 Defender Plan: Resource Manager

8.1.9.1 Ensure That Microsoft Defender for Resource Manager Is Set To 'On' (Automated)

Profile Applicability:

- Level 2

Description:

Microsoft Defender for Resource Manager scans incoming administrative requests to change your infrastructure from both CLI and the Azure portal.

Rationale:

Scanning resource requests lets you be alerted every time there is suspicious activity in order to prevent a security threat from being introduced.

Impact:

Enabling Microsoft Defender for Resource Manager requires enabling Microsoft Defender for your subscription. Both will incur additional charges.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, select **Environment Settings**.
3. Click on the subscription name.
4. Select the **Defender plans** blade.
5. Ensure **Status** is set to **On** for **Resource Manager**.

Audit from Azure CLI

Ensure the output of the below command is Standard

```
az security pricing show -n 'Arm' --query 'pricingTier'
```

Audit from PowerShell

```
Get-AzSecurityPricing -Name 'Arm' | Select-Object Name, PricingTier
```

Ensure the output of **PricingTier** is **Standard**

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [c3d20c29-b36d-48fe-808b-99a87530ad99](#) - **Name:** 'Azure Defender for Resource Manager should be enabled'

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Defender for Cloud**.
2. Under **Management**, select **Environment Settings**.
3. Click on the subscription name.
4. Select the **Defender plans** blade.
5. Select **On** under **Status** for **Resource Manager**.
6. Select **Save**.

Remediate from Azure CLI

Use the below command to enable Standard pricing tier for Defender for Resource Manager

```
az security pricing create -n 'Arm' --tier 'Standard'
```

Remediate from PowerShell

Use the below command to enable Standard pricing tier for Defender for Resource Manager

```
Set-AzSecurityPricing -Name 'Arm' -PricingTier 'Standard'
```

Default Value:

By default, Microsoft Defender for Resource Manager is not enabled.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/connect-azure-subscription>
2. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/defender-for-resource-manager-introduction>
3. <https://azure.microsoft.com/en-us/pricing/details/defender-for-cloud/>
4. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/alerts-overview>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-1-enable-threat-detection-capabilities>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.4 <u>Restrict Administrator Privileges to Dedicated Administrator Accounts</u> Restrict administrator privileges to dedicated administrator accounts on enterprise assets. Conduct general computing activities, such as internet browsing, email, and productivity suite use, from the user's primary, non-privileged account.			
v8	7.5 <u>Perform Automated Vulnerability Scans of Internal Enterprise Assets</u> Perform automated vulnerability scans of internal enterprise assets on a quarterly, or more frequent, basis. Conduct both authenticated and unauthenticated scans, using a SCAP-compliant vulnerability scanning tool.			
v7	3.1 <u>Run Automated Vulnerability Scanning Tools</u> Utilize an up-to-date SCAP-compliant vulnerability scanning tool to automatically scan all systems on the network on a weekly or more frequent basis to identify all potential vulnerabilities on the organization's systems.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1210	TA0008	M1016, M1050

8.1.10 Ensure that Microsoft Defender for Cloud is Configured to Check VM Operating Systems for Updates (Automated)

Profile Applicability:

- Level 1

Description:

Ensure that the latest OS patches for all virtual machines are applied.

Rationale:

Windows and Linux virtual machines should be kept updated to:

- Address a specific bug or flaw
- Improve an OS or application's general stability
- Fix a security vulnerability

Microsoft Defender for Cloud retrieves a list of available security and critical updates from Windows Update or Windows Server Update Services (WSUS), depending on which service is configured on a Windows VM. The security center also checks for the latest updates in Linux systems. If a VM is missing a system update, the security center will recommend system updates be applied.

Impact:

Running Microsoft Defender for Cloud incurs additional charges for each resource monitored. Please see attached reference for exact charges per hour.

Audit:

Audit from Azure Portal

1. From Azure Home select the Portal Menu
2. Select **Microsoft Defender for Cloud**
3. Then the **Recommendations** blade
4. Ensure that there are no recommendations for **System updates should be installed on your machines (powered by Update Center)**

Alternatively, you can employ your own patch assessment and management tool to periodically assess, report and install the required security patches for your OS.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [f85bf3e0-d513-442e-89c3-1784ad63382b](#) - **Name:** 'System updates should be installed on your machines (powered by Update Center)'
- **Policy ID:** [bd876905-5b84-4f73-ab2d-2e7a7c4568d9](#) - **Name:** 'Machines should be configured to periodically check for missing system updates'

Remediation:

Follow Microsoft Azure documentation to apply security patches from the security center. Alternatively, you can employ your own patch assessment and management tool to periodically assess, report, and install the required security patches for your OS.

Default Value:

By default, patches are not automatically deployed.

References:

1. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-posture-vulnerability-management#pv-6-rapidly-and-automatically-remediate-vulnerabilities>
2. <https://azure.microsoft.com/en-us/pricing/details/defender-for-cloud/>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.3 <u>Perform Automated Operating System Patch Management</u> Perform operating system updates on enterprise assets through automated patch management on a monthly, or more frequent, basis.			
v7	3.4 <u>Deploy Automated Operating System Patch Management Tools</u> Deploy automated software update tools in order to ensure that the operating systems are running the most recent security updates provided by the software vendor.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1068	TA0004	M1051

8.1.11 Ensure that non-deprecated Microsoft Cloud Security Benchmark policies are not set to 'Disabled' (Manual)

Profile Applicability:

- Level 1

Description:

The Microsoft Cloud Security Benchmark (MCSB) is an Azure Policy initiative automatically assigned to all subscriptions that evaluates resource configurations against Microsoft security best practice recommendations. Policies set to a **Disabled** effect are not evaluated, removing visibility into the associated recommendation. Microsoft deprecates policies when controls are superseded or no longer applicable; deprecated policies are set to **Disabled** by default and require no corrective action. Non-deprecated policies should not be set to **Disabled** without a documented organizational exception.

Note: A policy is considered deprecated when its definition name is prefixed with **[Deprecated]**: in the Azure Policy portal and its **policyDeprecated** metadata property is **true**.

Rationale:

Disabling a non-deprecated MCSB policy prevents Microsoft Defender for Cloud from evaluating the associated control and surfacing relevant recommendations, creating an undetected gap in security posture visibility. Exceptions allowing non-deprecated MCSB policies to be disabled can be reasonably made but each exception should be documented and reconsidered periodically.

Examples of reasonable exceptions:

- A compensating control exists which cannot be evaluated by the policy
- A scope exemption exists for certain resources within the evaluated scope
- The policy effect conflicts with architecturally deliberate choices
- The resource type is not used in the evaluated scope

Deprecated policies set to **Disabled** reflect an intentional Microsoft lifecycle decision and are not a finding.

Impact:

MCSB policies default to an **Audit** or **AuditIfNotExists** effect, which evaluates but does not enforce recommendations. Re-enabling a non-deprecated disabled policy restores evaluation only; no enforcement action occurs unless the effect is subsequently changed to **Deny** or **DeployIfNotExists** by the administrator.

Audit:

Audit from Azure Portal

Perform the following steps for each Management Group or Subscription:

1. From Azure Home, select the Portal Menu.
2. Select **Microsoft Defender for Cloud**.
3. Under **Management**, select **Environment settings**.
4. Click on the appropriate Management Group or Subscription.
5. Click **Security policies** in the left column.
6. Click **Microsoft cloud security benchmark**.
7. Click **Add filter**, select **Effect**, check the **Disabled** box, and click **Apply**.
8. For each policy returned, confirm whether the definition is deprecated:
 - Deprecated definitions display a **[Deprecated]**: name prefix.
 - Deprecated policies set to **Disabled** are **expected and compliant**.
9. Verify that no non-deprecated policies are displayed with a **Disabled** effect.
10. Repeat steps 1–9 for each Management Group or Subscription.

Cross-reference each overridden policy against current documented exceptions and the Azure Policy definitions list to confirm deprecation status before treating the result as a finding.

Remediation:

No remediation is required for deprecated MCSB policies set to **Disabled**.

For non-deprecated policies found to be set to **Disabled**, restore the effect to its default value or document an approved organizational exception.

Remediate from Azure Portal

1. From Azure Home, select the Portal Menu.
2. Select **Microsoft Defender for Cloud**.
3. Under **Management**, select **Environment settings**.
4. Click on the appropriate Management Group or Subscription.
5. Click **Security policies** in the left column.
6. Click **Microsoft cloud security benchmark**.
7. Click **Add filter**, select **Effect**, check **Disabled**, and click **Apply**.
8. Confirm the displayed policy is not deprecated before proceeding.
9. Click the blue ellipsis ... to the right of the policy name and select **Manage effect and parameters**.
10. Under **Policy effect**, select **Audit**.
11. Click **Save**, then click **Refresh**.
12. Repeat steps 8–11 for each non-deprecated disabled policy.
13. Repeat steps 1–12 for each Management Group or Subscription requiring remediation.

Default Value:

The MCSB policy initiative is automatically assigned to all subscriptions with Microsoft Defender for Cloud. Of the **412** policies currently in the MCSB initiative, **0** have a default effect of **Disabled** and **0** are deprecated. Any policy found with a **Disabled** effect reflects an administrative override applied after the default assignment.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/security-policy-concept>
2. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/implement-security-recommendations>
3. <https://learn.microsoft.com/en-us/rest/api/policy/policy-assignments/get>
4. <https://learn.microsoft.com/en-us/rest/api/policy/policy-assignments/create>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-governance-strategy#gs-7-define-and-implement-logging-threat-detection-and-incident-response-strategy>
6. <https://learn.microsoft.com/en-us/azure/governance/policy/concepts/effect-basics>
7. <https://learn.microsoft.com/en-us/azure/governance/policy/tutorials/create-and-manage>
8. <https://aka.ms/policydefdeprecation>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v8	4.2 Establish and Maintain a Secure Configuration Process for Network Infrastructure Establish and maintain a secure configuration process for network devices. Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	5.1 Establish Secure Configurations Maintain documented, standard security configuration standards for all authorized operating systems and software.			
v7	5.5 Implement Automated Configuration Monitoring Systems Utilize a Security Content Automation Protocol (SCAP) compliant configuration monitoring system to verify all security configuration elements, catalog approved exceptions, and alert when unauthorized changes occur.			

8.1.12 Ensure That 'All users with the following roles' is Set to 'Owner' (Automated)

Profile Applicability:

- Level 1

Description:

Enable security alert emails to subscription owners.

Rationale:

Enabling security alert emails to subscription owners ensures that they receive security alert emails from Microsoft. This ensures that they are aware of any potential security issues and can mitigate the risk in a timely fashion.

Audit:

Audit from Azure Portal

1. From Azure Home select the Portal Menu
2. Select **Microsoft Defender for Cloud**
3. Under **Management**, select **Environment Settings**
4. Click on the appropriate Management Group, Subscription, or Workspace
5. Click on **Email notifications**
6. Ensure that **All users with the following roles** is set to **Owner**

Audit from Azure CLI

Ensure the command below returns state of **On** and that **Owner** appears in roles.

```
az account get-access-token --query  
"{subscription:subscription,accessToken:accessToken}" --out tsv | xargs -L1  
bash -c 'curl -X GET -H "Authorization: Bearer $1" -H "Content-Type:  
application/json"  
https://management.azure.com/subscriptions/$0/providers/Microsoft.Security/se  
curityContacts?api-version=2020-01-01-preview'| jq '.[] |  
select(.name=="default").properties.notificationsByRole'
```

Remediation:

Remediate from Azure Portal

1. From Azure Home select the Portal Menu
2. Select **Microsoft Defender for Cloud**
3. Under **Management**, select **Environment Settings**
4. Click on the appropriate Management Group, Subscription, or Workspace
5. Click on **Email notifications**
6. In the drop down of the **All users with the following roles** field select **Owner**
7. Click **Save**

Remediate from Azure CLI

Use the below command to set **Send email also to subscription owners** to **On**.

```
az account get-access-token --query  
"{subscription:subscription,accessToken:accessToken}" --out tsv | xargs -L1  
bash -c 'curl -X PUT -H "Authorization: Bearer $1" -H "Content-Type:  
application/json"  
https://management.azure.com/subscriptions/$0/providers/Microsoft.Security/se  
curityContacts/default1?api-version=2017-08-01-preview -d@"input.json"'
```

Where **input.json** contains the data below, replacing **validEmailAddress** with a single email address or multiple comma-separated email addresses:

```
{  
  "id":  
    "/subscriptions/<Your_Subscription_Id>/providers/Microsoft.Security/securityC  
ontacts/default1",  
  "name": "default1",  
  "type": "Microsoft.Security/securityContacts",  
  "properties": {  
    "email": "<validEmailAddress>",  
    "alertNotifications": "On",  
    "alertsToAdmins": "On",  
    "notificationsByRole": "Owner"  
  }  
}
```

Default Value:

By default, **Owner** is selected

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/configure-email-notifications>
2. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/security-contacts/list>
3. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/security-contacts>
4. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-incident-response#ir-2-preparation---setup-incident-notification>

Additional Information:

Excluding any entries in the input.json properties block disables the specific setting by default.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	17.2 <u>Establish and Maintain Contact Information for Reporting Security Incidents</u> Establish and maintain contact information for parties that need to be informed of security incidents. Contacts may include internal staff, third-party vendors, law enforcement, cyber insurance providers, relevant government agencies, Information Sharing and Analysis Center (ISAC) partners, or other stakeholders. Verify contacts annually to ensure that information is up-to-date.			
v7	19.5 <u>Maintain Contact Information For Reporting Security Incidents</u> Assemble and maintain information on third-party contact information to be used to report a security incident, such as Law Enforcement, relevant government departments, vendors, and ISAC partners.			

8.1.13 Ensure 'Additional email addresses' is Configured with a Security Contact Email (Automated)

Profile Applicability:

- Level 1

Description:

Microsoft Defender for Cloud emails the subscription owners whenever a high-severity alert is triggered for their subscription. You should provide a security contact email address as an additional email address.

Rationale:

Microsoft Defender for Cloud emails the Subscription Owner to notify them about security alerts. Adding your Security Contact's email address to the 'Additional email addresses' field ensures that your organization's Security Team is included in these alerts. This ensures that the proper people are aware of any potential compromise in order to mitigate the risk in a timely fashion.

Audit:

Audit from Azure Portal

1. From Azure Home select the Portal Menu.
2. Select **Microsoft Defender for Cloud**.
3. Under **Management**, select **Environment Settings**.
4. Click on the appropriate Management Group, Subscription, or Workspace.
5. Click on **Email notifications**.
6. Ensure that a valid security contact email address is listed in the **Additional email addresses** field.

Audit from Azure CLI

Ensure the output of the below command is not empty and is set with appropriate email ids:

```
az account get-access-token --query  
"{subscription:subscription,accessToken:accessToken}" --out tsv | xargs -L1  
bash -c 'curl -X GET -H "Authorization: Bearer $1" -H "Content-Type:  
application/json"  
https://management.azure.com/subscriptions/$0/providers/Microsoft.Security/se  
curityContacts?api-version=2020-01-01-preview' | jq '.[].[] |  
select(.name=="default")'|jq '.properties.emails'
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [4f4f78b8-e367-4b10-a341-d9a4ad5cf1c7](#) - **Name:** 'Subscriptions should have a contact email address for security issues'

Remediation:

Remediate from Azure Portal

1. From Azure Home select the Portal Menu.
2. Select **Microsoft Defender for Cloud**.
3. Under **Management**, select **Environment Settings**.
4. Click on the appropriate Management Group, Subscription, or Workspace.
5. Click on **Email notifications**.
6. Enter a valid security contact email address (or multiple addresses separated by commas) in the **Additional email addresses** field.
7. Click **Save**.

Remediate from Azure CLI

Use the below command to set **Security contact emails** to **On**.

```
az account get-access-token --query  
"{subscription:subscription,accessToken:accessToken}" --out tsv | xargs -L1  
bash -c 'curl -X PUT -H "Authorization: Bearer $1" -H "Content-Type:  
application/json"  
https://management.azure.com/subscriptions/$0/providers/Microsoft.Security/se  
curityContacts/default?api-version=2020-01-01-preview -d@"input.json"'
```

Where **input.json** contains the data below, replacing **validEmailAddress** with a single email address or multiple comma-separated email addresses:

```
{  
  "id":  
  "/subscriptions/<Your_Subscription_Id>/providers/Microsoft.Security/securityC  
ontacts/default",  
  "name": "default",  
  "type": "Microsoft.Security/securityContacts",  
  "properties": {  
    "email": "<validEmailAddress>",  
    "alertNotifications": "On",  
    "alertsToAdmins": "On"  
  }  
}
```

Default Value:

By default, there are no additional email addresses entered.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/configure-email-notifications>
2. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/security-contacts/list>
3. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/security-contacts>
4. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-incident-response#ir-2-preparation---setup-incident-notification>

Additional Information:

Excluding any entries in the `input.json` properties block disables the specific setting by default.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	17.2 Establish and Maintain Contact Information for Reporting Security Incidents Establish and maintain contact information for parties that need to be informed of security incidents. Contacts may include internal staff, third-party vendors, law enforcement, cyber insurance providers, relevant government agencies, Information Sharing and Analysis Center (ISAC) partners, or other stakeholders. Verify contacts annually to ensure that information is up-to-date.			
v7	19.5 Maintain Contact Information For Reporting Security Incidents Assemble and maintain information on third-party contact information to be used to report a security incident, such as Law Enforcement, relevant government departments, vendors, and ISAC partners.			

8.1.14 Ensure that 'Notify about alerts with the following severity (or higher)' is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Enables emailing security alerts to the subscription owner or other designated security contact.

Rationale:

Enabling security alert emails ensures that security alert emails are sent by Microsoft. This ensures that the right people are aware of any potential security issues and can mitigate the risk.

Impact:

Enabling security alert emails can cause alert fatigue, increasing the risk of missing important alerts. Select an appropriate severity level to manage notifications. Azure aims to reduce alert fatigue by limiting the daily email volume per severity level. Learn more: <https://learn.microsoft.com/en-us/azure/defender-for-cloud/configure-email-notifications#email-frequency>.

Audit:

Audit from Azure Portal

1. From Azure Home select the Portal Menu.
2. Select **Microsoft Defender for Cloud**.
3. Under **Management**, select **Environment settings**.
4. Click on the appropriate Subscription.
5. Click on **Email notifications**.
6. Under **Notification types**, ensure that the box next to **Notify about alerts with the following severity (or higher)** is checked, and an appropriate severity level is selected.
7. Repeat steps 1-6 for each Subscription.

Audit from Azure CLI

Including a Subscription ID at the `$0` in `/subscriptions/$0/providers`, ensure the below command returns `"state": "On"`, and that `"minimalSeverity"` is set to an appropriate severity level:

```
az account get-access-token --query  
"{subscription:subscription,accessToken:accessToken}" --out tsv | xargs -L1  
bash -c 'curl -X GET -H "Authorization: Bearer $1" -H "Content-Type:  
application/json"  
https://management.azure.com/subscriptions/$0/providers/Microsoft.Security/se  
curityContacts?api-version=2020-01-01-preview' | jq '.|.[] |  
select(.name=="default")'|jq '.properties.alertNotifications'
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [6e2593d9-add6-4083-9c9b-4b7d2188c899](#) - **Name:** 'Email notification for high severity alerts should be enabled'
- **Policy ID:** [0b15565f-aa9e-48ba-8619-45960f2c314d](#) - **Name:** 'Email notification to subscription owner for high severity alerts should be enabled'

Remediation:

Remediate from Azure Portal

1. From Azure Home select the Portal Menu.
2. Select **Microsoft Defender for Cloud**.
3. Under **Management**, select **Environment settings**.
4. Click on the appropriate Subscription.
5. Click on **Email notifications**.
6. Under **Notification types**, check box next to **Notify about alerts with the following severity (or higher)** and select an appropriate severity level from the drop-down menu.
7. Click **Save**.
8. Repeat steps 1-7 for each Subscription requiring remediation.

Remediate from Azure CLI

Use the below command to enable **Send email notification for high severity alerts**:

```
az account get-access-token --query  
"{subscription:subscription,accessToken:accessToken}" --out tsv | xargs -L1  
bash -c 'curl -X PUT -H "Authorization: Bearer $1" -H "Content-Type:  
application/json"  
https://management.azure.com/subscriptions/<$0>/providers/Microsoft.Security/  
securityContacts/default1?api-version=2017-08-01-preview -d"input.json"'
```

Where **input.json** contains the data below, replacing **validEmailAddress** with a single email address or multiple comma-separated email addresses:

```
{  
  "id":  
  "/subscriptions/<subscriptionId>/providers/Microsoft.Security/securityContact  
s/default",  
  "name": "default",  
  "type": "Microsoft.Security/securityContacts",  
  "properties": {  
    "email": "<validEmailAddress>",  
    "alertNotifications": "On",  
    "alertsToAdmins": "On"  
  }  
}
```

Default Value:

By default, subscription owners receive email notifications for high-severity alerts.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/configure-email-notifications>
2. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/security-contacts>
3. <https://learn.microsoft.com/en-us/rest/api/defenderforcloud/security-contacts/list>
4. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-incident-response#ir-2-preparation---setup-incident-notification>

Additional Information:

Excluding any entries in the **input.json** properties block disables the specific setting by default. This recommendation has been updated to reflect recent changes to Microsoft REST APIs for getting and updating security contact information.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	13.11 <u>Tune Security Event Alerting Thresholds</u> Tune security event alerting thresholds monthly, or more frequently.			●
v7	6.8 <u>Regularly Tune SIEM</u> On a regular basis, tune your SIEM system to better identify actionable events and decrease event noise.			●

8.1.15 Ensure that 'Notify about attack paths with the following risk level (or higher)' is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Enables emailing attack paths to the subscription owner or other designated security contact.

Rationale:

Enabling attack path emails ensures that attack path emails are sent by Microsoft. This ensures that the right people are aware of any potential security issues and can mitigate the risk.

Impact:

Enabling attack path emails can cause alert fatigue, increasing the risk of missing important alerts. Select an appropriate risk level to manage notifications. Azure aims to reduce alert fatigue by limiting the daily email volume per risk level. Learn more:

<https://learn.microsoft.com/en-us/azure/defender-for-cloud/configure-email-notifications#email-frequency>.

Audit:

Audit from Azure Portal

1. From Azure Home select the Portal Menu.
2. Select **Microsoft Defender for Cloud**.
3. Under **Management**, select **Environment settings**.
4. Click on the appropriate Subscription.
5. Click on **Email notifications**.
6. Under Notification types, ensure that the box next to **Notify about attack paths with the following risk level (or higher)** is checked, and an appropriate risk level is selected.
7. Repeat steps 1-6 for each Subscription.

Audit from Azure CLI

Including a Subscription ID at the **\$0** in **/subscriptions/\$0/providers**, ensure the below command returns **"sourceType": "AttackPath"**, and that **"minimalRiskLevel"** is set to an appropriate risk level:

```
az account get-access-token --query  
"{subscription:subscription,accessToken:accessToken}" --out tsv | xargs -L1  
bash -c 'curl -X GET -H "Authorization: Bearer $1" -H "Content-Type:  
application/json"  
https://management.azure.com/subscriptions/$0/providers/Microsoft.Security/se  
curityContacts?api-version=2023-12-01-preview' | jq '...[]'
```

Remediation:

Remediate from Azure Portal

1. From Azure Home select the Portal Menu.
2. Select **Microsoft Defender for Cloud**.
3. Under **Management**, select **Environment settings**.
4. Click on the appropriate Subscription.
5. Click on **Email notifications**.
6. Under Notification types, check the box next to **Notify about attack paths with the following risk level (or higher)**, and select an appropriate risk level from the drop-down menu.
7. Repeat steps 1-6 for each Subscription.

References:

1. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/configure-email-notifications>
2. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/how-to-manage-attack-path>
3. <https://learn.microsoft.com/en-us/azure/defender-for-cloud/concept-attack-path>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	13.11 Tune Security Event Alerting Thresholds Tune security event alerting thresholds monthly, or more frequently.			●
v7	6.8 Regularly Tune SIEM On a regular basis, tune your SIEM system to better identify actionable events and decrease event noise.			●

8.1.16 Ensure that Microsoft Defender External Attack Surface Monitoring (EASM) is Enabled (Manual)

Profile Applicability:

- Level 2

Description:

An organization's attack surface is the collection of assets with a public network identifier or URI that an external threat actor can see or access from outside your cloud. It is the set of points on the boundary of a system, a system element, system component, or an environment where an attacker can try to enter, cause an effect on, or extract data from, that system, system element, system component, or environment. The larger the attack surface, the harder it is to protect.

This tool can be configured to scan your organization's online infrastructure such as specified domains, hosts, CIDR blocks, and SSL certificates, and store them in an Inventory. Inventory items can be added, reviewed, approved, and removed, and may contain enrichments ("insights") and additional information collected from the tool's different scan engines and open-source intelligence sources.

A Defender EASM workspace will generate an Inventory of publicly exposed assets by crawling and scanning the internet using *Seeds* you provide when setting up the tool. Seeds can be FQDNs, IP CIDR blocks, and WHOIS records.

Defender EASM will generate Insights within 24-48 hours after Seeds are provided, and these insights include vulnerability data (CVEs), ports and protocols, and weak or expired SSL certificates that could be used by an attacker for reconnaissance or exploitation.

Results are classified High/Medium/Low and some of them include proposed mitigations.

Rationale:

This tool can monitor the externally exposed resources of an organization, provide valuable insights, and export these findings in a variety of formats (including CSV) for use in vulnerability management operations and red/purple team exercises.

Impact:

Microsoft Defender EASM workspaces are currently available as Azure Resources with a 30-day free trial period but can quickly accrue significant charges. The costs are calculated daily as (Number of "billable" inventory items) x (item cost per day).

Estimated cost is not provided within the tool, and users are strongly advised to contact their Microsoft sales representative for pricing and set a calendar reminder for the end of the trial period.

If the workspace is deleted by the last day of a free trial period, no charges are billed.

Audit:

Audit from Azure Portal

1. Go to **Microsoft Defender EASM**.
2. Ensure that at least one Microsoft Defender EASM workspace is listed.
3. Click the name of a workspace.
4. Ensure the workspace is configured appropriately for your environment and organization.
5. Repeat steps 3-4 for each workspace.

Remediation:

Remediate from Azure Portal

1. Go to **Microsoft Defender EASM**.
2. Click **+ Create**.
3. Under **Project details**, select a subscription.
4. Select or create a resource group.
5. Under **Instance details**, enter a name for the workspace.
6. Select a region.
7. Click **Review + create**.
8. Click **Create**.
9. Once the deployment has completed, go to **Microsoft Defender EASM**.
10. Click the workspace name.
11. Configure the workspace appropriately for your environment and organization.

Default Value:

Microsoft Defender EASM is an optional, paid Azure Resource that must be created and configured inside a Subscription and Resource Group.

References:

1. <https://learn.microsoft.com/en-us/azure/external-attack-surface-management/>
2. <https://learn.microsoft.com/en-us/azure/external-attack-surface-management/deploying-the-defender-easm-azure-resource>
3. <https://www.microsoft.com/en-us/security/blog/2022/08/02/microsoft-announces-new-solutions-for-threat-intelligence-and-attack-surface-management/>

Additional Information:

Microsoft added its Defender for External Attack Surface management (EASM) offering to Azure following its 2022 acquisition of EASM SaaS tool company RiskIQ.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	<u>7.6 Perform Automated Vulnerability Scans of Externally-Exposed Enterprise Assets</u> Perform automated vulnerability scans of externally-exposed enterprise assets using a SCAP-compliant vulnerability scanning tool. Perform scans on a monthly, or more frequent, basis.			
v7	<u>3.1 Run Automated Vulnerability Scanning Tools</u> Utilize an up-to-date SCAP-compliant vulnerability scanning tool to automatically scan all systems on the network on a weekly or more frequent basis to identify all potential vulnerabilities on the organization's systems.			

8.2 Microsoft Defender for IoT

8.2.1 Ensure That Microsoft Defender for IoT Hub Is Set To 'On' (Manual)

Profile Applicability:

- Level 2

Description:

Microsoft Defender for IoT acts as a central security hub for IoT devices within your organization.

Rationale:

IoT devices are very rarely patched and can be potential attack vectors for enterprise networks. Updating their network configuration to use a central security hub allows for detection of these breaches.

Impact:

Enabling Microsoft Defender for IoT will incur additional charges dependent on the level of usage.

Audit:

Audit from Azure Portal

1. Go to **IoT Hub**.
2. Select an **IoT Hub** to validate.
3. Select **Overview** in **Defender for IoT**.
4. The Threat prevention and Threat detection screen will appear, if **Defender for IoT** is Enabled.

Remediation:

Remediate from Azure Portal

1. Go to **IoT Hub**.
2. Select an **IoT Hub** to validate.
3. Select **Overview** in **Defender for IoT**.
4. Click on **Secure your IoT solution**, and complete the onboarding.

Default Value:

By default, Microsoft Defender for IoT is not enabled.

References:

1. <https://www.microsoft.com/en-us/security/business/endpoint-security/microsoft-defender-iot#overview>
2. <https://learn.microsoft.com/en-us/azure/defender-for-iot/>
3. <https://www.microsoft.com/en-us/security/business/endpoint-security/microsoft-defender-iot-pricing>
4. <https://learn.microsoft.com/en-us/security/benchmark/azure/baselines/microsoft-defender-for-iot-security-baseline>
5. <https://learn.microsoft.com/en-us/cli/azure/iot>
6. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-logging-threat-detection#lt-1-enable-threat-detection-capabilities>
7. <https://learn.microsoft.com/en-us/azure/defender-for-iot/device-builders/quickstart-onboard-iot-hub>

Additional Information:

There are additional configurations for Microsoft Defender for IoT that allow for types of deployments called hybrid or local. Both run on your physical infrastructure. These are complicated setups and are primarily outside of the scope of a purely Azure benchmark. Please see the references to consider these options for your organization.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	7.5 Perform Automated Vulnerability Scans of Internal Enterprise Assets Perform automated vulnerability scans of internal enterprise assets on a quarterly, or more frequent, basis. Conduct both authenticated and unauthenticated scans, using a SCAP-compliant vulnerability scanning tool.			
v8	13.6 Collect Network Traffic Flow Logs Collect network traffic flow logs and/or network traffic to review and alert upon from network devices.			
v7	3.1 Run Automated Vulnerability Scanning Tools Utilize an up-to-date SCAP-compliant vulnerability scanning tool to automatically scan all systems on the network on a weekly or more frequent basis to identify all potential vulnerabilities on the organization's systems.			

8.3 Key Vault

This section covers security best practice recommendations for Azure Key Vault.

IMPORTANT NOTE: Requisite Key Vault Permissions for Data Plane access

Each Key Vault can have one of two permissions models:

1. [Preferred] Azure Role-based access control (RBAC), or
2. Vault access policy (legacy)

Prior to auditing or remediating each recommendation, the account performing the procedure must be assigned appropriate permissions. These permissions must be set in advance for each key vault individually. Because some recommendations in this section require data plane access, the use of these permissions should be time-bound and scoped carefully. Privileged Identity Management (PIM) is a helpful tool for temporary activation of privileges and is highly recommended if available.

Where Data Plane access is required, the procedures of each recommendation in this section will include a suggested minimum role for Data Plane permissions required for Audit and Remediation.

WARNING: These permissions grant direct access to sensitive secrets, keys, and certificates – remove them immediately after use, as any delay creates unnecessary and significant risk of unauthorized access.

To set permissions for Azure RBAC Key Vaults:

1. Open the target Key Vault, then click **Access Control (IAM)**.
2. Click **Add role assignment** then search for and select the **<recommendation-provided Role Name>** then click **Next**.
3. Assign the role to an appropriately narrow scope to exercise least privilege (e.g. PIM-eligible role [preferred], Managed Identity, or Entra ID security group), then click **Review + assign**.

Key Vault RBAC Access Model Reference: <https://learn.microsoft.com/en-us/azure/key-vault/general/rbac-guide?tabs=azure-cli>

To set permissions for access policy Key Vaults (legacy):

1. Open the target Key Vault, then click **Access policies**.
2. Click the checkboxes according to the **<recommendation-provided Policy permissions>** then click **Next**.
3. Assign the permissions policy to an appropriately narrow principal to exercise least privilege (e.g. PIM-eligible role [preferred], Managed Identity, or Entra ID security group), click **Next** then click **Create**.

WARNING: These permissions grant direct access to sensitive secrets, keys, and certificates – remove them immediately after use, as any delay creates unnecessary and significant risk of unauthorized access.

Help us improve this Benchmark! If you notice a needed correction, want to provide feedback, or wish to contribute security best practice guidance please join our community and create a ticket, propose a change, or start a discussion so we can improve this guidance!

The CIS Microsoft Azure Community is here:
<https://workbench.cisecurity.org/communities/72>.

Resources for Azure Key Vault Service

Azure Product Page:

- <https://azure.microsoft.com/en-us/products/key-vault/>

Azure Key Vault service overview:

- <https://learn.microsoft.com/en-us/azure/key-vault/general/overview>

Microsoft Cloud Security Baseline for Azure Key Vault:

- <https://learn.microsoft.com/en-us/security/benchmark/azure/baselines/key-vault-security-baseline>

8.3.1 Ensure that the Expiration Date is Set for all Keys in Key Vaults using RBAC (Automated)

Profile Applicability:

- Level 1

Description:

Ensure that all Keys in Azure Key Vaults using Role Based Access Control (RBAC) have an expiration date set.

Note: The audit commands require data-plane access to each Key Vault in addition to the management-plane **Reader** role. Without data-plane access these commands return empty results silently.

Rationale:

The Azure Key Vault enables users to store and use cryptographic keys within the Microsoft Azure environment. The **exp** (expiration date) attribute identifies the expiration date on or after which the key MUST NOT be used for encryption of new data, wrapping of new keys, and signing. By default, keys never expire. It is thus recommended that keys be rotated in the key vault and set an explicit expiration date for all keys to help enforce the key rotation. This ensures that the keys cannot be used beyond their assigned lifetimes.

Impact:

Keys cannot be used beyond their assigned expiration dates respectively. Keys need to be rotated periodically wherever they are used.

Audit:

Requisite RBAC Permissions for Keys - read section overview!

Audit Role = **Key Vault Crypto User**

Audit from Azure Portal

1. Go to **Key vaults**.
2. For each Key vault, under **Objects** select, **Keys**.
3. In the main pane, ensure that the status of the key is **enabled**.
4. For each enabled key, ensure that an appropriate **Expiration date** is set.

Audit from Azure CLI

Get a list of all the key vaults in your Azure environment by running the following command:

```
az keyvault list
```

Then for each key vault listed ensure that the output of the below command contains Key ID (kid), enabled status as **true** and Expiration date (**expires**) is not **empty** or **null**:

```
az keyvault key list --vault-name <VaultName> --query ' [*].{"kid":kid,"enabled":attributes.enabled,"expires":attributes.expires}'
```

Audit from PowerShell

Retrieve a list of Azure Key Vaults:

```
Get-AzKeyVault
```

For each Key vault run the following command to determine which vaults are configured to use RBAC.

```
(Get-AzKeyVault -VaultName <Vault Name>).EnableRbacAuthorization
```

For each Key vault with the **EnableRbacAuthorization** setting set to **True**, run the following command.

```
Get-AzKeyVaultKey -VaultName <Vault Name> | Where-Object { $_.Enabled -and -not $_.Expires }
```

This returns keys where **Expires** setting is not configured with a value and the **Enabled** setting is set to **True**.

To audit at scale

Enumerates ALL Key Vaults in the CURRENT subscription context only. Does NOT cover the entire tenant unless you loop through all subscriptions. Uses ARM (**Get-AzResource**) to reliably read **enableRbacAuthorization**.

```
$vaults = Get-AzResource -ResourceType "Microsoft.KeyVault/vaults" | ForEach-Object {  
    $kv = Get-AzResource -ResourceId $_.ResourceId -ExpandProperties  
    if ($kv.Properties.enableRbacAuthorization -eq $true) {  
        [PSCustomObject]@{  
            VaultName = $kv.Name  
            ResourceGroup = $kv.ResourceGroupName  
            EnableRbacAuthorization = $kv.Properties.enableRbacAuthorization  
            SubscriptionId = $kv.SubscriptionId  
        }  
    }  
}
```

Retrieves ALL keys in EACH RBAC-enabled vault in the CURRENT subscription. This can be very heavy in large environments (no filtering at API level). Returns ONLY violations: enabled secrets WITHOUT expiration.


```
foreach ($vault in $vaults) {  
    Get-AzKeyVaultKey -VaultName $vault.VaultName |  
    Where-Object { $_.Enabled -eq $true -and $_.Expires -eq $null } |  
    Select-Object @{Name="VaultName";Expression={$vault.VaultName}}, Name,  
    Enabled, Expires
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [152b15f7-8e1f-4c1f-ab71-8c010ba5dbc0](#) - **Name:** 'Key Vault keys should have an expiration date'

Remediation:

Requisite RBAC Permissions for Keys - read section overview!

Remediation Role = **Key Vault Crypto Officer**

Remediate from Azure Portal

1. Go to **Key vaults**.
2. For each Key vault, under **Objects**, select **Keys**.
3. In the main pane, ensure that the status of the key(s) is **Enabled**.
4. Click on the Key without **Expiration date**.
5. Click the checkbox **Set expiration date** and set an appropriate **Expiration date** on the Key.
6. Click Apply.

Remediate from Azure CLI

Update the **Expiration date** for the key using the below command:

```
az keyvault key set-attributes --name <keyName> --vault-name <vaultName> --  
expires Y-m-d'T'H:M:S'Z'
```

Remediate from PowerShell

```
Set-AzKeyVaultKeyAttribute -VaultName <VaultName> -Name <KeyName> -Expires  
<DateTime>
```

Default Value:

By default, keys do not expire.

References:

1. <https://learn.microsoft.com/en-us/azure/key-vault/general/basic-concepts>
2. <https://learn.microsoft.com/en-us/azure/key-vault/general/about-keys-secrets-certificates>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-data-protection#dp-6-use-a-secure-key-management-process>
4. <https://learn.microsoft.com/en-us/powershell/module/az.keyvault/set-azkeyvaultkeyattribute>
5. <https://learn.microsoft.com/en-us/azure/key-vault/general/rbac-guide?tabs=azure-cli#azure-built-in-roles-for-key-vault-data-plane-operations>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.1 Establish and Maintain a Data Management Process Establish and maintain a data management process. In the process, address data sensitivity, data owner, handling of data, data retention limits, and disposal requirements, based on sensitivity and retention standards for the enterprise. Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v8	6.2 Establish an Access Revoking Process Establish and follow a process, preferably automated, for revoking access to enterprise assets, through disabling accounts immediately upon termination, rights revocation, or role change of a user. Disabling accounts, instead of deleting accounts, may be necessary to preserve audit trails.			
v7	16.7 Establish Process for Revoking Access Establish and follow an automated process for revoking system access by disabling accounts immediately upon termination or change of responsibilities of an employee or contractor. Disabling these accounts, instead of deleting accounts, allows preservation of audit trails.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1552	TA0006	M1047

8.3.2 Ensure that the Expiration Date is set for All Keys in Key Vaults using access policies (legacy) (Automated)

Profile Applicability:

- Level 1

Description:

Ensure that all Keys in Azure Key Vaults with access policies have an expiration date set.

Note: Using access policies is a legacy method. For improved security, use the Role-Based Access Control (RBAC) permission model instead of access policies when managing Azure Key Vault. RBAC restricts permission management to only the **Owner** and **User Access Administrator** roles, ensuring a clear separation between security and administrative tasks. With the Access Policy permission model, users with the **Contributor**, **Key Vault Contributor**, or any role that includes **Microsoft.KeyVault/vaults/write** permissions can grant themselves data plane access by configuring a Key Vault access policy. This can result in unauthorized access and management of your key vaults, keys, secrets, and certificates. To reduce this risk, limit **Contributor** role access to key vaults when using the Access Policy model.

Rationale:

Azure Key Vault enables users to store and use cryptographic keys within the Microsoft Azure environment. The **exp** (expiration date) attribute identifies the expiration date on or after which the key MUST NOT be used for a cryptographic operation. By default, keys never expire. It is thus recommended that keys be rotated in the key vault and set an explicit expiration date for all keys. This ensures that the keys cannot be used beyond their assigned lifetimes.

Impact:

Keys cannot be used beyond their assigned expiration dates respectively. Keys need to be rotated periodically wherever they are used.

Audit:

Requisite Access Policy Permissions for Keys - read section overview!

Audit Permissions = Keys: **List**, **Get**

Audit from Azure Portal

1. Go to **Key vaults**.
2. For each Key vault, under **Objects**, select **Keys**.
3. In the main pane, ensure that the status of the key is **Enabled**.
4. For each enabled key, ensure that an appropriate **Expiration date** is set.

Audit from Azure CLI

Get a list of all the key vaults in your Azure environment by running the following command:

```
az keyvault list
```

For each key vault, ensure that the output of the below command contains Key ID (kid), enabled status as **true** and Expiration date (expires) is not empty or null:

```
az keyvault key list --vault-name <KEYVAULTNAME> --query '[*].{"kid":kid,"enabled":attributes.enabled,"expires":attributes.expires}'
```

Audit from PowerShell

Retrieve a list of Azure Key vaults:

```
Get-AzKeyVault
```

For each Key vault, run the following command to determine which vaults are configured to not use RBAC:

```
Get-AzKeyVault -VaultName <Vault Name>
```

For each Key vault with the **EnableRbacAuthorization** setting set to **False** or empty, run the following command.

```
Get-AzKeyVaultKey -VaultName <Vault Name>
```

Make sure the **Expires** setting is configured with a value as appropriate wherever the **Enabled** setting is set to **True**.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [152b15f7-8e1f-4c1f-ab71-8c010ba5dbc0](#) - **Name:** 'Key Vault keys should have an expiration date'

Remediation:

Requisite Access Policy Permissions for Keys - read section overview!

Remediation Permissions = Keys: **List**, **Get**, **Update**

Remediate from Azure Portal

1. Go to **Key vaults**.
2. For each Key vault, under **Objects**, select **Keys**.
3. In the main pane, ensure that the status of the key(s) is **Enabled**.
4. Click on the secret(s) without Expiration date.

5. Click on the **current version** key
6. Click the checkbox **Set expiration date** and set an appropriate **Expiration date** on the key.
7. Click Apply.

Remediate from Azure CLI

Update the **Expiration date** for the key using the below command:

```
az keyvault key set-attributes --name <keyName> --vault-name <vaultName> --expires Y-m-d'T'H:M:S'Z'
```

Remediate from PowerShell

```
Set-AzKeyVaultKeyAttribute -VaultName <Vault Name> -Name <Key Name> -Expires <DateTime>
```

Default Value:

By default, keys do not expire.

References:

1. <https://learn.microsoft.com/en-us/azure/key-vault/general/basic-concepts>
2. <https://learn.microsoft.com/en-us/azure/key-vault/general/about-keys-secrets-certificates>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-data-protection#dp-6-use-a-secure-key-management-process>
4. <https://learn.microsoft.com/en-us/powershell/module/az.keyvault/set-azkeyvaultkeyattribute>
5. <https://learn.microsoft.com/en-us/azure/key-vault/general/assign-access-policy>
6. <https://learn.microsoft.com/en-us/rest/api/keyvault/#key-vault-operations>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.1 Establish and Maintain a Data Management Process Establish and maintain a data management process. In the process, address data sensitivity, data owner, handling of data, data retention limits, and disposal requirements, based on sensitivity and retention standards for the enterprise. Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v8	6.2 Establish an Access Revoking Process Establish and follow a process, preferably automated, for revoking access to enterprise assets, through disabling accounts immediately upon termination, rights revocation, or role change of a user. Disabling accounts, instead of deleting accounts, may be necessary to preserve audit trails.			

Controls Version	Control	IG 1	IG 2	IG 3
v7	16.7 <u>Establish Process for Revoking Access</u> Establish and follow an automated process for revoking system access by disabling accounts immediately upon termination or change of responsibilities of an employee or contractor . Disabling these accounts, instead of deleting accounts, allows preservation of audit trails.		●	●

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1552	TA0006	M1047

8.3.3 Ensure that the Expiration Date is set for All Secrets in Key Vaults using RBAC (Automated)

Profile Applicability:

- Level 1

Description:

Ensure that all Secrets in Azure Key Vaults using Role Based Access Control (RBAC) have an expiration date set.

Note: The audit commands require data-plane access to each Key Vault in addition to the management-plane **Reader** role. Without data-plane access these commands return empty results silently.

Rationale:

The Azure Key Vault enables users to store and keep secrets within the Microsoft Azure environment. Secrets in the Azure Key Vault are octet sequences with a maximum size of 25k bytes each. The **exp** (expiration date) attribute identifies the expiration date on or after which the secret MUST NOT be used. By default, secrets never expire. It is thus recommended to rotate secrets in the key vault and set an explicit expiration date for all secrets. This ensures that the secrets cannot be used beyond their assigned lifetimes.

Impact:

Secrets cannot be used beyond their assigned expiry date respectively. Secrets need to be rotated periodically wherever they are used.

Audit:

Requisite RBAC Permissions for Secrets - read section overview!

Audit Role = **Key Vault Secrets User**

Audit from Azure Portal

1. Go to **Key vaults**.
2. For each Key vault, under **Objects**, select **Secrets**.
3. In the main pane, ensure that the status of the secret is **Enabled**.
4. For each enabled secret, ensure that an appropriate **Expiration date** is set.

Audit from Azure CLI

Ensure that the output of the below command contains ID (id), enabled status as **true** and Expiration date (**expires**) is not **empty** or **null**:

```
az keyvault secret list --vault-name <KEYVAULTNAME> --query '[*].{"kid":kid,"enabled":attributes.enabled,"expires":attributes.expires}'
```

Audit from PowerShell

Retrieve a list of Key Vaults:

```
Get-AzKeyVault
```

For each Key vault, run the following command to determine which vaults are configured to use RBAC:

```
(Get-AzKeyVault -VaultName <Vault Name>).EnableRbacAuthorization
```

For each Key vault with the **EnableRbacAuthorization** setting set to **True**, run the following command:

```
Get-AzKeyVaultSecret -VaultName <Vault Name> | Where-Object { $_.Enabled -and -not $_.Expires }
```

This returns secrets where **Expires** setting is not configured with a value and the **Enabled** setting is set to **True**.

To audit at scale

Enumerates ALL Key Vaults in the CURRENT subscription context only. Does NOT cover the entire tenant unless you loop through all subscriptions. Uses ARM (**Get-AzResource**) to reliably read **enableRbacAuthorization**.

```
$vaults = Get-AzResource -ResourceType "Microsoft.KeyVault/vaults" | ForEach-Object {
    $kv = Get-AzResource -ResourceId $_.ResourceId -ExpandProperties
    if ($kv.Properties.enableRbacAuthorization -eq $true) {
        [PSCustomObject]@{
            VaultName = $kv.Name
            ResourceGroup = $kv.ResourceGroupName
            EnableRbacAuthorization = $kv.Properties.enableRbacAuthorization
            SubscriptionId = $kv.SubscriptionId
        }
    }
}
```

Retrieves ALL secrets in EACH RBAC-enabled vault in the CURRENT subscription. This can be very heavy in large environments (no filtering at API level). Returns ONLY violations: enabled secrets WITHOUT expiration.

```
foreach ($vault in $vaults) {
    Get-AzKeyVaultSecret -VaultName $vault.VaultName |
    Where-Object { $_.Enabled -eq $true -and $_.Expires -eq $null } |
    Select-Object @{Name="VaultName";Expression={$vault.VaultName}}, Name,
    Enabled, Expires
}
```


Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [98728c90-32c7-4049-8429-847dc0f4fe37](#) - **Name:** 'Key Vault secrets should have an expiration date'

Remediation:

Requisite RBAC Permissions for Secrets - read section overview!

Remediation Role = **Key Vault Secrets Officer**

Remediate from Azure Portal

1. Go to **Key vaults**.
2. For each Key vault, under **Objects**, select **Secrets**.
3. In the main pane, ensure that the status of the secret(s) is **Enabled**.
4. Click on the secret(s) without **Expiration date**.
5. Click on the **current version** secret
6. Click the checkbox **Set expiration date** and set an appropriate **Expiration date** on the secret.
7. Click Apply.

Remediate from Azure CLI

Update the Expiration date for the secret using the below command:

```
az keyvault secret set-attributes --name <secret_name> --vault-name  
<vault_name> --expires Y-m-d'T'H:M:S'Z'
```

Remediate from PowerShell

```
Set-AzKeyVaultSecretAttribute -VaultName <vault_name> -Name <secret_name> -  
Expires <date_time>
```

Default Value:

By default, secrets do not expire.

References:

1. <https://learn.microsoft.com/en-us/azure/key-vault/general/basic-concepts>
2. <https://learn.microsoft.com/en-us/azure/key-vault/general/about-keys-secrets-certificates>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-data-protection#dp-6-use-a-secure-key-management-process>
4. <https://learn.microsoft.com/en-us/powershell/module/az.keyvault/set-azkeyvaultsecretattribute>
5. <https://learn.microsoft.com/en-us/azure/key-vault/general/rbac-guide?tabs=azure-cli#azure-built-in-roles-for-key-vault-data-plane-operations>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.1 <u>Establish and Maintain a Data Management Process</u> Establish and maintain a data management process. In the process, address data sensitivity, data owner, handling of data, data retention limits, and disposal requirements, based on sensitivity and retention standards for the enterprise. Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v8	6.2 <u>Establish an Access Revoking Process</u> Establish and follow a process, preferably automated, for revoking access to enterprise assets, through disabling accounts immediately upon termination, rights revocation, or role change of a user. Disabling accounts, instead of deleting accounts, may be necessary to preserve audit trails.			
v7	16.7 <u>Establish Process for Revoking Access</u> Establish and follow an automated process for revoking system access by disabling accounts immediately upon termination or change of responsibilities of an employee or contractor . Disabling these accounts, instead of deleting accounts, allows preservation of audit trails.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1552	TA0006	M1047

8.3.4 Ensure that the Expiration Date is set for All Secrets in Key Vaults using access policies (legacy) (Automated)

Profile Applicability:

- Level 1

Description:

Ensure that all Secrets in Azure Key Vaults with access policies have an expiration date set.

Note: Using access policies is a legacy method. For improved security, use the Role-Based Access Control (RBAC) permission model instead of access policies when managing Azure Key Vault. RBAC restricts permission management to only the **Owner** and **User Access Administrator** roles, ensuring a clear separation between security and administrative tasks. With the Access Policy permission model, users with the **Contributor**, **Key Vault Contributor**, or any role that includes **Microsoft.KeyVault/vaults/write** permissions can grant themselves data plane access by configuring a Key Vault access policy. This can result in unauthorized access and management of your key vaults, keys, secrets, and certificates. To reduce this risk, limit **Contributor** role access to key vaults when using the Access Policy model.

Rationale:

The Azure Key Vault enables users to store and keep secrets within the Microsoft Azure environment. Secrets in the Azure Key Vault are octet sequences with a maximum size of 25k bytes each. The **exp** (expiration date) attribute identifies the expiration date on or after which the secret MUST NOT be used. By default, secrets never expire. It is thus recommended to rotate secrets in the key vault and set an explicit expiration date for all secrets. This ensures that the secrets cannot be used beyond their assigned lifetimes.

Impact:

Secrets cannot be used beyond their assigned expiry date respectively. Secrets need to be rotated periodically wherever they are used.

Audit:

Requisite Access Policy Permissions for Secrets - read section overview!

Audit Permissions = Secrets: **List**, **Get**

Audit from Azure Portal

1. Go to **Key vaults**.
2. For each Key vault, under **Objects**, select **Secrets**.
3. In the main pane, ensure that the status of the secret(s) is **Enabled**.
4. Ensure the **Expiration date** column is not empty.

Audit from Azure CLI

Get a list of all the key vaults in your Azure environment by running the following command:

```
az keyvault list
```

For each key vault, ensure that the output of the below command contains ID (id), enabled status as **true** and Expiration date (expires) is not empty or null:

```
az keyvault secret list --vault-name <KEYVALUTNAME> --query '[*].{"kid":kid,"enabled":attributes.enabled,"expires":attributes.expires}'
```

Audit from PowerShell

Retrieve a list of Key vaults:

```
Get-AzKeyVault
```

For each Key vault run the following command to determine which vaults are configured to use RBAC:

```
Get-AzKeyVault -VaultName <Vault Name>
```

For each Key Vault with the **EnableRbacAuthorization** setting set to **False** or empty, run the following command.

```
Get-AzKeyVaultSecret -VaultName <Vault Name>
```

Make sure the **Expires** setting is configured with a value as appropriate wherever the **Enabled** setting is set to **True**.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [98728c90-32c7-4049-8429-847dc0f4fe37](#) - **Name:** 'Key Vault secrets should have an expiration date'

Remediation:

Requisite Access Policy Permissions for Secrets - read section overview!

Remediation Permissions = Secrets: **List**, **Get**, **Set**

Remediate from Azure Portal

1. Go to **Key vaults**.
2. For each Key vault, click on **Secrets**.
3. In the main pane, ensure that the status of the secret(s) is **Enabled**.
4. Click on the secret(s) without Expiration date.

5. Click on the **current version** secret
6. Click the checkbox **Set expiration date** and set an appropriate **Expiration date** on the secret.
7. Click Apply.

Remediate from Azure CLI

Update the **Expiration date** for the secret using the below command:

```
az keyvault secret set-attributes --name <secret_name> --vault-name <vault_name> --expires Y-m-d'T'H:M:S'Z'
```

Remediate from PowerShell

For each Key vault with the **EnableRbacAuthorization** setting set to **False** or empty, run the following command.

```
Set-AzKeyVaultSecret -VaultName <vault_name> -Name <secret_name> -Expires <date_time>
```

Default Value:

By default, secrets do not expire.

References:

1. <https://learn.microsoft.com/en-us/azure/key-vault/general/basic-concepts>
2. <https://learn.microsoft.com/en-us/azure/key-vault/general/about-keys-secrets-certificates>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-data-protection#dp-6-use-a-secure-key-management-process>
4. <https://learn.microsoft.com/en-us/powershell/module/az.keyvault/set-azkeyvaultsecret>
5. <https://learn.microsoft.com/en-us/azure/key-vault/general/assign-access-policy>
6. <https://learn.microsoft.com/en-us/rest/api/keyvault/#key-vault-operations>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.1 Establish and Maintain a Data Management Process Establish and maintain a data management process. In the process, address data sensitivity, data owner, handling of data, data retention limits, and disposal requirements, based on sensitivity and retention standards for the enterprise. Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			

Controls Version	Control	IG 1	IG 2	IG 3
v8	6.2 Establish an Access Revoking Process Establish and follow a process, preferably automated, for revoking access to enterprise assets, through disabling accounts immediately upon termination, rights revocation, or role change of a user. Disabling accounts, instead of deleting accounts, may be necessary to preserve audit trails.			
v7	16.7 Establish Process for Revoking Access Establish and follow an automated process for revoking system access by disabling accounts immediately upon termination or change of responsibilities of an employee or contractor . Disabling these accounts, instead of deleting accounts, allows preservation of audit trails.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1552	TA0006	M1047

8.3.5 Ensure 'Purge protection' is Set to 'Enabled' (Automated)

Profile Applicability:

- Level 1

Description:

Key vaults contain object keys, secrets, and certificates. Deletion of a key vault can cause immediate data loss or loss of security functions (authentication, validation, verification, non-repudiation, etc.) supported by the key vault objects.

It is recommended the key vault be made recoverable by enabling the "purge protection" function. This is to prevent the loss of encrypted data, including storage accounts, SQL databases, and/or dependent services provided by key vault objects (keys, secrets, certificates, etc.).

NOTE: In February 2025, Microsoft enabled soft delete protection on all key vaults. Users can no longer opt out of or turn off soft delete.

WARNING: A current limitation is that role assignments disappear when a key vault is deleted. All role assignments will need to be recreated after recovery.

Rationale:

Users may accidentally run delete/purge commands on a key vault, or an attacker or malicious user may do so deliberately in order to cause disruption. Deleting or purging a key vault leads to immediate data loss, as keys encrypting data and secrets/certificates allowing access/services will become inaccessible.

Enabling purge protection ensures that even if a key vault is deleted, the key vault and its objects remain recoverable during the configurable retention period. If no action is taken, the key vault and its objects will be purged once the retention period elapses.

Impact:

Once purge protection is enabled for a key vault, it cannot be disabled.

Audit:

Requisite Permissions - read section overview!

Data-plane access not required. Can be audited and remediated with Azure Resource Manager permissions.

Audit from Azure Portal

1. Go to **Key Vaults**.
2. Click the name of a key vault.
3. Under **Settings**, click **Properties**.

4. Next to **Purge protection**, ensure that **Enable purge protection (enforce a mandatory retention period for deleted vaults and vault objects)** is selected.
5. Repeat steps 1-4 for each key vault.

Audit from Azure CLI

Run the following command to list key vaults:

```
az resource list --query "[?type=='Microsoft.KeyVault/vaults']"
```

For each key vault, run the following command to get the purge protection setting:

```
az resource show --resource-group <resource-group> --name <key-vault> --resource-type "Microsoft.KeyVault/vaults" --query properties.enablePurgeProtection
```

Ensure that **true** is returned.

Audit from PowerShell

Run the following command to list key vaults:

```
Get-AzKeyVault
```

For each key vault, run the following command to get the key vault details:

```
Get-AzKeyVault -ResourceGroupName <resource-group> -VaultName <key-vault>
```

Ensure **Purge Protection Enabled?** is set to **True**.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [0b60c0b2-2dc2-4e1c-b5c9-abbed971de53](#) - **Name:** 'Key vaults should have deletion protection enabled'

Remediation:

Requisite Permissions - read section overview!

Data-plane access not required. Can be audited and remediated with Azure Resource Manager permissions.

Note: Once enabled, purge protection cannot be disabled.

Remediate from Azure Portal

1. Go to **Key Vaults**.
2. Click the name of a key vault.

3. Under **Settings**, click **Properties**.
4. Select the radio button next to **Enable purge protection (enforce a mandatory retention period for deleted vaults and vault objects)**.
5. Click **Save**.
6. Repeat steps 1-5 for each key vault requiring remediation.

Remediate from Azure CLI

For each key vault requiring remediation, run the following command to enable purge protection:

```
az resource update --resource-group <resource-group> --name <key-vault> --resource-type "Microsoft.KeyVault/vaults" --set properties.enablePurgeProtection=true
```

Remediate from PowerShell

For each key vault requiring remediation, run the following command to enable purge protection:

```
Update-AzKeyVault -ResourceGroupName <resource-group> -VaultName <key-vault> -EnablePurgeProtection
```

Default Value:

Purge protection is disabled by default.

References:

1. <https://learn.microsoft.com/en-us/azure/key-vault/general/key-vault-recovery>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-governance-strategy#gs-8-define-and-implement-backup-and-recovery-strategy>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-data-protection#dp-8-ensure-security-of-key-and-certificate-repository>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	11.1 Establish and Maintain a Data Recovery Process Establish and maintain a data recovery process. In the process, address the scope of data recovery activities, recovery prioritization, and the security of backup data. Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	10.2 Perform Complete System Backups Ensure that each of the organization's key systems are backed up as a complete system, through processes such as imaging, to enable the quick recovery of an entire system.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1485	TA0040	M1053

8.3.6 Ensure that Role Based Access Control for Azure Key Vault is Enabled (Automated)

Profile Applicability:

- Level 2

Description:

The recommended way to access Key Vaults is to use the Azure Role-Based Access Control (RBAC) permissions model.

Azure RBAC is an authorization system built on Azure Resource Manager that provides fine-grained access management of Azure resources. It allows users to manage Key, Secret, and Certificate permissions. It provides one place to manage all permissions across all key vaults.

Rationale:

The new RBAC permissions model for Key Vaults enables a much finer grained access control for key vault secrets, keys, certificates, etc., than the vault access policy. This in turn will permit the use of privileged identity management over these roles, thus securing the key vaults with JIT Access management.

Impact:

Implementation needs to be properly designed from the ground up, as this is a fundamental change to the way key vaults are accessed/managed. Changing permissions to key vaults will result in loss of service as permissions are re-applied. For the least amount of downtime, map your current groups and users to their corresponding permission needs.

Audit:

Requisite Permissions - read section overview!

Data-plane access not required. Can be audited and remediated with Azure Resource Manager permissions.

Audit from Azure Portal

1. From Azure Home open the Portal Menu in the top left corner.
2. Select **Key Vaults**.
3. Select a Key Vault to audit.
4. Under **Settings** select **Access configuration**.
5. Ensure the Permission Model radio button is set to **Azure role-based access control**.

Audit from Azure CLI

Run the following command for each Key Vault in each Resource Group:

```
az keyvault show --resource-group <resource_group> --name <vault_name>
```

Ensure the **enableRbacAuthorization** setting is set to **true** within the output of the above command.

Audit from PowerShell

Run the following PowerShell command:

```
Get-AzKeyVault -Vaultname <vault_name> -ResourceGroupName <resource_group>
```

Ensure the **Enabled For RBAC Authorization** setting is set to **True**

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~Definitions

- **Policy ID:** [12d4fa5e-1f9f-4c21-97a9-b99b3c6611b5](#) - **Name:** 'Azure Key Vault should use RBAC permission model'

Remediation:

Requisite Permissions - read section overview!

Data-plane access not required. Can be audited and remediated with Azure Resource Manager permissions.

Remediate from Azure Portal

Key Vaults can be configured to use **Azure role-based access control** on creation.

For existing Key Vaults:

1. From Azure Home open the Portal Menu in the top left corner
2. Select **Key Vaults**
3. Select a Key Vault to audit
4. Under **Settings** select **Access configuration**
5. Set the Permission model radio button to **Azure role-based access control**, taking note of the warning message
6. Click **Save**
7. Select **Access Control (IAM)**
8. Select the **Role Assignments** tab
9. Reapply permissions as needed to groups or users

Remediate from Azure CLI

To enable RBAC Authorization for each Key Vault, run the following Azure CLI command:

```
az keyvault update --resource-group <resource_group> --name <vault_name> --enable-rbac-authorization true
```

Remediate from PowerShell

To enable RBAC authorization on each Key Vault, run the following PowerShell command:

```
Update-AzKeyVault -ResourceGroupName <resource_group> -VaultName <vault_name> -EnableRbacAuthorization $True
```

Default Value:

The default value for Access control in Key Vaults is Vault Policy.

References:

1. <https://learn.microsoft.com/en-us/azure/key-vault/general/rbac-migration#vault-access-policy-to-azure-rbac-migration-steps>
2. <https://learn.microsoft.com/en-us/azure/role-based-access-control/role-assignments-portal>
3. <https://learn.microsoft.com/en-us/azure/role-based-access-control/overview>
4. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-data-protection#dp-8-ensure-security-of-key-and-certificate-repository>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.	●	●	●
v8	6.8 <u>Define and Maintain Role-Based Access Control</u> Define and maintain role-based access control, through determining and documenting the access rights necessary for each role within the enterprise to successfully carry out its assigned duties. Perform access control reviews of enterprise assets to validate that all privileges are authorized, on a recurring schedule at a minimum annually, or more frequently.			●
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.	●	●	●

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1552	TA0006	M1047

8.3.7 Ensure Public Network Access is Disabled (Automated)

Profile Applicability:

- Level 1

Description:

Disable public network access to prevent exposure to the internet and reduce the risk of unauthorized access. Use private endpoints to securely manage access within trusted networks.

When a private endpoint is configured on a key vault, connections from Azure resources within the same subnet will use its private IP address. However, network traffic from the public internet can still connect to the key vault's public endpoint (mykeyvault.vault.azure.net) using its public IP address unless public network access is disabled.

Disabling public network access removes the vault's public endpoint from Azure public DNS, reducing its exposure to the public internet. With a private endpoint configured, network traffic will use the vault's private endpoint IP address for all requests (mykeyvault.vault.privatelink.azure.net).

Rationale:

Disabling public network access improves security by ensuring that a service is not exposed on the public internet.

Removing a point of interconnection from the internet edge to your key vault can strengthen the network security boundary of your system and reduce the risk of exposing the control plane or vault objects to untrusted clients.

Although Azure resources are never truly isolated from the public internet, disabling the public endpoint removes a line of sight from the public internet and increases the effort required for an attack.

Impact:

NOTE: Prior to disabling public network access, it is strongly recommended that, for each key vault, either:

- virtual network integration is completed

OR

- private endpoints/links are set up as described in "**Ensure Private Endpoints are used to access Azure Key Vault.**"

Disabling public network access restricts access to the service. This enhances security but will require the configuration of a virtual network and/or private endpoints for any services or users needing access within trusted networks.

Audit:

Requisite Permissions - read section overview!

Data-plane access not required. Can be audited and remediated with Azure Resource Manager permissions.

Audit from Azure Portal

1. Go to **Key vaults**.
2. Click the name of a key vault.
3. Under **Settings**, click **Networking**.
4. Under **Firewalls and virtual networks**, ensure that **Allow access from:** is set to **Disable public access**.
5. Repeat steps 1-4 for each key vault.

Audit from Azure CLI

Run the following command to list key vaults:

```
az keyvault list
```

For each key vault, run the following command to get the public network access setting:

```
az keyvault show --resource-group <resource-group> --name <key-vault> --query properties.publicNetworkAccess
```

Ensure that "Disabled" is returned.

Audit from PowerShell

Run the following command to list key vaults:

```
Get-AzKeyVault
```

Run the following command to get the key vault in a resource group with a given name:

```
$vault = Get-AzKeyVault -ResourceGroupName <resource-group> -Name <key-vault>
```

Run the following command to get the public network access setting for the key vault:

```
$vault.PublicNetworkAccess
```

Ensure that **Disabled** is returned.

Repeat for each key vault.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [405c5871-3e91-4644-8a63-58e19d68ff5b](#) - **Name:** 'Azure Key Vault should disable public network access'

Remediation:

Requisite Permissions - read section overview!

Data-plane access not required. Can be audited and remediated with Azure Resource Manager permissions.

Remediate from Azure Portal

1. Go to **Key vaults**.
2. Click the name of a key vault.
3. Under **Settings**, click **Networking**.
4. Under **Firewalls and virtual networks**, next to **Allow access from:**, click the radio button next to **Disable public access**.
5. Click **Apply**.
6. Repeat steps 1-5 for each key vault requiring remediation.

Remediate from Azure CLI

For each key vault requiring remediation, run the following command to disable public network access:

```
az keyvault update --resource-group <resource-group> --name <key-vault> --public-network-access Disabled
```

Remediate from PowerShell

For each key vault requiring remediation, run the following command to disable public network access:

```
Update-AzKeyVault -ResourceGroupName <resource-group> -VaultName <vault-name> -PublicNetworkAccess "Disabled"
```

Default Value:

Public network access is enabled by default.

References:

1. <https://learn.microsoft.com/en-us/azure/key-vault/general/network-security>
2. <https://learn.microsoft.com/en-us/azure/key-vault/general/private-link-service>

Additional Information:

This recommendation is based on the Common Reference Recommendation **Ensure public network access is Disabled**, from the **Common Reference Recommendations > Networking > Virtual Networks (VNETs)** section.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1530	TA0009	M1037

8.3.8 Ensure Private Endpoints are Used to Access Azure Key Vault (Automated)

Profile Applicability:

- Level 2

Description:

Use private endpoints to allow clients and services to securely access data located over a network via an encrypted Private Link. To do this, the private endpoint uses an IP address from the VNet for each service. Network traffic between disparate services securely traverses encrypted over the VNet. This VNet can also link addressing space, extending your network and accessing resources on it. Similarly, it can be a tunnel through public networks to connect remote infrastructures together. This creates further security through segmenting network traffic and preventing outside sources from accessing it.

Private endpoints will secure network traffic from Azure Key Vault to the resources requesting secrets and keys.

Rationale:

Securing traffic between services through encryption protects the data from easy interception and reading.

Private endpoints will keep network requests to Azure Key Vault limited to the endpoints attached to the resources that are whitelisted to communicate with each other. Assigning the Key Vault to a network without an endpoint will allow other resources on that network to view all traffic from the Key Vault to its destination. In spite of the complexity in configuration, this is recommended for high security secrets.

Impact:

If an Azure Virtual Network is not implemented correctly, this may result in the loss of critical network traffic.

Private endpoints are charged per hour of use. Refer to <https://azure.microsoft.com/en-us/pricing/details/private-link/> and <https://azure.microsoft.com/en-us/pricing/calculator/> to estimate potential costs.

Audit:

Requisite Permissions - read section overview!

Data-plane access not required. Can be audited and remediated with Azure Resource Manager permissions.

Audit from Azure Portal

1. From Azure Home open the Portal Menu in the top left.
2. Select **Key Vaults**.
3. Select a Key Vault to audit.
4. Select **Networking** in the left column.
5. Select **Private endpoint connections** from the top row.
6. View if there is an endpoint attached.

Audit from Azure CLI

Run the following command within a subscription for each Key Vault you wish to audit.

```
az keyvault show --name <keyVaultName>
```

Ensure that **privateEndpointConnections** is not **null**.

Audit from PowerShell

Run the following command within a subscription for each Key Vault you wish to audit.

```
Get-AzPrivateEndpointConnection -PrivateLinkResourceId  
'/subscriptions/<subscriptionNumber>/resourceGroups/<resourceGroup>/providers  
/Microsoft.KeyVault/vaults/<keyVaultName>/'
```

Ensure that the response contains details of a private endpoint.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~Definitions

- **Policy ID:** [a6abeaec-4d90-4a02-805f-6b26c4d3fbe9](#) - **Name:** 'Azure Key Vaults should use private link'

Remediation:

Requisite Permissions - read section overview!

Data-plane access not required. Can be audited and remediated with Azure Resource Manager permissions.

Please see the additional information about the requirements needed before starting this remediation procedure.

Remediate from Azure Portal

1. From Azure Home open the Portal Menu in the top left.
2. Select **Key Vaults**.
3. Select a Key Vault to audit.

4. Select **Networking** in the left column.
5. Select **Private endpoint connections** from the top row.
6. Select **+ Create**.
7. Select the subscription the Key Vault is within, and other desired configuration.
8. Select **Next**.
9. For resource type select **Microsoft.KeyVault/vaults**.
10. Select the Key Vault to associate the Private Endpoint with.
11. Select **Next**.
12. In the **Virtual Networking** field, select the network to assign the Endpoint.
13. Select other configuration options as desired, including an existing or new application security group.
14. Select **Next**.
15. Select the private DNS the Private Endpoints will use.
16. Select **Next**.
17. Optionally add **Tags**.
18. Select **Next : Review + Create**.
19. Review the information and select **Create**. Follow the Audit Procedure to determine if it has successfully applied.
20. Repeat steps 3-19 for each Key Vault.

Remediate from Azure CLI

1. To create an endpoint, run the following command:

```
az network private-endpoint create --resource-group <resourceGroup> --vnet-name <vnetName> --subnet <subnetName> --name <PrivateEndpointName> --private-connection-resource-id "/subscriptions/<AZURE SUBSCRIPTION ID>/resourceGroups/<resourceGroup>/providers/Microsoft.KeyVault/vaults/<keyVaultName>" --group-ids vault --connection-name <privateLinkConnectionName> --location <azureRegion> --manual-request
```

2. To manually approve the endpoint request, run the following command:

```
az keyvault private-endpoint-connection approve --resource-group <resourceGroup> --vault-name <keyVaultName> -name <privateLinkName>
```

3. Determine the Private Endpoint's IP address to connect the Key Vault to the Private DNS you have previously created:
4. Look for the property `networkInterfaces` then `id`; the value must be placed in the variable `<privateEndpointNIC>` within step 7.

```
az network private-endpoint show -g <resourceGroupName> -n <privateEndpointName>
```

5. Look for the property `networkInterfaces` then `id`; the value must be placed on `<privateEndpointNIC>` in step 7.

```
az network nic show --ids <privateEndpointName>
```

6. Create a Private DNS record within the DNS Zone you created for the Private Endpoint:

```
az network private-dns record-set a add-record -g <resourceGroupName> -z "privatelink.vaultcore.azure.net" -n <keyVaultName> -a <privateEndpointNIC>
```

7. nslookup the private endpoint to determine if the DNS record is correct:

```
nslookup <keyVaultName>.vault.azure.net  
nslookup <keyVaultName>.privatelink.vaultcore.azure.net
```

Default Value:

By default, Private Endpoints are not created for services.

References:

1. <https://learn.microsoft.com/en-us/azure/private-link/private-endpoint-overview>
2. <https://learn.microsoft.com/en-us/azure/storage/common/storage-private-endpoints>
3. <https://azure.microsoft.com/en-us/pricing/details/private-link/>
4. <https://learn.microsoft.com/en-us/azure/key-vault/general/private-link-service>
5. <https://learn.microsoft.com/en-us/azure/virtual-network/quick-create-portal>
6. <https://learn.microsoft.com/en-us/azure/private-link/tutorial-private-endpoint-storage-portal>
7. <https://learn.microsoft.com/en-us/azure/bastion/bastion-overview>
8. <https://learn.microsoft.com/en-us/azure/dns/private-dns-getstarted-cli#create-an-additional-dns-record>
9. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-data-protection#dp-8-ensure-security-of-key-and-certificate-repository>

Additional Information:

This recommendation assumes that you have created a Resource Group containing a Virtual Network that the services are already associated with and configured private DNS. A Bastion on the virtual network is also required, and the service to which you are connecting must already have a Private Endpoint. For information concerning the installation of these services, please see the attached documentation.

Microsoft's own documentation lists the requirements as: A Key Vault. An Azure virtual network. A subnet in the virtual network. Owner or contributor permissions for both the Key Vault and the virtual network.

This recommendation is based on the Common Reference Recommendation **Ensure Private Endpoints are used to access {service}**, from the **Common Reference Recommendations > Networking > Private Endpoints** section.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	12.2 <u>Establish and Maintain a Secure Network Architecture</u> Establish and maintain a secure network architecture. A secure network architecture must address segmentation, least privilege, and availability, at a minimum.			
v7	14.1 <u>Segment the Network Based on Sensitivity</u> Segment the network based on the label or classification level of the information stored on the servers, locate all sensitive information on separated Virtual Local Area Networks (VLANs).			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1537	TA0010	M1037

8.3.9 Ensure Automatic Key Rotation is Enabled within Azure Key Vault (Automated)

Profile Applicability:

- Level 2

Description:

Automated cryptographic key rotation in Key Vault allows users to configure Key Vault to automatically generate a new key version at a specified frequency. A key rotation policy can be defined for each individual key.

Rationale:

Automatic key rotation reduces risk by ensuring that keys are rotated without manual intervention.

Azure and NIST recommend that keys be rotated every two years or less. Refer to 'Table 1: Suggested cryptoperiods for key types' on page 46 of the following document for more information:

<https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-57pt1r5.pdf>.

Impact:

There is an additional cost for each scheduled key rotation.

Audit:

Requisite Permissions - read section overview!

RBAC Role = **Key Vault Crypto Officer**

Access policy permission = Rotation Policy Operations: **Get Rotation Policy**

Audit from Azure Portal

1. Go to **Key Vaults**.
2. Select a Key Vault.
3. Under **Objects**, select **Keys**.
4. Select a key.
5. From the top row, select **Rotation policy**.
6. Ensure **Enable auto rotation** is set to **Enabled**.
7. Ensure the **Rotation time** is set to an appropriate value.
8. Repeat steps 1-7 for each Key Vault and Key.

Audit from Azure CLI

Run the following command:

```
az keyvault key rotation-policy show --vault-name <vault-name> --name <key-name>
```

Ensure that the response contains a **lifetimeAction** of **Rotate** and that **timeAfterCreate** is set to an appropriate value.

Audit from PowerShell

Run the following command:

```
Get-AzKeyVaultKeyRotationPolicy -VaultName <vault-name> -Name <key-name>
```

Ensure that the response contains a **LifetimeAction** of **Rotate** and that **TimeAfterCreate** is set to an appropriate value.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [d8cf8476-a2ec-4916-896e-992351803c44](#) - **Name:** 'Keys should have a rotation policy ensuring that their rotation is scheduled within the specified number of days after creation.'

Remediation:

Requisite Permissions - read section overview!

RBAC Role = **Key Vault Crypto Officer**

Access policy permission = Rotation Policy Operations: **Set Rotation Policy**

Note: Azure CLI and PowerShell use the ISO8601 duration format for time spans. The format is **P<timespanInISO8601Format>(Y,M,D)**. The leading P is required and is referred to as **period**. The **(Y,M,D)** are for the duration of Year, Month, and Day, respectively. A time frame of 2 years, 2 months, 2 days would be **P2Y2M2D**. For Azure CLI and PowerShell, it is easiest to supply the policy flags in a **.json file**, for example:

```
{
  "lifetimeActions": [
    {
      "trigger": {
        "timeAfterCreate": "P<timespanInISO8601Format>(Y,M,D)",
        "timeBeforeExpiry" : null
      },
      "action": {
        "type": "Rotate"
      }
    },
    {
      "trigger": {
        "timeBeforeExpiry" : "P<timespanInISO8601Format>(Y,M,D)"
      },
      "action": {
        "type": "Notify"
      }
    }
  ],
  "attributes": {
    "expiryTime": "P<timespanInISO8601Format>(Y,M,D)"
  }
}
```

Remediate from Azure Portal

1. Go to **Key Vaults**.
2. Select a Key Vault.
3. Under **Objects**, select **Keys**.
4. Select a key.
5. From the top row, select **Rotation policy**.
6. Select an appropriate **Expiry time**.
7. Set **Enable auto rotation** to **Enabled**.
8. Set an appropriate **Rotation option** and **Rotation time**.
9. Optionally, set a **Notification time**.
10. Click **Save**.
11. Repeat steps 1-10 for each Key Vault and Key.

Remediate from Azure CLI

Run the following command for each key to enable automatic rotation:

```
az keyvault key rotation-policy update --vault-name <vault-name> --name <key-name> --value <path/to/policy.json>
```

Remediate from PowerShell

Run the following command for each key to enable automatic rotation:

```
Set-AzKeyVaultKeyRotationPolicy -VaultName <vault-name> -Name <key-name> -  
PolicyPath <path/to/policy.json>
```

Default Value:

By default, automatic key rotation is not enabled.

References:

1. <https://learn.microsoft.com/en-us/azure/key-vault/keys/how-to-configure-key-rotation>
2. <https://learn.microsoft.com/en-us/azure/storage/common/customer-managed-keys-overview#update-the-key-version>
3. <https://learn.microsoft.com/en-us/azure/virtual-machines/windows/disks-enable-customer-managed-keys-powershell#set-up-an-azure-key-vault-and-diskencryptionset-optional-with-automatic-key-rotation>
4. <https://azure.microsoft.com/en-us/updates?id=public-preview-automatic-key-rotation-of-customermanaged-keys-for-encrypting-azure-managed-disks>
5. <https://learn.microsoft.com/en-us/cli/azure/keyvault/key/rotation-policy>
6. <https://learn.microsoft.com/en-us/powershell/module/az.keyvault/set-azkeyvaultkeyrotationpolicy>
7. <https://learn.microsoft.com/en-us/kusto/query/scalar-data-types/timespan>
8. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-data-protection#dp-6-use-a-secure-key-management-process>
9. <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-57pt1r5.pdf>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	6.2 Establish an Access Revoking Process Establish and follow a process, preferably automated, for revoking access to enterprise assets, through disabling accounts immediately upon termination, rights revocation, or role change of a user. Disabling accounts, instead of deleting accounts, may be necessary to preserve audit trails.			
v7	16.7 Establish Process for Revoking Access Establish and follow an automated process for revoking system access by disabling accounts immediately upon termination or change of responsibilities of an employee or contractor. Disabling these accounts, instead of deleting accounts, allows preservation of audit trails.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1552	TA0006	M1041

8.3.10 Ensure that Azure Key Vault Managed HSM is Used when Required (Manual)

Profile Applicability:

- Level 2

Description:

Azure Key Vault Managed HSM is a fully managed, highly available, single-tenant cloud service that safeguards cryptographic keys using FIPS 140-2 Level 3 validated HSMs.

Note: While an automated assessment procedure exists for this recommendation, the assessment status remains manual, as this recommendation to use Managed HSM applies only to scenarios where specific regulatory and compliance requirements mandate the use of a dedicated hardware security module.

Rationale:

Managed HSM is a fully managed, highly available, single-tenant service that ensures FIPS 140-2 Level 3 compliance. It provides centralized key management, isolated access control, and private endpoints for secure access. Integrated with Azure services, it supports migration from Key Vault, ensures data residency, and offers monitoring and auditing for enhanced security.

Impact:

Managed HSM incurs a cost per month for each actively used HSM-protected key, depending on the key type and quantity. Each key version is billed separately. Additionally, there is an hourly usage fee per Managed HSM pool. Refer to <https://azure.microsoft.com/en-us/pricing/details/key-vault/> to estimate potential costs.

Audit:

Requisite Permissions - read section overview!

Data-plane access not required. Can be audited and remediated with Azure Resource Manager permissions.

Audit from Azure CLI

Run the following command to list key vaults:

```
az keyvault list --query [*].[name,type]
```

Ensure that at least one key vault with type **Microsoft.KeyVault/managedHSMs** exists.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [587c79fe-dd04-4a5e-9d0b-f89598c7261b](#) - **Name:** 'Keys should be backed by a hardware security module (HSM)'

Remediation:

Requisite Permissions - read section overview!

Data-plane access not required. Can be audited and remediated with Azure Resource Manager permissions.

Remediate from Azure CLI

Run the following command to set **oid** to be the **OID** of the signed-in user:

```
$oid = az ad signed-in-user show --query id -o tsv
```

Alternatively, prepare a space-separated list of OIDs to be provided as the **administrators** of the HSM.

Run the following command to create a Managed HSM:

```
az keyvault create --resource-group <resource-group> --hsm-name <hsm-name> --retention-days <retention-days> --administrators $oid
```

The command can take several minutes to complete.

After the HSM has been created, it must be activated before it can be used. Activation requires providing a minimum of three and a maximum of ten RSA key pairs, as well as the minimum number of keys required to decrypt the security domain (called a quorum).

OpenSSL can be used to generate the self-signed certificates, for example:

```
openssl req -newkey rsa:2048 -nodes -keyout cert_1.key -x509 -days 365 -out cert_1.cer
```

Run the following command to download the security domain and activate the Managed HSM:

```
az keyvault security-domain download --hsm-name <managed-hsm> --sd-wrapping-keys <key-1> <key-2> <key-3> --sd-quorum <quorum> --security-domain-file <managed-hsm-security-domain>.json
```

Store the security domain file and the RSA key pairs securely. They will be required for disaster recovery or for creating another Managed HSM that shares the same security domain so that the two can share keys.

The Managed HSM will now be in an active state and ready for use.

References:

1. <https://learn.microsoft.com/en-us/azure/security/fundamentals/key-management-choose>
2. <https://learn.microsoft.com/en-us/azure/key-vault/managed-hsm/overview>
3. <https://azure.microsoft.com/en-us/pricing/details/key-vault/>
4. <https://learn.microsoft.com/en-us/azure/key-vault/managed-hsm/quick-create-cli>
5. <https://learn.microsoft.com/en-us/cli/azure/keyvault>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.12 Segment Data Processing and Storage Based on Sensitivity Segment data processing and storage based on the sensitivity of the data. Do not process sensitive data on enterprise assets intended for lower sensitivity data.			
v7	2.10 Physically or Logically Segregate High Risk Applications Physically or logically segregated systems should be used to isolate and run software that is required for business operations but incur higher risk for the organization.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1552	TA0006	M1041

8.3.11 Ensure Certificate 'Validity Period (in months)' is Less Than or Equal to '12' (Automated)

Profile Applicability:

- Level 1

Description:

Restrict the validity period of certificates stored in Azure Key Vault to 12 months or less.

Rationale:

Limiting certificate validity reduces the risk of misuse if compromised and helps ensure timely renewal, improving security and reliability.

Impact:

Minor administrative effort required to ensure certificate renewal and lifecycle management.

Audit:

Requisite Permissions - read section overview!

RBAC Role = **Key Vault Certificates Officer**

Access policy permissions = Certificates: **List**, **Get**

Audit from Azure Portal

1. Go to **Key vaults**.
2. Click the name of a key vault.
3. Under **Objects**, click **Certificates**.
4. Click the name of a certificate.
5. Click **Issuance Policy**.
6. Ensure that **Validity Period (in months)** is set to 12 or less.
7. Repeat steps 1-6 for each key vault and certificate.

Audit from Azure CLI

Run the following command to list key vaults:

```
az keyvault list
```

For each key vault, run the following command to list certificates:

```
az keyvault certificate list --vault-name <key-vault-name>
```

For each certificate, run the following command to get the certificate policy's **validityInMonths** setting:


```
az keyvault certificate show --id <certificate-id> --query  
policy.x509CertificateProperties.validityInMonths
```

Ensure that 12 or less is returned.

Audit from PowerShell

Run the following command to list key vaults:

```
Get-AzKeyVault
```

Run the following command to get the key vault with a given name:

```
$vault = Get-AzKeyVault -Name <key-vault-name>
```

Run the following command to list certificates in the key vault:

```
Get-AzKeyVaultCertificate -VaultName $vault.VaultName
```

Run the following command to get the policy of a certificate with a given name:

```
$certificate = Get-AzKeyVaultCertificatePolicy -VaultName $vault.VaultName -  
Name <certificate-name>
```

Run the following command to get the certificate policy's **ValidityInMonths** setting:

```
$certificate.ValidityInMonths
```

Ensure that 12 or less is returned.

Repeat for each key vault and certificate.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [0a075868-4c26-42ef-914c-5bc007359560](#) - **Name:** 'Certificates should have the specified maximum validity period'

Remediation:

Requisite Permissions - read section overview!

RBAC Role = **Key Vault Certificates Officer**

Access policy permissions = Certificates: **List, Get, Update**

Remediate from Azure Portal

1. Go to **Key vaults**.
2. Click the name of a key vault.

3. Under **Objects**, click **Certificates**.
4. Click the name of a certificate.
5. Click **Issuance Policy**.
6. Set **Validity Period (in months)** to an integer between 1 and 12, inclusive.
7. Click **Save**.
8. Repeat steps 1-7 for each key vault and certificate requiring remediation.

Remediate from PowerShell

For each certificate requiring remediation, run the following command to set **ValidityInMonths** to an integer between 1 and 12, inclusive:

```
Set-AzKeyVaultCertificatePolicy -VaultName $vault.VaultName -Name
<certificate-name> -ValidityInMonths <validity-in-months>
```

Default Value:

Validity Period (in months) is set to 12 by default.

References:

1. <https://learn.microsoft.com/en-us/azure/key-vault/certificates/about-certificates>
2. <https://learn.microsoft.com/en-us/cli/azure/keyvault>
3. <https://learn.microsoft.com/en-us/powershell/module/az.keyvault>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.6 Securely Manage Enterprise Assets and Software Securely manage enterprise assets and software. Example implementations include managing configuration through version-controlled-infrastructure-as-code and accessing administrative interfaces over secure network protocols, such as Secure Shell (SSH) and Hypertext Transfer Protocol Secure (HTTPS). Do not use insecure management protocols, such as Telnet (Teletype Network) and HTTP, unless operationally essential.			
v7	16.1 Maintain an Inventory of Authentication Systems Maintain an inventory of each of the organization's authentication systems, including those located onsite or at a remote service provider.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1649	TA0006	M1042

8.4 Azure Bastion

8.4.1 Ensure an Azure Bastion Host Exists (Automated)

Profile Applicability:

- Level 2

Description:

The Azure Bastion service allows secure remote access to Azure Virtual Machines over the Internet without exposing remote access protocol ports and services directly to the Internet. The Azure Bastion service provides this access using TLS over 443/TCP, and subscribes to hardened configurations within an organization's Azure Active Directory service.

Rationale:

The Azure Bastion service allows organizations a more secure means of accessing Azure Virtual Machines over the Internet without assigning public IP addresses to those Virtual Machines. The Azure Bastion service provides Remote Desktop Protocol (RDP) and Secure Shell (SSH) access to Virtual Machines using TLS within a web browser, thus preventing organizations from opening up 3389/TCP and 22/TCP to the Internet on Azure Virtual Machines. Additional benefits of the Bastion service includes Multi-Factor Authentication, Conditional Access Policies, and any other hardening measures configured within Azure Active Directory using a central point of access.

Impact:

The Azure Bastion service incurs additional costs and requires a specific virtual network configuration. The **Standard** tier offers additional configuration options compared to the **Basic** tier and may incur additional costs for those added features.

Audit:

Audit from Azure Portal

1. Click on **Bastions**
2. Ensure there is at least one **Bastion** host listed under the **Name** column

Audit from Azure CLI

```
az network bastion list --subscription <subscription ID>
```

Ensure the output of the above command is not empty.

Audit from PowerShell

Retrieve the **Bastion** host(s) information for a specific **Resource Group**

```
Get-AzBastion -ResourceGroupName <resource group name>
```

Ensure the output of the above command is not empty.

Remediation:

Remediate from Azure Portal

1. Click on **Bastions**
2. Select the **Subscription**
3. Select the **Resource group**
4. Type a **Name** for the new Bastion host
5. Select a **Region**
6. Choose **Standard** next to **Tier**
7. Use the slider to set the **Instance count**
8. Select the **Virtual network** or **Create new**
9. Select the **Subnet** named **AzureBastionSubnet**. Create a **Subnet** named **AzureBastionSubnet** using a **/26** CIDR range if it doesn't already exist.
10. Select the appropriate **Public IP address** option.
11. If **Create new** is selected for the **Public IP address** option, provide a **Public IP address name**.
12. If **Use existing** is selected for **Public IP address** option, select an IP address from **Choose public IP address**
13. Click **Next: Tags >**
14. Configure the appropriate **Tags**
15. Click **Next: Advanced >**
16. Select the appropriate **Advanced** options
17. Click **Next: Review + create >**
18. Click **Create**

Remediate from Azure CLI

```
az network bastion create --location <location> --name <name of bastion host>
--public-ip-address <public IP address name or ID> --resource-group <resource
group name or ID> --vnet-name <virtual network containing subnet called
"AzureBastionSubnet"> --scale-units <integer> --sku Standard [--disable-copy-
paste true|false] [--enable-ip-connect true|false] [--enable-tunneling
true|false]
```

Remediate from PowerShell

Create the appropriate **Virtual network** settings and **Public IP Address** settings.

```
$subnetName = "AzureBastionSubnet"
$subnet = New-AzVirtualNetworkSubnetConfig -Name $subnetName -AddressPrefix
<IP address range in CIDR notation making sure to use a /26>
$virtualNet = New-AzVirtualNetwork -Name <virtual network name> -
ResourceGroupName <resource group name> -Location <location> -AddressPrefix
<IP address range in CIDR notation> -Subnet $subnet
$publicip = New-AzPublicIpAddress -ResourceGroupName <resource group name> -
Name <public IP address name> -Location <location> -AllocationMethod Dynamic
-Sku Standard
```

Create the **Azure Bastion** service using the information within the created variables from above.

```
New-AzBastion -ResourceGroupName <resource group name> -Name <bastion name> -  
PublicIpAddress $publicip -VirtualNetwork $virtualNet -Sku "Standard" -  
ScaleUnit <integer>
```

Default Value:

By default, the Azure Bastion service is not configured.

References:

1. <https://learn.microsoft.com/en-us/azure/bastion/bastion-overview#sku>
2. <https://learn.microsoft.com/en-us/powershell/module/az.network/get-azbastion?view=azps-9.2.0>
3. <https://learn.microsoft.com/en-us/cli/azure/network/bastion?view=azure-cli-latest>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	12.1 <u>Ensure Network Infrastructure is Up-to-Date</u> Ensure network infrastructure is kept up-to-date. Example implementations include running the latest stable release of software and/or using currently supported network-as-a-service (NaaS) offerings. Review software versions monthly, or more frequently, to verify software support.			
v8	13.4 <u>Perform Traffic Filtering Between Network Segments</u> Perform traffic filtering between network segments, where appropriate.			
v7	9.2 <u>Ensure Only Approved Ports, Protocols and Services Are Running</u> Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			
v7	11.2 <u>Document Traffic Configuration Rules</u> All configuration rules that allow traffic to flow through network devices should be documented in a configuration management system with a specific business reason for each rule, a specific individual's name responsible for that business need, and an expected duration of the need.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1557	TA0006	M1035

8.5 Ensure Azure DDoS Network Protection is Enabled on Virtual Networks (Automated)

Profile Applicability:

- Level 2

Description:

Azure DDoS Network Protection defends resources in virtual networks against distributed denial-of-service (DDoS) attacks.

While an automated assessment procedure exists for this recommendation, the assessment status remains manual. Determining the appropriateness of enabling Azure DDoS Network Protection depends on the context and requirements of each organization and environment.

Rationale:

Virtual networks and resources are protected against attacks, helping to ensure reliability and availability for critical workloads.

Impact:

Azure DDoS Network Protection incurs a significant fixed monthly charge, with additional charges if more than 100 public IP resources are protected. Careful consideration and analysis should be applied before enabling DDoS protection. Refer to <https://azure.microsoft.com/en-us/pricing/details/ddos-protection> for detailed pricing information.

Audit:

Audit from Azure Portal

1. Go to **Virtual networks**.
2. Click the name of a virtual network.
3. Under **Settings**, click **DDoS protection**.
4. Ensure **DDoS Network Protection** is set to **Enable**.
5. Repeat steps 1-4 for each virtual network.

Audit from Azure CLI

Run the following command to list virtual networks:

```
az network vnet list
```

For each virtual network, run the following command to get the DDoS protection setting:

```
az network vnet show --resource-group <resource-group> --name <virtual-network> --query enableDdosProtection
```

Ensure **true** is returned.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

<https://portal.azure.com/#view/Microsoft Azure Policy/PolicyMenuBlade/~ /Definitions>

- **Policy ID:** [a7aca53f-2ed4-4466-a25e-0b45ade68efd](#) - **Name:** 'Azure DDoS Protection should be enabled'

Remediation:

Remediate from Azure Portal

1. Go to **Virtual networks**.
2. Click the name of a virtual network.
3. Under **Settings**, click **DDoS protection**.
4. Next to **DDoS Network Protection**, click **Enable**.
5. Provide a DDoS protection plan resource ID, or select a DDoS protection plan from the drop-down menu.
6. Click **Save**.
7. Repeat steps 1-6 for each virtual network requiring remediation.

Remediate from Azure CLI

For each virtual network requiring remediation, run the following command to enable DDoS protection:

```
az network vnet update --resource-group <resource-group> --name <virtual-network> --ddos-protection true --ddos-protection-plan <ddos-protection-plan>
```

Default Value:

DDoS protection is disabled by default.

References:

1. <https://learn.microsoft.com/en-us/azure/ddos-protection/ddos-protection-overview>
2. <https://learn.microsoft.com/en-us/azure/ddos-protection/manage-ddos-protection>
3. <https://azure.microsoft.com/en-us/pricing/details/ddos-protection>
4. <https://learn.microsoft.com/en-us/cli/azure/network/vnet>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	13.8 <u>Deploy a Network Intrusion Prevention Solution</u> Deploy a network intrusion prevention solution, where appropriate. Example implementations include the use of a Network Intrusion Prevention System (NIPS) or equivalent CSP service.			●
v7	12.7 <u>Deploy Network-Based Intrusion Prevention Systems</u> Deploy network-based Intrusion Prevention Systems (IPS) to block malicious network traffic at each of the organization's network boundaries.			●

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1498	TA0040	M1037

9 Storage Services

SERVICE CATEGORY BENCHMARK AVAILABLE

- "CIS Microsoft Azure Storage Services Benchmark"

To better understand the relationship between the Foundations Benchmark and Services Benchmarks, please read the "Introduction" section of this document.

PARTIAL RELOCATION - BE ADVISED!

Some recommendations previously covered in the CIS Microsoft Azure Foundations Benchmark (this document) related to Storage services have been relocated to a Benchmark titled **CIS Microsoft Azure Storage Services Benchmark**. This Storage Services section now provides a **foundational set** of secure configuration recommendations for products from Azure Product Directory's "Storage" category of services.

After applying foundational recommendations, take inventory of the entire set of Storage Services in use by your organization, then look to the Benchmarks section of the CIS Microsoft Azure Community (<https://workbench.cisecurity.org/communities/72>) for defense-in-depth guidance for those specific services in the **CIS Microsoft Azure Storage Services Benchmark**.

Services addressed in the **CIS Microsoft Azure Storage Services Benchmark**:

- Archive Storage
- Azure Backup
- Azure Blob Storage
- Azure Confidential Ledger
- Azure Container Storage
- Azure Data Box
- Azure Data Lake Storage
- Azure Data Share
- Azure Disk Storage
- Azure Elastic SAN
- Azure Files
- Azure Managed Lustre
- Azure NetApp Files
- Azure Storage Actions
- Queue Storage
- Storage Accounts
- Storage Explorer

Azure Product Directory Reference: <https://azure.microsoft.com/en-us/products#storage>

FEEDBACK REQUEST: Is there a specific service or recommendation in this section that you'd like to see addressed or improved? Let us know by making a ticket or starting a discussion in the CIS Microsoft Azure Community (<https://workbench.cisecurity.org/communities/72>).

9.1 Azure Files

This section covers security best practice recommendations for Azure Files.

Help us improve this Benchmark! If you notice a needed correction, want to provide feedback, or wish to contribute security best practice guidance please join our community and create a ticket, propose a change, or start a discussion so we can improve this guidance!

The CIS Microsoft Azure Community is here:
<https://workbench.cisecurity.org/communities/72>.

Resources for Azure Files

Azure Product Page:

- <https://azure.microsoft.com/en-us/products/storage/files/>

Azure Files service overview:

- <https://learn.microsoft.com/en-us/azure/storage/files/storage-files-introduction>

Microsoft Cloud Security Baseline for Azure File Sync:

- <https://learn.microsoft.com/en-us/security/benchmark/azure/baselines/azure-file-sync-security-baseline>

9.1.1 Ensure Soft Delete for Azure File Shares is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Azure Files offers soft delete for file shares, allowing you to easily recover your data when it is mistakenly deleted by an application or another storage account user.

Rationale:

Important data could be accidentally deleted or removed by a malicious actor. With soft delete enabled, the data is retained for the defined retention period before permanent deletion, allowing for recovery of the data.

Impact:

When a file share is soft-deleted, the used portion of the storage is charged for the indicated soft-deleted period. All other meters are not charged unless the share is restored.

Audit:

Audit from Azure Portal

1. Go to **Storage Accounts**.
2. For each storage account with file shares, under **Data storage**, click on **File shares**.
3. Under **File share settings**, ensure the value for **Soft delete** shows a number of days between 1 and 365, inclusive.

Audit from Azure CLI

Run the following command to list storage accounts:

```
az storage account list
```

Run the following command to determine if a storage account has file shares:

```
az storage share list --account-name <storage-account>
```

For each storage account with file shares, run the following command:

```
az storage account file-service-properties show --resource-group <resource-group> --account-name <storage-account>
```

Ensure that under **shareDeleteRetentionPolicy**, **enabled** is set to **true**, and **days** is set to an appropriate value between 1 and 365, inclusive.

Audit from PowerShell

Run the following command to list storage accounts:

```
Get-AzStorageAccount -ResourceGroupName <resource-group>
```

With a storage account context set, run the following command to determine if a storage account has file shares:

```
Get-AzStorageShare
```

For each storage account with file shares, run the following command:

```
Get-AzStorageFileServiceProperty -ResourceGroupName <resource-group> -  
AccountName <storage-account>
```

Ensure that **ShareDeleteRetentionPolicy.Enabled** is set to **True** and **ShareDeleteRetentionPolicy.Days** is set to an appropriate value between 1 and 365, inclusive.

Remediation:

Remediate from Azure Portal

1. Go to **Storage Accounts**.
2. For each storage account with file shares, under **Data storage**, click **File shares**.
3. Under **File share settings**, click the value next to **Soft delete**.
4. Under **Soft delete for all file shares**, click the toggle to set it to **Enabled**.
5. Under **Retention policies**, set an appropriate number of days to retain soft deleted data between 1 and 365, inclusive.
6. Click **Save**.

Remediate from Azure CLI

For each storage account requiring remediation, run the following command to enable soft delete for file shares and set an appropriate number of days for deleted data to be retained, between 1 and 365, inclusive:

```
az storage account file-service-properties update --account-name <storage-  
account> --enable-delete-retention true --delete-retention-days <retention-  
days>
```

Remediate from PowerShell

For each storage account requiring remediation, run the following command to enable soft delete for file shares and set an appropriate number of days for deleted data to be retained, between 1 and 365, inclusive:

```
Update-AzStorageFileServiceProperty -ResourceGroupName <resource-group> -  
AccountName <storage-account> -EnableShareDeleteRetentionPolicy $true -  
ShareRetentionDays <retention-days>
```

Default Value:

Soft delete is enabled by default at the storage account file share setting level.

References:

1. <https://learn.microsoft.com/en-us/azure/storage/files/storage-files-enable-soft-delete>
2. <https://learn.microsoft.com/en-us/cli/azure/storage/account/file-service-properties>
3. <https://learn.microsoft.com/en-us/powershell/module/az.storage/get-azstoragefileproperty>
4. <https://learn.microsoft.com/en-us/powershell/module/az.storage/update-azstoragefileproperty>
5. <https://learn.microsoft.com/en-us/azure/storage/files/storage-files-prevent-file-share-deletion>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	11.1 Establish and Maintain a Data Recovery Process Establish and maintain a data recovery process. In the process, address the scope of data recovery activities, recovery prioritization, and the security of backup data. Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	10.4 Ensure Protection of Backups Ensure that backups are properly protected via physical security or encryption when they are stored, as well as when they are moved across the network. This includes remote backups and cloud services.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1485	TA0040	M1053

9.1.2 Ensure 'SMB protocol version' is Set to 'SMB 3.1.1' or Higher for SMB file shares (Automated)

Profile Applicability:

- Level 1

Description:

Ensure that SMB file shares are configured to use the latest supported SMB protocol version. Keeping the SMB protocol updated helps mitigate risks associated with older SMB versions, which may contain vulnerabilities and lack essential security controls.

Rationale:

Using the latest supported SMB protocol version enhances the security of SMB file shares by preventing the exploitation of known vulnerabilities in outdated SMB versions.

Impact:

Using the latest SMB protocol version may impact client compatibility.

Audit:

Audit from Azure Portal

1. Go to **Storage accounts**.
2. Click the name of a storage account.
3. Under **Data storage**, click **File shares**.
4. Under **File share settings**, click the link next to **Security**.
5. Under **SMB protocol versions**, ensure that **SMB3.1.1** is the only checked protocol version.
6. Repeat steps 1-5 for each storage account.

Audit from Azure CLI

Run the following command to list storage accounts:

```
az storage account list
```

For each storage account, run the following command:

```
az storage account file-service-properties show --resource-group <resource-group> --account-name <storage-account>
```

Ensure that under **protocolSettings > smb, versions** is set to **SMB3.1.1**; only.

Audit from PowerShell

Run the following command to list storage accounts:

```
Get-AzStorageAccount
```

Run the following command to get the file service properties for a storage account in a resource group with a given name:

```
$storageaccountfileservice = Get-AzStorageFileServiceProperty -  
ResourceGroupName <resource-group> -AccountName <storage-account>
```

Run the following command to get the SMB protocol version setting:

```
$storageaccountfileservice.ProtocolSettings.Smb.Versions
```

Ensure that the command returns **SMB3.1.1** only.

Repeat for each storage account.

Remediation:

Remediate from Azure Portal

1. Go to **Storage accounts**.
2. Click the name of a storage account.
3. Under **Data storage**, click **File shares**.
4. Under **File share settings**, click the link next to **Security**.
5. If **Profile** is set to **Maximum compatibility**, click the drop-down menu and select **Maximum security** or **Custom**.
6. If selecting **Custom**, under **SMB protocol versions**, uncheck the boxes next to **SMB 2.1** and **SMB 3.0**.
7. Click **Save**.
8. Repeat steps 1-7 for each storage account requiring remediation.

Remediate from Azure CLI

For each storage account requiring remediation, run the following command to set the SMB protocol version:

```
az storage account file-service-properties update --resource-group <resource-group> --account-name <storage-account> --versions SMB3.1.1
```

Remediate from PowerShell

For each storage account requiring remediation, run the following command to set the SMB protocol version:

```
Update-AzStorageFileServiceProperty -ResourceGroupName <resource-group> -  
StorageAccountName <storage-account> -SmbProtocolVersion SMB3.1.1
```

Default Value:

By default, all SMB versions are allowed.

References:

1. <https://learn.microsoft.com/en-us/azure/well-architected/service-guides/azure-files#recommendations-for-smb-file-shares>
2. <https://learn.microsoft.com/en-us/azure/storage/files/files-smb-protocol#smb-security-settings>
3. <https://learn.microsoft.com/en-us/cli/azure/storage/account/file-service-properties>
4. <https://learn.microsoft.com/en-us/powershell/module/az.storage/get-azstoragefileproperty>
5. <https://learn.microsoft.com/en-us/powershell/module/az.storage/update-azstoragefileproperty>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	12.6 Use of Secure Network Management and Communication Protocols Use secure network management and communication protocols (e.g., 802.1X, Wi-Fi Protected Access 2 (WPA2) Enterprise or greater).			
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1210	TA0008	M1042

9.1.3 Ensure 'SMB channel encryption' is Set to 'AES-256-GCM' or Higher for SMB file shares (Automated)

Profile Applicability:

- Level 1

Description:

Implement SMB channel encryption with AES-256-GCM for SMB file shares to ensure data confidentiality and integrity in transit. This method offers strong protection against eavesdropping and man-in-the-middle attacks, safeguarding sensitive information.

Rationale:

AES-256-GCM encryption enhances the security of data transmitted over SMB channels by safeguarding it from unauthorized interception and tampering.

Impact:

Using the AES-256-GCM SMB channel encryption may impact client compatibility.

Audit:

Audit from Azure Portal

1. Go to **Storage accounts**.
2. Click the name of a storage account.
3. Under **Data storage**, click **File shares**.
4. Under **File share settings**, click the link next to **Security**.
5. Under **SMB channel encryption**, ensure that **AES-256-GCM**, or higher, is the only checked SMB channel encryption setting.
6. Repeat steps 1-5 for each storage account.

Audit from Azure CLI

Run the following command to list storage accounts:

```
az storage account list
```

For each storage account, run the following command:

```
az storage account file-service-properties show --resource-group <resource-group> --account-name <storage-account>
```

Ensure that under **protocolSettings > smb**, **channelEncryption** is set to **AES-256-GCM**, or higher, only.

Audit from PowerShell

Run the following command to list storage accounts:

```
Get-AzStorageAccount
```

Run the following command to get the file service properties for a storage account in a resource group with a given name:

```
$storageaccountfileservice = Get-AzStorageFileServiceProperty -  
ResourceGroupName <resource-group> -AccountName <storage-account>
```

Run the following command to get the SMB channel encryption setting:

```
$storageaccountfileservice.ProtocolSettings.Smb.ChannelEncryption
```

Ensure that the command returns **AES-256-GCM**, or higher, only.

Repeat for each storage account.

Remediation:

Remediate from Azure Portal

1. Go to **Storage accounts**.
2. Click the name of a storage account.
3. Under **Data storage**, click **File shares**.
4. Under **File share settings**, click the link next to **Security**.
5. If **Profile** is set to **Maximum compatibility**, click the drop-down menu and select **Maximum security** or **Custom**.
6. If selecting **Custom**, under **SMB channel encryption**, uncheck the boxes next to **AES-128-CCM** and **AES-128-GCM**.
7. Click **Save**.
8. Repeat steps 1-7 for each storage account requiring remediation.

Remediate from Azure CLI

For each storage account requiring remediation, run the following command to set the SMB channel encryption:

```
az storage account file-service-properties update --resource-group <resource-group> --account-name <storage-account> --channel-encryption AES-256-GCM
```

Remediate from PowerShell

For each storage account requiring remediation, run the following command to set the SMB channel encryption:

```
Update-AzStorageFileServiceProperty -ResourceGroupName <resource-group> -  
StorageAccountName <storage-account> -SmbChannelEncryption AES-256-GCM
```

Default Value:

By default, the following SMB channel encryption algorithms are allowed:

- AES-128-CCM
- AES-128-GCM
- AES-256-GCM

References:

1. <https://learn.microsoft.com/en-us/azure/well-architected/service-guides/azure-files#recommendations-for-smb-file-shares>
2. <https://learn.microsoft.com/en-us/azure/storage/files/files-smb-protocol?tabs=azure-portal#smb-security-settings>
3. <https://learn.microsoft.com/en-us/cli/azure/storage/account/file-service-properties>
4. <https://learn.microsoft.com/en-us/powershell/module/az.storage/get-azstoragefileproperty>
5. <https://learn.microsoft.com/en-us/powershell/module/az.storage/update-azstoragefileproperty>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.10 <u>Encrypt Sensitive Data in Transit</u> Encrypt sensitive data in transit. Example implementations can include: Transport Layer Security (TLS) and Open Secure Shell (OpenSSH).			
v7	14.4 <u>Encrypt All Sensitive Information in Transit</u> Encrypt all sensitive information in transit.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1210	TA0008	M1042

9.2 Azure Blob Storage

This section covers security best practice recommendations for Azure Blob Storage. Azure Blob Storage is a core storage service type for Azure Storage Accounts. Azure Data Lake services depend on the Azure Blob Service.

NOTE: If your organization is using Shared Access Signature (SAS) tokens, please review the CIS Microsoft Azure Storage Services Benchmark for best practice guidance on the configuration and use of those tokens.

Help us improve this Benchmark! If you notice a needed correction, want to provide feedback, or wish to contribute security best practice guidance please join our community and create a ticket, propose a change, or start a discussion so we can improve this guidance!

The CIS Microsoft Azure Community is here:
<https://workbench.cisecurity.org/communities/72>.

Resources for Azure Blob Storage

Azure Product Page:

- <https://azure.microsoft.com/en-us/products/storage/blobs/>

Azure Blob Storage service overview:

- <https://learn.microsoft.com/en-us/azure/storage/blobs/storage-blobs-overview>

Microsoft Cloud Security Baseline for Storage:

- <https://learn.microsoft.com/en-us/security/benchmark/azure/baselines/storage-security-baseline>

9.2.1 Ensure That Soft Delete for Blobs on Azure Blob Storage Storage Accounts is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Blobs in Azure storage accounts may contain sensitive or personal data, such as ePHI or financial information. Data that is erroneously modified or deleted by an application or a user can lead to data loss or unavailability.

It is recommended that soft delete be enabled on Azure storage accounts with blob storage to allow for the preservation and recovery of data when blobs or blob snapshots are deleted.

Rationale:

Blobs can be deleted incorrectly. An attacker or malicious user may do this deliberately in order to cause disruption. Deleting an Azure storage blob results in immediate data loss. Enabling this configuration for Azure storage accounts ensures that even if blobs are deleted from the storage account, the blobs are recoverable for a specific period of time, which is defined in the "Retention policies," ranging from 7 to 365 days.

Impact:

All soft-deleted data is billed at the same rate as active data. Additional costs may be incurred for deleted blobs until the soft delete period ends and the data is permanently removed.

Audit:

Audit from Azure Portal

1. Go to **Storage Accounts**.
2. For each Storage Account with blob storage, under **Data management**, go to **Data protection**.
3. Ensure that **Enable soft delete for blobs** is checked.
4. Ensure that the retention period is a sufficient length for your organization.

Audit from Azure CLI

Run the following command to list storage accounts:

```
az storage account list
```

Run the following command to determine if a storage account has containers:

```
az storage container list --account-name <storage-account>
```

For each storage account with containers, ensure that the output of the below command contains **"enabled": true** and **days** is not **null**:

```
az storage blob service-properties delete-policy show --account-name  
<storage-account>
```

Remediation:

Remediate from Azure Portal

1. Go to **Storage Accounts**.
2. For each Storage Account with blob storage, under **Data management**, go to **Data protection**.
3. Check the box next to **Enable soft delete for blobs**.
4. Set the retention period to a sufficient length for your organization.
5. Click **Save**.

Remediate from Azure CLI

For each storage account requiring remediation, run the following command to enable soft delete for blobs:

```
az storage blob service-properties delete-policy update --days-retained  
<retention-days> --account-name <storage-account> --enable true
```

Default Value:

Soft delete for blob storage is **enabled** by default on storage accounts created via the Azure Portal, and **disabled** by default on storage accounts created via Azure CLI or PowerShell.

References:

1. <https://learn.microsoft.com/en-us/azure/storage/blobs/soft-delete-blob-overview>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	11.1 <u>Establish and Maintain a Data Recovery Process</u> Establish and maintain a data recovery process. In the process, address the scope of data recovery activities, recovery prioritization, and the security of backup data. Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	10.4 <u>Ensure Protection of Backups</u> Ensure that backups are properly protected via physical security or encryption when they are stored, as well as when they are moved across the network. This includes remote backups and cloud services.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1485	TA0040	M1053

9.2.2 Ensure that Soft Delete for Containers on Azure Blob Storage Storage Accounts is Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Containers in Azure storage accounts may contain sensitive or personal data, such as ePHI or financial information. Data that is erroneously modified or deleted by an application or a user can lead to data loss or unavailability.

It is recommended that soft delete for containers be enabled on Azure storage accounts with blob storage to allow for the preservation and recovery of data when containers are deleted.

Rationale:

Containers can be deleted incorrectly. An attacker or malicious user may do this deliberately in order to cause disruption. Deleting a container results in immediate data loss. Enabling this configuration for Azure storage accounts ensures that even if containers are deleted from the storage account, the containers are recoverable for a specific period of time, which is defined in the "Retention policies," ranging from 7 to 365 days.

Impact:

All soft-deleted data is billed at the same rate as active data. Additional costs may be incurred for deleted containers until the soft delete period ends and the data is permanently removed.

Audit:

Audit from Azure Portal

1. Go to **Storage accounts**.
2. For each Storage Account with blob storage, under **Data management**, go to **Data protection**.
3. Ensure that **Enable soft delete for containers** is checked.
4. Ensure that the retention period is a sufficient length for your organization.

Audit from Azure CLI

Run the following command to list storage accounts:

```
az storage account list
```

Run the following command to determine if a storage account has containers:

```
az storage container list --account-name <storage-account>
```

For each storage account with containers, run the following command to get the retention settings:

```
az storage account blob-service-properties show --resource-group <resource-group> --account-name <storage-account> --query containerDeleteRetentionPolicy
```

Ensure that **enabled** is set to **true** and **days** is not **null**.

Remediation:

Remediate from Azure Portal

1. Go to **Storage accounts**.
2. For each Storage Account with blob storage, under **Data management**, go to **Data protection**.
3. Check the box next to **Enable soft delete for containers**.
4. Set the retention period to a sufficient length for your organization.
5. Click **Save**.

Remediate from Azure CLI

Run the following command to update container retention:

```
az storage account blob-service-properties update --resource-group <resource-group> --account-name <storage-account> --enable-container-delete-retention true --container-delete-retention-days <retention-days>
```

Default Value:

Soft delete for containers is **enabled** by default on storage accounts created via the Azure Portal, and **disabled** by default on storage accounts created via Azure CLI or PowerShell.

References:

1. <https://learn.microsoft.com/en-us/azure/storage/blobs/soft-delete-container-overview>
2. <https://learn.microsoft.com/en-us/azure/storage/blobs/soft-delete-container-enable>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	11.1 Establish and Maintain a Data Recovery Process Establish and maintain a data recovery process. In the process, address the scope of data recovery activities, recovery prioritization, and the security of backup data. Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	10.4 Ensure Protection of Backups Ensure that backups are properly protected via physical security or encryption when they are stored, as well as when they are moved across the network. This includes remote backups and cloud services.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1485	TA0040	M1053

9.2.3 Ensure 'Versioning' is Set to 'Enabled' on Azure Blob Storage Storage Accounts (Automated)

Profile Applicability:

- Level 2

Description:

Enabling blob versioning allows for the automatic retention of previous versions of objects. With blob versioning enabled, earlier versions of a blob are accessible for data recovery in the event of modifications or deletions.

Rationale:

Blob versioning safeguards data integrity and enables recovery by retaining previous versions of stored objects, facilitating quick restoration from accidental deletion, modification, or malicious activity.

Impact:

Enabling blob versioning for a storage account creates a new version with each write operation to a blob, which can increase storage costs. To control these costs, a lifecycle management policy can be applied to automatically delete older versions.

Audit:

Audit from Azure Portal

1. Go to **Storage accounts**.
2. Click the name of a storage account with blob storage.
3. In the **Overview** page, on the **Properties** tab, under **Blob service**, ensure **Versioning** is set to **Enabled**.
4. Repeat steps 1-3 for each storage account with blob storage.

Audit from Azure CLI

Run the following command to list storage accounts:

```
az storage account list
```

Run the following command to determine if a storage account has containers:

```
az storage container list --account-name <storage-account>
```

For each storage account with containers, ensure that the output of the below command contains **"isVersioningEnabled": true**:

```
az storage account blob-service-properties show --account-name <storage-account>
```

Audit from PowerShell

Run the following command to list storage accounts:

```
Get-AzStorageAccount
```

Run the following command to create an Azure Storage context for a storage account:

```
$context = New-AzStorageContext -StorageAccountName <storage-account>
```

Run the following command to list containers for the storage account:

```
Get-AzStorageContainer -Context $context
```

If the storage account has containers, run the following command to get the blob service properties of the storage account:

```
$account = Get-AzStorageBlobServiceProperty -ResourceGroupName <resource-group> -AccountName <storage-account>
```

Run the following command to get the blob versioning setting for the storage account:

```
$account.IsVersioningEnabled
```

Ensure that the command returns **True**.

Repeat for each storage account.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

<https://portal.azure.com/#view/Microsoft Azure Policy/PolicyMenuBlade/~ /Definitions>

- **Policy ID:** [c36a325b-ae04-4863-ad4f-19c6678f8e08](#) - **Name:** 'Configure your Storage account to enable blob versioning'

Remediation:

Remediate from Azure Portal

1. Go to **Storage accounts**.
2. Click the name of a storage account with blob storage.
3. In the **Overview** page, on the **Properties** tab, under **Blob service**, click **Disabled** next to **Versioning**.
4. Under **Tracking**, check the box next to **Enable versioning for blobs**.
5. Select the radio button next to **Keep all versions** or **Delete versions after (in days)**.
6. If selecting to delete versions, enter a number of in the box after which to delete blob versions.
7. Click **Save**.
8. Repeat steps 1-7 for each storage account with blob storage.

Remediate from Azure CLI

For each storage account requiring remediation, run the following command to enable blob versioning:

```
az storage account blob-service-properties update --account-name <storage-account> --enable-versioning true
```

Remediate from PowerShell

For each storage account requiring remediation, run the following command to enable blob versioning:

```
Update-AzStorageBlobServiceProperty -ResourceGroupName <resource-group> -StorageAccountName <storage-account> -IsVersioningEnabled $true
```

Default Value:

Blob versioning is disabled by default on storage accounts.

References:

1. <https://learn.microsoft.com/en-us/cli/azure/storage/account>
2. <https://learn.microsoft.com/en-us/cli/azure/storage/account/blob-service-properties>
3. <https://learn.microsoft.com/en-us/powershell/module/az.storage/get-azstorageaccount>
4. <https://learn.microsoft.com/en-us/powershell/module/az.storage/new-azstoragecontext>
5. <https://learn.microsoft.com/en-us/powershell/module/az.storage/get-azstoragecontainer>
6. <https://learn.microsoft.com/en-us/powershell/module/az.storage/get-azstorageblobserviceproperty>
7. <https://learn.microsoft.com/en-us/powershell/module/az.storage/update-azstorageblobserviceproperty>
8. <https://learn.microsoft.com/en-us/azure/storage/blobs/versioning-overview>
9. <https://learn.microsoft.com/en-us/azure/storage/blobs/lifecycle-management-overview>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	11.1 <u>Establish and Maintain a Data Recovery Process</u> Establish and maintain a data recovery process. In the process, address the scope of data recovery activities, recovery prioritization, and the security of backup data. Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			

Controls Version	Control	IG 1	IG 2	IG 3
v7	10.1 <u>Ensure Regular Automated Back Ups</u> Ensure that all system data is automatically backed up on regular basis.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1485	TA0040	M1053

9.3 Storage Accounts

This section covers security best practice recommendations for Storage Accounts in Azure.

The recommendations in this section apply to the Storage Account, but not to the Storage Services which may be running on that account. Use the Storage Account recommendations as a starting place for securing the account, then proceed to apply the recommendations from the storage services section(s) that are relevant to the storage services running on your account.

Storage Accounts are a family of account types that support different Storage Services. The Storage Account types and their supported services follow:

- **Standard general-purpose v2** supported services: Blob Storage (including Data Lake Storage), Queue Storage, Table Storage, and Azure Files.
- **Premium block blobs** supported services: Blob Storage (including Data Lake Storage)
- **Premium file shares** supported services: Azure Files
- **Premium page blobs** supported services: Page blobs only

Help us improve this Benchmark! If you notice a needed correction, want to provide feedback, or wish to contribute security best practice guidance please join our community and create a ticket, propose a change, or start a discussion so we can improve this guidance!

The CIS Microsoft Azure Community is here:
<https://workbench.cisecurity.org/communities/72>.

Resources for Storage Accounts

Azure Product page:

- <https://azure.microsoft.com/en-us/products/category/storage/>

Azure Storage Account overview:

- <https://learn.microsoft.com/en-us/azure/storage/common/storage-account-overview>

Microsoft Cloud Security Baseline for Storage:

- <https://learn.microsoft.com/en-us/security/benchmark/azure/baselines/storage-security-baseline>

9.3.1 Secrets and Keys

9.3.1.1 Ensure That 'Enable key rotation reminders' is Enabled for Each Storage Account (Automated)

Profile Applicability:

- Level 1

Description:

Access Keys authenticate application access requests to data contained in Storage Accounts. A periodic rotation of these keys is recommended to ensure that potentially compromised keys cannot result in a long-term exploitable credential. The "Rotation Reminder" is an automatic reminder feature for a manual procedure.

Rationale:

Reminders such as those generated by this recommendation will help maintain a regular and healthy cadence for activities which improve the overall efficacy of a security program.

Cryptographic key rotation periods will vary depending on your organization's security requirements and the type of data which is being stored in the Storage Account. For example, PCI DSS mandates that cryptographic keys be replaced or rotated 'regularly,' and advises that keys for static data stores be rotated every 'few months.'

For the purposes of this recommendation, 90 days will be prescribed for the reminder. Review and adjustment of the 90 day period is recommended, and may even be necessary. Your organization's security requirements should dictate the appropriate setting.

Impact:

This recommendation only creates a periodic reminder to regenerate access keys. Regenerating access keys can affect services in Azure as well as the organization's applications that are dependent on the storage account. All clients that use the access key to access the storage account must be updated to use the new key.

Audit:

Audit from Azure Portal

1. Go to **Storage Accounts**
2. For each Storage Account, under **Security + networking**, go to **Access keys**
3. If the button **Edit rotation reminder** is displayed, the Storage Account is compliant. Click **Edit rotation reminder** and review the **Remind me every** field for a desirable periodic setting that fits your security program's needs. If the button **Set rotation reminder** is displayed, the Storage Account is not compliant.

Audit from Powershell

```
$rgName = <resource group name for the storage>
$accountName = <storage account name>
$account = Get-AzStorageAccount -ResourceGroupName $rgName -Name $accountName

Write-Output $accountName ->
Write-Output "Expiration Reminder set to:
$(($account.KeyPolicy.KeyExpirationPeriodInDays) Days"
Write-Output "Key1 Last Rotated:
$(($account.KeyCreationTime.Key1.ToShortDateString()))"
Write-Output "Key2 Last Rotated:
$(($account.KeyCreationTime.Key2.ToShortDateString()))"
```

Key rotation is recommended if the creation date for any key is empty.

If the reminder is set, the period in days will be returned. The recommended period is 90 days.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [044985bb-afe1-42cd-8a36-9d5d42424537](#) - **Name:** 'Storage account keys should not be expired'

Remediation:

Remediate from Azure Portal

1. Go to **Storage Accounts**
2. For each Storage Account that is not compliant, under **Security + networking**, go to **Access keys**
3. Click **Set rotation reminder**
4. Check **Enable key rotation reminders**
5. In the **Send reminders** field select **Custom**, then set the **Remind me every** field to **90** and the period drop down to **Days**
6. Click **Save**

Remediate from Powershell

```
$rgName = <resource group name for the storage>
$accountName = <storage account name>
$account = Get-AzStorageAccount -ResourceGroupName $rgName -Name $accountName
if ($account.KeyCreationTime.Key1 -eq $null -or $account.KeyCreationTime.Key2
-eq $null){
    Write-output ("You must regenerate both keys at least once before
setting expiration policy")
} else {
    $account = Set-AzStorageAccount -ResourceGroupName $rgName -Name
$accountName -KeyExpirationPeriodInDay 90
}
$account.KeyPolicy.KeyExpirationPeriodInDays
```

Default Value:

By default, Key rotation reminders are not configured.

References:

1. <https://learn.microsoft.com/en-us/azure/storage/common/storage-account-keys-manage>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-privileged-access#pa-1-separate-and-limit-highly-privilegedadministrative-users>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-identity-management#im-3-manage-application-identities-securely-and-automatically>
4. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-governance-strategy#gs-6-define-and-implement-identity-and-privileged-access-strategy>
5. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-identity-management#im-8-restrict-the-exposure-of-credentials-and-secrets>
6. <https://www.pcidssguide.com/pci-dss-key-rotation-requirements/>
7. <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-57pt1r5.pdf>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.1 Establish and Maintain a Secure Configuration Process Establish and maintain a secure configuration process for enterprise assets (end-user devices, including portable and mobile, non-computing/IoT devices, and servers) and software (operating systems and applications). Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	11.3 Use Automated Tools to Verify Standard Device Configurations and Detect Changes Compare all network device configuration against approved security configurations defined for each network device in use and alert when any deviations are discovered.			

9.3.1.2 Ensure That Storage Account Access keys are Periodically Regenerated (Automated)

Profile Applicability:

- Level 1

Description:

For increased security, regenerate storage account access keys periodically.

Rationale:

When a storage account is created, Azure generates two 512-bit storage access keys which are used for authentication when the storage account is accessed. Rotating these keys periodically ensures that any inadvertent access or exposure does not result from the compromise of these keys.

Cryptographic key rotation periods will vary depending on your organization's security requirements and the type of data which is being stored in the Storage Account. For example, PCI DSS mandates that cryptographic keys be replaced or rotated 'regularly,' and advises that keys for static data stores be rotated every 'few months.'

For the purposes of this recommendation, 90 days will be prescribed for the reminder. Review and adjustment of the 90 day period is recommended, and may even be necessary. Your organization's security requirements should dictate the appropriate setting.

Impact:

Regenerating access keys can affect services in Azure as well as the organization's applications that are dependent on the storage account. All clients who use the access key to access the storage account must be updated to use the new key.

Audit:

Audit from Azure Portal

1. Go to **Storage Accounts**.
2. For each Storage Account, under **Security + networking**, go to **Access keys**.
3. Review the date and days in the **Last rotated** field for **each** key.

If the **Last rotated** field indicates a number or days greater than 90 [or greater than your organization's period of validity], the key should be rotated.

Audit from Azure CLI

1. Get a list of storage accounts

```
az storage account list --subscription <subscription-id>
```

Make a note of **id**, **name** and **resourceGroup**.

2. For every storage account make sure that key is regenerated in the past 90 days.

```
az monitor activity-log list --namespace Microsoft.Storage --offset 90d --  
query "[?contains(authorization.action, 'regenerateKey')]" --resource-id  
<resource id>
```

The output should contain

```
"authorization"/"scope": <your_storage_account> AND "authorization"/"action":  
"Microsoft.Storage/storageAccounts/regeneratekey/action" AND  
"status"/"localizedValue": "Succeeded" "status"/"Value": "Succeeded"
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [044985bb-afe1-42cd-8a36-9d5d42424537](#) - **Name:** 'Storage account keys should not be expired'

Remediation:

Remediate from Azure Portal

1. Go to **Storage Accounts**.
2. For each Storage Account with outdated keys, under **Security + networking**, go to **Access keys**.
3. Click **Rotate key** next to the outdated key, then click **Yes** to the prompt confirming that you want to regenerate the access key.

After Azure regenerates the Access Key, you can confirm that **Access keys** reflects a **Last rotated** date of **(0 days ago)**.

Default Value:

By default, access keys are not regenerated periodically.

References:

1. <https://learn.microsoft.com/en-us/azure/storage/common/storage-account-keys-manage>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-privileged-access#pa-1-separate-and-limit-highly-privileged-administrative-users>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-identity-management#im-2-protect-identity-and-authentication-systems>
4. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-governance-strategy#gs-6-define-and-implement-identity-and-privileged-access-strategy>
5. <https://www.pcidssguide.com/pci-dss-key-rotation-requirements/>
6. <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-57pt1r5.pdf>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.6 Securely Manage Enterprise Assets and Software Securely manage enterprise assets and software. Example implementations include managing configuration through version-controlled-infrastructure-as-code and accessing administrative interfaces over secure network protocols, such as Secure Shell (SSH) and Hypertext Transfer Protocol Secure (HTTPS). Do not use insecure management protocols, such as Telnet (Teletype Network) and HTTP, unless operationally essential.			
v8	6.2 Establish an Access Revoking Process Establish and follow a process, preferably automated, for revoking access to enterprise assets, through disabling accounts immediately upon termination, rights revocation, or role change of a user. Disabling accounts, instead of deleting accounts, may be necessary to preserve audit trails.			
v7	16.7 Establish Process for Revoking Access Establish and follow an automated process for revoking system access by disabling accounts immediately upon termination or change of responsibilities of an employee or contractor. Disabling these accounts, instead of deleting accounts, allows preservation of audit trails.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1098	TA0003	M1026

9.3.1.3 Ensure 'Allow storage account key access' for Azure Storage Accounts is 'Disabled' (Automated)

Profile Applicability:

- Level 1

Description:

Every secure request to an Azure Storage account must be authorized. By default, requests can be authorized with either Microsoft Entra credentials or by using the account access key for Shared Key authorization.

Rationale:

Microsoft Entra ID provides superior security and ease of use compared to Shared Key and is recommended by Microsoft. To require clients to use Microsoft Entra ID for authorizing requests, you can disallow requests to the storage account that are authorized with Shared Key.

Impact:

When you disallow Shared Key authorization for a storage account, any requests to the account that are authorized with Shared Key, including shared access signatures (SAS), will be denied. Client applications that currently access the storage account using the Shared Key will no longer function.

Audit:

Audit from Azure Portal

1. Go to **Storage accounts**.
2. Click on a storage account.
3. Under **Settings**, click **Configuration**.
4. Under **Allow storage account key access**, ensure that the radio button next to **Disabled** is selected.
5. Repeat steps 1-4 for each storage account.

Audit from Azure CLI

Run the following command to list storage accounts:

```
az storage account list
```

For each storage account, run the following command:

```
az storage account show --resource-group <resource-group> --name <storage-account>
```

Ensure that **allowSharedKeyAccess** is set to **false**.

Audit from PowerShell

Run the following command to list storage accounts:

```
Get-AzStorageAccount
```

Run the following command to get the storage account in a resource group with a given name:

```
$storageAccount = Get-AzStorageAccount -ResourceGroupName <resource-group> -  
Name <storage-account>
```

Run the following command to get the shared key access setting for the storage account:

```
$storageAccount.allowSharedKeyAccess
```

Ensure that the command returns **False**.

Repeat for each storage account.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [8c6a50c6-9ffd-4ae7-986f-5fa6111f9a54](#) - **Name:** 'Storage accounts should prevent shared key access'

Remediation:

Remediate from Azure Portal

1. Go to **Storage accounts**.
2. Click on a storage account.
3. Under **Settings**, click **Configuration**.
4. Under **Allow storage account key access**, click the radio button next to **Disabled**.
5. Click **Save**.
6. Repeat steps 1-5 for each storage account requiring remediation.

Remediate from Azure CLI

For each storage account requiring remediation, run the following command to disallow shared key authorization:

```
az storage account update --resource-group <resource-group> --name <storage-  
account> --allow-shared-key-access false
```

Remediate from PowerShell

For each storage account requiring remediation, run the following command to disallow shared key authorization:

```
Set-AzStorageAccount -ResourceGroupName <resource-group> -Name <storage-account> -AllowSharedKeyAccess $false
```

Default Value:

The AllowSharedKeyAccess property of a storage account is not set by default and does not return a value until you explicitly set it. The storage account permits requests that are authorized with the Shared Key when the property value is **null** or when it is **true**.

References:

1. <https://learn.microsoft.com/en-us/azure/storage/common/shared-key-authorization-prevent>
2. <https://learn.microsoft.com/en-us/cli/azure/storage/account>
3. <https://learn.microsoft.com/en-us/powershell/module/az.storage/get-azstorageaccount>
4. <https://learn.microsoft.com/en-us/powershell/module/az.storage/set-azstorageaccount>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	6.2 <u>Establish an Access Revoking Process</u> Establish and follow a process, preferably automated, for revoking access to enterprise assets, through disabling accounts immediately upon termination, rights revocation, or role change of a user. Disabling accounts, instead of deleting accounts, may be necessary to preserve audit trails.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

Controls Version	Control	IG 1	IG 2	IG 3
v7	16.7 <u>Establish Process for Revoking Access</u> Establish and follow an automated process for revoking system access by disabling accounts immediately upon termination or change of responsibilities of an employee or contractor . Disabling these accounts, instead of deleting accounts, allows preservation of audit trails.		●	●

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1530	TA0009	M1022

9.3.2 Networking

9.3.2.1 Ensure Private Endpoints are Used to Access Storage Accounts (Automated)

Profile Applicability:

- Level 2

Description:

Use private endpoints for your Azure Storage accounts to allow clients and services to securely access data located over a network via an encrypted Private Link. To do this, the private endpoint uses an IP address from the VNet for each service. Network traffic between disparate services securely traverses encrypted over the VNet. This VNet can also link addressing space, extending your network and accessing resources on it. Similarly, it can be a tunnel through public networks to connect remote infrastructures together. This creates further security through segmenting network traffic and preventing outside sources from accessing it.

Rationale:

Securing traffic between services through encryption protects the data from easy interception and reading.

Impact:

If an Azure Virtual Network is not implemented correctly, this may result in the loss of critical network traffic.

Private endpoints are charged per hour of use. Refer to <https://azure.microsoft.com/en-us/pricing/details/private-link/> and <https://azure.microsoft.com/en-us/pricing/calculator/> to estimate potential costs.

Audit:

Audit from Azure Portal

1. Open the **Storage Accounts** blade.
2. For each listed Storage Account, perform the following check:
3. Under the **Security + networking** heading, click on **Networking**.
4. Click on the **Private endpoint connections** tab at the top of the networking window.
5. Ensure that for each VNet that the Storage Account must be accessed from, a unique Private Endpoint is deployed and the **Connection state** for each Private Endpoint is **Approved**.

Repeat the procedure for each Storage Account.

Audit from PowerShell

```
$storageAccount = Get-AzStorageAccount -ResourceGroup '<ResourceGroupName>' -  
Name '<storageaccountname>'

Get-AzPrivateEndpoint -ResourceGroup '<ResourceGroupName>' | Where-Object  
{$_PrivateLinkServiceConnectionsText -match $storageAccount.id}
```

If the results of the second command returns information, the Storage Account is using a Private Endpoint and complies with this Benchmark, otherwise if the results of the second command are empty, the Storage Account generates a finding.

Audit from Azure CLI

```
az storage account show --name '<storage account name>' --query  
"privateEndpointConnections[0].id"
```

If the above command returns data, the Storage Account complies with this Benchmark, otherwise if the results are empty, the Storage Account generates a finding.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [6edd7eda-6dd8-40f7-810d-67160c639cd9](#) - **Name:** 'Storage accounts should use private link'

Remediation:

Remediate from Azure Portal

1. Open the **Storage Accounts** blade
2. For each listed Storage Account, perform the following:
3. Under the **Security + networking** heading, click on **Networking**
4. Click on the **Private endpoint connections** tab at the top of the networking window
5. Click the **+ Private endpoint** button
6. In the **1 - Basics** tab/step:
 - **Enter a name** that will be easily recognizable as associated with the Storage Account (*Note: The "Network Interface Name" will be automatically completed, but you can customize it if needed.*)
 - Ensure that the **Region** matches the region of the Storage Account
 - Click **Next**
7. In the **2 - Resource** tab/step:
 - Select the **target sub-resource** based on what type of storage resource is being made available
 - Click **Next**
8. In the **3 - Virtual Network** tab/step:
 - Select the **Virtual network** that your Storage Account will be connecting to
 - Select the **Subnet** that your Storage Account will be connecting to
 - (Optional) Select other network settings as appropriate for your environment
 - Click **Next**
9. In the **4 - DNS** tab/step:
 - (Optional) Select other DNS settings as appropriate for your environment
 - Click **Next**
10. In the **5 - Tags** tab/step:
 - (Optional) Set any tags that are relevant to your organization
 - Click **Next**
11. In the **6 - Review + create** tab/step:
 - A validation attempt will be made and after a few moments it should indicate **Validation Passed** - if it does not pass, double-check your settings before beginning more in depth troubleshooting.
 - If validation has passed, click **Create** then wait for a few minutes for the scripted deployment to complete.

Repeat the above procedure for each Private Endpoint required within every Storage Account.

Remediate from PowerShell

```
$storageAccount = Get-AzStorageAccount -ResourceGroupName
'<ResourceGroupName>' -Name '<storageaccountname>'

$privateEndpointConnection = @{
    Name = 'connectionName'
    PrivateLinkServiceId = $storageAccount.Id
    GroupID =
"blob|blob_secondary|file|file_secondary|table|table_secondary|queue|queue_se
condary|web|web_secondary|dfs|dfs_secondary"
}

$privateLinkServiceConnection = New-AzPrivateLinkServiceConnection
@privateEndpointConnection

$virtualNetDetails = Get-AzVirtualNetwork -ResourceGroupName
'<ResourceGroupName>' -Name '<name>'

$privateEndpoint = @{
    ResourceGroupName = '<ResourceGroupName>'
    Name = '<PrivateEndpointName>'
    Location = '<location>'
    Subnet = $virtualNetDetails.Subnets[0]
    PrivateLinkServiceConnection =
$privateLinkServiceConnection
}

New-AzPrivateEndpoint @privateEndpoint
```

Remediate from Azure CLI

```
az network private-endpoint create --resource-group <ResourceGroupName> --
location <location> --name <private endpoint name> --vnet-name <VNET Name> --
subnet <subnet name> --private-connection-resource-id <storage account ID> --
connection-name <private link service connection name> --group-id
<blob|blob_secondary|file|file_secondary|table|table_secondary|queue|queue_se
condary|web|web_secondary|dfs|dfs_secondary>
```

Default Value:

By default, Private Endpoints are not created for Storage Accounts.

References:

1. <https://learn.microsoft.com/en-us/azure/storage/common/storage-private-endpoints>
2. <https://learn.microsoft.com/en-us/azure/virtual-network/virtual-networks-overview>
3. <https://learn.microsoft.com/en-us/azure/private-link/create-private-endpoint-portal>

4. <https://learn.microsoft.com/en-us/azure/private-link/create-private-endpoint-cli>
5. <https://learn.microsoft.com/en-us/azure/private-link/create-private-endpoint-powershell>
6. <https://learn.microsoft.com/en-us/azure/private-link/tutorial-private-endpoint-storage-portal>
7. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-network-security#ns-2-secure-cloud-native-services-with-network-controls>

Additional Information:

A NAT gateway is the recommended solution for outbound internet access.

This recommendation is based on the Common Reference Recommendation **Ensure Private Endpoints are used to access {service}**, from the **Common Reference Recommendations > Networking > Private Endpoints** section.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	12.2 <u>Establish and Maintain a Secure Network Architecture</u> Establish and maintain a secure network architecture. A secure network architecture must address segmentation, least privilege, and availability, at a minimum.			
v7	14.1 <u>Segment the Network Based on Sensitivity</u> Segment the network based on the label or classification level of the information stored on the servers, locate all sensitive information on separated Virtual Local Area Networks (VLANs).			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1537	TA0010	M1037

9.3.2.2 Ensure that 'Public Network Access' is 'Disabled' for Storage Accounts (Automated)

Profile Applicability:

- Level 1

Description:

Disable public network access to prevent exposure to the internet and reduce the risk of unauthorized access. Use private endpoints to securely manage access within trusted networks.

Disallowing public network access for a storage account overrides the public access settings for individual containers in that storage account.

Rationale:

Disabling public network access improves security by ensuring that a storage account is not exposed on the public internet.

The default network configuration for a storage account permits a user with appropriate permissions to configure public network access to containers and blobs in a storage account. Keep in mind that public access to a container is always turned off by default and must be explicitly configured to permit anonymous requests. It grants read-only access to these resources without sharing the account key, and without requiring a shared access signature. It is recommended not to provide public network access to storage accounts until, and unless, it is strongly desired. A shared access signature token or Azure AD RBAC should be used for providing controlled and timed access to blob containers.

Impact:

NOTE: Prior to disabling public network access, it is strongly recommended that, for each storage account, either:

- virtual network integration is completed

OR

- private endpoints/links are set up as described in **"Ensure Private Endpoints are used to access Storage Accounts."**

Disabling public network access restricts direct access to the service. This enhances security but will require the configuration of a virtual network and/or private endpoints for any services or users needing access within trusted networks.

Access will have to be managed using shared access signatures or via Azure AD RBAC.

Audit:

Audit from Azure Portal

1. Go to **Storage Accounts**.
2. For each storage account, under the **Security + networking** section, click **Networking**.
3. Ensure the **Public network access** setting is set to **Disabled**.

Audit from Azure CLI

Ensure **publicNetworkAccess** is **Disabled**

```
az storage account show --name <storage-account> --resource-group <resource-group> --query "{publicNetworkAccess:publicNetworkAccess}"
```

Audit from PowerShell

For each Storage Account, ensure **PublicNetworkAccess** is **Disabled**

```
Get-AzStorageAccount -Name <storage account name> -ResourceGroupName <resource group name> |select PublicNetworkAccess
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~Definitions

- **Policy ID:** [b2982f36-99f2-4db5-8eff-283140c09693](#) - **Name:** 'Storage accounts should disable public network access'

Remediation:

Remediate from Azure Portal

First, follow Microsoft documentation and create shared access signature tokens for your blob containers. Then,

1. Go to **Storage Accounts**.
2. For each storage account, under the **Security + networking** section, click **Networking**.
3. Set **Public network access** to **Disabled**.
4. Click **Save**.

Remediate from Azure CLI

Set 'Public Network Access' to **Disabled** on the storage account

```
az storage account update --name <storage-account> --resource-group  
<resource-group> --public-network-access Disabled
```

Remediate from PowerShell

For each Storage Account, run the following to set the **PublicNetworkAccess** setting to **Disabled**

```
Set-AzStorageAccount -ResourceGroupName <resource group name> -Name <storage  
account name> -PublicNetworkAccess Disabled
```

Default Value:

By default, **Public Network Access** is set to **Enabled from all networks** for the Storage Account.

References:

1. <https://learn.microsoft.com/en-us/azure/storage/blobs/anonymous-read-access-configure>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-governance-strategy#gs-2-define-and-implement-enterprise-segmentationseparation-of-duties-strategy>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-network-security#ns-2-secure-cloud-native-services-with-network-controls>
4. <https://learn.microsoft.com/en-us/azure/storage/blobs/assign-azure-role-data-access>
5. <https://learn.microsoft.com/en-us/azure/storage/common/storage-network-security>

Additional Information:

This recommendation is based on the Common Reference Recommendation **Ensure public network access is Disabled**, from the **Common Reference Recommendations > Networking > Virtual Networks (VNETs)** section.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1530	TA0009	M1037

9.3.2.3 Ensure Default Network Access Rule for Storage Accounts is Set to Deny (Automated)

Profile Applicability:

- Level 1

Description:

Restricting default network access helps to provide a new layer of security, since storage accounts accept connections from clients on any network. To limit access to selected networks, the default action must be changed.

Rationale:

NOTE: This recommendation is only applicable if Public Network Access has not been disabled due to necessity or exception.

Storage accounts should be configured to deny access to traffic from all networks (including internet traffic). Access can be granted to traffic from specific Azure Virtual networks, allowing a secure network boundary for specific applications to be built. Access can also be granted to public internet IP address ranges to enable connections from specific internet or on-premises clients. When network rules are configured, only applications from allowed networks can access a storage account. When calling from an allowed network, applications continue to require proper authorization (a valid access key or SAS token) to access the storage account.

Impact:

All allowed networks will need to be whitelisted on each specific network, creating administrative overhead. This may result in loss of network connectivity, so do not turn on for critical resources during business hours.

Audit:

Audit from Azure Portal

1. Go to Storage Accounts.
2. For each storage account, under **Security + networking**, click **Networking**.
3. Click the **Firewalls and virtual networks** heading.
4. Ensure that **Public network access** is not set to **Enabled from all networks**.

Audit from Azure CLI

Ensure **defaultAction** is not set to **Allow**.

```
az storage account list --query '[*].networkRuleSet'
```

Audit from PowerShell

```
Connect-AzAccount
Set-AzContext -Subscription <subscription ID>
Get-AzStorageAccountNetworkRuleset -ResourceGroupName <resource group> -Name
<storage account name> | Select-Object DefaultAction
```

PowerShell Result - Non-Compliant

```
DefaultAction      : Allow
```

PowerShell Result - Compliant

```
DefaultAction      : Deny
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [34c877ad-507e-4c82-993e-3452a6e0ad3c](#) - **Name:** 'Storage accounts should restrict network access'
- **Policy ID:** [2a1a9cdf-e04d-429a-8416-3bfb72a1b26f](#) - **Name:** 'Storage accounts should restrict network access using virtual network rules'

Remediation:

Remediate from Azure Portal

1. Go to **Storage Accounts**.
2. For each storage account, under **Security + networking**, click **Networking**.
3. Click the **Firewalls and virtual networks** heading.
4. Set **Public network access** to **Enabled from selected virtual networks and IP addresses**.
5. Add rules to allow traffic from specific networks and IP addresses.
6. Click **Save**.

Remediate from Azure CLI

Use the below command to update **default-action** to **Deny**.

```
az storage account update --name <StorageAccountName> --resource-group
<resourceGroupName> --default-action Deny
```

Default Value:

By default, Storage Accounts will accept connections from clients on any network.

References:

1. <https://learn.microsoft.com/en-us/azure/storage/common/storage-network-security>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-governance-strategy#gs-2-define-and-implement-enterprise-segmentationseparation-of-duties-strategy>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-network-security#ns-2-secure-cloud-native-services-with-network-controls>

Additional Information:

This recommendation is based on the Common Reference Recommendation **Ensure Network Access Rules are set to Deny-by-default**, from the **Common Reference Recommendations > Networking > Virtual Networks (VNETs)** section.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	12.2 Establish and Maintain a Secure Network Architecture Establish and maintain a secure network architecture. A secure network architecture must address segmentation, least privilege, and availability, at a minimum.			
v7	13.3 Monitor and Block Unauthorized Network Traffic Deploy an automated tool on network perimeters that monitors for unauthorized transfer of sensitive information and blocks such transfers while alerting information security professionals.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1530	TA0009	M1037

9.3.3 Identity and Access Management

9.3.3.1 Ensure that 'Default to Microsoft Entra authorization in the Azure portal' is Set to 'Enabled' (Automated)

Profile Applicability:

- Level 1

Description:

When this property is enabled, the Azure portal authorizes requests to blobs, files, queues, and tables with Microsoft Entra ID by default.

Rationale:

Microsoft Entra ID provides superior security and ease of use over Shared Key.

Audit:

Audit from Azure Portal

1. Go to **Storage accounts**.
2. Click the name of a storage account.
3. Under **Settings**, click **Configuration**.
4. Ensure that **Default to Microsoft Entra authorization in the Azure portal** is set to **Enabled**.
5. Repeat steps 1-4 for each storage account.

Audit from Azure CLI

Run the following command to get the **name** and **defaultToOAuthAuthentication** setting for each storage account:

```
az storage account list --query [*].[name,defaultToOAuthAuthentication]
```

Ensure that **true** is returned for each storage account.

Remediation:

Remediate from Azure Portal

1. Go to **Storage accounts**.
2. Click the name of a storage account.
3. Under **Settings**, click **Configuration**.
4. Under **Default to Microsoft Entra authorization in the Azure portal**, click the radio button next to **Enabled**.
5. Click **Save**.
6. Repeat steps 1-5 for each storage account requiring remediation.

Remediate from Azure CLI

For each storage account requiring remediation, run the following command to enable **defaultToOAuthAuthentication**:

```
az storage account update --resource-group <resource-group> --name <storage-account> --set defaultToOAuthAuthentication=true
```

Default Value:

By default, **defaultToOAuthAuthentication** is disabled.

References:

1. <https://learn.microsoft.com/en-us/azure/storage/blobs/authorize-data-operations-portal#default-to-microsoft-entra-authorization-in-the-azure-portal>
2. <https://learn.microsoft.com/en-us/cli/azure/storage/account>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1530	TA0009	M1022

9.3.4 Ensure that 'Secure transfer required' is Set to 'Enabled' (Automated)

Profile Applicability:

- Level 1

Description:

Enable data encryption in transit.

Rationale:

The secure transfer option enhances the security of a storage account by only allowing requests to the storage account by a secure connection. For example, when calling REST APIs to access storage accounts, the connection must use HTTPS. Any requests using HTTP will be rejected when 'secure transfer required' is enabled. When using the Azure files service, connection without encryption will fail, including scenarios using SMB 2.1, SMB 3.0 without encryption, and some flavors of the Linux SMB client. Because Azure storage doesn't support HTTPS for custom domain names, this option is not applied when using a custom domain name.

Audit:

Audit from Azure Portal

1. Go to **Storage Accounts**.
2. For each storage account, under **Settings**, click **Configuration**.
3. Ensure that **Secure transfer required** is set to **Enabled**.

Audit from Azure CLI

Use the below command to ensure the **Secure transfer required** is enabled for all the **Storage Accounts** by ensuring the output contains **true** for each of the **Storage Accounts**.

```
az storage account list --query "[*].[name,enableHttpsTrafficOnly]"
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [404c3081-a854-4457-ae30-26a93ef643f9](#) - **Name:** 'Secure transfer to storage accounts should be enabled'

Remediation:

Remediate from Azure Portal

1. Go to **Storage Accounts**.
2. For each storage account, under **Settings**, click **Configuration**.
3. Set **Secure transfer required** to **Enabled**.
4. Click **Save**.

Remediate from Azure CLI

Use the below command to enable **Secure transfer required** for a **Storage Account**

```
az storage account update --name <storageAccountName> --resource-group <resourceGroupName> --https-only true
```

Default Value:

By default, **Secure transfer required** is set to **Disabled**.

References:

1. <https://learn.microsoft.com/en-us/azure/storage/blobs/security-recommendations>
2. <https://learn.microsoft.com/en-us/cli/azure/storage/account>
3. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-data-protection#dp-3-encrypt-sensitive-data-in-transit>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.10 <u>Encrypt Sensitive Data in Transit</u> Encrypt sensitive data in transit. Example implementations can include: Transport Layer Security (TLS) and Open Secure Shell (OpenSSH).			
v7	14.4 <u>Encrypt All Sensitive Information in Transit</u> Encrypt all sensitive information in transit.			

9.3.5 Ensure 'Allow trusted Microsoft services to access this resource' is Enabled for Storage Account Access (Automated)

Profile Applicability:

- Level 2

Description:

NOTE: This recommendation assumes that the **Public network access** parameter is set to **Enabled from selected virtual networks and IP addresses**. Please ensure the prerequisite recommendation has been implemented before proceeding:

- Ensure Default Network Access Rule for Storage Accounts is Set to Deny

If the **Allow trusted Microsoft services to access this resource** exception is enabled, the following services are granted access to the storage account:

- Azure Backup,
- Azure Data Box,
- Azure DevTest Labs,
- Azure Event Grid,
- Azure Event Hubs,
- Azure File Sync,
- Azure HDInsight,
- Azure Import/Export,
- Azure Monitor,
- Azure Networking Services, and
- Azure Site Recovery (when registered in the subscription).

Rationale:

NOTE: If none of the services listed in the Description are in use in your environment, you may wish to make an exception to this recommendation and disable 'Allow trusted Microsoft services to access this resource.'

Turning on firewall rules for a storage account will block access to incoming requests for data, including from other Azure services.

Some Azure services that interact with storage accounts operate from networks that can't be granted access through network rules. To help this type of service work as intended, allow the set of trusted Azure services to bypass the network rules. These services will then use strong authentication to access the storage account.

Impact:

This creates authentication credentials for services that need access to storage resources so that services will no longer need to communicate via network request. There may be a temporary loss of communication as you set each Storage Account. It is recommended to not do this on mission-critical resources during business hours.

Audit:

Audit from Azure Portal

1. Go to **Storage Accounts**.
2. For each storage account, under **Security + networking**, click **Networking**.
3. Under the **Public access** tab, look for the **Resource settings: Virtual networks, IP Addresses, and exceptions** heading, then click the **View** button.
4. In the **Resource Access settings** pane, review the **Exceptions** header and ensure that **Allow trusted Microsoft services to access this resource** is checked.

Audit from Azure CLI

Ensure **bypass** contains **AzureServices**

```
az storage account list --query '[*].networkRuleSet'
```

Audit from PowerShell

```
Connect-AzAccount  
Set-AzContext -Subscription <subscription ID>  
Get-AzStorageAccountNetworkRuleSet -ResourceGroupName <resource group> -Name  
<storage account name> |Select-Object Bypass
```

If the response from the above command is **None**, the storage account configuration is out of compliance with this check. If the response is **AzureServices**, the storage account configuration is in compliance with this check.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~/Definitions

- **Policy ID:** [c9d007d0-c057-4772-b18c-01e546713bcd](#) - **Name:** 'Storage accounts should allow access from trusted Microsoft services'

Remediation:

Remediate from Azure Portal

1. Go to **Storage Accounts**.
2. For each storage account, under **Security + networking**, click **Networking**.
3. Under the **Public access** tab, look for the **Resource settings: Virtual networks, IP Addresses, and exceptions** heading, then click the **View** button.
4. In the **Resource Access settings** pane, go to the **Exceptions** header and check the box next to **Allow trusted Microsoft services to access this resource**.
5. Click **Save**.

Remediate from Azure CLI

Use the below command to update **bypass** to **Azure services**.

```
az storage account update --name <StorageAccountName> --resource-group <resourceGroupName> --bypass AzureServices
```

Default Value:

By default, Storage Accounts will accept connections from clients on any network.

References:

1. <https://learn.microsoft.com/en-us/azure/storage/common/storage-network-security>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-network-security#ns-2-secure-cloud-native-services-with-network-controls>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 Configure Data Access Control Lists Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v8	13.5 Manage Access Control for Remote Assets Manage access control for assets remotely connecting to enterprise resources. Determine amount of access to enterprise resources based on: up-to-date anti-malware software installed, configuration compliance with the enterprise's secure configuration process, and ensuring the operating system and applications are up-to-date.			

Controls Version	Control	IG 1	IG 2	IG 3
v7	13.3 Monitor and Block Unauthorized Network Traffic Deploy an automated tool on network perimeters that monitors for unauthorized transfer of sensitive information and blocks such transfers while alerting information security professionals.			●

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1530	TA0009	M1037

9.3.6 Ensure the 'Minimum TLS version' for Storage Accounts is Set to 'Version 1.2' (Automated)

Profile Applicability:

- Level 1

Description:

In some cases, Azure Storage sets the minimum TLS version to be version 1.0 by default. TLS 1.0 is a legacy version and has known vulnerabilities. This minimum TLS version can be configured to be later protocols such as TLS 1.2.

Rationale:

TLS 1.0 has known vulnerabilities and has been replaced by later versions of the TLS protocol. Continued use of this legacy protocol affects the security of data in transit.

Impact:

When set to TLS 1.2 all requests must leverage this version of the protocol. Applications leveraging legacy versions of the protocol will fail.

Audit:

Audit from Azure Portal

1. Go to **Storage Accounts**.
2. For each storage account, under **Settings**, click **Configuration**.
3. Ensure that the **Minimum TLS version** is set to **Version 1.2**.

Audit from Azure CLI

Get a list of all storage accounts and their resource groups

```
az storage account list | jq '.[ ] | {name, resourceGroup}'
```

Then query the minimumTLSVersion field

```
az storage account show \
  --name <storage-account> \
  --resource-group <resource-group> \
  --query minimumTlsVersion \
  --output tsv
```

Audit from PowerShell

To get the minimum TLS version, run the following command:

```
(Get-AzStorageAccount -Name <STORAGEACCOUNTNAME> -ResourceGroupName  
<RESOURCEGROUPNAME>).MinimumTlsVersion
```

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [fe83a0eb-a853-422d-aac2-1bffd182c5d0](#) - **Name:** 'Storage accounts should have the specified minimum TLS version'

Remediation:

Remediate from Azure Portal

1. Go to **Storage Accounts**.
2. For each storage account, under **Settings**, click **Configuration**.
3. Set the **Minimum TLS version** to **Version 1.2**.
4. Click **Save**.

Remediate from Azure CLI

```
az storage account update \  
  --name <storage-account> \  
  --resource-group <resource-group> \  
  --min-tls-version TLS1_2
```

Remediate from PowerShell

To set the minimum TLS version, run the following command:

```
Set-AzStorageAccount -AccountName <STORAGEACCOUNTNAME> \  
  -ResourceGroupName <RESOURCEGROUPNAME> \  
  -MinimumTlsVersion TLS1_2
```

Default Value:

If a storage account is created through the portal, the MinimumTlsVersion property for that storage account will be set to TLS 1.2.

If a storage account is created through PowerShell or CLI, the MinimumTlsVersion property for that storage account will not be set, and defaults to TLS 1.0.

References:

1. <https://learn.microsoft.com/en-us/azure/storage/common/transport-layer-security-configure-minimum-version>
2. <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-data-protection#dp-3-encrypt-sensitive-data-in-transit>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.10 <u>Encrypt Sensitive Data in Transit</u> Encrypt sensitive data in transit. Example implementations can include: Transport Layer Security (TLS) and Open Secure Shell (OpenSSH).			
v7	14.4 <u>Encrypt All Sensitive Information in Transit</u> Encrypt all sensitive information in transit.			

9.3.7 Ensure 'Cross Tenant Replication' is Not Enabled (Automated)

Profile Applicability:

- Level 1

Description:

Cross Tenant Replication in Azure allows data to be replicated across multiple Azure tenants. While this feature can be beneficial for data sharing and availability, it also poses a significant security risk if not properly managed. Unauthorized data access, data leakage, and compliance violations are potential risks. Disabling Cross Tenant Replication ensures that data is not inadvertently replicated across different tenant boundaries without explicit authorization.

Rationale:

Disabling Cross Tenant Replication minimizes the risk of unauthorized data access and ensures that data governance policies are strictly adhered to. This control is especially critical for organizations with stringent data security and privacy requirements, as it prevents the accidental sharing of sensitive information.

Impact:

Disabling Cross Tenant Replication may affect data availability and sharing across different Azure tenants. Ensure that this change aligns with your organizational data sharing and availability requirements.

Audit:

Audit from Azure Portal

1. Go to **Storage Accounts**.
2. For each storage account, under **Data management**, click **Object replication**.
3. Click **Advanced settings**.
4. Ensure **Allow cross-tenant replication** is not checked.

Audit from Azure CLI

```
az storage account list --query "[*].[name,allowCrossTenantReplication]"
```

The value of **false** should be returned for each storage account listed.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [92a89a79-6c52-4a7e-a03f-61306fc49312](#) - **Name:** 'Storage accounts should prevent cross tenant object replication'

Remediation:

Remediate from Azure Portal

1. Go to **Storage Accounts**.
2. For each storage account, under **Data management**, click **Object replication**.
3. Click **Advanced settings**.
4. Uncheck **Allow cross-tenant replication**.
5. Click **OK**.

Remediate from Azure CLI

Replace the information within <> with appropriate values:

```
az storage account update --name <storageAccountName> --resource-group  
<resourceGroupName> --allow-cross-tenant-replication false
```

Default Value:

For new storage accounts created after Dec 15, 2023 cross tenant replication is not enabled.

References:

1. <https://learn.microsoft.com/en-us/azure/storage/blobs/object-replication-prevent-cross-tenant-policies>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	13.4 <u>Only Allow Access to Authorized Cloud Storage or Email Providers</u> Only allow access to authorized cloud storage or email providers.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1565.001	TA1020	M1022

9.3.8 Ensure that 'Allow Blob Anonymous Access' is Set to 'Disabled' (Automated)

Profile Applicability:

- Level 1

Description:

The Azure Storage setting 'Allow Blob Anonymous Access' (aka "allowBlobPublicAccess") controls whether anonymous access is allowed for blob data in a storage account. When this property is set to True, it enables public read access to blob data, which can be convenient for sharing data but may carry security risks. When set to False, it disallows public access to blob data, providing a more secure storage environment.

Rationale:

If "Allow Blob Anonymous Access" is enabled, blobs can be accessed by adding the blob name to the URL to see the contents. An attacker can enumerate a blob using methods, such as brute force, and access them.

Exfiltration of data by brute force enumeration of items from a storage account may occur if this setting is set to 'Enabled'.

Impact:

Additional consideration may be required for exceptional circumstances where elements of a storage account require public accessibility. In these circumstances, it is highly recommended that all data stored in the public facing storage account be reviewed for sensitive or potentially compromising data, and that sensitive or compromising data is never stored in these storage accounts.

Audit:

Audit from Azure Portal

1. Go to **Storage Accounts**.
2. For each storage account, under **Settings**, click **Configuration**.
3. Ensure **Allow Blob Anonymous Access** is set to **Disabled**.

Audit from Azure CLI

For every storage account in scope:

```
az storage account show --name "<yourStorageAccountName>" --query allowBlobPublicAccess
```

Ensure that every storage account in scope returns **false** for the "allowBlobPublicAccess" setting.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~./Definitions

- **Policy ID:** [4fa4b6c0-31ca-4c0d-b10d-24b96f62a751](#) - **Name:** 'Storage account public access should be disallowed'

Remediation:

Remediate from Azure Portal

1. Go to **Storage Accounts**.
2. For each storage account, under **Settings**, click **Configuration**.
3. Set **Allow Blob Anonymous Access** to **Disabled**.
4. Click **Save**.

Remediate from Powershell

For every storage account in scope, run the following:

```
$storageAccount = Get-AzStorageAccount -ResourceGroupName  
"<yourResourceGroup>" -Name "<yourStorageAccountName>"  
$storageAccount.AllowBlobPublicAccess = $false  
Set-AzStorageAccount -InputObject $storageAccount
```

Default Value:

Disabled

References:

1. <https://learn.microsoft.com/en-us/azure/storage/blobs/anonymous-read-access-prevent?tabs=portal>
2. <https://learn.microsoft.com/en-us/azure/storage/blobs/anonymous-read-access-prevent?source=recommendations&tabs=portal>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 <u>Configure Data Access Control Lists</u> Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.			
v7	14.6 <u>Protect Information through Access Control Lists</u> Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1567	TA0010	M1057

9.3.9 Ensure Azure Resource Manager Delete Locks are Applied to Azure Storage Accounts (Manual)

Profile Applicability:

- Level 1

Description:

Azure Resource Manager *CannotDelete (Delete)* locks can prevent users from accidentally or maliciously deleting a storage account. This feature ensures that while the Storage account can still be modified or used, deletion of the Storage account resource requires removal of the lock by a user with appropriate permissions.

This feature is a protective control for the availability of data. By ensuring that a storage account or its parent resource group cannot be deleted without first removing the lock, the risk of data loss is reduced.

While an automated assessment procedure exists for this recommendation, the assessment status remains manual. Determining storage accounts that require CannotDelete locks depends on the context and requirements of each organization and environment.

Rationale:

Applying a *Delete* lock on storage accounts protects the availability of data by preventing the accidental or unauthorized deletion of the entire storage account. It is a fundamental protective control that can prevent data loss

Impact:

- Prevents the deletion of the Storage account Resource entirely.
- Prevents the deletion of the parent Resource Group containing the locked Storage account resource.
- Does not prevent other control plane operations, including modification of configurations, network settings, containers, and access.
- Does not prevent deletion of containers or other objects within the storage account.

Audit:

Audit from Azure Portal

1. Navigate to the storage account in the Azure portal.
2. For each storage account, under **Settings**, click **Locks**.
3. Ensure that a **Delete** lock exists on the storage account.

Audit from Azure CLI

```
az lock list --resource-group <resource-group> \  
             --resource-name <storage-account> \  
             --resource-type "Microsoft.Storage/storageAccounts"
```

Audit from PowerShell

```
Get-AzResourceLock -ResourceGroupName <RESOURCEGROUPNAME> \  
                  -ResourceName <STORAGEACCOUNTNAME> \  
                  -ResourceType "Microsoft.Storage/storageAccounts"
```

Audit from Azure Policy

There is currently no built-in Microsoft policy to audit resource locks on storage accounts. Custom and community policy definitions can check for the existence of a “Microsoft.Authorization/locks” resource with an AuditIfNotExists effect.

Remediation:

Remediate from Azure Portal

1. Navigate to the storage account in the Azure portal.
2. Under the **Settings** section, select **Locks**.
3. Select **Add**.
4. Provide a Name, and choose **Delete** for the type of lock.
5. Add a note about the lock if desired.

Remediate from Azure CLI

Replace the information within <> with appropriate values:

```
az lock create --name <lock> \  
              --resource-group <resource-group> \  
              --resource <storage-account> \  
              --lock-type CanNotDelete \  
              --resource-type Microsoft.Storage/storageAccounts
```

Remediate from PowerShell

Replace the information within <> with appropriate values:

```
New-AzResourceLock -LockLevel CanNotDelete \  
                  -LockName <lock> \  
                  -ResourceName <storage-account> \  
                  -ResourceType Microsoft.Storage/storageAccounts \  
                  -ResourceGroupName <resource-group>
```

Default Value:

By default, no locks are applied to Azure resources, including storage accounts. Locks must be manually configured after resource creation.

References:

1. <https://learn.microsoft.com/en-us/azure/storage/common/lock-account-resource>
2. <https://learn.microsoft.com/en-us/azure/azure-resource-manager/management/lock-resources>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3 Data Protection Develop processes and technical controls to identify, classify, securely handle, retain, and dispose of data.			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1485, T1490	TA0040	

9.3.10 Ensure Azure Resource Manager ReadOnly Locks are Considered for Azure Storage Accounts (Manual)

Profile Applicability:

- Level 2

Description:

Adding an Azure Resource Manager **ReadOnly** lock can prevent users from accidentally or maliciously deleting a storage account, modifying its properties and containers, or creating access assignments. The lock must be removed before the storage account can be deleted or updated. It provides more protection than a **CannotDelete**-type of resource manager lock.

This feature prevents **POST** operations on a storage account and containers to the Azure Resource Manager control plane, *management.azure.com*. Blocked operations include **listKeys** which prevents clients from obtaining the account shared access keys.

Microsoft does not recommend **ReadOnly** locks for storage accounts with Azure Files and Table service containers.

This Azure Resource Manager REST API documentation (spec) provides information about the control plane **POST** operations for *Microsoft.Storage* resources.

While an automated assessment procedure exists for this recommendation, the assessment status remains manual. Determining storage accounts that require ReadOnly locks depends on the context and requirements of each organization and environment.

Rationale:

Applying a **ReadOnly** lock on storage accounts protects the confidentiality and availability of data by preventing the accidental or unauthorized deletion of the entire storage account and modification of the account, container properties, or access permissions. It can offer enhanced protection for blob and queue workloads with tradeoffs in usability and compatibility for clients using account shared access keys.

Impact:

- Prevents the deletion of the Storage account Resource entirely.
- Prevents the deletion of the parent Resource Group containing the locked Storage account resource.
- Prevents clients from obtaining the storage account shared access keys using a **listKeys** operation.
- Requires Entra credentials to access blob and queue data in the Portal.
- Data in Azure Files or the Table service may be inaccessible to clients using the account shared access keys.

- Prevents modification of account properties, network settings, containers, and RBAC assignments.
- Does not prevent access using existing account shared access keys issued to clients.
- Does not prevent deletion of containers or other objects within the storage account.

Audit:

Audit from Azure Portal

1. Navigate to the storage account in the Azure portal.
2. For each storage account, under **Settings**, click **Locks**.
3. Ensure that a **ReadOnly** lock exists on the storage account.

Audit from Azure CLI

```
az lock list --resource-group <resource-group> \
             --resource-name <storage-account> \
             --resource-type "Microsoft.Storage/storageAccounts"
```

Audit from PowerShell

```
Get-AzResourceLock -ResourceGroupName <RESOURCEGROUPNAME> `
                  -ResourceName <STORAGEACCOUNTNAME> `
                  -ResourceType "Microsoft.Storage/storageAccounts"
```

Audit from Azure Policy

There is currently no built-in Microsoft policy to audit resource locks on storage accounts. Custom and community policy definitions can check for the existence of a “Microsoft.Authorization/locks” resource with an AuditIfNotExists effect.

Remediation:

Remediate from Azure Portal

1. Navigate to the storage account in the Azure portal.
2. Under the **Settings** section, select **Locks**.
3. Select **Add**.
4. Provide a Name, and choose **ReadOnly** for the type of lock.
5. Add a note about the lock if desired.

Remediate from Azure CLI

Replace the information within <> with appropriate values:

```
az lock create --name <lock> \
  --resource-group <resource-group> \
  --resource <storage-account> \
  --lock-type ReadOnly \
  --resource-type Microsoft.Storage/storageAccounts
```

Remediate from PowerShell

Replace the information within <> with appropriate values:

```
New-AzResourceLock -LockLevel ReadOnly `
  -LockName <lock> `
  -ResourceName <storage-account> `
  -ResourceType Microsoft.Storage/storageAccounts `
  -ResourceGroupName <resource-group>
```

Default Value:

By default, no locks are applied to Azure resources, including storage accounts. Locks must be manually configured after resource creation.

References:

1. <https://learn.microsoft.com/en-us/azure/storage/common/lock-account-resource>
2. <https://learn.microsoft.com/en-us/azure/azure-resource-manager/management/lock-resources>
3. <https://github.com/Azure/azure-rest-api-specs/tree/main/specification/storage>

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3 Data Protection Develop processes and technical controls to identify, classify, securely handle, retain, and dispose of data.			

9.3.11 Ensure Redundancy is Set to 'geo-redundant storage (GRS)' on Critical Azure Storage Accounts (Automated)

Profile Applicability:

- Level 2

Description:

Geo-redundant storage (GRS) in Azure replicates data three times within the primary region using locally redundant storage (LRS) and asynchronously copies it to a secondary region hundreds of miles away. This setup ensures high availability and resilience by providing 16 nines (99.99999999999999%) durability over a year, safeguarding data against regional outages.

Rationale:

Enabling GRS protects critical data from regional failures by maintaining a copy in a geographically separate location. This significantly reduces the risk of data loss, supports business continuity, and meets high availability requirements for disaster recovery.

NOTE: In recent discussions the CIS Microsoft Azure Community has discussed revising this recommendation to ensure other geographically redundant storage options are considered. While the recommendation remains unchanged in this publication, other Geographically redundant settings (RA-GRS, GZRS, RA-GZRS) should be considered legitimate alternatives.

Impact:

Enabling geo-redundant storage on Azure storage accounts increases costs due to cross-region data replication.

Audit:

Audit from Azure Portal

1. Go to **Storage accounts**.
2. Click on a storage account.
3. Under **Data management**, click **Redundancy**.
4. Ensure that **Redundancy** is set to **Geo-redundant storage (GRS)**.
5. Repeat steps 1-4 for each storage account.

Audit from Azure CLI

Run the following command to list storage accounts:

```
az storage account list
```

For each storage account, run the following command:

```
az storage account show --resource-group <resource-group> --name <storage-account>
```

Under **sku**, ensure that **name** is set to **Standard_GRS**.

Audit from PowerShell

Run the following command to list storage accounts:

```
Get-AzStorageAccount
```

Run the following command to get the storage account in a resource group with a given name:

```
$storageAccount = Get-AzStorageAccount -ResourceGroupName <resource-group> -Name <storage-account>
```

Run the following command to get the redundancy setting for the storage account:

```
$storageAccount.SKU.Name
```

Ensure that the command returns **Standard_GRS**.

Repeat for each storage account.

Audit from Azure Policy

If referencing a digital copy of this Benchmark, clicking a Policy ID will open a link to the associated Policy definition in Azure.

If referencing a printed copy, you can search Policy IDs from this URL:

https://portal.azure.com/#view/Microsoft_Azure_Policy/PolicyMenuBlade/~Definitions

- **Policy ID:** [bf045164-79ba-4215-8f95-f8048dc1780b](#) - **Name:** 'Geo-redundant storage should be enabled for Storage Accounts'

Remediation:

Remediate from Azure Portal

1. Go to **Storage accounts**.
2. Click on a storage account.
3. Under **Data management**, click **Redundancy**.
4. From the **Redundancy** drop-down menu, select **Geo-redundant storage (GRS)**.
5. Click **Save**.
6. Repeat steps 1-5 for each storage account requiring remediation.

Remediate from Azure CLI

For each storage account requiring remediation, run the following command to enable geo-redundant storage:

```
az storage account update --resource-group <resource-group> --name <storage-account> --sku Standard_GRS
```

Remediate from PowerShell

For each storage account requiring remediation, run the following command to enable geo-redundant storage:

```
Set-AzStorageAccount -ResourceGroupName <resource-group> -Name <storage-account> -SkuName "Standard_GRS"
```

Default Value:

When creating a storage account in the Azure Portal, the default redundancy setting is geo-redundant storage (GRS). Using the Azure CLI, the default is read-access geo-redundant storage (RA-GRS). In PowerShell, a redundancy level must be explicitly specified during account creation.

References:

1. <https://learn.microsoft.com/en-us/azure/storage/common/storage-redundancy>
2. <https://learn.microsoft.com/en-us/azure/storage/common/redundancy-migration>
3. <https://learn.microsoft.com/en-us/cli/azure/storage/account?view=azure-cli-latest#az-storage-account-update>
4. <https://learn.microsoft.com/en-us/powershell/module/az.storage/set-azstorageaccount>
5. <https://learn.microsoft.com/en-us/azure/storage/common/storage-disaster-recovery-guidance>

Additional Information:

When choosing the best redundancy option, weigh the trade-offs between lower costs and higher availability. Key factors to consider include:

- The method of data replication within the primary region.
- The replication of data from a primary to a geographically distant secondary region for protection against regional disasters (geo-replication).
- The necessity for read access to replicated data in the secondary region during an outage in the primary region (geo-replication with read access).

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	11.1 <u>Establish and Maintain a Data Recovery Process</u> Establish and maintain a data recovery process. In the process, address the scope of data recovery activities, recovery prioritization, and the security of backup data. Review and update documentation annually, or when significant enterprise changes occur that could impact this Safeguard.			
v7	10 <u>Data Recovery Capabilities</u> Data Recovery Capabilities			

MITRE ATT&CK Mappings:

Techniques / Sub-techniques	Tactics	Mitigations
T1485	TA0040	M1053

Appendix: Summary Table

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
1	Introduction		
1.1	CIS Microsoft Azure Foundations Benchmarks		
1.2	CIS Microsoft Azure Service Category Benchmarks		
1.3	Multiple Methods of Audit and Remediation		
1.4	Automating with Azure Policy		
2	Analytics Services		
2.1	Azure Databricks		
2.1.1	Ensure that Azure Databricks is deployed in a customer-managed virtual network (VNet) (Automated)	☐	☐
2.1.2	Ensure that Network Security Groups are Configured for Databricks Subnets (Automated)	☐	☐
2.1.3	Ensure that Traffic is Encrypted Between Cluster Worker Nodes (Manual)	☐	☐
2.1.4	Ensure that Users and Groups are Synced from Microsoft Entra ID to Azure Databricks (Manual)	☐	☐
2.1.5	Ensure that Unity Catalog is Configured for Azure Databricks (Manual)	☐	☐
2.1.6	Ensure that Usage is Restricted and Expiry is Enforced for Databricks Personal Access Tokens (Manual)	☐	☐
2.1.7	Ensure that Diagnostic Log Delivery is Configured for Azure Databricks (Automated)	☐	☐
2.1.8	Ensure Critical Data in Azure Databricks is Encrypted with Customer-managed Keys (CMK) (Manual)	☐	☐
2.1.9	Ensure 'No Public IP' is Set to 'Enabled' (Automated)	☐	☐

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
2.1.10	Ensure 'Allow Public Network Access' is set to 'Disabled' (Automated)	☐	☐
2.1.11	Ensure Private Endpoints are used to access Azure Databricks workspaces (Automated)	☐	☐
2.1.12	Ensure Azure Databricks groups are reviewed periodically (Manual)	☐	☐
3	Compute Services		
3.1	Virtual Machines		
3.1.1	Ensure only MFA Enabled Identities can Access Privileged Virtual Machine (Manual)	☐	☐
4	Database Services (reference)		
5	Identity Services		
5.1	Security Defaults (Per-User MFA)		
5.1.1	Ensure that 'security defaults' is Enabled in Microsoft Entra ID (Automated)	☐	☐
5.1.2	Ensure that 'Require Multifactor Authentication to register or join devices with Microsoft Entra' is set to 'Yes' (Manual)	☐	☐
5.1.3	Ensure that 'multifactor authentication' is 'enabled' For All Users (Automated)	☐	☐
5.1.4	Ensure that 'Allow users to remember multifactor authentication on devices they trust' is Disabled (Manual)	☐	☐
5.2	Conditional Access		
5.3	Periodic Identity Reviews		
5.3.1	Ensure that Azure Admin Accounts Are Not Used for Daily Operations (Manual)	☐	☐

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
5.3.2	Ensure that Guest Users are Reviewed on a Regular Basis (Manual)	☐	☐
5.3.3	Ensure That Use of the 'User Access Administrator' Role is Restricted (Automated)	☐	☐
5.3.4	Ensure that All 'Privileged' Role Assignments are Periodically Reviewed (Manual)	☐	☐
5.3.5	Ensure Disabled User Accounts do not Have Read, Write, or Owner Permissions (Manual)	☐	☐
5.3.6	Ensure 'Tenant Creator' Role Assignments are Periodically Reviewed (Manual)	☐	☐
5.3.7	Ensure All Non-privileged Role Assignments are Periodically Reviewed (Manual)	☐	☐
5.4	Ensure that No Custom Subscription Administrator Roles Exist (Automated)	☐	☐
5.5	Ensure that a Custom Role is Assigned Permissions for Administering Resource Locks (Manual)	☐	☐
5.6	Ensure that 'Subscription leaving Microsoft Entra tenant' and 'Subscription entering Microsoft Entra tenant' is set to 'Permit no one' (Manual)	☐	☐
5.7	Ensure there are between 2 and 3 Subscription Owners (Automated)	☐	☐
6	Management and Governance Services		
6.1	Logging and Monitoring		
6.1.1	Configuring Diagnostic Settings		
6.1.1.1	Ensure that a 'Diagnostic Setting' Exists for Subscription Activity Logs (Automated)	☐	☐
6.1.1.2	Ensure Diagnostic Setting Captures Appropriate Categories (Automated)	☐	☐

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
6.1.1.3	Ensure the Storage Account Containing the Container with Activity Logs is Encrypted with Customer-managed Key (CMK) (Manual)	☐	☐
6.1.1.4	Ensure that Logging for Azure Key Vault is 'Enabled' (Automated)	☐	☐
6.1.1.5	Ensure that Network Security Group Flow Logs are Captured and Sent to Log Analytics (Manual)	☐	☐
6.1.1.6	Ensure that Virtual Network Flow Logs are Captured and Sent to Log Analytics (Manual)	☐	☐
6.1.1.7	Ensure that a Microsoft Entra Diagnostic Setting Exists to Send Microsoft Graph Activity Logs to an Appropriate Destination (Manual)	☐	☐
6.1.1.8	Ensure that a Microsoft Entra Diagnostic Setting Exists to Send Microsoft Entra Activity Logs to an Appropriate Destination (Manual)	☐	☐
6.1.1.9	Ensure that Intune Logs are Captured and Sent to Log Analytics (Manual)	☐	☐
6.1.2	Monitoring Using Activity Log Alerts		
6.1.2.1	Ensure that Activity Log Alert Exists for Create Policy Assignment (Automated)	☐	☐
6.1.2.2	Ensure that Activity Log Alert exists for Delete Policy Assignment (Automated)	☐	☐
6.1.2.3	Ensure that Activity Log Alert Exists for Create or Update Network Security Group (Automated)	☐	☐
6.1.2.4	Ensure that Activity Log Alert Exists for Delete Network Security Group (Automated)	☐	☐
6.1.2.5	Ensure that Activity Log Alert Exists for Create or Update Security Solution (Automated)	☐	☐

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
6.1.2.6	Ensure that Activity Log Alert Exists for Delete Security Solution (Automated)	☐	☐
6.1.2.7	Ensure that Activity Log Alert Exists for Create or Update SQL Server Firewall Rule (Automated)	☐	☐
6.1.2.8	Ensure that Activity Log Alert Exists for Delete SQL Server Firewall Rule (Automated)	☐	☐
6.1.2.9	Ensure that Activity Log Alert Exists for Create or Update Public IP Address rule (Automated)	☐	☐
6.1.2.10	Ensure that Activity Log Alert Exists for Delete Public IP Address rule (Automated)	☐	☐
6.1.2.11	Ensure that an Activity Log Alert Exists for Service Health (Automated)	☐	☐
6.1.3	Configuring Application Insights		
6.1.3.1	Ensure Application Insights are Configured (Automated)	☐	☐
6.1.4	Ensure that Azure Monitor Resource Logging is Enabled for All Services that Support it (Manual)	☐	☐
6.1.5	Ensure Basic, Free, and Consumption SKUs are not used on Production artifacts requiring monitoring and SLA (Manual)	☐	☐
6.2	Ensure that Resource Locks are set for Mission-Critical Azure Resources (Manual)	☐	☐
7	Networking Services		
7.1	Ensure that RDP Access from the Internet is Evaluated and Restricted (Automated)	☐	☐
7.2	Ensure that SSH Access from the Internet is Evaluated and Restricted (Automated)	☐	☐
7.3	Ensure that UDP Port Access from the Internet is Evaluated and Restricted (Automated)	☐	☐

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
7.4	Ensure that HTTP(S) Access from the Internet is Evaluated and Restricted (Automated)	☐	☐
7.5	Ensure that Network Security Group Flow Log Retention Days is Set to Greater than or equal to 90 (Automated)	☐	☐
7.6	Ensure that Network Watcher is 'Enabled' for Azure Regions That are in Use (Automated)	☐	☐
7.7	Ensure that Public IP Addresses are Evaluated on a Periodic Basis (Manual)	☐	☐
7.8	Ensure that Virtual Network Flow Log Retention Days is Set to Greater than or Equal to 90 (Automated)	☐	☐
7.9	Ensure 'Authentication type' is Set to 'Azure Active Directory' only for Azure VPN Gateway Point-to-Site Configuration (Automated)	☐	☐
7.10	Ensure Azure Web Application Firewall (WAF) is Enabled on Azure Application Gateway (Automated)	☐	☐
7.11	Ensure Subnets Are Associated with Network Security Groups (Automated)	☐	☐
7.12	Ensure the SSL Policy's 'Min protocol version' is Set to 'TLSv1_2' or Higher on Azure Application Gateway (Automated)	☐	☐
7.13	Ensure 'HTTP2' is Set to 'Enabled' on Azure Application Gateway (Automated)	☐	☐
7.14	Ensure Request Body Inspection is Enabled in Azure Web Application Firewall policy on Azure Application Gateway (Automated)	☐	☐
7.15	Ensure Bot Protection is Enabled in Azure Web Application Firewall Policy on Azure Application Gateway (Automated)	☐	☐
7.16	Ensure Azure Network Security Perimeter is Used to Secure Azure Platform-as-a-service Resources (Manual)	☐	☐

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
8	Security Services		
8.1	Microsoft Defender for Cloud		
8.1.1	Microsoft Cloud Security Posture Management (CSPM)		
8.1.1.1	Ensure Microsoft Defender CSPM is Set to 'On' (Automated)	☐	☐
8.1.2	Defender Plan: APIs		
8.1.2.1	Ensure Microsoft Defender for APIs is Set to 'On' (Automated)	☐	☐
8.1.3	Defender Plan: Servers		
8.1.3.1	Ensure that Defender for Servers is Set to 'On' (Automated)	☐	☐
8.1.3.2	Ensure that 'Vulnerability assessment for machines' Component Status is set to 'On' (Manual)	☐	☐
8.1.3.3	Ensure that 'Endpoint protection' Component Status is set to 'On' (Automated)	☐	☐
8.1.3.4	Ensure that 'Agentless scanning for machines' Component Status is Set to 'On' (Manual)	☐	☐
8.1.3.5	Ensure that 'File Integrity Monitoring' Component Status is Set to 'On' (Manual)	☐	☐
8.1.4	Defender Plan: Containers		
8.1.4.1	Ensure That Microsoft Defender for Containers Is Set To 'On' (Automated)	☐	☐
8.1.5	Defender Plan: Storage		
8.1.5.1	Ensure That Microsoft Defender for Storage Is Set To 'On' (Automated)	☐	☐

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
8.1.5.2	Ensure Advanced Threat Protection Alerts for Storage Accounts Are Monitored (Manual)	☐	☐
8.1.6	Defender Plan: App Service		
8.1.6.1	Ensure That Microsoft Defender for App Services Is Set To 'On' (Automated)	☐	☐
8.1.7	Defender Plan: Databases		
8.1.7.1	Ensure That Microsoft Defender for Azure Cosmos DB Is Set To 'On' (Automated)	☐	☐
8.1.7.2	Ensure That Microsoft Defender for Open-Source Relational Databases Is Set To 'On' (Automated)	☐	☐
8.1.7.3	Ensure That Microsoft Defender for (Managed Instance) Azure SQL Databases Is Set To 'On' (Automated)	☐	☐
8.1.7.4	Ensure That Microsoft Defender for SQL Servers on Machines Is Set To 'On' (Automated)	☐	☐
8.1.8	Defender Plan: Key Vault		
8.1.8.1	Ensure That Microsoft Defender for Key Vault Is Set To 'On' (Automated)	☐	☐
8.1.9	Defender Plan: Resource Manager		
8.1.9.1	Ensure That Microsoft Defender for Resource Manager Is Set To 'On' (Automated)	☐	☐
8.1.10	Ensure that Microsoft Defender for Cloud is Configured to Check VM Operating Systems for Updates (Automated)	☐	☐
8.1.11	Ensure that non-deprecated Microsoft Cloud Security Benchmark policies are not set to 'Disabled' (Manual)	☐	☐
8.1.12	Ensure That 'All users with the following roles' is Set to 'Owner' (Automated)	☐	☐

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
8.1.13	Ensure 'Additional email addresses' is Configured with a Security Contact Email (Automated)	☐	☐
8.1.14	Ensure that 'Notify about alerts with the following severity (or higher)' is Enabled (Automated)	☐	☐
8.1.15	Ensure that 'Notify about attack paths with the following risk level (or higher)' is Enabled (Automated)	☐	☐
8.1.16	Ensure that Microsoft Defender External Attack Surface Monitoring (EASM) is Enabled (Manual)	☐	☐
8.2	Microsoft Defender for IoT		
8.2.1	Ensure That Microsoft Defender for IoT Hub Is Set To 'On' (Manual)	☐	☐
8.3	Key Vault		
8.3.1	Ensure that the Expiration Date is Set for all Keys in Key Vaults using RBAC (Automated)	☐	☐
8.3.2	Ensure that the Expiration Date is set for All Keys in Key Vaults using access policies (legacy) (Automated)	☐	☐
8.3.3	Ensure that the Expiration Date is set for All Secrets in Key Vaults using RBAC (Automated)	☐	☐
8.3.4	Ensure that the Expiration Date is set for All Secrets in Key Vaults using access policies (legacy) (Automated)	☐	☐
8.3.5	Ensure 'Purge protection' is Set to 'Enabled' (Automated)	☐	☐
8.3.6	Ensure that Role Based Access Control for Azure Key Vault is Enabled (Automated)	☐	☐
8.3.7	Ensure Public Network Access is Disabled (Automated)	☐	☐
8.3.8	Ensure Private Endpoints are Used to Access Azure Key Vault (Automated)	☐	☐

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
8.3.9	Ensure Automatic Key Rotation is Enabled within Azure Key Vault (Automated)	☐	☐
8.3.10	Ensure that Azure Key Vault Managed HSM is Used when Required (Manual)	☐	☐
8.3.11	Ensure Certificate 'Validity Period (in months)' is Less Than or Equal to '12' (Automated)	☐	☐
8.4	Azure Bastion		
8.4.1	Ensure an Azure Bastion Host Exists (Automated)	☐	☐
8.5	Ensure Azure DDoS Network Protection is Enabled on Virtual Networks (Automated)	☐	☐
9	Storage Services		
9.1	Azure Files		
9.1.1	Ensure Soft Delete for Azure File Shares is Enabled (Automated)	☐	☐
9.1.2	Ensure 'SMB protocol version' is Set to 'SMB 3.1.1' or Higher for SMB file shares (Automated)	☐	☐
9.1.3	Ensure 'SMB channel encryption' is Set to 'AES-256-GCM' or Higher for SMB file shares (Automated)	☐	☐
9.2	Azure Blob Storage		
9.2.1	Ensure That Soft Delete for Blobs on Azure Blob Storage Storage Accounts is Enabled (Automated)	☐	☐
9.2.2	Ensure that Soft Delete for Containers on Azure Blob Storage Storage Accounts is Enabled (Automated)	☐	☐
9.2.3	Ensure 'Versioning' is Set to 'Enabled' on Azure Blob Storage Storage Accounts (Automated)	☐	☐
9.3	Storage Accounts		

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
9.3.1	Secrets and Keys		
9.3.1.1	Ensure That 'Enable key rotation reminders' is Enabled for Each Storage Account (Automated)	☐	☐
9.3.1.2	Ensure That Storage Account Access keys are Periodically Regenerated (Automated)	☐	☐
9.3.1.3	Ensure 'Allow storage account key access' for Azure Storage Accounts is 'Disabled' (Automated)	☐	☐
9.3.2	Networking		
9.3.2.1	Ensure Private Endpoints are Used to Access Storage Accounts (Automated)	☐	☐
9.3.2.2	Ensure that 'Public Network Access' is 'Disabled' for Storage Accounts (Automated)	☐	☐
9.3.2.3	Ensure Default Network Access Rule for Storage Accounts is Set to Deny (Automated)	☐	☐
9.3.3	Identity and Access Management		
9.3.3.1	Ensure that 'Default to Microsoft Entra authorization in the Azure portal' is Set to 'Enabled' (Automated)	☐	☐
9.3.4	Ensure that 'Secure transfer required' is Set to 'Enabled' (Automated)	☐	☐
9.3.5	Ensure 'Allow trusted Microsoft services to access this resource' is Enabled for Storage Account Access (Automated)	☐	☐
9.3.6	Ensure the 'Minimum TLS version' for Storage Accounts is Set to 'Version 1.2' (Automated)	☐	☐
9.3.7	Ensure 'Cross Tenant Replication' is Not Enabled (Automated)	☐	☐
9.3.8	Ensure that 'Allow Blob Anonymous Access' is Set to 'Disabled' (Automated)	☐	☐

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
9.3.9	Ensure Azure Resource Manager Delete Locks are Applied to Azure Storage Accounts (Manual)	☐	☐
9.3.10	Ensure Azure Resource Manager ReadOnly Locks are Considered for Azure Storage Accounts (Manual)	☐	☐
9.3.11	Ensure Redundancy is Set to 'geo-redundant storage (GRS)' on Critical Azure Storage Accounts (Automated)	☐	☐

Appendix: CIS Controls v7 IG 1 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
2.1.2	Ensure that Network Security Groups are Configured for Databricks Subnets	☐	☐
2.1.7	Ensure that Diagnostic Log Delivery is Configured for Azure Databricks	☐	☐
2.1.9	Ensure 'No Public IP' is Set to 'Enabled'	☐	☐
2.1.10	Ensure 'Allow Public Network Access' is set to 'Disabled'	☐	☐
5.3.1	Ensure that Azure Admin Accounts Are Not Used for Daily Operations	☐	☐
5.3.2	Ensure that Guest Users are Reviewed on a Regular Basis	☐	☐
5.3.6	Ensure 'Tenant Creator' Role Assignments are Periodically Reviewed	☐	☐
5.5	Ensure that a Custom Role is Assigned Permissions for Administering Resource Locks	☐	☐
5.6	Ensure that 'Subscription leaving Microsoft Entra tenant' and 'Subscription entering Microsoft Entra tenant' is set to 'Permit no one'	☐	☐
6.1.1.7	Ensure that a Microsoft Entra Diagnostic Setting Exists to Send Microsoft Graph Activity Logs to an Appropriate Destination	☐	☐
6.1.1.8	Ensure that a Microsoft Entra Diagnostic Setting Exists to Send Microsoft Entra Activity Logs to an Appropriate Destination	☐	☐
6.1.1.9	Ensure that Intune Logs are Captured and Sent to Log Analytics	☐	☐
6.1.3.1	Ensure Application Insights are Configured	☐	☐
6.1.5	Ensure Basic, Free, and Consumption SKUs are not used on Production artifacts requiring monitoring and SLA	☐	☐
6.2	Ensure that Resource Locks are set for Mission-Critical Azure Resources	☐	☐

Recommendation		Set Correctly	
		Yes	No
7.6	Ensure that Network Watcher is 'Enabled' for Azure Regions That are in Use	☐	☐
7.7	Ensure that Public IP Addresses are Evaluated on a Periodic Basis	☐	☐
7.11	Ensure Subnets Are Associated with Network Security Groups	☐	☐
7.13	Ensure 'HTTP2' is Set to 'Enabled' on Azure Application Gateway	☐	☐
7.16	Ensure Azure Network Security Perimeter is Used to Secure Azure Platform-as-a-service Resources	☐	☐
8.1.10	Ensure that Microsoft Defender for Cloud is Configured to Check VM Operating Systems for Updates	☐	☐
8.1.11	Ensure that non-deprecated Microsoft Cloud Security Benchmark policies are not set to 'Disabled'	☐	☐
8.1.12	Ensure That 'All users with the following roles' is Set to 'Owner'	☐	☐
8.1.13	Ensure 'Additional email addresses' is Configured with a Security Contact Email	☐	☐
8.3.5	Ensure 'Purge protection' is Set to 'Enabled'	☐	☐
8.3.6	Ensure that Role Based Access Control for Azure Key Vault is Enabled	☐	☐
8.3.7	Ensure Public Network Access is Disabled	☐	☐
9.1.1	Ensure Soft Delete for Azure File Shares is Enabled	☐	☐
9.2.1	Ensure That Soft Delete for Blobs on Azure Blob Storage Storage Accounts is Enabled	☐	☐
9.2.2	Ensure that Soft Delete for Containers on Azure Blob Storage Storage Accounts is Enabled	☐	☐
9.2.3	Ensure 'Versioning' is Set to 'Enabled' on Azure Blob Storage Storage Accounts	☐	☐
9.3.1.3	Ensure 'Allow storage account key access' for Azure Storage Accounts is 'Disabled'	☐	☐
9.3.2.2	Ensure that 'Public Network Access' is 'Disabled' for Storage Accounts	☐	☐
9.3.3.1	Ensure that 'Default to Microsoft Entra authorization in the Azure portal' is Set to 'Enabled'	☐	☐

Recommendation		Set Correctly	
		Yes	No
9.3.8	Ensure that 'Allow Blob Anonymous Access' is Set to 'Disabled'	☐	☐

Appendix: CIS Controls v7 IG 2 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
2.1.1	Ensure that Azure Databricks is deployed in a customer-managed virtual network (VNet)	☐	☐
2.1.2	Ensure that Network Security Groups are Configured for Databricks Subnets	☐	☐
2.1.3	Ensure that Traffic is Encrypted Between Cluster Worker Nodes	☐	☐
2.1.4	Ensure that Users and Groups are Synced from Microsoft Entra ID to Azure Databricks	☐	☐
2.1.5	Ensure that Unity Catalog is Configured for Azure Databricks	☐	☐
2.1.6	Ensure that Usage is Restricted and Expiry is Enforced for Databricks Personal Access Tokens	☐	☐
2.1.7	Ensure that Diagnostic Log Delivery is Configured for Azure Databricks	☐	☐
2.1.9	Ensure 'No Public IP' is Set to 'Enabled'	☐	☐
2.1.10	Ensure 'Allow Public Network Access' is set to 'Disabled'	☐	☐
2.1.11	Ensure Private Endpoints are used to access Azure Databricks workspaces	☐	☐
3.1.1	Ensure only MFA Enabled Identities can Access Privileged Virtual Machine	☐	☐
5.1.1	Ensure that 'security defaults' is Enabled in Microsoft Entra ID	☐	☐
5.1.2	Ensure that 'Require Multifactor Authentication to register or join devices with Microsoft Entra' is set to 'Yes'	☐	☐
5.1.3	Ensure that 'multifactor authentication' is 'enabled' For All Users	☐	☐
5.1.4	Ensure that 'Allow users to remember multifactor authentication on devices they trust' is Disabled	☐	☐
5.3.1	Ensure that Azure Admin Accounts Are Not Used for Daily Operations	☐	☐

Recommendation		Set Correctly	
		Yes	No
5.3.2	Ensure that Guest Users are Reviewed on a Regular Basis	☐	☐
5.3.3	Ensure That Use of the 'User Access Administrator' Role is Restricted	☐	☐
5.3.4	Ensure that All 'Privileged' Role Assignments are Periodically Reviewed	☐	☐
5.3.5	Ensure Disabled User Accounts do not Have Read, Write, or Owner Permissions	☐	☐
5.3.6	Ensure 'Tenant Creator' Role Assignments are Periodically Reviewed	☐	☐
5.3.7	Ensure All Non-privileged Role Assignments are Periodically Reviewed	☐	☐
5.4	Ensure that No Custom Subscription Administrator Roles Exist	☐	☐
5.5	Ensure that a Custom Role is Assigned Permissions for Administering Resource Locks	☐	☐
5.6	Ensure that 'Subscription leaving Microsoft Entra tenant' and 'Subscription entering Microsoft Entra tenant' is set to 'Permit no one'	☐	☐
5.7	Ensure there are between 2 and 3 Subscription Owners	☐	☐
6.1.1.1	Ensure that a 'Diagnostic Setting' Exists for Subscription Activity Logs	☐	☐
6.1.1.2	Ensure Diagnostic Setting Captures Appropriate Categories	☐	☐
6.1.1.4	Ensure that Logging for Azure Key Vault is 'Enabled'	☐	☐
6.1.1.5	Ensure that Network Security Group Flow Logs are Captured and Sent to Log Analytics	☐	☐
6.1.1.6	Ensure that Virtual Network Flow Logs are Captured and Sent to Log Analytics	☐	☐
6.1.1.7	Ensure that a Microsoft Entra Diagnostic Setting Exists to Send Microsoft Graph Activity Logs to an Appropriate Destination	☐	☐
6.1.1.8	Ensure that a Microsoft Entra Diagnostic Setting Exists to Send Microsoft Entra Activity Logs to an Appropriate Destination	☐	☐

Recommendation		Set Correctly	
		Yes	No
6.1.1.9	Ensure that Intune Logs are Captured and Sent to Log Analytics	☐	☐
6.1.2.1	Ensure that Activity Log Alert Exists for Create Policy Assignment	☐	☐
6.1.2.2	Ensure that Activity Log Alert exists for Delete Policy Assignment	☐	☐
6.1.2.3	Ensure that Activity Log Alert Exists for Create or Update Network Security Group	☐	☐
6.1.2.4	Ensure that Activity Log Alert Exists for Delete Network Security Group	☐	☐
6.1.2.5	Ensure that Activity Log Alert Exists for Create or Update Security Solution	☐	☐
6.1.2.6	Ensure that Activity Log Alert Exists for Delete Security Solution	☐	☐
6.1.2.7	Ensure that Activity Log Alert Exists for Create or Update SQL Server Firewall Rule	☐	☐
6.1.2.8	Ensure that Activity Log Alert Exists for Delete SQL Server Firewall Rule	☐	☐
6.1.2.9	Ensure that Activity Log Alert Exists for Create or Update Public IP Address rule	☐	☐
6.1.2.10	Ensure that Activity Log Alert Exists for Delete Public IP Address rule	☐	☐
6.1.2.11	Ensure that an Activity Log Alert Exists for Service Health	☐	☐
6.1.3.1	Ensure Application Insights are Configured	☐	☐
6.1.4	Ensure that Azure Monitor Resource Logging is Enabled for All Services that Support it	☐	☐
6.1.5	Ensure Basic, Free, and Consumption SKUs are not used on Production artifacts requiring monitoring and SLA	☐	☐
6.2	Ensure that Resource Locks are set for Mission-Critical Azure Resources	☐	☐
7.1	Ensure that RDP Access from the Internet is Evaluated and Restricted	☐	☐
7.2	Ensure that SSH Access from the Internet is Evaluated and Restricted	☐	☐

Recommendation		Set Correctly	
		Yes	No
7.3	Ensure that UDP Port Access from the Internet is Evaluated and Restricted	☐	☐
7.4	Ensure that HTTP(S) Access from the Internet is Evaluated and Restricted	☐	☐
7.5	Ensure that Network Security Group Flow Log Retention Days is Set to Greater than or equal to 90	☐	☐
7.6	Ensure that Network Watcher is 'Enabled' for Azure Regions That are in Use	☐	☐
7.7	Ensure that Public IP Addresses are Evaluated on a Periodic Basis	☐	☐
7.8	Ensure that Virtual Network Flow Log Retention Days is Set to Greater than or Equal to 90	☐	☐
7.9	Ensure 'Authentication type' is Set to 'Azure Active Directory' only for Azure VPN Gateway Point-to-Site Configuration	☐	☐
7.11	Ensure Subnets Are Associated with Network Security Groups	☐	☐
7.12	Ensure the SSL Policy's 'Min protocol version' is Set to 'TLSv1_2' or Higher on Azure Application Gateway	☐	☐
7.13	Ensure 'HTTP2' is Set to 'Enabled' on Azure Application Gateway	☐	☐
7.16	Ensure Azure Network Security Perimeter is Used to Secure Azure Platform-as-a-service Resources	☐	☐
8.1.1.1	Ensure Microsoft Defender CSPM is Set to 'On'	☐	☐
8.1.2.1	Ensure Microsoft Defender for APIs is Set to 'On'	☐	☐
8.1.3.1	Ensure that Defender for Servers is Set to 'On'	☐	☐
8.1.3.2	Ensure that 'Vulnerability assessment for machines' Component Status is set to 'On'	☐	☐
8.1.3.3	Ensure that 'Endpoint protection' Component Status is set to 'On'	☐	☐
8.1.3.4	Ensure that 'Agentless scanning for machines' Component Status is Set to 'On'	☐	☐
8.1.3.5	Ensure that 'File Integrity Monitoring' Component Status is Set to 'On'	☐	☐
8.1.4.1	Ensure That Microsoft Defender for Containers Is Set To 'On'	☐	☐

Recommendation		Set Correctly	
		Yes	No
8.1.5.1	Ensure That Microsoft Defender for Storage Is Set To 'On'	☐	☐
8.1.5.2	Ensure Advanced Threat Protection Alerts for Storage Accounts Are Monitored	☐	☐
8.1.6.1	Ensure That Microsoft Defender for App Services Is Set To 'On'	☐	☐
8.1.7.1	Ensure That Microsoft Defender for Azure Cosmos DB Is Set To 'On'	☐	☐
8.1.7.2	Ensure That Microsoft Defender for Open-Source Relational Databases Is Set To 'On'	☐	☐
8.1.7.3	Ensure That Microsoft Defender for (Managed Instance) Azure SQL Databases Is Set To 'On'	☐	☐
8.1.7.4	Ensure That Microsoft Defender for SQL Servers on Machines Is Set To 'On'	☐	☐
8.1.8.1	Ensure That Microsoft Defender for Key Vault Is Set To 'On'	☐	☐
8.1.9.1	Ensure That Microsoft Defender for Resource Manager Is Set To 'On'	☐	☐
8.1.10	Ensure that Microsoft Defender for Cloud is Configured to Check VM Operating Systems for Updates	☐	☐
8.1.11	Ensure that non-deprecated Microsoft Cloud Security Benchmark policies are not set to 'Disabled'	☐	☐
8.1.12	Ensure That 'All users with the following roles' is Set to 'Owner'	☐	☐
8.1.13	Ensure 'Additional email addresses' is Configured with a Security Contact Email	☐	☐
8.1.16	Ensure that Microsoft Defender External Attack Surface Monitoring (EASM) is Enabled	☐	☐
8.2.1	Ensure That Microsoft Defender for IoT Hub Is Set To 'On'	☐	☐
8.3.1	Ensure that the Expiration Date is Set for all Keys in Key Vaults using RBAC	☐	☐
8.3.2	Ensure that the Expiration Date is set for All Keys in Key Vaults using access policies (legacy)	☐	☐
8.3.3	Ensure that the Expiration Date is set for All Secrets in Key Vaults using RBAC	☐	☐

Recommendation		Set Correctly	
		Yes	No
8.3.4	Ensure that the Expiration Date is set for All Secrets in Key Vaults using access policies (legacy)	☐	☐
8.3.5	Ensure 'Purge protection' is Set to 'Enabled'	☐	☐
8.3.6	Ensure that Role Based Access Control for Azure Key Vault is Enabled	☐	☐
8.3.7	Ensure Public Network Access is Disabled	☐	☐
8.3.8	Ensure Private Endpoints are Used to Access Azure Key Vault	☐	☐
8.3.9	Ensure Automatic Key Rotation is Enabled within Azure Key Vault	☐	☐
8.3.11	Ensure Certificate 'Validity Period (in months)' is Less Than or Equal to '12'	☐	☐
8.4.1	Ensure an Azure Bastion Host Exists	☐	☐
9.1.1	Ensure Soft Delete for Azure File Shares is Enabled	☐	☐
9.1.2	Ensure 'SMB protocol version' is Set to 'SMB 3.1.1' or Higher for SMB file shares	☐	☐
9.1.3	Ensure 'SMB channel encryption' is Set to 'AES-256-GCM' or Higher for SMB file shares	☐	☐
9.2.1	Ensure That Soft Delete for Blobs on Azure Blob Storage Storage Accounts is Enabled	☐	☐
9.2.2	Ensure that Soft Delete for Containers on Azure Blob Storage Storage Accounts is Enabled	☐	☐
9.2.3	Ensure 'Versioning' is Set to 'Enabled' on Azure Blob Storage Storage Accounts	☐	☐
9.3.1.1	Ensure That 'Enable key rotation reminders' is Enabled for Each Storage Account	☐	☐
9.3.1.2	Ensure That Storage Account Access keys are Periodically Regenerated	☐	☐
9.3.1.3	Ensure 'Allow storage account key access' for Azure Storage Accounts is 'Disabled'	☐	☐
9.3.2.1	Ensure Private Endpoints are Used to Access Storage Accounts	☐	☐
9.3.2.2	Ensure that 'Public Network Access' is 'Disabled' for Storage Accounts	☐	☐

Recommendation		Set Correctly	
		Yes	No
9.3.3.1	Ensure that 'Default to Microsoft Entra authorization in the Azure portal' is Set to 'Enabled'	☐	☐
9.3.4	Ensure that 'Secure transfer required' is Set to 'Enabled'	☐	☐
9.3.6	Ensure the 'Minimum TLS version' for Storage Accounts is Set to 'Version 1.2'	☐	☐
9.3.7	Ensure 'Cross Tenant Replication' is Not Enabled	☐	☐
9.3.8	Ensure that 'Allow Blob Anonymous Access' is Set to 'Disabled'	☐	☐

Appendix: CIS Controls v7 IG 3 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
2.1.1	Ensure that Azure Databricks is deployed in a customer-managed virtual network (VNet)	☐	☐
2.1.2	Ensure that Network Security Groups are Configured for Databricks Subnets	☐	☐
2.1.3	Ensure that Traffic is Encrypted Between Cluster Worker Nodes	☐	☐
2.1.4	Ensure that Users and Groups are Synced from Microsoft Entra ID to Azure Databricks	☐	☐
2.1.5	Ensure that Unity Catalog is Configured for Azure Databricks	☐	☐
2.1.6	Ensure that Usage is Restricted and Expiry is Enforced for Databricks Personal Access Tokens	☐	☐
2.1.7	Ensure that Diagnostic Log Delivery is Configured for Azure Databricks	☐	☐
2.1.8	Ensure Critical Data in Azure Databricks is Encrypted with Customer-managed Keys (CMK)	☐	☐
2.1.9	Ensure 'No Public IP' is Set to 'Enabled'	☐	☐
2.1.10	Ensure 'Allow Public Network Access' is set to 'Disabled'	☐	☐
2.1.11	Ensure Private Endpoints are used to access Azure Databricks workspaces	☐	☐
3.1.1	Ensure only MFA Enabled Identities can Access Privileged Virtual Machine	☐	☐
5.1.1	Ensure that 'security defaults' is Enabled in Microsoft Entra ID	☐	☐
5.1.2	Ensure that 'Require Multifactor Authentication to register or join devices with Microsoft Entra' is set to 'Yes'	☐	☐
5.1.3	Ensure that 'multifactor authentication' is 'enabled' For All Users	☐	☐
5.1.4	Ensure that 'Allow users to remember multifactor authentication on devices they trust' is Disabled	☐	☐

Recommendation		Set Correctly	
		Yes	No
5.3.1	Ensure that Azure Admin Accounts Are Not Used for Daily Operations	☐	☐
5.3.2	Ensure that Guest Users are Reviewed on a Regular Basis	☐	☐
5.3.3	Ensure That Use of the 'User Access Administrator' Role is Restricted	☐	☐
5.3.4	Ensure that All 'Privileged' Role Assignments are Periodically Reviewed	☐	☐
5.3.5	Ensure Disabled User Accounts do not Have Read, Write, or Owner Permissions	☐	☐
5.3.6	Ensure 'Tenant Creator' Role Assignments are Periodically Reviewed	☐	☐
5.3.7	Ensure All Non-privileged Role Assignments are Periodically Reviewed	☐	☐
5.4	Ensure that No Custom Subscription Administrator Roles Exist	☐	☐
5.5	Ensure that a Custom Role is Assigned Permissions for Administering Resource Locks	☐	☐
5.6	Ensure that 'Subscription leaving Microsoft Entra tenant' and 'Subscription entering Microsoft Entra tenant' is set to 'Permit no one'	☐	☐
5.7	Ensure there are between 2 and 3 Subscription Owners	☐	☐
6.1.1.1	Ensure that a 'Diagnostic Setting' Exists for Subscription Activity Logs	☐	☐
6.1.1.2	Ensure Diagnostic Setting Captures Appropriate Categories	☐	☐
6.1.1.3	Ensure the Storage Account Containing the Container with Activity Logs is Encrypted with Customer-managed Key (CMK)	☐	☐
6.1.1.4	Ensure that Logging for Azure Key Vault is 'Enabled'	☐	☐
6.1.1.5	Ensure that Network Security Group Flow Logs are Captured and Sent to Log Analytics	☐	☐
6.1.1.6	Ensure that Virtual Network Flow Logs are Captured and Sent to Log Analytics	☐	☐

Recommendation		Set Correctly	
		Yes	No
6.1.1.7	Ensure that a Microsoft Entra Diagnostic Setting Exists to Send Microsoft Graph Activity Logs to an Appropriate Destination	☐	☐
6.1.1.8	Ensure that a Microsoft Entra Diagnostic Setting Exists to Send Microsoft Entra Activity Logs to an Appropriate Destination	☐	☐
6.1.1.9	Ensure that Intune Logs are Captured and Sent to Log Analytics	☐	☐
6.1.2.1	Ensure that Activity Log Alert Exists for Create Policy Assignment	☐	☐
6.1.2.2	Ensure that Activity Log Alert exists for Delete Policy Assignment	☐	☐
6.1.2.3	Ensure that Activity Log Alert Exists for Create or Update Network Security Group	☐	☐
6.1.2.4	Ensure that Activity Log Alert Exists for Delete Network Security Group	☐	☐
6.1.2.5	Ensure that Activity Log Alert Exists for Create or Update Security Solution	☐	☐
6.1.2.6	Ensure that Activity Log Alert Exists for Delete Security Solution	☐	☐
6.1.2.7	Ensure that Activity Log Alert Exists for Create or Update SQL Server Firewall Rule	☐	☐
6.1.2.8	Ensure that Activity Log Alert Exists for Delete SQL Server Firewall Rule	☐	☐
6.1.2.9	Ensure that Activity Log Alert Exists for Create or Update Public IP Address rule	☐	☐
6.1.2.10	Ensure that Activity Log Alert Exists for Delete Public IP Address rule	☐	☐
6.1.2.11	Ensure that an Activity Log Alert Exists for Service Health	☐	☐
6.1.3.1	Ensure Application Insights are Configured	☐	☐
6.1.4	Ensure that Azure Monitor Resource Logging is Enabled for All Services that Support it	☐	☐
6.1.5	Ensure Basic, Free, and Consumption SKUs are not used on Production artifacts requiring monitoring and SLA	☐	☐

Recommendation		Set Correctly	
		Yes	No
6.2	Ensure that Resource Locks are set for Mission-Critical Azure Resources	☐	☐
7.1	Ensure that RDP Access from the Internet is Evaluated and Restricted	☐	☐
7.2	Ensure that SSH Access from the Internet is Evaluated and Restricted	☐	☐
7.3	Ensure that UDP Port Access from the Internet is Evaluated and Restricted	☐	☐
7.4	Ensure that HTTP(S) Access from the Internet is Evaluated and Restricted	☐	☐
7.5	Ensure that Network Security Group Flow Log Retention Days is Set to Greater than or equal to 90	☐	☐
7.6	Ensure that Network Watcher is 'Enabled' for Azure Regions That are in Use	☐	☐
7.7	Ensure that Public IP Addresses are Evaluated on a Periodic Basis	☐	☐
7.8	Ensure that Virtual Network Flow Log Retention Days is Set to Greater than or Equal to 90	☐	☐
7.9	Ensure 'Authentication type' is Set to 'Azure Active Directory' only for Azure VPN Gateway Point-to-Site Configuration	☐	☐
7.10	Ensure Azure Web Application Firewall (WAF) is Enabled on Azure Application Gateway	☐	☐
7.11	Ensure Subnets Are Associated with Network Security Groups	☐	☐
7.12	Ensure the SSL Policy's 'Min protocol version' is Set to 'TLSv1_2' or Higher on Azure Application Gateway	☐	☐
7.13	Ensure 'HTTP2' is Set to 'Enabled' on Azure Application Gateway	☐	☐
7.14	Ensure Request Body Inspection is Enabled in Azure Web Application Firewall policy on Azure Application Gateway	☐	☐
7.15	Ensure Bot Protection is Enabled in Azure Web Application Firewall Policy on Azure Application Gateway	☐	☐
7.16	Ensure Azure Network Security Perimeter is Used to Secure Azure Platform-as-a-service Resources	☐	☐

Recommendation		Set Correctly	
		Yes	No
8.1.1.1	Ensure Microsoft Defender CSPM is Set to 'On'	☐	☐
8.1.2.1	Ensure Microsoft Defender for APIs is Set to 'On'	☐	☐
8.1.3.1	Ensure that Defender for Servers is Set to 'On'	☐	☐
8.1.3.2	Ensure that 'Vulnerability assessment for machines' Component Status is set to 'On'	☐	☐
8.1.3.3	Ensure that 'Endpoint protection' Component Status is set to 'On'	☐	☐
8.1.3.4	Ensure that 'Agentless scanning for machines' Component Status is Set to 'On'	☐	☐
8.1.3.5	Ensure that 'File Integrity Monitoring' Component Status is Set to 'On'	☐	☐
8.1.4.1	Ensure That Microsoft Defender for Containers Is Set To 'On'	☐	☐
8.1.5.1	Ensure That Microsoft Defender for Storage Is Set To 'On'	☐	☐
8.1.5.2	Ensure Advanced Threat Protection Alerts for Storage Accounts Are Monitored	☐	☐
8.1.6.1	Ensure That Microsoft Defender for App Services Is Set To 'On'	☐	☐
8.1.7.1	Ensure That Microsoft Defender for Azure Cosmos DB Is Set To 'On'	☐	☐
8.1.7.2	Ensure That Microsoft Defender for Open-Source Relational Databases Is Set To 'On'	☐	☐
8.1.7.3	Ensure That Microsoft Defender for (Managed Instance) Azure SQL Databases Is Set To 'On'	☐	☐
8.1.7.4	Ensure That Microsoft Defender for SQL Servers on Machines Is Set To 'On'	☐	☐
8.1.8.1	Ensure That Microsoft Defender for Key Vault Is Set To 'On'	☐	☐
8.1.9.1	Ensure That Microsoft Defender for Resource Manager Is Set To 'On'	☐	☐
8.1.10	Ensure that Microsoft Defender for Cloud is Configured to Check VM Operating Systems for Updates	☐	☐
8.1.11	Ensure that non-deprecated Microsoft Cloud Security Benchmark policies are not set to 'Disabled'	☐	☐

Recommendation		Set Correctly	
		Yes	No
8.1.12	Ensure That 'All users with the following roles' is Set to 'Owner'	☐	☐
8.1.13	Ensure 'Additional email addresses' is Configured with a Security Contact Email	☐	☐
8.1.14	Ensure that 'Notify about alerts with the following severity (or higher)' is Enabled	☐	☐
8.1.15	Ensure that 'Notify about attack paths with the following risk level (or higher)' is Enabled	☐	☐
8.1.16	Ensure that Microsoft Defender External Attack Surface Monitoring (EASM) is Enabled	☐	☐
8.2.1	Ensure That Microsoft Defender for IoT Hub Is Set To 'On'	☐	☐
8.3.1	Ensure that the Expiration Date is Set for all Keys in Key Vaults using RBAC	☐	☐
8.3.2	Ensure that the Expiration Date is set for All Keys in Key Vaults using access policies (legacy)	☐	☐
8.3.3	Ensure that the Expiration Date is set for All Secrets in Key Vaults using RBAC	☐	☐
8.3.4	Ensure that the Expiration Date is set for All Secrets in Key Vaults using access policies (legacy)	☐	☐
8.3.5	Ensure 'Purge protection' is Set to 'Enabled'	☐	☐
8.3.6	Ensure that Role Based Access Control for Azure Key Vault is Enabled	☐	☐
8.3.7	Ensure Public Network Access is Disabled	☐	☐
8.3.8	Ensure Private Endpoints are Used to Access Azure Key Vault	☐	☐
8.3.9	Ensure Automatic Key Rotation is Enabled within Azure Key Vault	☐	☐
8.3.10	Ensure that Azure Key Vault Managed HSM is Used when Required	☐	☐
8.3.11	Ensure Certificate 'Validity Period (in months)' is Less Than or Equal to '12'	☐	☐
8.4.1	Ensure an Azure Bastion Host Exists	☐	☐
8.5	Ensure Azure DDoS Network Protection is Enabled on Virtual Networks	☐	☐

Recommendation		Set Correctly	
		Yes	No
9.1.1	Ensure Soft Delete for Azure File Shares is Enabled	☐	☐
9.1.2	Ensure 'SMB protocol version' is Set to 'SMB 3.1.1' or Higher for SMB file shares	☐	☐
9.1.3	Ensure 'SMB channel encryption' is Set to 'AES-256-GCM' or Higher for SMB file shares	☐	☐
9.2.1	Ensure That Soft Delete for Blobs on Azure Blob Storage Storage Accounts is Enabled	☐	☐
9.2.2	Ensure that Soft Delete for Containers on Azure Blob Storage Storage Accounts is Enabled	☐	☐
9.2.3	Ensure 'Versioning' is Set to 'Enabled' on Azure Blob Storage Storage Accounts	☐	☐
9.3.1.1	Ensure That 'Enable key rotation reminders' is Enabled for Each Storage Account	☐	☐
9.3.1.2	Ensure That Storage Account Access keys are Periodically Regenerated	☐	☐
9.3.1.3	Ensure 'Allow storage account key access' for Azure Storage Accounts is 'Disabled'	☐	☐
9.3.2.1	Ensure Private Endpoints are Used to Access Storage Accounts	☐	☐
9.3.2.2	Ensure that 'Public Network Access' is 'Disabled' for Storage Accounts	☐	☐
9.3.2.3	Ensure Default Network Access Rule for Storage Accounts is Set to Deny	☐	☐
9.3.3.1	Ensure that 'Default to Microsoft Entra authorization in the Azure portal' is Set to 'Enabled'	☐	☐
9.3.4	Ensure that 'Secure transfer required' is Set to 'Enabled'	☐	☐
9.3.5	Ensure 'Allow trusted Microsoft services to access this resource' is Enabled for Storage Account Access	☐	☐
9.3.6	Ensure the 'Minimum TLS version' for Storage Accounts is Set to 'Version 1.2'	☐	☐
9.3.7	Ensure 'Cross Tenant Replication' is Not Enabled	☐	☐
9.3.8	Ensure that 'Allow Blob Anonymous Access' is Set to 'Disabled'	☐	☐

Appendix: CIS Controls v7 Unmapped Recommendations

Recommendation		Set Correctly	
		Yes	No
2.1.12	Ensure Azure Databricks groups are reviewed periodically	☐	☐
9.3.9	Ensure Azure Resource Manager Delete Locks are Applied to Azure Storage Accounts	☐	☐
9.3.10	Ensure Azure Resource Manager ReadOnly Locks are Considered for Azure Storage Accounts	☐	☐

Appendix: CIS Controls v8 IG 1 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
2.1.2	Ensure that Network Security Groups are Configured for Databricks Subnets	☐	☐
2.1.6	Ensure that Usage is Restricted and Expiry is Enforced for Databricks Personal Access Tokens	☐	☐
2.1.7	Ensure that Diagnostic Log Delivery is Configured for Azure Databricks	☐	☐
2.1.9	Ensure 'No Public IP' is Set to 'Enabled'	☐	☐
2.1.10	Ensure 'Allow Public Network Access' is set to 'Disabled'	☐	☐
3.1.1	Ensure only MFA Enabled Identities can Access Privileged Virtual Machine	☐	☐
5.1.1	Ensure that 'security defaults' is Enabled in Microsoft Entra ID	☐	☐
5.1.2	Ensure that 'Require Multifactor Authentication to register or join devices with Microsoft Entra' is set to 'Yes'	☐	☐
5.1.3	Ensure that 'multifactor authentication' is 'enabled' For All Users	☐	☐
5.1.4	Ensure that 'Allow users to remember multifactor authentication on devices they trust' is Disabled	☐	☐
5.3.1	Ensure that Azure Admin Accounts Are Not Used for Daily Operations	☐	☐
5.3.2	Ensure that Guest Users are Reviewed on a Regular Basis	☐	☐
5.3.5	Ensure Disabled User Accounts do not Have Read, Write, or Owner Permissions	☐	☐
5.3.6	Ensure 'Tenant Creator' Role Assignments are Periodically Reviewed	☐	☐
5.4	Ensure that No Custom Subscription Administrator Roles Exist	☐	☐
5.5	Ensure that a Custom Role is Assigned Permissions for Administering Resource Locks	☐	☐

Recommendation		Set Correctly	
		Yes	No
5.6	Ensure that 'Subscription leaving Microsoft Entra tenant' and 'Subscription entering Microsoft Entra tenant' is set to 'Permit no one'	☐	☐
5.7	Ensure there are between 2 and 3 Subscription Owners	☐	☐
6.1.1.7	Ensure that a Microsoft Entra Diagnostic Setting Exists to Send Microsoft Graph Activity Logs to an Appropriate Destination	☐	☐
6.1.1.8	Ensure that a Microsoft Entra Diagnostic Setting Exists to Send Microsoft Entra Activity Logs to an Appropriate Destination	☐	☐
6.1.1.9	Ensure that Intune Logs are Captured and Sent to Log Analytics	☐	☐
6.1.3.1	Ensure Application Insights are Configured	☐	☐
6.1.5	Ensure Basic, Free, and Consumption SKUs are not used on Production artifacts requiring monitoring and SLA	☐	☐
6.2	Ensure that Resource Locks are set for Mission-Critical Azure Resources	☐	☐
7.5	Ensure that Network Security Group Flow Log Retention Days is Set to Greater than or equal to 90	☐	☐
7.7	Ensure that Public IP Addresses are Evaluated on a Periodic Basis	☐	☐
7.8	Ensure that Virtual Network Flow Log Retention Days is Set to Greater than or Equal to 90	☐	☐
7.11	Ensure Subnets Are Associated with Network Security Groups	☐	☐
7.13	Ensure 'HTTP2' is Set to 'Enabled' on Azure Application Gateway	☐	☐
8.1.3.1	Ensure that Defender for Servers is Set to 'On'	☐	☐
8.1.3.3	Ensure that 'Endpoint protection' Component Status is set to 'On'	☐	☐
8.1.9.1	Ensure That Microsoft Defender for Resource Manager Is Set To 'On'	☐	☐
8.1.10	Ensure that Microsoft Defender for Cloud is Configured to Check VM Operating Systems for Updates	☐	☐

Recommendation		Set Correctly	
		Yes	No
8.1.11	Ensure that non-deprecated Microsoft Cloud Security Benchmark policies are not set to 'Disabled'	☐	☐
8.1.12	Ensure That 'All users with the following roles' is Set to 'Owner'	☐	☐
8.1.13	Ensure 'Additional email addresses' is Configured with a Security Contact Email	☐	☐
8.3.1	Ensure that the Expiration Date is Set for all Keys in Key Vaults using RBAC	☐	☐
8.3.2	Ensure that the Expiration Date is set for All Keys in Key Vaults using access policies (legacy)	☐	☐
8.3.3	Ensure that the Expiration Date is set for All Secrets in Key Vaults using RBAC	☐	☐
8.3.4	Ensure that the Expiration Date is set for All Secrets in Key Vaults using access policies (legacy)	☐	☐
8.3.5	Ensure 'Purge protection' is Set to 'Enabled'	☐	☐
8.3.6	Ensure that Role Based Access Control for Azure Key Vault is Enabled	☐	☐
8.3.7	Ensure Public Network Access is Disabled	☐	☐
8.3.9	Ensure Automatic Key Rotation is Enabled within Azure Key Vault	☐	☐
8.3.11	Ensure Certificate 'Validity Period (in months)' is Less Than or Equal to '12'	☐	☐
8.4.1	Ensure an Azure Bastion Host Exists	☐	☐
9.1.1	Ensure Soft Delete for Azure File Shares is Enabled	☐	☐
9.2.1	Ensure That Soft Delete for Blobs on Azure Blob Storage Storage Accounts is Enabled	☐	☐
9.2.2	Ensure that Soft Delete for Containers on Azure Blob Storage Storage Accounts is Enabled	☐	☐
9.2.3	Ensure 'Versioning' is Set to 'Enabled' on Azure Blob Storage Storage Accounts	☐	☐
9.3.1.1	Ensure That 'Enable key rotation reminders' is Enabled for Each Storage Account	☐	☐
9.3.1.2	Ensure That Storage Account Access keys are Periodically Regenerated	☐	☐

Recommendation		Set Correctly	
		Yes	No
9.3.1.3	Ensure 'Allow storage account key access' for Azure Storage Accounts is 'Disabled'	☐	☐
9.3.2.2	Ensure that 'Public Network Access' is 'Disabled' for Storage Accounts	☐	☐
9.3.3.1	Ensure that 'Default to Microsoft Entra authorization in the Azure portal' is Set to 'Enabled'	☐	☐
9.3.5	Ensure 'Allow trusted Microsoft services to access this resource' is Enabled for Storage Account Access	☐	☐
9.3.7	Ensure 'Cross Tenant Replication' is Not Enabled	☐	☐
9.3.8	Ensure that 'Allow Blob Anonymous Access' is Set to 'Disabled'	☐	☐
9.3.11	Ensure Redundancy is Set to 'geo-redundant storage (GRS)' on Critical Azure Storage Accounts	☐	☐

Appendix: CIS Controls v8 IG 2 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
2.1.1	Ensure that Azure Databricks is deployed in a customer-managed virtual network (VNet)	☐	☐
2.1.2	Ensure that Network Security Groups are Configured for Databricks Subnets	☐	☐
2.1.3	Ensure that Traffic is Encrypted Between Cluster Worker Nodes	☐	☐
2.1.4	Ensure that Users and Groups are Synced from Microsoft Entra ID to Azure Databricks	☐	☐
2.1.5	Ensure that Unity Catalog is Configured for Azure Databricks	☐	☐
2.1.6	Ensure that Usage is Restricted and Expiry is Enforced for Databricks Personal Access Tokens	☐	☐
2.1.7	Ensure that Diagnostic Log Delivery is Configured for Azure Databricks	☐	☐
2.1.8	Ensure Critical Data in Azure Databricks is Encrypted with Customer-managed Keys (CMK)	☐	☐
2.1.9	Ensure 'No Public IP' is Set to 'Enabled'	☐	☐
2.1.10	Ensure 'Allow Public Network Access' is set to 'Disabled'	☐	☐
2.1.11	Ensure Private Endpoints are used to access Azure Databricks workspaces	☐	☐
3.1.1	Ensure only MFA Enabled Identities can Access Privileged Virtual Machine	☐	☐
5.1.1	Ensure that 'security defaults' is Enabled in Microsoft Entra ID	☐	☐
5.1.2	Ensure that 'Require Multifactor Authentication to register or join devices with Microsoft Entra' is set to 'Yes'	☐	☐
5.1.3	Ensure that 'multifactor authentication' is 'enabled' For All Users	☐	☐
5.1.4	Ensure that 'Allow users to remember multifactor authentication on devices they trust' is Disabled	☐	☐

Recommendation		Set Correctly	
		Yes	No
5.3.1	Ensure that Azure Admin Accounts Are Not Used for Daily Operations	☐	☐
5.3.2	Ensure that Guest Users are Reviewed on a Regular Basis	☐	☐
5.3.5	Ensure Disabled User Accounts do not Have Read, Write, or Owner Permissions	☐	☐
5.3.6	Ensure 'Tenant Creator' Role Assignments are Periodically Reviewed	☐	☐
5.4	Ensure that No Custom Subscription Administrator Roles Exist	☐	☐
5.5	Ensure that a Custom Role is Assigned Permissions for Administering Resource Locks	☐	☐
5.6	Ensure that 'Subscription leaving Microsoft Entra tenant' and 'Subscription entering Microsoft Entra tenant' is set to 'Permit no one'	☐	☐
5.7	Ensure there are between 2 and 3 Subscription Owners	☐	☐
6.1.1.1	Ensure that a 'Diagnostic Setting' Exists for Subscription Activity Logs	☐	☐
6.1.1.2	Ensure Diagnostic Setting Captures Appropriate Categories	☐	☐
6.1.1.3	Ensure the Storage Account Containing the Container with Activity Logs is Encrypted with Customer-managed Key (CMK)	☐	☐
6.1.1.4	Ensure that Logging for Azure Key Vault is 'Enabled'	☐	☐
6.1.1.5	Ensure that Network Security Group Flow Logs are Captured and Sent to Log Analytics	☐	☐
6.1.1.6	Ensure that Virtual Network Flow Logs are Captured and Sent to Log Analytics	☐	☐
6.1.1.7	Ensure that a Microsoft Entra Diagnostic Setting Exists to Send Microsoft Graph Activity Logs to an Appropriate Destination	☐	☐
6.1.1.8	Ensure that a Microsoft Entra Diagnostic Setting Exists to Send Microsoft Entra Activity Logs to an Appropriate Destination	☐	☐
6.1.1.9	Ensure that Intune Logs are Captured and Sent to Log Analytics	☐	☐

Recommendation		Set Correctly	
		Yes	No
6.1.2.1	Ensure that Activity Log Alert Exists for Create Policy Assignment	☐	☐
6.1.2.2	Ensure that Activity Log Alert exists for Delete Policy Assignment	☐	☐
6.1.2.3	Ensure that Activity Log Alert Exists for Create or Update Network Security Group	☐	☐
6.1.2.4	Ensure that Activity Log Alert Exists for Delete Network Security Group	☐	☐
6.1.2.5	Ensure that Activity Log Alert Exists for Create or Update Security Solution	☐	☐
6.1.2.6	Ensure that Activity Log Alert Exists for Delete Security Solution	☐	☐
6.1.2.7	Ensure that Activity Log Alert Exists for Create or Update SQL Server Firewall Rule	☐	☐
6.1.2.8	Ensure that Activity Log Alert Exists for Delete SQL Server Firewall Rule	☐	☐
6.1.2.9	Ensure that Activity Log Alert Exists for Create or Update Public IP Address rule	☐	☐
6.1.2.10	Ensure that Activity Log Alert Exists for Delete Public IP Address rule	☐	☐
6.1.2.11	Ensure that an Activity Log Alert Exists for Service Health	☐	☐
6.1.3.1	Ensure Application Insights are Configured	☐	☐
6.1.4	Ensure that Azure Monitor Resource Logging is Enabled for All Services that Support it	☐	☐
6.1.5	Ensure Basic, Free, and Consumption SKUs are not used on Production artifacts requiring monitoring and SLA	☐	☐
6.2	Ensure that Resource Locks are set for Mission-Critical Azure Resources	☐	☐
7.1	Ensure that RDP Access from the Internet is Evaluated and Restricted	☐	☐
7.2	Ensure that SSH Access from the Internet is Evaluated and Restricted	☐	☐
7.3	Ensure that UDP Port Access from the Internet is Evaluated and Restricted	☐	☐

Recommendation		Set Correctly	
		Yes	No
7.4	Ensure that HTTP(S) Access from the Internet is Evaluated and Restricted	☐	☐
7.5	Ensure that Network Security Group Flow Log Retention Days is Set to Greater than or equal to 90	☐	☐
7.6	Ensure that Network Watcher is 'Enabled' for Azure Regions That are in Use	☐	☐
7.7	Ensure that Public IP Addresses are Evaluated on a Periodic Basis	☐	☐
7.8	Ensure that Virtual Network Flow Log Retention Days is Set to Greater than or Equal to 90	☐	☐
7.9	Ensure 'Authentication type' is Set to 'Azure Active Directory' only for Azure VPN Gateway Point-to-Site Configuration	☐	☐
7.11	Ensure Subnets Are Associated with Network Security Groups	☐	☐
7.12	Ensure the SSL Policy's 'Min protocol version' is Set to 'TLSv1_2' or Higher on Azure Application Gateway	☐	☐
7.13	Ensure 'HTTP2' is Set to 'Enabled' on Azure Application Gateway	☐	☐
7.16	Ensure Azure Network Security Perimeter is Used to Secure Azure Platform-as-a-service Resources	☐	☐
8.1.1.1	Ensure Microsoft Defender CSPM is Set to 'On'	☐	☐
8.1.2.1	Ensure Microsoft Defender for APIs is Set to 'On'	☐	☐
8.1.3.1	Ensure that Defender for Servers is Set to 'On'	☐	☐
8.1.3.2	Ensure that 'Vulnerability assessment for machines' Component Status is set to 'On'	☐	☐
8.1.3.3	Ensure that 'Endpoint protection' Component Status is set to 'On'	☐	☐
8.1.3.4	Ensure that 'Agentless scanning for machines' Component Status is Set to 'On'	☐	☐
8.1.3.5	Ensure that 'File Integrity Monitoring' Component Status is Set to 'On'	☐	☐
8.1.4.1	Ensure That Microsoft Defender for Containers Is Set To 'On'	☐	☐
8.1.5.1	Ensure That Microsoft Defender for Storage Is Set To 'On'	☐	☐

Recommendation		Set Correctly	
		Yes	No
8.1.5.2	Ensure Advanced Threat Protection Alerts for Storage Accounts Are Monitored	☐	☐
8.1.6.1	Ensure That Microsoft Defender for App Services Is Set To 'On'	☐	☐
8.1.7.1	Ensure That Microsoft Defender for Azure Cosmos DB Is Set To 'On'	☐	☐
8.1.7.2	Ensure That Microsoft Defender for Open-Source Relational Databases Is Set To 'On'	☐	☐
8.1.7.3	Ensure That Microsoft Defender for (Managed Instance) Azure SQL Databases Is Set To 'On'	☐	☐
8.1.7.4	Ensure That Microsoft Defender for SQL Servers on Machines Is Set To 'On'	☐	☐
8.1.8.1	Ensure That Microsoft Defender for Key Vault Is Set To 'On'	☐	☐
8.1.9.1	Ensure That Microsoft Defender for Resource Manager Is Set To 'On'	☐	☐
8.1.10	Ensure that Microsoft Defender for Cloud is Configured to Check VM Operating Systems for Updates	☐	☐
8.1.11	Ensure that non-deprecated Microsoft Cloud Security Benchmark policies are not set to 'Disabled'	☐	☐
8.1.12	Ensure That 'All users with the following roles' is Set to 'Owner'	☐	☐
8.1.13	Ensure 'Additional email addresses' is Configured with a Security Contact Email	☐	☐
8.1.16	Ensure that Microsoft Defender External Attack Surface Monitoring (EASM) is Enabled	☐	☐
8.2.1	Ensure That Microsoft Defender for IoT Hub Is Set To 'On'	☐	☐
8.3.1	Ensure that the Expiration Date is Set for all Keys in Key Vaults using RBAC	☐	☐
8.3.2	Ensure that the Expiration Date is set for All Keys in Key Vaults using access policies (legacy)	☐	☐
8.3.3	Ensure that the Expiration Date is set for All Secrets in Key Vaults using RBAC	☐	☐
8.3.4	Ensure that the Expiration Date is set for All Secrets in Key Vaults using access policies (legacy)	☐	☐

Recommendation		Set Correctly	
		Yes	No
8.3.5	Ensure 'Purge protection' is Set to 'Enabled'	☐	☐
8.3.6	Ensure that Role Based Access Control for Azure Key Vault is Enabled	☐	☐
8.3.7	Ensure Public Network Access is Disabled	☐	☐
8.3.8	Ensure Private Endpoints are Used to Access Azure Key Vault	☐	☐
8.3.9	Ensure Automatic Key Rotation is Enabled within Azure Key Vault	☐	☐
8.3.10	Ensure that Azure Key Vault Managed HSM is Used when Required	☐	☐
8.3.11	Ensure Certificate 'Validity Period (in months)' is Less Than or Equal to '12'	☐	☐
8.4.1	Ensure an Azure Bastion Host Exists	☐	☐
9.1.1	Ensure Soft Delete for Azure File Shares is Enabled	☐	☐
9.1.2	Ensure 'SMB protocol version' is Set to 'SMB 3.1.1' or Higher for SMB file shares	☐	☐
9.1.3	Ensure 'SMB channel encryption' is Set to 'AES-256-GCM' or Higher for SMB file shares	☐	☐
9.2.1	Ensure That Soft Delete for Blobs on Azure Blob Storage Storage Accounts is Enabled	☐	☐
9.2.2	Ensure that Soft Delete for Containers on Azure Blob Storage Storage Accounts is Enabled	☐	☐
9.2.3	Ensure 'Versioning' is Set to 'Enabled' on Azure Blob Storage Storage Accounts	☐	☐
9.3.1.1	Ensure That 'Enable key rotation reminders' is Enabled for Each Storage Account	☐	☐
9.3.1.2	Ensure That Storage Account Access keys are Periodically Regenerated	☐	☐
9.3.1.3	Ensure 'Allow storage account key access' for Azure Storage Accounts is 'Disabled'	☐	☐
9.3.2.1	Ensure Private Endpoints are Used to Access Storage Accounts	☐	☐
9.3.2.2	Ensure that 'Public Network Access' is 'Disabled' for Storage Accounts	☐	☐

Recommendation		Set Correctly	
		Yes	No
9.3.2.3	Ensure Default Network Access Rule for Storage Accounts is Set to Deny	☐	☐
9.3.3.1	Ensure that 'Default to Microsoft Entra authorization in the Azure portal' is Set to 'Enabled'	☐	☐
9.3.4	Ensure that 'Secure transfer required' is Set to 'Enabled'	☐	☐
9.3.5	Ensure 'Allow trusted Microsoft services to access this resource' is Enabled for Storage Account Access	☐	☐
9.3.6	Ensure the 'Minimum TLS version' for Storage Accounts is Set to 'Version 1.2'	☐	☐
9.3.7	Ensure 'Cross Tenant Replication' is Not Enabled	☐	☐
9.3.8	Ensure that 'Allow Blob Anonymous Access' is Set to 'Disabled'	☐	☐
9.3.11	Ensure Redundancy is Set to 'geo-redundant storage (GRS)' on Critical Azure Storage Accounts	☐	☐

Appendix: CIS Controls v8 IG 3 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
2.1.1	Ensure that Azure Databricks is deployed in a customer-managed virtual network (VNet)	☐	☐
2.1.2	Ensure that Network Security Groups are Configured for Databricks Subnets	☐	☐
2.1.3	Ensure that Traffic is Encrypted Between Cluster Worker Nodes	☐	☐
2.1.4	Ensure that Users and Groups are Synced from Microsoft Entra ID to Azure Databricks	☐	☐
2.1.5	Ensure that Unity Catalog is Configured for Azure Databricks	☐	☐
2.1.6	Ensure that Usage is Restricted and Expiry is Enforced for Databricks Personal Access Tokens	☐	☐
2.1.7	Ensure that Diagnostic Log Delivery is Configured for Azure Databricks	☐	☐
2.1.8	Ensure Critical Data in Azure Databricks is Encrypted with Customer-managed Keys (CMK)	☐	☐
2.1.9	Ensure 'No Public IP' is Set to 'Enabled'	☐	☐
2.1.10	Ensure 'Allow Public Network Access' is set to 'Disabled'	☐	☐
2.1.11	Ensure Private Endpoints are used to access Azure Databricks workspaces	☐	☐
2.1.12	Ensure Azure Databricks groups are reviewed periodically	☐	☐
3.1.1	Ensure only MFA Enabled Identities can Access Privileged Virtual Machine	☐	☐
5.1.1	Ensure that 'security defaults' is Enabled in Microsoft Entra ID	☐	☐
5.1.2	Ensure that 'Require Multifactor Authentication to register or join devices with Microsoft Entra' is set to 'Yes'	☐	☐
5.1.3	Ensure that 'multifactor authentication' is 'enabled' For All Users	☐	☐

Recommendation		Set Correctly	
		Yes	No
5.1.4	Ensure that 'Allow users to remember multifactor authentication on devices they trust' is Disabled	☐	☐
5.3.1	Ensure that Azure Admin Accounts Are Not Used for Daily Operations	☐	☐
5.3.2	Ensure that Guest Users are Reviewed on a Regular Basis	☐	☐
5.3.3	Ensure That Use of the 'User Access Administrator' Role is Restricted	☐	☐
5.3.4	Ensure that All 'Privileged' Role Assignments are Periodically Reviewed	☐	☐
5.3.5	Ensure Disabled User Accounts do not Have Read, Write, or Owner Permissions	☐	☐
5.3.6	Ensure 'Tenant Creator' Role Assignments are Periodically Reviewed	☐	☐
5.3.7	Ensure All Non-privileged Role Assignments are Periodically Reviewed	☐	☐
5.4	Ensure that No Custom Subscription Administrator Roles Exist	☐	☐
5.5	Ensure that a Custom Role is Assigned Permissions for Administering Resource Locks	☐	☐
5.6	Ensure that 'Subscription leaving Microsoft Entra tenant' and 'Subscription entering Microsoft Entra tenant' is set to 'Permit no one'	☐	☐
5.7	Ensure there are between 2 and 3 Subscription Owners	☐	☐
6.1.1.1	Ensure that a 'Diagnostic Setting' Exists for Subscription Activity Logs	☐	☐
6.1.1.2	Ensure Diagnostic Setting Captures Appropriate Categories	☐	☐
6.1.1.3	Ensure the Storage Account Containing the Container with Activity Logs is Encrypted with Customer-managed Key (CMK)	☐	☐
6.1.1.4	Ensure that Logging for Azure Key Vault is 'Enabled'	☐	☐
6.1.1.5	Ensure that Network Security Group Flow Logs are Captured and Sent to Log Analytics	☐	☐
6.1.1.6	Ensure that Virtual Network Flow Logs are Captured and Sent to Log Analytics	☐	☐

Recommendation		Set Correctly	
		Yes	No
6.1.1.7	Ensure that a Microsoft Entra Diagnostic Setting Exists to Send Microsoft Graph Activity Logs to an Appropriate Destination	☐	☐
6.1.1.8	Ensure that a Microsoft Entra Diagnostic Setting Exists to Send Microsoft Entra Activity Logs to an Appropriate Destination	☐	☐
6.1.1.9	Ensure that Intune Logs are Captured and Sent to Log Analytics	☐	☐
6.1.2.1	Ensure that Activity Log Alert Exists for Create Policy Assignment	☐	☐
6.1.2.2	Ensure that Activity Log Alert exists for Delete Policy Assignment	☐	☐
6.1.2.3	Ensure that Activity Log Alert Exists for Create or Update Network Security Group	☐	☐
6.1.2.4	Ensure that Activity Log Alert Exists for Delete Network Security Group	☐	☐
6.1.2.5	Ensure that Activity Log Alert Exists for Create or Update Security Solution	☐	☐
6.1.2.6	Ensure that Activity Log Alert Exists for Delete Security Solution	☐	☐
6.1.2.7	Ensure that Activity Log Alert Exists for Create or Update SQL Server Firewall Rule	☐	☐
6.1.2.8	Ensure that Activity Log Alert Exists for Delete SQL Server Firewall Rule	☐	☐
6.1.2.9	Ensure that Activity Log Alert Exists for Create or Update Public IP Address rule	☐	☐
6.1.2.10	Ensure that Activity Log Alert Exists for Delete Public IP Address rule	☐	☐
6.1.2.11	Ensure that an Activity Log Alert Exists for Service Health	☐	☐
6.1.3.1	Ensure Application Insights are Configured	☐	☐
6.1.4	Ensure that Azure Monitor Resource Logging is Enabled for All Services that Support it	☐	☐
6.1.5	Ensure Basic, Free, and Consumption SKUs are not used on Production artifacts requiring monitoring and SLA	☐	☐

Recommendation		Set Correctly	
		Yes	No
6.2	Ensure that Resource Locks are set for Mission-Critical Azure Resources	☐	☐
7.1	Ensure that RDP Access from the Internet is Evaluated and Restricted	☐	☐
7.2	Ensure that SSH Access from the Internet is Evaluated and Restricted	☐	☐
7.3	Ensure that UDP Port Access from the Internet is Evaluated and Restricted	☐	☐
7.4	Ensure that HTTP(S) Access from the Internet is Evaluated and Restricted	☐	☐
7.5	Ensure that Network Security Group Flow Log Retention Days is Set to Greater than or equal to 90	☐	☐
7.6	Ensure that Network Watcher is 'Enabled' for Azure Regions That are in Use	☐	☐
7.7	Ensure that Public IP Addresses are Evaluated on a Periodic Basis	☐	☐
7.8	Ensure that Virtual Network Flow Log Retention Days is Set to Greater than or Equal to 90	☐	☐
7.9	Ensure 'Authentication type' is Set to 'Azure Active Directory' only for Azure VPN Gateway Point-to-Site Configuration	☐	☐
7.10	Ensure Azure Web Application Firewall (WAF) is Enabled on Azure Application Gateway	☐	☐
7.11	Ensure Subnets Are Associated with Network Security Groups	☐	☐
7.12	Ensure the SSL Policy's 'Min protocol version' is Set to 'TLSv1_2' or Higher on Azure Application Gateway	☐	☐
7.13	Ensure 'HTTP2' is Set to 'Enabled' on Azure Application Gateway	☐	☐
7.14	Ensure Request Body Inspection is Enabled in Azure Web Application Firewall policy on Azure Application Gateway	☐	☐
7.15	Ensure Bot Protection is Enabled in Azure Web Application Firewall Policy on Azure Application Gateway	☐	☐
7.16	Ensure Azure Network Security Perimeter is Used to Secure Azure Platform-as-a-service Resources	☐	☐

Recommendation		Set Correctly	
		Yes	No
8.1.1.1	Ensure Microsoft Defender CSPM is Set to 'On'	☐	☐
8.1.2.1	Ensure Microsoft Defender for APIs is Set to 'On'	☐	☐
8.1.3.1	Ensure that Defender for Servers is Set to 'On'	☐	☐
8.1.3.2	Ensure that 'Vulnerability assessment for machines' Component Status is set to 'On'	☐	☐
8.1.3.3	Ensure that 'Endpoint protection' Component Status is set to 'On'	☐	☐
8.1.3.4	Ensure that 'Agentless scanning for machines' Component Status is Set to 'On'	☐	☐
8.1.3.5	Ensure that 'File Integrity Monitoring' Component Status is Set to 'On'	☐	☐
8.1.4.1	Ensure That Microsoft Defender for Containers Is Set To 'On'	☐	☐
8.1.5.1	Ensure That Microsoft Defender for Storage Is Set To 'On'	☐	☐
8.1.5.2	Ensure Advanced Threat Protection Alerts for Storage Accounts Are Monitored	☐	☐
8.1.6.1	Ensure That Microsoft Defender for App Services Is Set To 'On'	☐	☐
8.1.7.1	Ensure That Microsoft Defender for Azure Cosmos DB Is Set To 'On'	☐	☐
8.1.7.2	Ensure That Microsoft Defender for Open-Source Relational Databases Is Set To 'On'	☐	☐
8.1.7.3	Ensure That Microsoft Defender for (Managed Instance) Azure SQL Databases Is Set To 'On'	☐	☐
8.1.7.4	Ensure That Microsoft Defender for SQL Servers on Machines Is Set To 'On'	☐	☐
8.1.8.1	Ensure That Microsoft Defender for Key Vault Is Set To 'On'	☐	☐
8.1.9.1	Ensure That Microsoft Defender for Resource Manager Is Set To 'On'	☐	☐
8.1.10	Ensure that Microsoft Defender for Cloud is Configured to Check VM Operating Systems for Updates	☐	☐
8.1.11	Ensure that non-deprecated Microsoft Cloud Security Benchmark policies are not set to 'Disabled'	☐	☐

Recommendation		Set Correctly	
		Yes	No
8.1.12	Ensure That 'All users with the following roles' is Set to 'Owner'	☐	☐
8.1.13	Ensure 'Additional email addresses' is Configured with a Security Contact Email	☐	☐
8.1.14	Ensure that 'Notify about alerts with the following severity (or higher)' is Enabled	☐	☐
8.1.15	Ensure that 'Notify about attack paths with the following risk level (or higher)' is Enabled	☐	☐
8.1.16	Ensure that Microsoft Defender External Attack Surface Monitoring (EASM) is Enabled	☐	☐
8.2.1	Ensure That Microsoft Defender for IoT Hub Is Set To 'On'	☐	☐
8.3.1	Ensure that the Expiration Date is Set for all Keys in Key Vaults using RBAC	☐	☐
8.3.2	Ensure that the Expiration Date is set for All Keys in Key Vaults using access policies (legacy)	☐	☐
8.3.3	Ensure that the Expiration Date is set for All Secrets in Key Vaults using RBAC	☐	☐
8.3.4	Ensure that the Expiration Date is set for All Secrets in Key Vaults using access policies (legacy)	☐	☐
8.3.5	Ensure 'Purge protection' is Set to 'Enabled'	☐	☐
8.3.6	Ensure that Role Based Access Control for Azure Key Vault is Enabled	☐	☐
8.3.7	Ensure Public Network Access is Disabled	☐	☐
8.3.8	Ensure Private Endpoints are Used to Access Azure Key Vault	☐	☐
8.3.9	Ensure Automatic Key Rotation is Enabled within Azure Key Vault	☐	☐
8.3.10	Ensure that Azure Key Vault Managed HSM is Used when Required	☐	☐
8.3.11	Ensure Certificate 'Validity Period (in months)' is Less Than or Equal to '12'	☐	☐
8.4.1	Ensure an Azure Bastion Host Exists	☐	☐
8.5	Ensure Azure DDoS Network Protection is Enabled on Virtual Networks	☐	☐

Recommendation		Set Correctly	
		Yes	No
9.1.1	Ensure Soft Delete for Azure File Shares is Enabled	☐	☐
9.1.2	Ensure 'SMB protocol version' is Set to 'SMB 3.1.1' or Higher for SMB file shares	☐	☐
9.1.3	Ensure 'SMB channel encryption' is Set to 'AES-256-GCM' or Higher for SMB file shares	☐	☐
9.2.1	Ensure That Soft Delete for Blobs on Azure Blob Storage Storage Accounts is Enabled	☐	☐
9.2.2	Ensure that Soft Delete for Containers on Azure Blob Storage Storage Accounts is Enabled	☐	☐
9.2.3	Ensure 'Versioning' is Set to 'Enabled' on Azure Blob Storage Storage Accounts	☐	☐
9.3.1.1	Ensure That 'Enable key rotation reminders' is Enabled for Each Storage Account	☐	☐
9.3.1.2	Ensure That Storage Account Access keys are Periodically Regenerated	☐	☐
9.3.1.3	Ensure 'Allow storage account key access' for Azure Storage Accounts is 'Disabled'	☐	☐
9.3.2.1	Ensure Private Endpoints are Used to Access Storage Accounts	☐	☐
9.3.2.2	Ensure that 'Public Network Access' is 'Disabled' for Storage Accounts	☐	☐
9.3.2.3	Ensure Default Network Access Rule for Storage Accounts is Set to Deny	☐	☐
9.3.3.1	Ensure that 'Default to Microsoft Entra authorization in the Azure portal' is Set to 'Enabled'	☐	☐
9.3.4	Ensure that 'Secure transfer required' is Set to 'Enabled'	☐	☐
9.3.5	Ensure 'Allow trusted Microsoft services to access this resource' is Enabled for Storage Account Access	☐	☐
9.3.6	Ensure the 'Minimum TLS version' for Storage Accounts is Set to 'Version 1.2'	☐	☐
9.3.7	Ensure 'Cross Tenant Replication' is Not Enabled	☐	☐
9.3.8	Ensure that 'Allow Blob Anonymous Access' is Set to 'Disabled'	☐	☐
9.3.11	Ensure Redundancy is Set to 'geo-redundant storage (GRS)' on Critical Azure Storage Accounts	☐	☐

Appendix: CIS Controls v8 Unmapped Recommendations

Recommendation		Set Correctly	
		Yes	No
	No unmapped recommendations to CIS Controls v8	☐	☐

Appendix: Change History

Date	Version	Changes for this version
25-Sep-25	6.0.0	UPDATE - Ensure that 'Number of methods required to reset' is set to '2' - Default value was corrected to indicate '1' is default. (Ticket 24877)
24-Nov-25	6.0.0	UPDATE - Introduction - Multiple Methods [...] - AzPS Module Installation Link Updated (Ticket 26664)
25-Jan-26	6.0.0	UPDATE - Change All Recommendations to Title Case (Ticket 27078)
18-Feb-26	6.0.0	UPDATE - vNext Bump - Definition of "other vulnerable UDP-based services" (Ticket 23821)
20-Feb-26	6.0.0	UPDATE - Ensure that all 'privileged' role assignments are periodically reviewed - vNEXT Bump - Preview - Consider duplicating this concept (not word-for-word) for "Privileged Access Administrator" role (Ticket 23837)
3-Apr-26	6.0.0	DELETE - Conditional Access - Ensure that 'trusted locations' are Defined (Migrated to M365 Foundations) (Ticket 27266)
3-Apr-26	6.0.0	DELETE - Conditional Access - Ensure that an Exclusionary Geographic Conditional Access Policy is Considered (Migrated to M365 Foundations) (Ticket 27267)
3-Apr-26	6.0.0	DELETE - Conditional Access - Ensure a Token Protection Conditional Access Policy is Considered (Migrated to M365 Foundations) (Ticket 27268)
3-Apr-26	6.0.0	DELETE - Identity Services - Ensure that 'Restrict non-admin users from creating tenants' is set to 'Yes' (Migrated to M365 Foundations) (Ticket 27353)
3-Apr-26	6.0.0	DELETE - Identity Services - Ensure that 'Number of methods required to reset' is set to '2' (Migrated to M365 Foundations) (Ticket 27256)
3-Apr-26	6.0.0	DELETE - Identity Services - Ensure that Account 'Lockout threshold' is Less than or Equal to '10' (Migrated to M365 Foundations) (Ticket 27257)
3-Apr-26	6.0.0	DELETE - Identity Services - Ensure that Account 'Lockout duration in seconds' is Greater than or Equal to '60' (Migrated to M365 Foundations) (Ticket 27258)

Date	Version	Changes for this version
3-Apr-26	6.0.0	DELETE - Identity Services - Ensure that a 'Custom banned password list' is set to 'Enforce' (Merged to M365 Foundations) (Ticket 27355)
3-Apr-26	6.0.0	DELETE - Identity Services - Ensure that 'Number of days before users are asked to re-confirm their authentication information' is not set to '0' (Ticket 27259)
3-Apr-26	6.0.0	DELETE - Identity Services - Ensure that 'Notify users on password resets?' is set to 'Yes' (Migrated to M365 Foundations) (Ticket 27260)
3-Apr-26	6.0.0	DELETE - Identity Services - Ensure that 'Notify all admins when other admins reset their password?' is set to 'Yes' (Migrated to M365 Foundations) (Ticket 27261)
3-Apr-26	6.0.0	DELETE - Identity Services - Ensure that 'User consent for applications' is set to 'Do not allow user consent' (Merged to M365 Foundations v6:5.1.5.1) (Ticket 27509)
3-Apr-26	6.0.0	DELETE - Identity Services - Ensure that 'User consent for applications' is set to 'Allow user consent for apps from verified publishers, for selected permissions' (Migrated to M365 Foundations) (Ticket 27510)
3-Apr-26	6.0.0	DELETE - Identity Services - Ensure that 'Restrict user ability to access groups features in My Groups' is set to 'Yes' (Migrated to M365 Foundations v7: #5.1.3.2) (Ticket 27262)
3-Apr-26	6.0.0	DELETE - Identity Services - Ensure that 'Owners can manage group membership requests in My Groups' is set to 'No' (Migrated to M365 Foundations v7: #5.1.3.3) (Ticket 27263)
3-Apr-26	6.0.0	DELETE - Identity Services - Ensure that 'Users can create Microsoft 365 groups in Azure portals, API or PowerShell' is set to 'No' (Migrated to M365 Foundations v7: #5.1.3.4) (Ticket 27264)
5-Apr-26	6.0.0	UPDATE - Management and Governance - Ensure that Resource Locks are Set for Mission-Critical Azure Resources - Powershell script added to evaluate multiple resources at a time (Ticket 27545)
5-Apr-26	6.0.0	ADD - Azure Databricks - Ensure Azure Groups are Reviewed Periodically (Ticket 27333)

Date	Version	Changes for this version
6-Apr-26	6.0.0	DELETE - Identity Services - Ensure that 'Users can register applications' is set to 'No' (Ticket 27550)
6-Apr-26	6.0.0	DELETE - Identity Services - Ensure that 'Guest users access restrictions' is set to 'Guest user access is restricted to properties and memberships of their own directory objects' (M365 Merge) (Ticket 27552)
6-Apr-26	6.0.0	DELETE - Identity Services - Ensure that 'Guest invite restrictions' is set to 'Only users assigned to specific admin roles [...]' or 'No one [...]' (M365 Merge) (Ticket 27554)
6-Apr-26	6.0.0	DELETE - Identity Services - Ensure that 'Restrict access to Microsoft Entra admin center' is set to 'Yes' (M365 Merge) (Ticket 27558)
6-Apr-26	6.0.0	DELETE - Identity Services - Ensure that 'Users can create security groups in Azure portals, API or PowerShell' is set to 'No' (M365 Merge) (Ticket 27560)
6-Apr-26	6.0.0	DELETE - Identity Services - Ensure Fewer than 5 Users have Global Administrator Assignment (M365 Merge) (Ticket 27562)
6-Apr-26	6.0.0	DELETE - Conditional Access - Ensure that an Exclusionary Device Code Flow Policy is Considered (Ticket 27566)
6-Apr-26	6.0.0	DELETE - Conditional Access - Ensure that a Multifactor Authentication Policy Exists for All Users (M365 Merge) (Ticket 27579)
6-Apr-26	6.0.0	DELETE - Conditional Access - Ensure that multifactor authentication is required to access Microsoft Admin Portals (M365 Merge) (Ticket 24425)
6-Apr-26	6.0.0	DELETE - Conditional Access - Ensure that multifactor authentication is required for Windows Azure Service Management API (M365 Merge) (Ticket 24424)
6-Apr-26	6.0.0	DELETE - Ensure that multifactor authentication is required for risky sign-ins (M365 Merge) (Ticket 24423)
6-Apr-26	6.0.0	DELETE - Ensure Passwordless Authentication Methods are Considered (M365 Merge) (Ticket 27581)
8-Apr-26	6.0.0	UPDATE - Identity Services - Ensure that 'security defaults' is Enabled in Microsoft Entra ID - Updated safeguard Mappings (Ticket 27606)

Date	Version	Changes for this version
8-Apr-26	6.0.0	UPDATE - Microsoft Defender - Ensure Microsoft Defender CSPM is Set to 'On' - CIS Safeguard Mappings updated (Ticket 27611)
9-Apr-26	6.0.0	UPDATE - Ensure that SKU Basic/Consumption is not Used on Artifacts that Need to be Monitored (Particularly for Production Workloads) - Improved title, updated description, rationale, impact, corrected procedures, and added refere (Ticket 27612)
14-Apr-26	6.0.0	UDPATE - Key Vault - Ensure that the Expiration Date is Set for all Keys in Key Vaults using RBAC - Added permissions requirements, prose and procedures updated and improved. (Ticket 27648)
14-Apr-26	6.0.0	UDPATE - Key Vault - Ensure that the Expiration Date is set for All Secrets in RBAC Key Vaults - Added permissions requirements, prose and procedures updated and improved. (Ticket 27610)
14-Apr-26	6.0.0	UPDATE - Storage Accounts - Ensure 'Allow trusted Microsoft services to access this resource' is Enabled for Storage Account Access - Title, description, and procedures updated to reflect UI changes. (Ticket 27383)
15-Apr-26	6.0.0	UPDATE - Key Vault - Ensure that the Expiration Date is set for All Secrets in Non-RBAC Key Vaults - Marked legacy, prose updated and improved, procedures updated and improved (Ticket 27609)
17-Apr-26	6.0.0	UPDATE - Microsoft Defender - Ensure that non-deprecated Microsoft Cloud Security Benchmark policies are not set to 'Disabled' - Prose rewritten to allow and describe reasonable use cases for 'disabled' policy (Ticket 26048)
17-Apr-26	6.0.0	UPDATE - Networking - Ensure the SSL Policy's 'Min protocol version' is Set to 'TLSv1_2' or Higher on Azure Application Gateway - Prose discusses deprecation of TLSv1_0 and TLSv1_1, and PQC capability of TLS v1.3 (Ticket 26843)
18-Apr-26	6.0.0	UPDATE - Network - Ensure 'Authentication type' is Set to 'Azure Active Directory' only for Azure VPN Gateway Point-to-Site - Added CLI syntax and updated status to reflect automation potential (Ticket 27480)

Date	Version	Changes for this version
18-Apr-26	6.0.0	DELETE - Ensure that logging for Azure AppService 'HTTP logs' is enabled - vNext Bump - Moved to CIS Microsoft Azure Compute Services Benchmark (Ticket 25481)
19-Apr-26	6.0.0	UPDATE - Key Vault - Ensure that the Expiration Date is set for All Keys in Key Vaults using access policies (legacy) - Updated title to reflect legacy status, added permissions details, prose improved (Ticket 27649)
19-Apr-26	6.0.0	DELETE - Ensure that 'multifactor authentication' is 'enabled' for all users (Ticket 24420)
28-Apr-25	5.0.0	UPDATE - Multiple Methods of Audit and Remediation - Remove "DEPRECATION WARNING" for MSONline and AzureAD cmdlet modules (Ticket 24419)
9-Jul-25	5.0.0	UPDATE - Ensure that diagnostic log delivery is configured for Azure Databricks - Add Azure Policy (Ticket 25371)
10-Jul-25	5.0.0	UPDATE - Ensure that SSH access from the Internet is evaluated and restricted - Add Resource Graph query to Audit Procedure (Ticket 23818)
10-Jul-25	5.0.0	UPDATE - Ensure that SSH access from the Internet is evaluated and restricted - Add Azure Portal and Azure CLI steps to Remediation Procedure (Ticket 22434)
10-Jul-25	5.0.0	UPDATE - Ensure that UDP access from the Internet is evaluated and restricted - Add Resource Graph query to Audit Procedure (Ticket 23820)
10-Jul-25	5.0.0	UPDATE - Ensure that UDP access from the Internet is evaluated and restricted - Add Azure Portal and Azure CLI steps to Remediation Procedure (Ticket 22435)
11-Jul-25	5.0.0	UPDATE - Ensure that HTTP(S) access from the Internet is evaluated and restricted - Add Resource Graph query to Audit Procedure (Ticket 23822)
11-Jul-25	5.0.0	UPDATE - Ensure that Network Security Group Flow Log retention period is 'greater than 90 days' - Remove DeployIfNotExists Azure Policy (Ticket 24487)
11-Jul-25	5.0.0	UPDATE - Ensure that Activity Log Alert exists for Create or Update Security Solution - Change Azure Policy (Ticket 24478)

Date	Version	Changes for this version
11-Jul-25	5.0.0	UPDATE - Ensure that Activity Log Alert exists for Delete Security Solution - Change Azure Policy (Ticket 24479)
11-Jul-25	5.0.0	UPDATE - Ensure that Activity Log Alert exists for Create or Update Public IP Address rule - Remove DeployIfNotExists Azure Policy (Ticket 24480)
11-Jul-25	5.0.0	UPDATE - Ensure that Activity Log Alert exists for Delete Public IP Address rule - Remove DeployIfNotExists Azure Policy (Ticket 24481)
11-Jul-25	5.0.0	UPDATE - Ensure the storage account containing the container with activity logs is encrypted with Customer Managed Key (CMK) - Align wi (Ticket 24591)
14-Jul-25	5.0.0	UPDATE - Ensure that a 'Diagnostic Setting' exists for Subscription Activity Logs - Replace incorrect Audit from PowerShell command (Ticket 25316)
14-Jul-25	5.0.0	UPDATE - Ensure that 'Allow Blob Anonymous Access' is set to 'Disabled' - Remove reference to classic deployment model retirement (Ticket 24436)
14-Jul-25	5.0.0	UPDATE - Ensure that 'Public Network Access' is 'Disabled' for storage accounts - Align with Common Reference Recommendation (Ticket 24594)
14-Jul-25	5.0.0	UPDATE - Ensure that 'Public Network Access' is 'Disabled' for storage accounts - Remove reference to classic deployment model retirement (Ticket 24435)
14-Jul-25	5.0.0	UPDATE - Ensure that 'Public Network Access' is 'Disabled' for storage accounts - Update Impact Statement to clarify additional network configuration (Ticket 25245)
14-Jul-25	5.0.0	UPDATE - Ensure an Azure Bastion Host Exists - Remove reference to Preview CLI module (Ticket 24434)
14-Jul-25	5.0.0	UPDATE - Ensure that Private Endpoints are Used for Azure Key Vault - Align with Common Reference Recommendation (Ticket 24596)
14-Jul-25	5.0.0	UPDATE - Ensure Public Network Access is disabled when using Private Endpoint - Align with Common Reference Recommendation (Ticket 24593)
15-Jul-25	5.0.0	UPDATE - Ensure critical data in Azure Databricks is encrypted with customer-managed keys (CMK) - Align with Common Reference Recommendation (Ticket 24590)

Date	Version	Changes for this version
15-Jul-25	5.0.0	UPDATE - Ensure public network access is Disabled - Update Impact Statement to clarify additional network configuration (Ticket 25244)
24-Jul-25	5.0.0	ADD - Azure Databricks - Ensure 'No Public IP' is set to 'Enabled' (Ticket 25436)
24-Jul-25	5.0.0	ADD - Azure Databricks - Ensure 'Allow Public Network Access' is set to 'Disabled' (Ticket 25445)
24-Jul-25	5.0.0	ADD - Azure Databricks - Ensure private endpoints are used to access Azure Databricks workspaces (Ticket 25451)
24-Jul-25	5.0.0	UPDATE - Ensure that network security groups are configured for Databricks subnets - Update Assessment Status from Manual to Automated (Ticket 24540)
24-Jul-25	5.0.0	UPDATE - Ensure that diagnostic log delivery is configured for Azure Databricks - Update Assessment Status from Manual to Automated (Ticket 24541)
24-Jul-25	5.0.0	UPDATE - Ensure that network security groups are configured for Databricks subnets - Update description and rationale impact to discuss Implicit vs Explicit Deny (Ticket 25520)
25-Jul-25	5.0.0	UPDATE - Encryption Key Management Overview - Specify role of CEK and DEK for Customer Managed Keys (Ticket 24227)
4-Aug-25	5.0.0	ADD - Periodic Identity Reviews - Ensure 'Tenant Creator' role assignments are periodically reviewed (Ticket 23838)
4-Aug-25	5.0.0	ADD - Periodic Identity Reviews - Ensure disabled user accounts do not have read, write, or owner permissions (Ticket 25465)
4-Aug-25	5.0.0	ADD - Identity Services - Ensure passwordless authentication methods are considered (Ticket 23836)
11-Aug-25	5.0.0	ADD - Periodic Identity Reviews - Ensure all non-privileged role assignments are periodically reviewed (Ticket 23842)
11-Aug-25	5.0.0	UPDATE - Ensure that Network Security Group Flow logs are captured and sent to Log Analytics - Update Description and Remediation Procedure to reflect NSG flow log status (Ticket 24427)

Date	Version	Changes for this version
14-Aug-25	5.0.0	UPDATE - Ensure that a 'Diagnostic Setting' exists for Subscription Activity Logs - Update Assessment Status from Manual to Automated (Ticket 24549)
14-Aug-25	5.0.0	ADD - Networking - Ensure 'Authentication type' is set to 'Azure Active Directory' only for Azure VPN Gateway point-to-site configuration (Ticket 25466)
14-Aug-25	5.0.0	ADD - Networking - Ensure Azure Web Application Firewall (WAF) is enabled on Azure Application Gateway (Ticket 25479)
14-Aug-25	5.0.0	ADD - Networking - Ensure subnets are associated with network security groups (Ticket 25482)
14-Aug-25	5.0.0	ADD - Networking - Ensure the SSL policy's 'Min protocol version' is set to 'TLSv1_2' or higher on Azure Application Gateway (Ticket 25706)
19-Aug-25	5.0.0	UPDATE - Ensure that Public IP addresses are Evaluated on a Periodic Basis - Update Description to clarify Assessment Status (Ticket 24553)
20-Aug-25	5.0.0	UPDATE- Ensure Azure Resource Manager Delete locks are applied to Azure Storage Accounts - Update Description to clarify Assessment Status (Ticket 24559)
20-Aug-25	5.0.0	UPDATE - Ensure Azure Resource Manager ReadOnly locks are considered for Azure Storage Accounts - Update Description to clarify Assessment Status (Ticket 24560)
20-Aug-25	5.0.0	ADD - Identity - Ensure there are between 2 and 3 subscription owners (Ticket 25453)
21-Aug-25	5.0.0	ADD - Networking - Ensure 'HTTP2' is set to 'Enabled' on Azure Application Gateway (Ticket 25729)
21-Aug-25	5.0.0	ADD - Networking - Ensure request body inspection is enabled in Azure Web Application Firewall policy on Azure Application Gateway (Ticket 25755)
21-Aug-25	5.0.0	ADD - Networking - Ensure bot protection is enabled in Azure Web Application Firewall policy on Azure Application Gateway (Ticket 25766)
28-Aug-25	5.0.0	ADD - Security Services - Ensure Azure DDoS Network Protection is enabled on virtual networks (Ticket 22724)
28-Aug-25	5.0.0	ADD - Microsoft Cloud Security Posture Management (CSPM) - Ensure Microsoft Defender CSPM is set to 'On' (Ticket 25497)

Date	Version	Changes for this version
28-Aug-25	5.0.0	ADD - Defender Plan: APIs - Ensure Microsoft Defender for APIs is set to 'On' (Ticket 25498)
28-Aug-25	5.0.0	UPDATE - Ensure that 'Vulnerability assessment for machines' component status is set to 'On' - Add Azure Policy (Ticket 24488)
28-Aug-25	5.0.0	UPDATE - Ensure that 'Endpoint protection' component status is set to 'On' - Update Assessment Status from Manual to Automated (Ticket 24554)
28-Aug-25	5.0.0	UPDATE - Ensure That Microsoft Defender for (Managed Instance) Azure SQL Databases Is Set To 'On' - Add Azure Policy (Ticket 25491)
28-Aug-25	5.0.0	UPDATE - Ensure that 'Notify about alerts with the following severity (or higher)' is enabled - Add Azure Policy (Ticket 25384)
28-Aug-25	5.0.0	DELETE - [LEGACY] Ensure That Microsoft Defender for DNS Is Set To 'On' - Defender for DNS plan is deprecated (Ticket 24959)
28-Aug-25	5.0.0	ADD - Key Vault - Ensure certificate 'Validity Period (in months)' is less than or equal to '12' (Ticket 25490)
28-Aug-25	5.0.0	UPDATE - Ensure the Key Vault is Recoverable - Update recommendation to focus on purge protection (Ticket 24433)
28-Aug-25	5.0.0	UPDATE - Key Vault - Ensure Public Network Access is Disabled - Updated prose to reflect need for private endpoints and align with similar recommendations disabling public network access (Ticket 24974)
28-Aug-25	5.0.0	UPDATE - Ensure that Azure Key Vault Managed HSM is used when required - Update Description to clarify Assessment Status (Ticket 24555)
28-Aug-25	5.0.0	UPDATE - Ensure that Azure Key Vault Managed HSM is used when required - Add Azure Policy (Ticket 24508)
4-Sep-25	5.0.0	UPDATE - Ensure that soft delete for containers on Azure Blob Storage storage accounts is Enabled - reword for clarity and move to Blob Storage section (Ticket 24922)
4-Sep-25	5.0.0	UPDATE - Ensure that 'Enable key rotation reminders' is enabled for each Storage Account - Update Assessment Status from Manual to Automated (Ticket 24556)

Date	Version	Changes for this version
4-Sep-25	5.0.0	UPDATE - Ensure that Storage Account access keys are periodically regenerated - Update Assessment Status from Manual to Automated (Ticket 24557)
15-Sep-25	5.0.0	UPDATE - Ensure that Azure Monitor Resource Logging is Enabled for All Services that Support it - Add Azure Policy for Application Gateway (Ticket 25754)
19-Sep-25	5.0.0	UPDATE - Ensure that 'security defaults' is enabled in Microsoft Entra ID - ADD Azure CLI, Powershell (Ticket 25546)
19-Sep-25	5.0.0	UPDATE - Ensure that 'multifactor authentication' is 'enabled' for all users - Update Assessment Status from Manual to Automated (Ticket 24542)
19-Sep-25	5.0.0	ADD - Conditional Access - Ensure a Token Protection Conditional Access policy is considered (Ticket 24089)
19-Sep-25	5.0.0	UPDATE - Ensure that guest users are reviewed on a regular basis - Update Description to clarify Assessment Status (Ticket 24546)
24-Sep-25	5.0.0	UPDATE - Ensure that Network Security Group Flow Log retention period is 'greater than 90 days' - Align wording in Title, Audit Procedure, Remediation Procedure (Ticket 25029)
24-Sep-25	5.0.0	UPDATE - Ensure that UDP access from the Internet is evaluated and restricted - Add Azure Policy (Ticket 24485)
24-Sep-25	5.0.0	UPDATE - Ensure that HTTP(S) access from the Internet is evaluated and restricted - Add Azure Policy (Ticket 24486)
25-Sep-25	5.0.0	UPDATE - Ensure that 'trusted locations' are defined - Added reasoning for 'manual' automation status. (Ticket 24543)
25-Sep-25	5.0.0	UPDATE - Ensure that 'Guest invite restrictions' is set to 'Only users assigned to specific admin roles [...]' or 'No one [...]' - Updated title and content to indicate that 'No One [...]' is an acceptable setting (Ticket 24943)
26-Sep-25	5.0.0	UPDATE - Ensure fewer than 5 users have global administrator assignment - Rationale improved to advise on the use of stronger authentication methods (Ticket 25521)

Date	Version	Changes for this version
26-Sep-25	5.0.0	UPDATE - Ensure Default Network Access Rule for Storage Accounts is Set to Deny - Added a note to explain that this recommendation is applicable only if Public Network Access is not "Disabled" (Ticket 22278)
26-Sep-25	5.0.0	UPDATE - Ensure 'Allow Azure services on the trusted services list to access this storage account' is Enabled for Storage Account Access - Added note to rationale about reasonable exception. (Ticket 22615)
29-Sep-25	5.0.0	UPDATE - Ensure that diagnostic log delivery is configured for Azure Databricks - Diagnostic Settings for Databricks updated to use "Accounts" and "Filesystems" categories (Ticket 24955)
29-Sep-25	5.0.0	ADD - Security - Ensure Advanced Threat Protection Alerts for Storage Accounts Are Monitored (Ticket 25980)
29-Sep-25	5.0.0	UPDATE - Ensure that Activity Log Alert exists for Create Policy Assignment - Removed 'level' filter from remediation procedures. (Ticket 23613)
30-Sep-25	5.0.0	ADD - Automating using Azure Policy - Intro section added for background and resources aiding the use of Azure Policy (Ticket 22910)
23-Mar-25	4.0.0	ADD - Ensure that a Microsoft Entra diagnostic setting exists to send Microsoft Entra activity logs to an appropriate destination (Ticket 24299)
14-Mar-25	4.0.0	ADD - Ensure that a Microsoft Entra diagnostic setting exists to send Microsoft Graph activity logs to an appropriate destination (Ticket 20484)
12-Mar-25	4.0.0	ADD - Ensure that Azure admin accounts are not used for daily operations (Ticket 22179)
23-Mar-25	4.0.0	ADD - Ensure that Azure Databricks is deployed in a customer-managed virtual network (VNet) (Ticket 24302)
5-Mar-25	4.0.0	ADD - Ensure that Azure Key Vault Managed HSM is used when required (Ticket 17196)
23-Mar-25	4.0.0	ADD - Ensure that data at rest and in transit is encrypted in Azure Databricks using customer managed keys (CMK) (Ticket 24312)
23-Mar-25	4.0.0	ADD - Ensure that 'Default to Microsoft Entra authorization in the Azure portal' is set to 'Enabled' (Ticket 20792)

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23-Mar-25	4.0.0	ADD - Ensure that diagnostic log delivery is configured for Azure Databricks (Ticket 24311)
5-Mar-25	4.0.0	ADD - Ensure that encryption at host is enabled (Ticket 20793)
23-Mar-25	4.0.0	ADD - Ensure that Intune logs are captured and sent to Log Analytics (Ticket 18602)
23-Mar-25	4.0.0	ADD - Ensure that network security groups are configured for Databricks subnets (Ticket 24303)
14-Mar-25	4.0.0	ADD - Ensure that 'Notify about attack paths with the following risk level (or higher)' is enabled (Ticket 23796)
23-Mar-25	4.0.0	ADD - Ensure that Public Network Access when using Private Endpoint is disabled (Ticket 24314)
23-Mar-25	4.0.0	ADD - Ensure that traffic is encrypted between cluster worker nodes (Ticket 24304)
23-Mar-25	4.0.0	ADD - Ensure that Unity Catalog is configured for Azure Databricks (Ticket 24309)
23-Mar-25	4.0.0	ADD - Ensure that usage is restricted and expiry is enforced for Databricks personal access tokens (Ticket 24310)
5-Mar-25	4.0.0	ADD - Ensure that use of the 'User Access Administrator' role is restricted (Ticket 19007)
23-Mar-25	4.0.0	ADD - Ensure that users and groups are synced from Microsoft Entra ID to Azure Databricks (Ticket 24308)
23-Mar-25	4.0.0	ADD - Ensure that virtual network flow log retention days is set to greater than or equal to 90 (Ticket 24313)
14-Mar-25	4.0.0	ADD - Ensure that virtual network flow logs are captured and sent to Log Analytics (Ticket 22595)
12-Feb-25	4.0.0	DELETE - Ensure that a multifactor authentication policy exists for administrative groups (Ticket 22337)
20-Feb-25	4.0.0	DELETE - Ensure that 'Agentless container vulnerability assessment' component status is 'On' (Ticket 23436)
20-Feb-25	4.0.0	DELETE - Ensure that 'Agentless discovery for Kubernetes' component status 'On' (Ticket 23435)
20-Feb-25	4.0.0	DELETE - Ensure that Auto provisioning of 'Log Analytics agent for Azure VMs' is Set to 'On' (Ticket 22569)

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20-Feb-25	4.0.0	DELETE - Ensure that Microsoft Defender for Cloud Apps integration with Microsoft Defender for Cloud is Selected (Ticket 22726)
10-Feb-25	4.0.0	DELETE - Ensure that 'multifactor authentication' is 'enabled' for all non-privileged users - MFA for privileged accounts is now default, MFA recommendations have merged to 'all users' (Ticket 23810)
23-Mar-25	4.0.0	MOVED - ALL (28 of 28) Database Recommendations to CIS Microsoft Azure Database Services Benchmark (Ticket 24317)
23-Mar-25	4.0.0	MOVED - MOST (29 of 30) Compute Recommendations moved to CIS Microsoft Azure Compute Services Benchmark (Ticket 24316)
23-Mar-25	4.0.0	MOVED - MOST (35 of 56) Storage Recommendations moved to CIS Microsoft Azure Storage Services Benchmark (Ticket 24318)
10-Mar-25	4.0.0	REMOVE - Ensure that 'Enable Infrastructure Encryption' for Each Storage Account in Azure Storage is Set to 'enabled' - Move to Storage Services Benchmark (Ticket 22275)
19-Mar-25	4.0.0	UPDATE - Ensure that Microsoft Defender for Cloud is configured to check VM operating systems for updates - Added policy that checks status of Update Center (Ticket 22062)
4-Feb-25	4.0.0	UPDATE - All Identity Section Recommendations - Recommendation Names Updated for Consistency (Ticket 23344)
6-Mar-25	4.0.0	UPDATE - Ensure an Azure Bastion Host Exists - Update MITRE Mappings (Ticket 22525)
7-Mar-25	4.0.0	UPDATE - Ensure Application Insights are Configured - Update MITRE Mappings (Ticket 22524)
20-Feb-25	4.0.0	UPDATE - Ensure automatic key rotation is enabled within Azure Key Vault - Added rationale to indicate origin of 'two years' maximum key lifetime duration (Ticket 22721)
11-Mar-25	4.0.0	UPDATE - Ensure Azure Key Vaults are Used to Store Secrets - Update Audit from Azure CLI (Ticket 23087)
11-Mar-25	4.0.0	UPDATE - Ensure only MFA enabled identities can access privileged Virtual Machine - Add instructions for Conditional Access (Ticket 22753)

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27-Feb-25	4.0.0	UPDATE - Ensure Private Endpoints are used to access Cosmos DB accounts - Add Audit from CLI & PowerShell, Remediate from CLI, Align with Common Reference Recommendation (Ticket 22426)
6-Feb-25	4.0.0	UPDATE - Ensure that a custom role is assigned permissions for administering resource locks - Updated Procedures & Prose (Ticket 22451)
20-Feb-25	4.0.0	UPDATE - Ensure that App Service apps are configured to use managed identities - Nomenclature and language updated (Ticket 23768)
5-Mar-25	4.0.0	UPDATE - Ensure that Defender for Servers is set to 'On' - Update description to specify enabled components (Ticket 22728)
10-Mar-25	4.0.0	UPDATE - Ensure that 'Enable key rotation reminders' is enabled for each Storage Account - Added supportive Azure Policy (Ticket 24135)
6-Mar-25	4.0.0	UPDATE - Ensure that Endpoint Protection for all Virtual Machines is installed - Replace deprecated Azure Policy (Ticket 23642)
13-Mar-25	4.0.0	UPDATE - Ensure that logging for Azure AppService 'HTTP logs' is enabled - Add Policy (Ticket 18601)
20-Feb-25	4.0.0	UPDATE - Ensure that Microsoft Defender External Attack Surface Monitoring (EASM) is enabled - Procedures improved (Ticket 22422)
20-Feb-25	4.0.0	UPDATE - Ensure That Microsoft Defender for Containers Is Set To 'On' - Updated to reflect multiple extensions related to the primary product and simplified to one recommendation instead of three (Ticket 21755)
5-Mar-25	4.0.0	UPDATE - Ensure That Microsoft Defender for Storage Is Set To 'On' - Replaced Deprecated Azure Policy (Ticket 23567)
10-Feb-25	4.0.0	UPDATE - Ensure that 'multifactor authentication' is 'enabled' for all users - MFA for privileged accounts is now default, MFA recommendations have merged to 'all users' (Ticket 23809)
11-Mar-25	4.0.0	UPDATE - Ensure that Network Security Group Flow Log retention period is 'greater than 90 days' - Retirement Notice added: NSG Flow Logs to be Deprecated in 2027 (Ticket 23636)

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5-Mar-25	4.0.0	UPDATE - Ensure that 'Notify about alerts with the following severity (or higher)' is enabled - Remove severity requirement, add Impact Statement (Ticket 23639)
13-Feb-25	4.0.0	UPDATE - Ensure that 'Public Network Access' is set to 'Selected Networks' - CLI & Powershell updated, Title Updated (Ticket 22425)
14-Mar-25	4.0.0	UPDATE - Ensure that RDP access from the Internet is evaluated and restricted - Added a resource graph query (Ticket 23817)
14-Mar-25	4.0.0	UPDATE - Ensure that RDP access from the Internet is evaluated and restricted - Refine Audit and Remediation Procedures (Ticket 19629)
6-Mar-25	4.0.0	UPDATE - Ensure that SKU Basic/Consumption is not used on artifacts that need to be monitored (Particularly for Production Workloads) - Add detail to Remediation Procedure (Ticket 22432)
20-Feb-25	4.0.0	UPDATE - Ensure the Key Vault is Recoverable - Removed obsolete policy for soft delete which is checked with existing policy object (Ticket 23589)
27-Feb-25	4.0.0	UPDATE - Ensure the 'Minimum Inbound TLS Version' for apps is set to '1.2' or higher - Prose changed to reflect "1.2 or higher" and policy added to disable TLS 1.0 and 1.1 (Ticket 23033)
27-Feb-25	4.0.0	UPDATE - Ensure Trusted Launch is enabled on Virtual Machines - Add Audit from CLI & PowerShell, Remediate from CLI & PowerShell (Ticket 23089)
27-Feb-25	4.0.0	UPDATE - Ensure Trusted Launch is enabled on Virtual Machines - Update Azure Policy (Ticket 23633)
5-Feb-25	4.0.0	UPDATE - Included Management Group in Audit and Remediation Steps for "Ensure that Microsoft Cloud Security Benchmark policies are not set to 'Disabled'" (Ticket 22727)
17-Mar-25	4.0.0	UPDATE - Multiple Methods of Audit and Remediation - Updated Authenticating with PowerShell to use method other than device code (Ticket 24226)
2-Sep-24	4.0.0	UPDATE - Update MySQL Database recommendations to apply to Azure Database for MySQL - Flexible Servers - Updated recommendations to apply to flexible servers (Ticket 22455)

